

LINE: Queueing Analysis Algorithms

Example Manual – Java Edition

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Contents

1	About this manual	1
2	Basic models	2
2.1	Open queueing networks	2
2.1.1	mam_transient_mapmap1	2
2.1.2	oqn_basic	3
2.1.3	oqn_bmmapt	5
2.1.4	oqn_cs_routing	5
2.1.5	oqn_fourqueues	7
2.1.6	oqn_mapt	9
2.1.7	oqn_mapt_service	9
2.1.8	oqn_mmapt	10
2.1.9	oqn_multichain_cs	10
2.1.10	oqn_nhpp	11
2.1.11	oqn_online	12
2.1.12	oqn_trace_driven	13
2.1.13	oqn_vsinks	13
2.2	Closed queueing networks	15
2.2.1	cqn_bas_blocking	15
2.2.2	cqn_bcmp_theorem	15
2.2.3	cqn_lcfs_multiclass	17
2.2.4	cqn_mmpp2_service	17
2.2.5	cqn_multichain_cs	18
2.2.6	cqn_multiserver	18
2.2.7	cqn_multiserver_nc	20
2.2.8	cqn_online	20
2.2.9	cqn_repairmen	21
2.2.10	cqn_repairmen_multi	22
2.2.11	cqn_scheduling_dps	24
2.2.12	cqn_threeclass_hyperl	25
2.2.13	cqn_twoclass_erl	26
2.2.14	cqn_twoclass_hyperl	27
2.2.15	cqn_twoqueues_multi	28
2.3	Mixed networks	29
2.3.1	mqn_basic	29
2.3.2	mqn_multichain_cs	30
2.3.3	mqn_multiserver_fcfs	32
2.3.4	mqn_multiserver_ps	33
2.3.5	mqn_singleserver_fcfs	35

2.3.6	mqn_singleserver_ps	37
2.4	Class switching	38
2.4.1	cs_implicit	39
2.4.2	cs_multi_diamond	39
2.4.3	cs_single_diamond	40
2.4.4	cs_transient_class	41
2.5	Priority models	43
2.5.1	prio_hol_closed	43
2.5.2	prio_hol_open	45
2.5.3	prio_identical	46
2.5.4	prio_psprio	47
2.6	Fork-join networks	48
2.6.1	fj_asymm	49
2.6.2	fj_basic_closed	50
2.6.3	fj_basic_nesting	51
2.6.4	fj_basic_open	52
2.6.5	fj_complex_serial	54
2.6.6	fj_cs_multi_visits	55
2.6.7	fj_cs_postfork	56
2.6.8	fj_cs_prefork	57
2.6.9	fj_deep_nesting	58
2.6.10	fj_delays	60
2.6.11	fj_mixed_openclosed	61
2.6.12	fj_nojoin	63
2.6.13	fj_route_overlap	64
2.6.14	fj_serialfjs_closed	66
2.6.15	fj_serialfjs_open	67
2.6.16	fj_threebranches	68
2.6.17	fj_tiny_closed	70
2.6.18	fj_twoclasses_forked	70
2.6.19	fj_variable_fanout	72
2.7	Layered queueing networks	73
2.7.1	lqn_basic	73
2.7.2	lqn_bpmn	74
2.7.3	lqn_bpmn_trace	76
2.7.4	lqn_flatph	78
2.7.5	lqn_fork_open_arrival	79
2.7.6	lqn_init	80
2.7.7	lqn_jsq	81
2.7.8	lqn_moment3	82
2.7.9	lqn_multi_solvers	82
2.7.10	lqn_ofbiz	84
2.7.11	lqn_open_arrival	84
2.7.12	lqn_paramident	86
2.7.13	lqn_rrobin	86
2.7.14	lqn_serial	87
2.7.15	lqn_server_pools	88
2.7.16	lqn_setup	89
2.7.17	lqn_sockshop	90
2.7.18	lqn_srvnph	92

2.7.19	lqn_transient	93
2.7.20	lqn_twotasks	94
2.7.21	lqn_workflows	95
2.8	Cache models	97
2.8.1	cache_compare_replc	97
2.8.2	cache_itemsize_costcap	98
2.8.3	cache_mmap_rr_env	99
2.8.4	cache_replc_climb	100
2.8.5	cache_replc_fifo	101
2.8.6	cache_replc_hlru	102
2.8.7	cache_replc_lru	103
2.8.8	cache_replc_qlru	104
2.8.9	cache_replc_routing	105
2.8.10	cache_replc_rr	107
2.8.11	cache_rmf_transient	108
2.8.12	cachenet_tandem	109
2.8.13	retrieval_chain	110
2.8.14	retrieval_default	111
2.8.15	retrieval_ps	113
2.8.16	retrieval_routing	114
2.8.17	retrieval_simple	116
2.9	Cluster models	117
2.9.1	cl_basic	117
2.9.2	cl_closed	118
2.9.3	cl_compare	119
2.9.4	cl_multiclass	120
2.9.5	cl_scheduling	120
2.9.6	cl_stations	121
2.9.7	cl_sweep	121
2.9.8	dispatch_closed	122
2.9.9	dispatch_mixed	122
2.9.10	dispatch_open	123
2.10	Stochastic Petri nets	124
2.10.1	spn_basic_closed	124
2.10.2	spn_basic_open	125
2.10.3	spn_closed_fourplaces	126
2.10.4	spn_closed_twoplaces	127
2.10.5	spn_colored_gspn	128
2.10.6	spn_fluid_dae	128
2.10.7	spn_fourmodes	129
2.10.8	spn_inhibiting	130
2.10.9	spn_lpbounds	131
2.10.10	spn_open_sevenplaces	132
2.10.11	spn_pareto_service	133
2.10.12	spn_productform_nc	134
2.10.13	spn_queueing_place	134
2.10.14	spn_twomodes	135
2.10.15	test_spn_nrm_open	136
2.11	Workflow models	137
2.11.1	wf_aph	137

2.11.2	wf_branch	137
2.11.3	wf_complex	137
2.11.4	wf_erlang	137
2.11.5	wf_loop	138
2.11.6	wf_parallel	138
2.11.7	wf_serial	138
3	Advanced models	139
3.1	Load-dependent stations	139
3.1.1	fes_aggregation	139
3.1.2	fes_single_class	140
3.1.3	ld_class_dependence	141
3.1.4	ld_fes_multiclass	141
3.1.5	ld_fes_singleclass	143
3.1.6	ld_global_dependence	144
3.1.7	ld_joint_dependence	145
3.1.8	ld_multiserver_fcfs	145
3.1.9	ld_multiserver_ps	147
3.1.10	ld_multiserver_ps_twoclasses	148
3.1.11	ld_whittle_bandwidth	149
3.2	Finite capacity regions	150
3.2.1	fcr_constraints	150
3.2.2	fcr_lincon	151
3.2.3	fcr_lossn	153
3.2.4	fcr_mmlkdrop	154
3.2.5	fcr_mmlwaitq	155
3.2.6	fcr_oqndrop	156
3.2.7	fcr_oqnwaitq	157
3.3	State probabilities	158
3.3.1	statepr_aggr	158
3.3.2	statepr_aggr_large	159
3.3.3	statepr_allprobs_fcfs	160
3.3.4	statepr_allprobs_ps	161
3.3.5	statepr_sys_aggr	162
3.3.6	statepr_sys_aggr_large	163
3.3.7	statepr_sys_marg	164
3.4	Initial state and warm start	165
3.4.1	init_state_fcfs_exp	165
3.4.2	init_state_fcfs_nonexp	166
3.4.3	init_state_ps	167
3.4.4	ldes_warmstart	168
3.5	State-dependent routing	169
3.5.1	sdroute_closed	170
3.5.2	sdroute_jsq	171
3.5.3	sdroute_krzesinski	171
3.5.4	sdroute_multibranch	173
3.5.5	sdroute_open	173
3.5.6	sdroute_twoclasses_closed	174
3.6	Cyclic polling	175
3.6.1	polling_exhaustive_det	175

3.6.2	polling_exhaustive_exp	176
3.6.3	polling_gated	177
3.6.4	polling_klimited	178
3.7	Switchover times	179
3.7.1	switchover_basic	179
3.8	Response time distribution	180
3.8.1	cdf_respt_closed	180
3.8.2	cdf_respt_closed_threeclasses	181
3.8.3	cdf_respt_distrib	182
3.8.4	cdf_respt_open_twoclasses	183
3.8.5	cdf_respt_populations	185
3.9	Busy periods	186
3.9.1	busyp_subnetwork	186
3.10	Stochastic network calculus	187
3.10.1	snc_delay_quantile	188
3.10.2	snc_tandem_multiclass	189
3.11	Random environments	190
3.11.1	example_mapqn2renv	190
3.11.2	renv_basic	191
3.11.3	renv_container_terminal	192
3.11.4	renv_fourstages_repairmen	193
3.11.5	renv_genqn	194
3.11.6	renv_lqn_twostages	195
3.11.7	renv_map_fallback	195
3.11.8	renv_node_breakdown	196
3.11.9	renv_rotterdam_blending	197
3.11.10	renv_threestages_repairmen	197
3.11.11	renv_twostages_repairmen	198
3.12	Layered cache-queueing models	200
3.12.1	lcq_async_cache	200
3.12.2	lcq_singlehost	201
3.12.3	lcq_threehosts	202
3.13	Pass and swap	203
3.13.1	pas_closed_tandem_fig6	203
3.13.2	pas_compatibility_5class	204
3.13.3	pas_cyclic_ctmc_vs_bruteforce	204
3.13.4	pas_cyclic_normconst_bruteforce	205
3.13.5	pas_mmk	206
3.13.6	pas_nc_sampling	207
3.13.7	pas_saturation	207
3.13.8	pas_selfloop	208
3.14	Reward models	209
3.14.1	rewardModel_aggregation	209
3.14.2	rewardModel_mmk	210
3.14.3	rewardModel_multiclass	211
3.14.4	rewardModel_templates	212
3.15	Agent models	213
3.15.1	ag_closed_network	213
3.15.2	ag_gnetwork	214
3.15.3	ag_jackson_network	215

3.15.4	ag_multiclass_closed	216
3.15.5	ag_tandem_open	218
3.15.6	ag_tandem_phasetype	219
3.16	Large-scale models	220
3.16.1	largescale_tbi	220
3.17	Custom solvers	221
3.17.1	solver_custom	221
3.17.2	solver_custom_analyzer	221
4	Model gallery	222
4.1	Reading the names	222
4.2	The catalogue	223
4.2.1	gallery_aphm1	223
4.2.2	gallery_cache_lru	224
4.2.3	gallery_cache_routing	225
4.2.4	gallery_coxm1	226
4.2.5	gallery_cqn	226
4.2.6	gallery_cqn_multiclass	227
4.2.7	gallery_detm1	227
4.2.8	gallery_dm1	228
4.2.9	gallery_erldk	228
4.2.10	gallery_erlerl1	229
4.2.11	gallery_erlerl1_reentrant	229
4.2.12	gallery_erlm1	229
4.2.13	gallery_erlm1_ps	231
4.2.14	gallery_erlm1_reentrant	231
4.2.15	gallery_erlm1ps	232
4.2.16	gallery_fcr	233
4.2.17	gallery_fj_closed	233
4.2.18	gallery_fj_open	234
4.2.19	gallery_fj_quorum	235
4.2.20	gallery_gamm1	236
4.2.21	gallery_hyperl1_feedback	237
4.2.22	gallery_hyperl1_reentrant	237
4.2.23	gallery_hyperlk	238
4.2.24	gallery_hyphyp1_linear	239
4.2.25	gallery_hyphyp1_reentrant	239
4.2.26	gallery_hyphyp1_tandem	240
4.2.27	gallery_hypm1	241
4.2.28	gallery_hypm1_reentrant	241
4.2.29	gallery_lqn_basic	242
4.2.30	gallery_lqn_random	243
4.2.31	gallery_lqn_workflows	245
4.2.32	gallery_lukumar_reentrant	246
4.2.33	gallery_mapm1	246
4.2.34	gallery_mapmk	247
4.2.35	gallery_mdk	247
4.2.36	gallery_merl1	247
4.2.37	gallery_merl1_linear	249
4.2.38	gallery_merl1_reentrant	249

4.2.39	gallery_merl1_tandem	250
4.2.40	gallery_merlk	251
4.2.41	gallery_mhyp1	251
4.2.42	gallery_mhyp1_linear	252
4.2.43	gallery_mhyp1_reentrant	252
4.2.44	gallery_mhyp1_tandem	253
4.2.45	gallery_mhypk	254
4.2.46	gallery_mm1	254
4.2.47	gallery_mm1_feedback	255
4.2.48	gallery_mm1_linear	255
4.2.49	gallery_mm1_multiclass	255
4.2.50	gallery_mm1_prio	257
4.2.51	gallery_mm1_ps	258
4.2.52	gallery_mm1_ps_feedback	258
4.2.53	gallery_mm1_ps_multiclass	259
4.2.54	gallery_mm1_ps_reentrant	259
4.2.55	gallery_mm1_reentrant	260
4.2.56	gallery_mm1_tandem	261
4.2.57	gallery_mm1_tandem_multiclass	262
4.2.58	gallery_mm1k	263
4.2.59	gallery_mmap1	263
4.2.60	gallery_mmap1_multiclass	263
4.2.61	gallery_mmapk	264
4.2.62	gallery_mmk	264
4.2.63	gallery_mpar1	264
4.2.64	gallery_multitier	266
4.2.65	gallery_multitier_storage	267
4.2.66	gallery_parm1	268
4.2.67	gallery_qn_random	269
4.2.68	gallery_renv_breakdown	269
4.2.69	gallery_repairmen	270
4.2.70	gallery_replayerm1	270
4.2.71	gallery_um1	271
5	Discrete-time models	273
5.1	The slotted models	273
5.1.1	dt_bernoulli_loaddep	273
5.1.2	dt_cycle	274
5.1.3	dt_cycle_loaddep	275
5.1.4	dt_cycle_multiclass	276
5.1.5	dt_geogeol	277
5.1.6	dt_geogeol_loss	277
6	Solver-specific examples	279
6.0.1	ctmc_spn_mm1	279
6.0.2	ctmc_spn_vs_nc_closed_qn	279
6.0.3	ctmc_tandem_mm1	280

7	Model interchange	282
7.1	The fixtures	282
7.1.1	json_balking_retrial	282
7.1.2	json_classcap_droprule	283
7.1.3	json_fcr_details	283
7.1.4	json_hetero_servers	283
7.1.5	json_join_deadline	283
7.1.6	json_signal_classes	285
8	Inference and estimation	286
8.1	The estimators	286
8.1.1	est_ekf_changepoint	286
8.1.2	est_ekf_closed	287
8.1.3	est_ekf_open	287
8.1.4	est_erps_closed	287
8.1.5	est_erps_open	289
8.1.6	est_mcmc_closed	289
8.1.7	est_mle_open	289
8.1.8	est_qmle_closed	290
8.1.9	est_trace_closed	290
8.1.10	est_ubo_changepoint	290
8.1.11	est_ubo_closed	292
8.1.12	est_ubo_open	292
8.1.13	est_ubo_variants_closed	292
8.1.14	est_ubr_changepoint	293
8.1.15	est_ubr_closed	293
8.1.16	est_ubr_open	293
8.1.17	est_vi_closed	295
9	Optimization	297
9.1	Sizing an M/M/c queue	297
9.2	Exact sizing by bisection	298
9.3	Continuous service rate tuning	298
9.4	Load balancing across heterogeneous servers	299
9.5	Population sizing in a closed network	299
9.6	Robust sizing under workload scenarios	300
9.7	Cost-performance tradeoff frontier	300
9.8	Decomposing a multi-variable problem	301
9.9	The models behind the tutorials	301
9.9.1	opt_bisection_sizing	301
9.9.2	opt_decomposition	302
9.9.3	opt_load_balancing	303
9.9.4	opt_lqn_host_demand	304
9.9.5	opt_pareto_frontier	306
9.9.6	opt_population_sizing	307
9.9.7	opt_robust_sizing	308
9.9.8	opt_sensitivity_report	309
9.9.9	opt_server_sizing	309
9.9.10	opt_service_rate	310

Chapter 1

About this manual

This manual documents the worked-example inventory of LINE, one edition per codebase. An entry is runnable only when this edition identifies a source implementation and does not mark the required operation as unsupported: the `jline.examples` package tree of `jline.jar`, and each chapter follows the organization of that tree. Script names agree across the four codebases, so an example can be read side by side between editions; where a codebase does not carry a family at all, the chapter says so rather than leaving the reader to infer it.

The reference material on model classes, solvers and options is in the LINE user manual, and the guided tour and the introductory tutorials are in the LINE primer. Neither is repeated here: this manual starts where the primer stops, at the point where one already knows how to build a model and wants to see the feature one needs.

Every model in the suite is written in the same four blocks, and it is worth learning to read them:

1. the nodes: stations, sources, sinks and the special nodes;
2. the classes and their arrival and service distributions;
3. the topology, as a routing matrix or a serial routing shorthand;
4. the solution: one or more solvers, and the metrics read from them.

The chapters are:

- *Basic models* (Chapter 2): the elementary model families, one per node or class feature.
- *Advanced models* (Chapter 3): features that leave the product-form setting.
- *Model gallery* (Chapter 4): a catalogue of small named models, in Kendall notation where one applies.
- *Discrete-time models* (Chapter 5): models on a slotted time scale.
- *Solver-specific examples* (Chapter 6): scripts written around one solver rather than one model.
- *Model interchange* (Chapter 7): the JSON model format and its round-trip fixtures.
- *Inference and estimation* (Chapter 8): fitting model parameters to measurements.
- *Optimization* (Chapter 9): design problems on queueing networks.

Each chapter opens with a worked example and then gives one section per script in the reference inventory. A complete entry contains a topology figure, a feature table, source code, and captured output. These components are generated from the harvested model, the implementation in the named language tree, and an actual run.

Chapter 2

Basic models

The basic examples cover one model family each: a node type, a class type or a scheduling feature, in the smallest model that shows it. They live under `jline.examples.java.basic`, one `*Model` class holding the models and one `*Examples` class solving them, and the script names agree across the four codebases, so a model can be compared line by line between editions.

2.1 Open queueing networks

An open network has an unbounded population: jobs enter at a `Source`, visit stations according to the routing matrix and leave at a `Sink`. Stability requires the utilization of every station to stay below one, which is a property of the arrival rates and the demands, not something the model declares.

The archetype is `oqn_basic`: a Poisson stream feeds a Delay station with hyperexponential think time and an FCFS queue with exponential service. Four blocks build it – nodes, classes and their distributions, the routing matrix, and the solve – and every model in this manual follows the same four blocks.

```
Network model = new Network("myModel");
// Block 1: nodes
Delay node1 = new Delay(model, "Delay");
Queue node2 = new Queue(model, "Queue1", SchedStrategy.FCFS);
Source node3 = new Source(model, "Source");
Sink node4 = new Sink(model, "Sink");
// Block 2: classes
OpenClass jobclass1 = new OpenClass(model, "Class1", 0);
node1.setService(jobclass1, new HyperExp(0.5, 3.0, 10.0));
node2.setService(jobclass1, new Exp(1.0));
node3.setArrival(jobclass1, new Exp(0.1));
// Block 3: topology
RoutingMatrix P = model.initRoutingMatrix();
P.set(jobclass1, jobclass1, new Matrix("[0,1,0,0; 0,0,0,1; 1,0,0,0; 0,0,0,0]"));
model.link(P);
// Block 4: solution
new MVA(model).getAvgTable().print();
```

The scripts of this family are:

2.1.1 `mam_transient_mapmap1`

Transient analysis of a MAP/MAP/1 queue started empty, through the Laplace-domain transient QBD method that `getTranAvg` auto-selects when both the arrival and the service process are correlated. The example is built and solved as follows.

```
Network model = OpenModel.mam_transient_mapmap1();
```

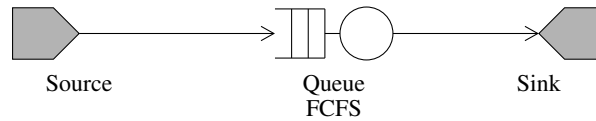


Figure 2.1: Topology of mam_transient_mapmap1: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: MAP
Service	Queue – Class1: MAP

Table 2.1: Features of mam_transient_mapmap1.

```

// The horizon is named once and printed from here, NOT from
// SolverResult.t: nothing in jline.solvers.mam populates that field, so
// reading it is a null dereference. initStateTwins, the other Java
// transient example, avoids it for the same reason. The other three
// codebases print the curve's own last time, which is this horizon.
final double horizon = 40.0;
MAM solver = new MAM(model, "timespan", new double[]{0, horizon});
solver.getTranAvg();
SolverResult res = solver.getResults();

// Station 1 is the Queue; the Source carries no transient curve.
//
// A TRANSIENT CURVE IS TWO COLUMNS, [metric, t], so the last VALUE is
// (rows-1, 0) and not the last ELEMENT: `get(nelem-1)` walks the flat
// row-major array and lands on the last TIME instead. This example
// printed E[N]=U=Tput=40.00000 -- the horizon, three times -- and the
// discrepancy had never been seen because the run before it never
// finished (see Solver_mam_transient_qbd's capacity test). The reference
// reads `QNT{2,1}.metric(end)`, which is this column.
Matrix qt = res.QNT[1][0];
Matrix ut = res.UNT[1][0];
Matrix tt = res.TNT[1][0];
int last = qt.getNumRows() - 1;
System.out.println("MAP/MAP/1 transient (rho=0.6), start empty:");

    ... (7 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

// does on init_state_*'s SteadyStateQLen line. All four codebases print
// the same three labels so the cells line up.
System.out.printf("TranEndQLen[MAM/Queue]: %.6f\n", qt.get(last, 0));
System.out.printf("TranEndUtil[MAM/Queue]: %.6f\n", ut.get(last, 0));
System.out.printf("TranEndTput[MAM/Queue]: %.6f\n", tt.get(last, 0));
pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.1.2 oqn_basic

Source, Delay and an FCFS queue with one open class and a hyperexponential think time, solved with every solver that admits the model.

The example is built and solved as follows.

```

public static void oqn_basic() throws Exception {
    Network model = OpenModel.oqn_basic();

    // Create array of different solvers to compare
    CTMC solverCTMC = new CTMC(model, "keep", false, "cutoff", 10);

```

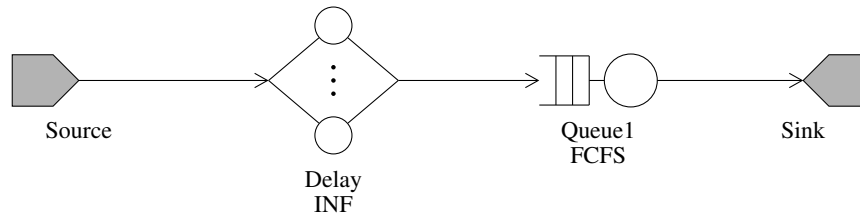


Figure 2.2: Topology of oqn_basic: an open network over 2 queueing stations.

Nodes	4: Source, Queue, Delay, Sink
Classes	1: Class1 (open)
Scheduling	Delay: INF; Queue1: FCFS
Arrivals	Source – Class1: Exp(0.1)
Service	Delay – Class1: HyperExp(0.5,0.5;3,10) Queue1 – Class1: Exp(1)
Solvers	CTMC, FLD, MVA, MAM, NC, JMT, SSA, LDES

Table 2.2: Features of oqn_basic.

```

JMT solverJMT = new JMT(model, "seed", 23000, "verbose", VerboseLevel.STD, "keep", false)
;
SSA solverSSA = new SSA(model, "seed", 23000, "verbose", VerboseLevel.STD, "samples",
10000);
FLD solverFluid = new FLD(model);
MVA solverMVA = new MVA(model);
NC solverNC = new NC(model);

// Execute each solver and display results
try {
    solverCTMC.getAvgTable().print();
} catch (Exception e) {
    System.out.println("Solver failed: " + e.getMessage());
}

try {
    solverJMT.getAvgTable().print();
} catch (Exception e) {
    System.out.println("Solver failed: " + e.getMessage());
}

try {
    solverSSA.getAvgTable().print();

... (28 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

try {
    new LDES(model, "seed", 23000, "samples", 200000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("Solver failed: " + e.getMessage());
}

pauseForUser();

```

Running this edition prints

```

WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 10 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.021667 0.021667 0.21667 0.21667 0.1 0.1
Queue1 Class1 0.11111 0.1 1.1111 1.1111 0.1 0.1
Source Class1 0 0 0 0 0 0.1
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.022679 0.022679 0.21483 0.21483 0.099621 0.099607
Queue1 Class1 0.11332 0.10222 1.1075 1.1075 0.099607 0.10001
Source Class1 0 0 0 0 0 0.099621

```

```

SSA analysis [method: default/nrm; type: approximate, randomized; lang: java; env: python] completed in <t
>s. Iterations: 10000.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.021778 0.021778 0.21365 0.21365 0.1 0.10194
Queue1 Class1 0.12018 0.10537 1.1406 1.1406 0.10194 0.10537

... (22 captured-output lines omitted; rerun the source example for the full output)

LDES events: 4000 8000 12000 16000 20000 24000 28000 32000 36000 40000 44000 48000 52000
56000 60000 64000 68000 72000 76000 80000 84000 88000 92000 96000 100000 104000 108000
112000 116000 120000 124000 128000 132000 136000 140000 144000 148000 152000 156000 160000 164000
168000 172000 176000 180000 184000 188000 192000 196000 200000 200000
LDES analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 200000.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.021586 0.021697 0.21538 0.21538 0.1 0.10014
Queue1 Class1 0.11144 0.10026 1.1112 1.1112 0.10014 0.10014
Source Class1 0 0 0 0 0 0.1

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 8 solvers the script runs (CTMC, JMT, SSA, FLD, MVA, NC, MAM, LDES), which is the point of the example: the same model read by methods of different cost and different exactness.

2.1.3 oqn_bmmapt

A BATCH, MARKED, time-inhomogeneous MAP arrival stream: a block is indexed by segment, mark and batch size, so the schedule moves the class mix and the batch size together while the epoch rate stays put. The job rate therefore differs from the epoch rate, and per class within a segment.

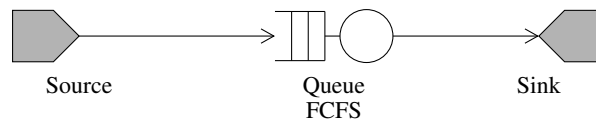


Figure 2.3: Topology of oqn_bmmapt: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Premium (open), Economy (open)
Scheduling	Queue: FCFS
Arrivals	Source – Premium: BMMAPt(True); Economy: BMMAPt(True)
Service	Queue – Premium: Exp(16); Economy: Exp(16)
Solvers	LDES

Table 2.3: Features of oqn_bmmapt.

The example is built and solved as follows.

```

new LDES(OpenBMMAPtModel.example(), "seed", 23000, "samples", 400000)
    .getAvgTable().print();
pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.1.4 oqn_cs_routing

Three open classes over two PS queues, two of which switch class at a ClassSwitch node; the topology is imported from a JSIMgraph file.

The example is built and solved as follows.

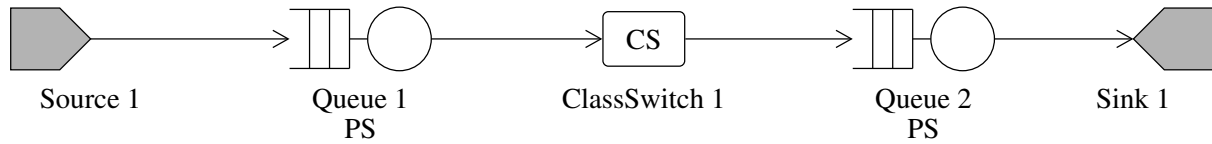


Figure 2.4: Topology of `oqn_cs_routing`: an open network over 2 queueing stations with a class-switch node.

Nodes	5: Source, Queue $\times$ 2, ClassSwitch, Sink
Classes	3: Class A (open), Class B (open), Class C (open)
Scheduling	Queue 1: PS; Queue 2: PS
Arrivals	Source 1 – Class A: Exp(2); Class B: Exp(1); Class C: Disabled
Service	Queue 1 – Class A: Exp(5); Class B: Exp(3.333); Class C: Exp(3) Queue 2 – Class A: Exp(1); Class B: Exp(1); Class C: Exp(6.667)
Solvers	CTMC, FLD, MVA, MAM, NC, JMT, SSA, LDES

Table 2.4: Features of `oqn_cs_routing`.

```

Network model = OpenModel.oqn_cs_routing();

// Create multiple solvers as in notebook
CTMC solverCTMC = new CTMC(model, "keep", true, "verbose", 1, "cutoff", new int[][]{{1,1,0},
    {3,3,0}, {0,0,3}});
FLD solverFLD = new FLD(model, "keep", true, "verbose", 1);
MVA solverMVA = new MVA(model, "keep", true, "verbose", 1);
MAM solverMAM = new MAM(model, "keep", true, "verbose", 1);
NC solverNC = new NC(model, "keep", true, "verbose", 1);
JMT solverJMT = new JMT(model, "keep", true, "verbose", 1, "seed", 23000, "samples", 100000);
SSA solverSSA = new SSA(model, "keep", true, "verbose", 1, "seed", 23000, "samples", 1000000);
;
LDES solverLDES = new LDES(model, "keep", true, "verbose", 1, "seed", 23000, "samples",
    100000);

Object[] solvers = {solverCTMC, solverFluid, solverMVA, solverMAM, solverNC, solverJMT,
    solverSSA,
    solverLDES};

// Execute all solvers and collect results
for (int i = 0; i < solvers.length; i++) {
    try {
        if (solvers[i] instanceof CTMC) {
            ((CTMC) solvers[i]).getAvgTable().print();
        } else if (solvers[i] instanceof FLD) {
            ((FLD) solvers[i]).getAvgTable().print();
        } else if (solvers[i] instanceof MVA) {
            ((MVA) solvers[i]).getAvgTable().print();
        } else if (solvers[i] instanceof MAM) {
            ((MAM) solvers[i]).getAvgTable().print();
        }

        ... (8 source lines omitted; full implementation: jar/src/main/java/jline/examples/)
    }
} catch (Exception e) {
    System.out.println("Solver failed: " + e.getMessage());
}
}
pauseForUser();

```

Running this edition prints

```

WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 1 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Source 1 Class A 0 0 0 0 0 1.8159

```

```

Source 1  Class B      0      0      0      0      0  0.94121
Queue 1   Class A    0.8844  0.36317  0.48704  0.32469  1.8159  1.8159
Queue 1   Class B    0.71089  0.28236  0.75529  0.25176  0.94121  0.94121
Queue 2   Class C    0.61834  0.41356  0.22427  0.22427  2.7571  2.7571
Fluid analysis [method: minnormal; type: approximate, deterministic; lang: java; env: python] completed in
<t>s.
Fluid analysis [method: default/minnormal; type: approximate, deterministic; lang: java; env: python]
completed in <t>s.
Station  JobClass    QLen    Util    RespT    ResidT    ArvR    Tput
Source 1  Class A      0      0      0      0      0      2
Source 1  Class B      0      0      0      0      0      1
Queue 1   Class A    0.99361  0.4    0.4968  0.3312    2      2
Queue 1   Class B    0.7452   0.3    0.7452  0.2484    1      1
Queue 2   Class C    0.61888  0.45   0.20629  0.20629   3      3

... (38 captured-output lines omitted; rerun the source example for the full output)

Station  JobClass    QLen    Util    RespT    ResidT    ArvR    Tput
Source 1  Class A      0      0      0      0      0      2
Source 1  Class B      0      0      0      0      0      1
Queue 1   Class A    1.3086  0.39838  0.65708  0.43805   2    1.9901
Queue 1   Class B    0.96972  0.29936  0.9717  0.3239    1    0.99703
Queue 2   Class C    0.81451  0.44816  0.2726  0.2726   2.9871  2.9868

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 8 solvers the script runs (CTMC, FLD, MVA, MAM, NC, JMT, SSA, LDES), which is the point of the example: the same model read by methods of different cost and different exactness.

2.1.5 oqn_fourqueues

A larger multiclass network over four stations mixing PS and FCFS scheduling.

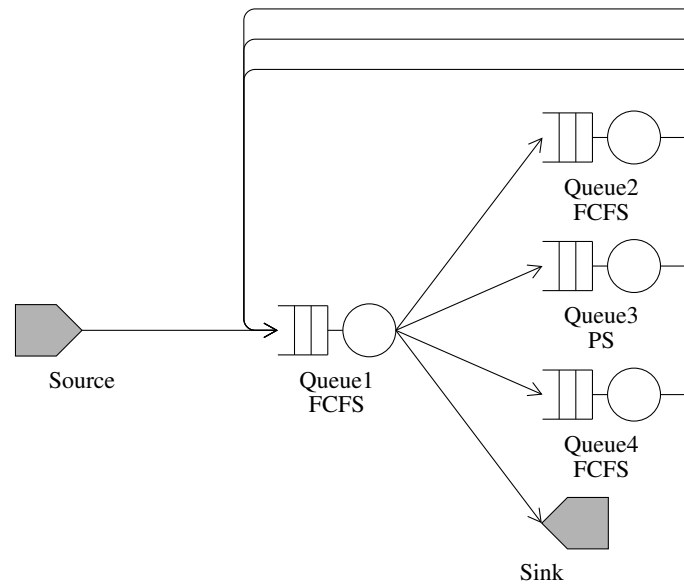


Figure 2.5: Topology of oqn_fourqueues: an open network over 4 queueing stations.

The example is built and solved as follows.

```

Network model = OpenModel.oqn_fourqueues();

// Create multiple solvers as in MATLAB version (some commented out in MATLAB)
CTMC solverCTMC = new CTMC(model, "seed", 23000, "cutoff", 1);
// FLD solverFluid = new FLD(model, "seed", 23000); // Commented out in MATLAB
MVA solverMVA = new MVA(model, "seed", 23000);
MAM solverMAM = new MAM(model, "seed", 23000);
JMT solverJMT = new JMT(model, "seed", 23000, "samples", 1000000);

```


Nodes	6: Source, Queue $\times$ 4, Sink
Classes	3: Class1 (open), Class2 (open), Class3 (open)
Scheduling	Queue1: FCFS; Queue2: FCFS; Queue3: PS; Queue4: FCFS
Arrivals	Source – Class1: Exp(0.2); Class2: Exp(0.125); Class3: Exp(0.1429)
Service	Queue1 – Class1: Exp(3.333); Class2: Exp(2); Class3: Exp(1.667) Queue2 – Class1: Exp(0.9091); Class2: Exp(0.7692); Class3: Exp(0.6667) Queue3 – Class1: Exp(0.5); Class2: Exp(0.4762); Class3: Exp(0.5263) Queue4 – Class1: Exp(0.6667); Class2: Exp(1.111); Class3: Exp(0.4348)
Solvers	CTMC, MVA, MAM, JMT, LDES

Table 2.5: Features of oqn_fourqueues.

```
// SSA solverSSA = new SSA(model, "seed", 23000); // Commented out in MATLAB
// NC solverNC = new NC(model, "seed", 23000); // Commented out in MATLAB

Object[] solvers = {solverCTMC, solverMVA, solverMAM, solverJMT};
String[] solverNames = {"CTMC", "MVA", "MAM", "JMT"};

// Execute all solvers and collect results
for (int i = 0; i < solvers.length; i++) {
    try {
        if (solvers[i] instanceof CTMC) {
            ((CTMC) solvers[i]).getAvgTable().print();
        } else if (solvers[i] instanceof MVA) {
            ((MVA) solvers[i]).getAvgTable().print();
        } else if (solvers[i] instanceof MAM) {
            ((MAM) solvers[i]).getAvgTable().print();
        } else if (solvers[i] instanceof JMT) {
            ((JMT) solvers[i]).getAvgTable().print();
        }
    } catch (Exception e) {
        System.out.println("Solver failed: " + e.getMessage());
    }
}
pauseForUser();
```

Running this edition prints

```
WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 1 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Source Class1 0 0 0 0 0 0.078166
Source Class2 0 0 0 0 0 0.061126
Source Class3 0 0 0 0 0 0.059853
Queue1 Class1 0.14622 0.093799 0.46767 1.8707 0.31266 0.31266
Queue1 Class2 0.15681 0.12225 0.64133 2.5653 0.24451 0.24451
Queue1 Class3 0.16999 0.14365 0.71004 2.8402 0.23941 0.23941
Queue2 Class1 0.10897 0.085983 1.394 1.394 0.078166 0.078166
Queue2 Class2 0.095956 0.079464 1.5698 1.5698 0.061126 0.061126
Queue2 Class3 0.10304 0.08978 1.7216 1.7216 0.059853 0.059853
Queue3 Class1 0.20295 0.15633 2.5963 2.5963 0.078166 0.078166
Queue3 Class2 0.17126 0.12837 2.8018 2.8018 0.061126 0.061126
Queue3 Class3 0.15414 0.11372 2.5753 2.5753 0.059853 0.059853
Queue4 Class1 0.15103 0.11725 1.9322 1.9322 0.078166 0.078166

... (48 captured-output lines omitted; rerun the source example for the full output)

Queue3 Class1 5.9986 0.39722 28.928 28.928 0.19922 0.19922
Queue3 Class2 3.8567 0.26258 30.078 30.078 0.12464 0.12464
Queue3 Class3 3.9759 0.27255 26.901 26.901 0.14254 0.14267
Queue4 Class1 1.3583 0.29939 6.6818 6.6818 0.19939 0.19939
Queue4 Class2 0.72756 0.1126 5.7367 5.7367 0.12507 0.1246
Queue4 Class3 1.0722 0.32813 7.4165 7.4165 0.14274 0.14286

[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 4 solvers the script runs (CTMC, MVA, MAM, JMT), which is the point of the example: the same model read by methods of different cost and different exactness.

2.1.6 oqn_mapt

A time-inhomogeneous MAP arrival stream and the Ko-Pender fluid and diffusion limits returned by the `kp` fluid method.

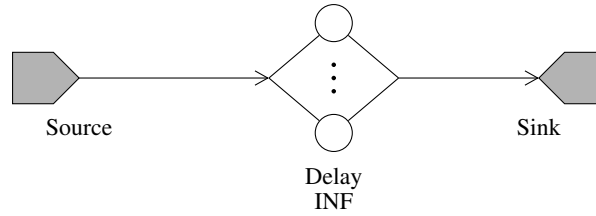


Figure 2.6: Topology of `oqn_mapt`: an open network over 1 queueing station.

Nodes	3: Source, Delay, Sink
Classes	1: OpenClass (open)
Scheduling	Delay: INF
Arrivals	Source – OpenClass: MAPt
Service	Delay – OpenClass: Exp(2)
Solvers	FLD

Table 2.6: Features of `oqn_mapt`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.1.7 oqn_mapt_service

A time-inhomogeneous MAP as the SERVICE process of a finite-buffer queue: the phase walk is taken from the service start instant and the modulating phase carries across successive services, so the stream is correlated as well as non-stationary.

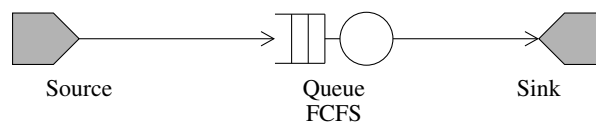


Figure 2.7: Topology of `oqn_mapt_service`: an open network over 1 queueing station.

The example is built and solved as follows.

```
new LDES (OpenMAPtServiceModel.oqn_mapt_service(),
    "seed", 1234, "samples", 1000000).getAvgTable().print();

System.out.println("\nconstant schedule vs ordinary MAP (must agree):");
new LDES (OpenMAPtServiceModel.constantSchedule(),
    "seed", 1234, "samples", 400000).getAvgTable().print();
new LDES (OpenMAPtServiceModel.homogeneousReference(),
    "seed", 1234, "samples", 400000).getAvgTable().print();
pauseForUser();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

Nodes	3: Source, Queue, Sink
Classes	1: OpenClass (open)
Scheduling	Queue: FCFS
Arrivals	Source – OpenClass: Exp(10)
Service	Queue – OpenClass: MAPt
Features	finite buffer 5 at Queue
Solvers	LDES

Table 2.7: Features of oqn_mapt_service.

2.1.8 oqn_mmapt

A MARKED time-inhomogeneous MAP arrival stream: one modulating chain drives every marked class, the mark selects the class of the arriving job, and the schedule moves the class MIX while leaving the aggregate rate flat.

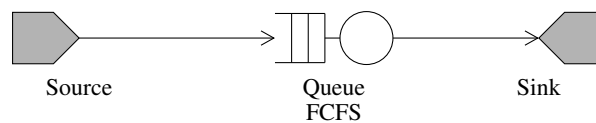


Figure 2.8: Topology of oqn_mmapt: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Morning (open), Evening (open)
Scheduling	Queue: FCFS
Arrivals	Source – Morning: MMAPt(True); Evening: MMAPt(True)
Service	Queue – Morning: Exp(8); Evening: Exp(8)
Solvers	LDES

Table 2.8: Features of oqn_mmapt.

The example is built and solved as follows.

```
MMAPt arrival = OpenMMAPtModel.arrival();
System.out.printf("Aggregate arrival rate over the cycle: %.4f%n",
    arrival.getTimeAverageRate());
double[] lam = arrival.getTimeAverageMarkRates();
System.out.printf("Per-mark rates over the cycle:      %.4f %.4f "
    + "(they sum to the aggregate)%n", lam[0], lam[1]);
System.out.println("The cycle average is a 2:2 split, so a run that shows anything "
    + "else is reading\nthe marks from the segment in force rather than from "
    + "the average.\n");

// LDES simulates the marked schedule directly: the mark is decided by
// WHICH block's transition fired, inside the same competing-transitions
// draw that ends the interval, so the labelling costs no extra randomness.
new LDES(OpenMMAPtModel.example(), "seed", 23000, "samples", 400000)
    .getAvgTable().print();
pauseForUser();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.1.9 oqn_multichain_cs

Five classes grouped into three chains by class switching, the smallest model in which chains and classes differ.

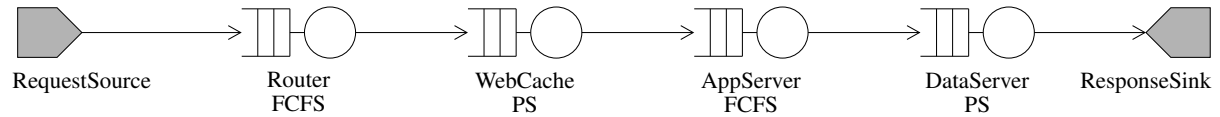


Figure 2.9: Topology of oqn_multichain_cs: an open network over 4 queueing stations.

Nodes	6: Source, Queue $\times$ 4, Sink
Classes	5: InteractiveA (open), InteractiveB (open), BatchA (open), BatchB (open), RealTime (open)
Scheduling	Router: FCFS; WebCache: PS; AppServer: FCFS; DataServer: PS
Arrivals	RequestSource – InteractiveA: Exp(0.3); InteractiveB: Exp(0.1); BatchA: Exp(0.15); BatchB: Exp(0.12); RealTime: Exp(0.2)
Service	Router – InteractiveA: Exp(6.667); InteractiveB: Exp(5.556); BatchA: Exp(5); BatchB: Exp(4.545); RealTime: Exp(10) WebCache – InteractiveA: Exp(3.333); InteractiveB: Exp(2.857); BatchA: Exp(1); BatchB: Exp(0.8333); RealTime: Exp(5) AppServer – InteractiveA: Exp(2); InteractiveB: Exp(1.667); BatchA: Exp(0.6667); BatchB: Exp(0.5556); RealTime: Exp(3.333) DataServer – InteractiveA: Exp(2.5); InteractiveB: Exp(2.222); BatchA: Exp(1.25); BatchB: Exp(1); RealTime: Exp(4)
Solvers	MVA, FLD, SSA

Table 2.9: Features of oqn_multichain_cs.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.1.10 oqn_nhpp

A cyclic non-homogeneous Poisson arrival stream, whose rate is a periodic function of time.

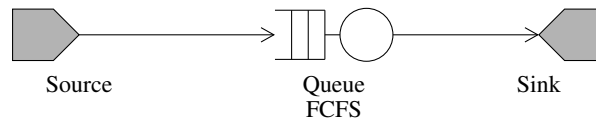


Figure 2.10: Topology of oqn_nhpp: an open network over 1 queueing station.

The example is built and solved as follows.

```
Network model = OpenNHPPModel.oqn_nhpp();

try {
    new LDES(model, "seed", 1234, "samples", 100000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("LDES failed: " + e.getMessage());
}

try {
    new FLD(model, "timespan", new double[]{0, 12}, "verbose", 0).getAvgTable().print();
} catch (Exception e) {
    System.out.println("FLD failed: " + e.getMessage());
}

pauseForUser();
```

Running this edition prints

```
LDDES events: 2000 4000 6000 8000 10000 12000 14000 16000 18000 20000 22000 24000 26000
28000 30000 32000 34000 36000 38000 40000 42000 44000 46000 48000 50000 52000 54000
56000 58000 60000 62000 64000 66000 68000 70000 72000 74000 76000 78000 80000 82000
84000 86000 88000 90000 92000 94000 96000 98000 100000 100000
LDDES analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 100000.
```

Nodes	3: Source, Queue, Sink
Classes	1: OpenClass (open)
Scheduling	Queue: FCFS
Arrivals	Source – OpenClass: NHPP
Service	Queue – OpenClass: Exp(10)
Solvers	LDES

Table 2.10: Features of oqn_nhpp.

Station	JobClass	QLen	Util	RespT	ResidT	ArvR	Tput
Source	OpenClass	0	0	0	0	0	3.6667
Queue	OpenClass	0.7226	0.3662	0.19751	0.19751	3.6667	3.6581
Fluid analysis [method: closing; type: approximate, deterministic; lang: java; env: python] completed in < t>s.							
Fluid analysis [method: closing; type: approximate, deterministic; lang: java; env: python] completed in < t>s.							
Station	JobClass	QLen	Util	RespT	ResidT	ArvR	Tput
Source	OpenClass	0	0	0	0	0	3.6667
Queue	OpenClass	0.4	0.4	0.1	0	3.6667	4
[Running in non-interactive mode, continuing...]							

The rows repeat for each of the 2 solvers the script runs (LDES, FLD), which is the point of the example: the same model read by methods of different cost and different exactness.

2.1.11 oqn_online

The same tandem written in one line from the arrival, demand and think time matrices, through `Network.tandemPsInf`.

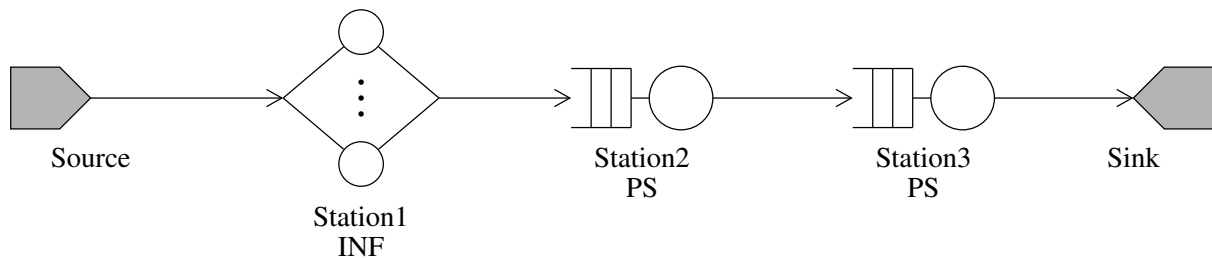


Figure 2.11: Topology of oqn_online: an open network over 3 queueing stations.

Nodes	5: Source, Queue $\times$ 2, Delay, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Station1: INF; Station2: PS; Station3: PS
Arrivals	Source – Class1: Exp(0.02); Class2: Exp(0.04)
Service	Station1 – Class1: Exp(0.01099); Class2: Exp(0.01087) Station2 – Class1: Exp(0.1); Class2: Exp(0.2) Station3 – Class1: Exp(0.2); Class2: Exp(0.1111)
Solvers	MVA

Table 2.11: Features of oqn_online.

The example is built and solved as follows.

```
Network model = OpenModel.oqn_online();

// Create solver as in MATLAB version
MVA solver = new MVA(model);
solver.getAvgTable().print();

pauseForUser();
```

Running this edition prints

```
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 44.
Station  JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Source   Class1         0     0         0         0     0   0.02
Source   Class2         0     0         0         0     0   0.04
Station1 Class1       1.82   1.82        91        91   0.02  0.02
Station1 Class2       3.68   3.68       92       92   0.04  0.04
Station2 Class1    0.33333   0.2   16.667   16.667   0.02  0.02
Station2 Class2    0.33333   0.2   8.3333  8.3333   0.04  0.04
Station3 Class1    0.18518   0.1   9.2592   9.2592   0.02  0.02
Station3 Class2    0.66667   0.36  16.667   16.667   0.04  0.04

[Running in non-interactive mode, continuing...]
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

2.1.12 oqn_trace_driven

Arrivals and service times replayed from a measurement trace with Replayer.

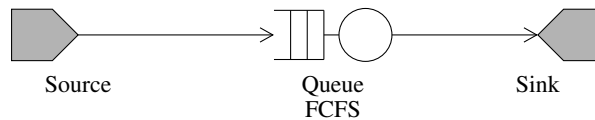


Figure 2.12: Topology of oqn_trace_driven: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: OpenClass (open)
Scheduling	Queue: FCFS
Arrivals	Source – OpenClass: Exp(1)
Service	Queue – OpenClass: Replayer(trace)
Solvers	JMT, LDES

Table 2.12: Features of oqn_trace_driven.

The example is built and solved as follows.

```
Network model = OpenModel.oqn_trace_driven();

try {
    new JMT(model, "seed", 23000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT failed: " + e.getMessage());
}

try {
    new LDES(model, "seed", 23000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("LDES failed: " + e.getMessage());
}

pauseForUser();
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

2.1.13 oqn_vsinks

Several sinks in one model, the virtual-sink idiom for measuring departure streams separately.

The example is built and solved as follows.

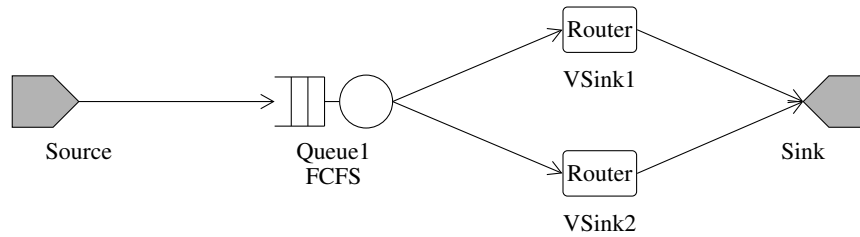


Figure 2.13: Topology of oqn_vsinks: an open network over 1 queueing station with a router.

Nodes	5: Source, Queue, Router $\times$ 2, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue1: FCFS
Arrivals	Source – Class1: Exp(1); Class2: Exp(1)
Service	Queue1 – Class1: Exp(100); Class2: Exp(100)
Solvers	MVA, MAM, NC

Table 2.13: Features of oqn_vsinks.

```

Network model = OpenModel.oqn_vsinks();

// Create solvers as in MATLAB version
MVA solverMVA = new MVA(model);
MAM solverMAM = new MAM(model);
NC solverNC = new NC(model);

// MVA solver results
solverMVA.getAvgTable().print();
solverMVA.getAvgNodeTable().print();

// MAM solver results
solverMAM.getAvgTable().print();
solverMAM.getAvgNodeTable().print();

// NC solver results
solverNC.getAvgTable().print();
solverNC.getAvgNodeTable().print();

pauseForUser();

```

Running this edition prints

```

MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 16.

```

Station	JobClass	QLen	Util	RespT	ResidT	ArvR	Tput
Source	Class1	0	0	0	0	0	1
Source	Class2	0	0	0	0	0	1
Queue1	Class1	0.010204	0.01	0.010204	0.010204	1	1
Queue1	Class2	0.010204	0.01	0.010204	0.010204	1	1
Node	JobClass	QLen	Util	RespT	ResidT	ArvR	Tput
Source	Class1	0	0	0	0	0	1
Source	Class2	0	0	0	0	0	1
Queue1	Class1	0.010204	0.01	0.010204	0.010204	1	1
Queue1	Class2	0.010204	0.01	0.010204	0.010204	1	1
Sink	Class1	0	0	0	0	1	0
Sink	Class2	0	0	0	0	1	0
VSink1	Class1	0	0	0	0	0.6	0.6
VSink1	Class2	0	0	0	0	0.1	0.1
VSink2	Class1	0	0	0	0	0.4	0.4

... (29 captured-output lines omitted; rerun the source example for the full output)

Sink	Class1	0	0	0	0	1	0
Sink	Class2	0	0	0	0	1	0
VSink1	Class1	0	0	0	0	0.6	0.6
VSink1	Class2	0	0	0	0	0.1	0.1
VSink2	Class1	0	0	0	0	0.4	0.4

```

VSink2  Class2      0      0      0      0  0.9  0.9
[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 3 solvers the script runs (MVA, MAM, NC), which is the point of the example: the same model read by methods of different cost and different exactness.

2.2 Closed queueing networks

A closed network holds a fixed population per class. Each closed class names a reference station, at which its visit ratio is one and through which its throughput is measured; the population is the third argument of the class constructor. A Delay station with think time Z and a queue with demand D is the interactive-system archetype, and the repairman model is its oldest form.

Because the population is finite, every metric is bounded and the network is stable by construction. What varies across the scripts below is the scheduling discipline, the service distribution and the number of servers, that is, exactly the assumptions under which a product-form solution does or does not hold.

2.2.1 cqn_bas_blocking

Blocking-after-service with finite buffers, where a completing job holds its server until the destination frees a slot.

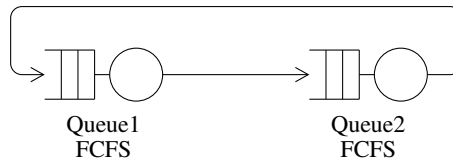


Figure 2.14: Topology of `cqn_bas_blocking`: a closed network over 2 queueing stations.

Nodes	2: Queue $\times$ 2
Classes	1: Class1 (closed, $N = 2$)
Scheduling	Queue1: FCFS; Queue2: FCFS
Service	Queue1 – Class1: Exp(1) Queue2 – Class1: Exp(0.8)
Features	drop rule {'Class1': 'blockingAfterService'} at Queue1; finite buffer 1 at Queue2
Solvers	MVA

Table 2.14: Features of `cqn_bas_blocking`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.2.2 cqn_bcmp_theorem

The product-form scheduling policies of the BCMP theorem (FCFS with class-independent exponential service, PS, LCFS-PR, INF) compared on one network.

The example is built and solved as follows.

```

// Test PS scheduling model
Network modelPS = ClosedModel.cqn_bcmp_theorem_ps();
try {
    CTMC solverPS = new CTMC(modelPS);
}

```

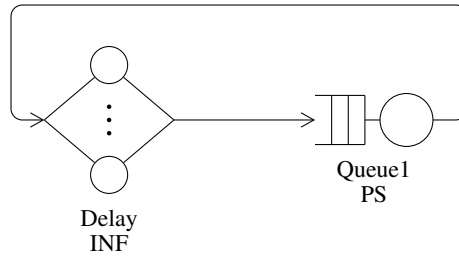



Figure 2.15: Topology of cqn_bcmp_theorem: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 2$), Class2 (closed, $N = 2$)
Scheduling	Delay: INF; Queue1: PS
Service	Delay – Class1: Erlang(3,2); Class2: HyperExp(0.5,0.5;3,10) Queue1 – Class1: Exp(1); Class2: Exp(1)
Solvers	CTMC

Table 2.15: Features of cqn_bcmp_theorem.

```

        solverPS.getAvgTable().print();
    } catch (Exception e) {
        System.out.println("Error solving PS model: " + e.getMessage());
    }

    // Test FCFS scheduling model
    Network modelFCFS = ClosedModel.cqn_bcmp_theorem_fcfs();
    try {
        CTMC solverFCFS = new CTMC(modelFCFS);
        solverFCFS.getAvgTable().print();
    } catch (Exception e) {
        System.out.println("Error solving FCFS model: " + e.getMessage());
    }

    // Test LCFS-PR scheduling model
    Network modelLCFSPR = ClosedModel.cqn_bcmp_theorem_lcfspr();
    CTMC solverLCFSPR = new CTMC(modelLCFSPR);
    solverLCFSPR.getAvgTable().print();

    pauseForUser();

```

Running this edition prints

```

CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay   Class1   0.30861 0.30861 0.66667 0.66667 0.46291 0.46291
Delay   Class2   0.11625 0.11625 0.21667 0.21667 0.53653 0.53653
Queue1  Class1   1.6914 0.46291 3.6538 3.6538 0.46291 0.46291
Queue1  Class2   1.8838 0.53653 3.511 3.511 0.53653 0.53653
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay   Class1   0.30861 0.30861 0.66667 0.66667 0.46291 0.46291
Delay   Class2   0.11625 0.11625 0.21667 0.21667 0.53653 0.53653
Queue1  Class1   1.6914 0.46291 3.6538 3.6538 0.46291 0.46291
Queue1  Class2   1.8838 0.53653 3.511 3.511 0.53653 0.53653
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay   Class1   0.30861 0.30861 0.66667 0.66667 0.46291 0.46291
Delay   Class2   0.11625 0.11625 0.21667 0.21667 0.53653 0.53653
Queue1  Class1   1.6914 0.46291 3.6538 3.6538 0.46291 0.46291
Queue1  Class2   1.8838 0.53653 3.511 3.511 0.53653 0.53653
[Running in non-interactive mode, continuing...]

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by CTMC.

2.2.3 cqn_lcfs_multiclass

Multiclass LCFS and LCFS-PR scheduling, and the difference preemption makes.

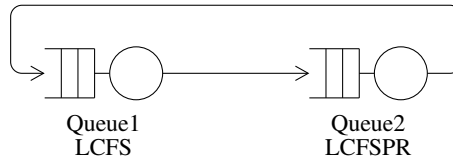


Figure 2.16: Topology of `cqn_lcfs_multiclass`: a closed network over 2 queueing stations.

Nodes	2: Queue $\times$ 2
Classes	3: Class1 (closed, $N = 1$), Class2 (closed, $N = 1$), Class3 (closed, $N = 1$)
Scheduling	Queue1: LCFS; Queue2: LCFS-PR
Service	Queue1 – Class1: Exp(1); Class2: Exp(0.3333); Class3: Exp(0.2) Queue2 – Class1: Exp(0.5); Class2: Exp(0.25); Class3: Exp(0.1667)
Solvers	MVA, CTMC

Table 2.16: Features of `cqn_lcfs_multiclass`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.2.4 cqn_mmpp2_service

MMPP(2) service times, a correlated service process that no product-form solver can represent exactly.

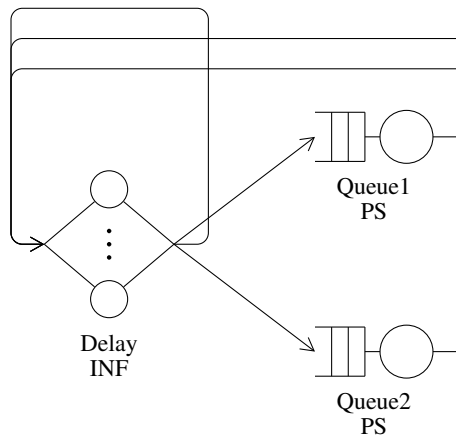


Figure 2.17: Topology of `cqn_mmpp2_service`: a closed network over 3 queueing stations.

The example is built and solved as follows.

```
Network model = ClosedModel.cqn_mmpp2_service();

// The reference pins the seed and takes the engine's own run length, then
// runs the discrete-event engine at the same seed.
show("JMT", () -> new JMT(model, "seed", 23000).getAvgTable());
show("LDES", () -> new LDES(model, "seed", 23000).getAvgTable());

pauseForUser();
```

Nodes	3: Queue $\times$ 2, Delay
Classes	2: Class1 (closed, $N = 1$), Class2 (closed, $N = 1$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS
Service	Delay – Class1: Erlang(3,2); Class2: HyperExp(0.5,0.5;3,10) Queue1 – Class1: HyperExp(0.1,0.9;1,10); Class2: MMPP2 Queue2 – Class1: HyperExp(0.1,0.9;1,10); Class2: Erlang(1,2)
Features	class switching on 3 links
Solvers	JMT, LDES

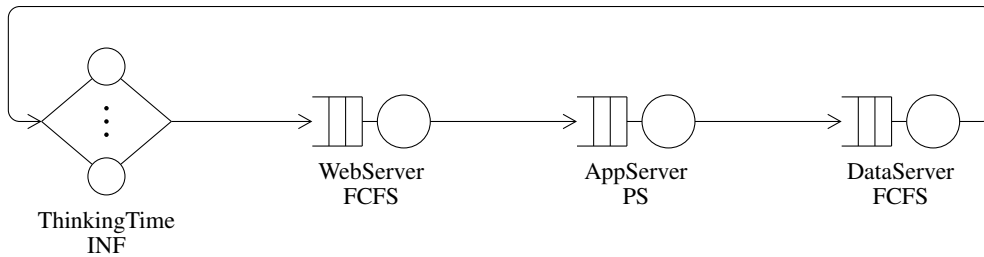
Table 2.17: Features of `cqn_mmpp2_service`.

Running this edition prints

```
jline.lang.Network@1f89ab83
```

2.2.5 `cqn_multichain_cs`

Five classes in three chains, the closed counterpart of `oqn_multichain_cs`.

Figure 2.18: Topology of `cqn_multichain_cs`: a closed network over 4 queuing stations.

Nodes	4: Queue $\times$ 3, Delay
Classes	5: HighPriorityA (closed, $N = 2$), HighPriorityB (closed, $N = 1$), RegularA (closed, $N = 3$), RegularB (closed, $N = 2$), Background (closed, $N = 2$)
Scheduling	ThinkingTime: INF; WebServer: FCFS; AppServer: PS; DataServer: FCFS
Service	ThinkingTime – HighPriorityA: Exp(0.6667); HighPriorityB: Exp(0.625); RegularA: Exp(0.5); RegularB: Exp(0.4762); Background: Exp(0.3333) WebServer – HighPriorityA: Exp(2.5); HighPriorityB: Exp(2.381); RegularA: Exp(2); RegularB: Exp(1.923); Background: Exp(1.25) AppServer – HighPriorityA: Exp(1.429); HighPriorityB: Exp(1.333); RegularA: Exp(1.111); RegularB: Exp(1.053); Background: Exp(0.8333) DataServer – HighPriorityA: Exp(2); HighPriorityB: Exp(1.923); RegularA: Exp(1.667); RegularB: Exp(1.613); Background: Exp(1)
Solvers	MVA, SSA

Table 2.18: Features of `cqn_multichain_cs`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.2.6 `cqn_multiserver`

A closed network with multiserver FCFS and PS stations, Erlang service and a class disabled at one station.

The example is built and solved as follows.

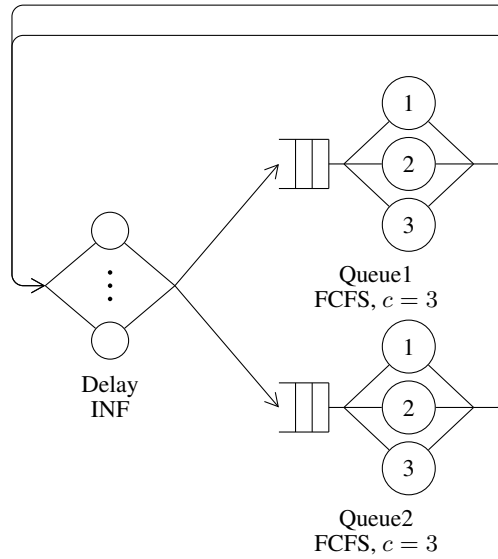


Figure 2.19: Topology of `cqn_multiserver`: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	4: Class1 (closed, $N = 2$), Class2 (closed, $N = 2$), Class3 (closed, $N = 2$), Class4 (closed, $N = 1$)
Scheduling	Delay: INF; Queue1: FCFS ($c = 3$); Queue2: FCFS ($c = 3$)
Service	Delay – Class1: Exp(1); Class2: Exp(1); Class3: Exp(10); Class4: Exp(1) Queue1 – Class1: Exp(1); Class2: Erlang(1,2); Class3: Exp(10); Class4: Exp(1) Queue2 – Class1: Disabled; Class2: Disabled; Class3: Erlang(1,2); Class4: Exp(1)
Solvers	MVA

Table 2.19: Features of `cqn_multiserver`.

```

Network model = ClosedModel.cqn_multiserver();
MVA solver = new MVA(model);

solver.getAvgTable().print();

pauseForUser();

```

Running this edition prints

```

MVA analysis [method: default/lin; type: approximate, deterministic; lang: java; env: python] completed in
<t>s. Iterations: 124.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.95207 0.95207 1 0.66667 0.95207 0.95207
Delay Class2 0.47603 0.47603 1 0.33333 0.47603 0.47603
Delay Class3 0.13232 0.13232 0.1 0.06667 1.3232 1.3232
Delay Class4 0.66159 0.66159 1 0.33333 0.66159 0.66159
Queue1 Class1 1.286 0.31736 1.3507 0.90046 0.95207 0.95207
Queue1 Class2 1.286 0.31736 2.7014 0.90046 0.47603 0.47603
Queue1 Class3 0.19955 0.03308 0.20108 0.10054 0.99239 0.99239
Queue1 Class4 1.3303 0.22053 2.0108 0.67026 0.66159 0.66159
Queue2 Class3 0.67621 0.22053 2.0442 0.3407 0.3308 0.3308

[Running in non-interactive mode, continuing...]

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

2.2.7 cqn_multiserver_nc

Delay plus a multiserver FCFS queue, solved by the normalizing constant method.

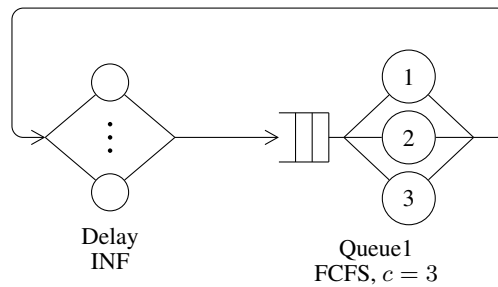


Figure 2.20: Topology of `cqn_multiserver_nc`: a closed network over 2 queueing stations.

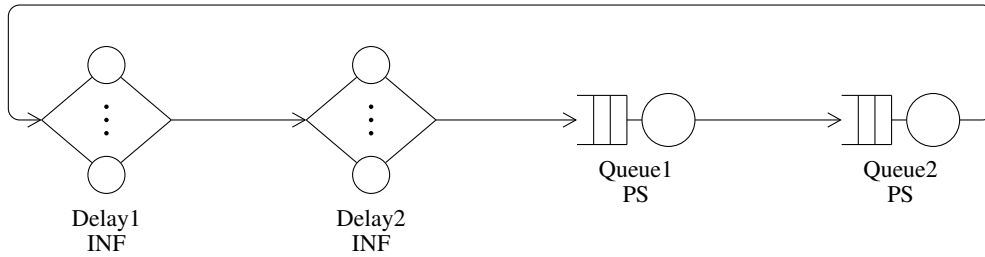
Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 5$)
Scheduling	Delay: INF; Queue1: FCFS ($c = 3$)
Service	Delay – Class1: Exp(1) Queue1 – Class1: Exp(1.25)
Solvers	CTMC, MVA, NC

Table 2.20: Features of `cqn_multiserver_nc`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.2.8 cqn_online

The closed counterpart of the one-line construction, from the demand and think time matrices. The example is built and solved as follows.

Figure 2.21: Topology of `cqn_online`: a closed network over 4 queueing stations.

Nodes	4: Queue $\times$ 2, Delay $\times$ 2
Classes	2: Class1 (closed, $N = 1$), Class2 (closed, $N = 2$)
Scheduling	Delay1: INF; Delay2: INF; Queue1: PS; Queue2: PS
Service	Delay1 – Class1: Exp(0.01099); Class2: Exp(0.01087) Delay2 – Class1: Exp(0.01075); Class2: Exp(0.01064) Queue1 – Class1: Exp(0.1); Class2: Exp(0.2) Queue2 – Class1: Exp(0.2); Class2: Exp(0.1111)
Solvers	MVA

Table 2.21: Features of `cqn_online`.

```

Network model = ClosedModel.cqn_online();
MVA solver = new MVA(model, "method", "exact");

solver.getAvgTable().print();

pauseForUser();

```

Running this edition prints

```

MVA analysis [method: exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 Class1 0.45505 0.45505 91 91 0.0050005 0.0050005
Delay1 Class2 0.91525 0.91525 92 92 0.0099484 0.0099484
Delay2 Class1 0.46505 0.46505 93 93 0.0050005 0.0050005
Delay2 Class2 0.93515 0.93515 94 94 0.0099484 0.0099484
Queue1 Class1 0.052561 0.050005 10.511 10.511 0.0050005 0.0050005
Queue1 Class2 0.053601 0.049742 5.3879 5.3879 0.0099484 0.0099484
Queue2 Class1 0.027348 0.025002 5.469 5.469 0.0050005 0.0050005
Queue2 Class2 0.096001 0.089535 9.6499 9.6499 0.0099484 0.0099484

[Running in non-interactive mode, continuing...]

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

2.2.9 `cqn_repairmen`

The machine repairman model: a Delay station of thinking machines and one FCFS repair queue.

Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 10$)
Scheduling	Delay: INF; Queue1: FCFS
Service	Delay – Class1: Exp(1) Queue1 – Class1: Exp(0.6667)
Solvers	CTMC, JMT, SSA, FLD, MVA, NC, MAM, LDES

Table 2.22: Features of `cqn_repairmen`.

The example is built and solved as follows.

```

Network model = ClosedModel.cqn_repairmen();

```

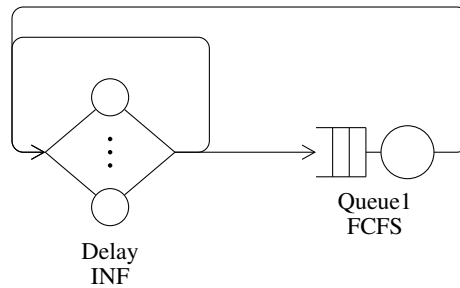


Figure 2.22: Topology of `cqn_repairmen`: a closed network over 2 queueing stations.

```
show("CTMC", () -> new CTMC(model).getAvgTable());
show("JMT", () -> new JMT(model, "seed", 23000).getAvgTable());
show("SSA", () -> new SSA(model, "seed", 23000, "samples", 5000).getAvgTable());
show("FLD", () -> new FLD(model).getAvgTable());
show("MVA", () -> new MVA(model).getAvgTable());
show("NC", () -> new NC(model, "method", "exact").getAvgTable());
show("MAM", () -> new MAM(model).getAvgTable());
show("LDES", () -> new LDES(model, "seed", 23000, "samples", 5000).getAvgTable());

pauseForUser();
```

Running this edition prints

```
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 2.222 2.222 1 1 2.222 2.222
Queue1 Class1 7.778 0.99991 11.668 3.5004 0.66661 0.66661
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 2.1909 2.1909 1.0081 1.0081 2.2124 2.2129
Queue1 Class1 7.7331 0.99977 11.573 3.472 0.67574 0.68022
SSA analysis [method: default/nrm; type: approximate, randomized; lang: java; env: python] completed in <t>
>s. Iterations: 5000.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 2.251 2.251 1 1 2.2422 2.251
Queue1 Class1 7.749 0.99977 11.626 3.4879 0.67529 0.66651
Fluid analysis [method: minnormal; type: approximate, deterministic; lang: java; env: python] completed in
<t>s.
Fluid analysis [method: default/minnormal; type: approximate, deterministic; lang: java; env: python]
completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
... (13 captured-output lines omitted; rerun the source example for the full output)
Queue1 Class1 7.778 0.99991 11.668 3.5004 0.66661 0.66661
LDES events: 100 200 300 400 500 600 700 800 900 1000 1100 1200 1300
1400 1500 1600 1700 1800 1900 2000 2100 2200 2300 2400 2500 2600 2700
2800 2900 3000 3100 3200 3300 3400 3500 3600 3700 3800 3900 4000 4100
4200 4300 4400 4500 4600 4700 4800 4900 5000
LDES analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 5000.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 2.1531 2.1849 0.99118 0.99118 2.1885 2.1849
Queue1 Class1 7.8469 1 11.808 3.5425 0.65547 0.65909

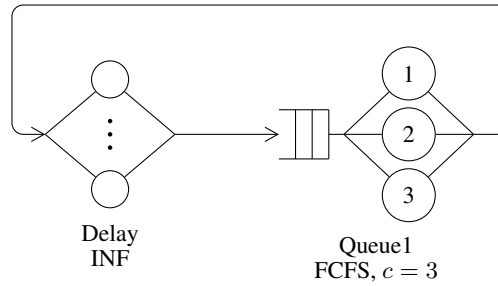
[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 8 solvers the script runs (CTMC, JMT, SSA, FLD, MVA, NC, MAM, LDES), which is the point of the example: the same model read by methods of different cost and different exactness.

2.2.10 `cqn_repairmen_multi`

The repairman model with several repair stations.

The example is built and solved as follows.

Figure 2.23: Topology of `cqn_repairmen_multi`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 4$), Class2 (closed, $N = 2$)
Scheduling	Delay: INF; Queue1: FCFS ($c = 3$)
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(1); Class2: Exp(10)
Solvers	CTMC, LQNS, MVA, NC

Table 2.23: Features of `cqn_repairmen_multi`.

```

Network model = ClosedModel.cqn_repairmen_multi();
show("CTMC", () -> new CTMC(model).getAvgTable());
// THE MULTISERVER RULE IS AN APPROXIMATION, not a setting: the golden holds
// the softmin table because that is the one the reference prints first.
show("MVA", () -> new MVA(model, "config.multiserver", "softmin").getAvgTable());
show("MVA", () -> new MVA(model, "config.multiserver", "seidmann").getAvgTable());
show("NC", () -> new NC(model).getAvgTable());

pauseForUser();

```

Running this edition prints

```

CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass   QLen    Util    RespT    ResidT    ArvR    Tput
Delay   Class1     1.9455   1.9455     1         1   1.9455   1.9455
Delay   Class2     1.6383   1.6383     1         1   1.6383   1.6383
Queue1  Class1     2.0545   0.64849   1.0561   1.0561   1.9455   1.9455
Queue1  Class2     0.36172  0.054609  0.22079  0.22079  1.6383   1.6383
MVA analysis [method: default/lin; type: approximate, deterministic; lang: java; env: python] completed in
<t>s. Iterations: 504.
Station JobClass   QLen    Util    RespT    ResidT    ArvR    Tput
Delay   Class1     2.4671   2.4671     1         1   2.4671   2.4671
Delay   Class2     1.3804   1.3804     1         1   1.3804   1.3804
Queue1  Class1     1.5329   0.82236   0.62136   0.62136  2.4671   2.4671
Queue1  Class2     0.61962  0.046013  0.44888   0.44888  1.3804   1.3804
MVA analysis [method: default/lin; type: approximate, deterministic; lang: java; env: python] completed in
<t>s. Iterations: 127.
Station JobClass   QLen    Util    RespT    ResidT    ArvR    Tput
Delay   Class1     1.7794   1.7794     1         1   1.7794   1.7794
Delay   Class2     1.2993   1.2993     1         1   1.2993   1.2993
Queue1  Class1     2.2206   0.59313   1.248     1.248   1.7794   1.7794
Queue1  Class2     0.7007   0.04331   0.53929   0.53929  1.2993   1.2993
NC analysis [method: default/comom; type: exact, deterministic; lang: java; env: python] completed in <t>s
.
Station JobClass   QLen    Util    RespT    ResidT    ArvR    Tput
Delay   Class1     1.7852   1.7852     1         1   1.7852   1.7852
Delay   Class2     1.7418   1.7418     1         1   1.7418   1.7418
Queue1  Class1     2.2148   0.58399   1.2406   1.2406   1.7852   1.7852
Queue1  Class2     0.2582   0.063355  0.14824   0.14824  1.7418   1.7418

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 3 solvers the script runs (CTMC, MVA, NC), which is the point of the example: the same model read by methods of different cost and different exactness.

2.2.11 cqn_scheduling_dps

Discriminatory processor sharing, with the class weights that PS ignores.

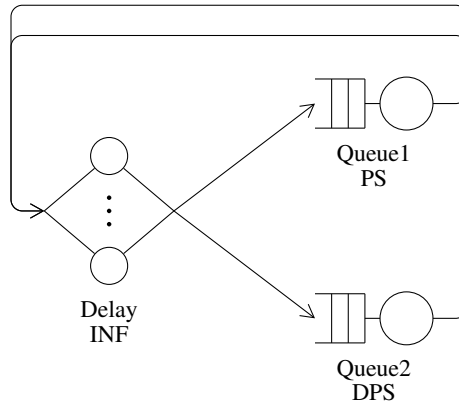


Figure 2.24: Topology of `cqn_scheduling_dps`: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	2: Class1 (closed, $N = 2$), Class2 (closed, $N = 1$)
Scheduling	Delay: INF; Queue1: PS; Queue2: DPS
Service	Delay – Class1: Exp(3); Class2: Exp(0.5) Queue1 – Class1: Exp(0.1); Class2: Exp(1) Queue2 – Class1: Exp(0.1); Class2: Exp(1)
Solvers	CTMC, JMT, FLD, MVA, LDES

Table 2.24: Features of `cqn_scheduling_dps`.

The example is built and solved as follows.

```
Network model = ClosedModel.cqn_scheduling_dps();
show("CTMC", () -> new CTMC(model).getAvgTable());
show("JMT", () -> new JMT(model, "seed", 23000, "samples", 10000).getAvgTable());
show("FLD", () -> new FLD(model).getAvgTable());
show("MVA", () -> new MVA(model).getAvgTable());
show("LDES", () -> new LDES(model, "seed", 23000, "samples", 10000).getAvgTable());

pauseForUser();
```

Running this edition prints

```
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.037231 0.037231 0.33333 0.33333 0.11169 0.11169
Delay Class2 0.58654 0.58654 2 2 0.29327 0.29327
Queue1 Class1 0.53462 0.33508 15.955 4.7864 0.033508 0.033508
Queue1 Class2 0.29844 0.20529 1.4537 1.0176 0.20529 0.20529
Queue2 Class1 1.4282 0.78186 18.266 12.786 0.078186 0.078186
Queue2 Class2 0.11502 0.087981 1.3074 0.39221 0.087981 0.087981
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.037452 0.037452 0.32388 0.32388 0.11054 0.11054
Delay Class2 0.59463 0.59463 2.0208 2.0208 0.29296 0.29168
Queue1 Class1 0.51061 0.34442 15.763 4.7288 0.033025 0.033031
Queue1 Class2 0.29395 0.19848 1.4077 0.98538 0.20436 0.20441
Queue2 Class1 1.4321 0.81316 17.991 12.594 0.078776 0.078876

... (21 captured-output lines omitted; rerun the source example for the full output)

Delay Class1 0.036927 0.036066 0.34062 0.34062 0.1082 0.1082
Delay Class2 0.58166 0.58427 1.9903 1.9903 0.29213 0.29213
Queue1 Class1 0.56762 0.36037 16.743 5.0229 0.032459 0.033902
Queue1 Class2 0.30045 0.2007 1.4656 1.0259 0.20449 0.20494
Queue2 Class1 1.3954 0.77226 18.692 13.084 0.075739 0.074296
```

```
Queue2  Class2      0.11789  0.090243  1.3508  0.40523  0.08764  0.0872
[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 5 solvers the script runs (CTMC, JMT, FLD, MVA, LDES), which is the point of the example: the same model read by methods of different cost and different exactness.

2.2.12 cqn_threeclass_hyperl

The three-class version, with multiserver stations.

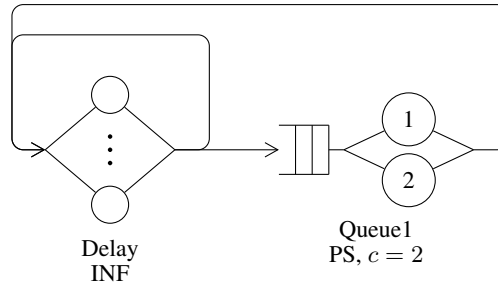


Figure 2.25: Topology of `cqn_threeclass_hyperl`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	3: Class1 (closed, $N = 2$), Class2 (closed, $N = 0$), Class3 (closed, $N = 1$)
Scheduling	Delay: INF; Queue1: PS ($c = 2$)
Service	Delay – Class1: Erlang(3,2); Class2: HyperExp(0.5,0.5;3,10); Class3: Exp(1) Queue1 – Class1: HyperExp(0.1,0.9;1,10); Class2: Exp(2); Class3: Exp(3)
Solvers	CTMC, JMT, SSA, FLD, MVA, NC, MAM, LDES

Table 2.25: Features of `cqn_threeclass_hyperl`.

The example is built and solved as follows.

```
Network model = ClosedModel.cqn_threeclass_hyperl();
show("CTMC", () -> new CTMC(model).getAvgTable());
show("JMT", () -> new JMT(model, "seed", 23000, "samples", 5000).getAvgTable());
show("SSA", () -> new SSA(model, "seed", 23000, "samples", 5000).getAvgTable());
show("FLD", () -> new FLD(model).getAvgTable());
show("MVA", () -> new MVA(model).getAvgTable());
show("NC", () -> new NC(model, "method", "exact").getAvgTable());
show("MAM", () -> new MAM(model).getAvgTable());
show("LDES", () -> new LDES(model, "seed", 23000, "samples", 5000).getAvgTable());

pauseForUser();
```

Running this edition prints

```
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 1.1235 1.1235 0.66667 0.39683 1.6853 1.6853
Delay Class2 0.2483 0.2483 0.21667 0.087698 1.146 1.146
Delay Class3 0.74132 0.74132 1 1 0.74132 0.74132
Queue1 Class1 0.033246 0.01601 0.19727 0.011742 0.16853 0.16853
Queue1 Class2 0.59492 0.2865 0.51913 0.21012 1.146 1.146
Queue1 Class3 0.25868 0.12355 0.34894 0.34894 0.74132 0.74132
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 1.1301 1.1301 0.66008 0.3929 1.7071 1.7092
Delay Class2 0.24356 0.24356 0.20999 0.084996 1.1618 1.1618
Delay Class3 0.73801 0.73801 1.0024 1.0024 0.74033 0.74028
Queue1 Class1 0.038038 0.014299 0.20157 0.011999 0.16551 0.16612
Queue1 Class2 0.60099 0.30343 0.51428 0.20816 1.1618 1.1622
```

```
... (46 captured-output lines omitted; rerun the source example for the full output)
```

```
Delay   Class1    1.1251    1.1302    0.66335    0.39485    1.668    1.6953
Delay   Class2    0.24232    0.24423    0.21479    0.086939   1.1503    1.1272
Delay   Class3    0.73684    0.71338    1.029     1.029     0.71338    0.71338
Queue1   Class1    0.032402   0.015879   0.1947    0.011589   0.16953    0.16642
Queue1   Class2    0.60015    0.2898     0.53241    0.2155     1.1272     1.1261
Queue1   Class3    0.26316    0.12628    0.36833    0.36833    0.71338    0.71338
```

```
[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 8 solvers the script runs (CTMC, JMT, SSA, FLD, MVA, NC, MAM, LDES), which is the point of the example: the same model read by methods of different cost and different exactness.

2.2.13 cqn_twoclass_erl

Two classes with Erlang service over PS and INF stations, exercising the RAND, round-robin and weighted round-robin routing policies.

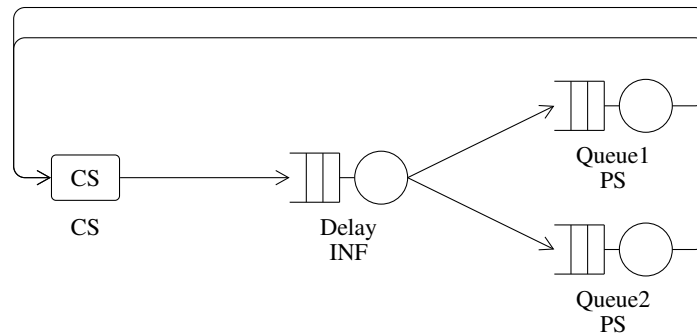


Figure 2.26: Topology of `cqn_twoclass_erl`: a closed network over 3 queueing stations with a class-switch node.

Nodes	4: Queue $\times$ 3, ClassSwitch
Classes	2: Class1 (closed, $N = 15$), Class2 (closed, $N = 5$)
Scheduling	Queue1: PS; Queue2: PS; Delay: INF
Service	Queue1 – Class1: Exp(0.6667); Class2: Erlang(1.333,2) Queue2 – Class1: Erlang(1.333,2); Class2: Exp(0.6667) Delay – Class1: Exp(1); Class2: Exp(1)
Features	class switching on 3 links
Solvers	JMT, LDES

Table 2.26: Features of `cqn_twoclass_erl`.

The example is built and solved as follows.

```
Network model = ClosedModel.cqn_twoclass_erl();
show("JMT", () -> new JMT(model, "seed", 23000).getAvgTable());
show("LDES", () -> new LDES(model, "seed", 23000).getAvgTable());

pauseForUser();
```

Running this edition prints

```
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Queue1 Class1 0.90422 0.4215 3.0843 0.77107 0.28506 0.28397
Queue1 Class2 0.5383 0.28473 2.8414 0.47356 0.19131 0.18984
Queue2 Class1 7.4591 0.44033 26.209 6.5522 0.287 0.28698
```

```

Queue2  Class2      9.92  0.55965  26.615  8.8718  0.3808  0.37815
Delay   Class1      0.57561 0.57561  1.0061  0.50304 0.56853 0.56405
Delay   Class2      0.54055 0.54055  0.97771 0.48885 0.56807 0.56532
LDES events: 4000 8000 12000 16000 20000 24000 28000 32000 36000 40000 44000 48000 52000
              56000 60000 64000 68000 72000 76000 80000 84000 88000 92000 96000 100000 104000 108000
              112000 116000 120000 124000 128000 132000 136000 140000 144000 148000 152000 156000 160000 164000
              168000 172000 176000 180000 184000 188000 192000 196000 200000 200000
LDES analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 200000.
Station JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
Queue1  Class1    0.9522 0.43593 3.3372 0.83431 0.28629 0.28629
Queue1  Class2    0.58817 0.28755 3.0874 0.51456 0.19086 0.19086
Queue2  Class1    7.3688 0.42701 25.739 6.4347 0.28629 0.2863
Queue2  Class2    9.9439 0.57299 26.037 8.6791 0.38172 0.3817
Delay   Class1    0.57314 0.57258 1.0008 0.50038 0.57257 0.57258
Delay   Class2    0.57374 0.57258 1.0021 0.50104 0.57259 0.57258

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, LDES), which is the point of the example: the same model read by methods of different cost and different exactness.

2.2.14 cqn_twoclass_hyperl

Two classes with hyperexponential and Erlang service on PS stations.

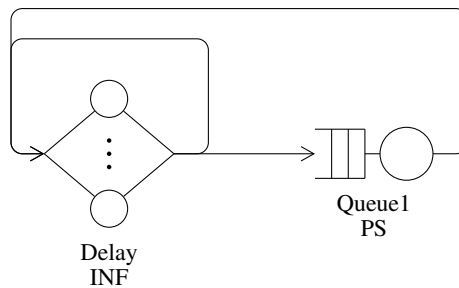


Figure 2.27: Topology of cqn_twoclass_hyperl: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 2$), Class2 (closed, $N = 2$)
Scheduling	Delay: INF; Queue1: PS
Service	Delay – Class1: Erlang(3,2); Class2: HyperExp(0.5,0.5;3,10) Queue1 – Class1: HyperExp(0.1,0.9;1,10); Class2: Exp(1)
Solvers	CTMC, JMT, SSA, FLD, MVA, NC, MAM, LDES

Table 2.27: Features of cqn_twoclass_hyperl.

The example is built and solved as follows.

```

Network model = ClosedModel.cqn_twoclass_hyperl();
show("CTMC", () -> new CTMC(model).getAvgTable());
show("JMT", () -> new JMT(model, "seed", 23000, "samples", 5000).getAvgTable());
show("SSA", () -> new SSA(model, "seed", 23000, "samples", 5000).getAvgTable());
show("FLD", () -> new FLD(model).getAvgTable());
show("MVA", () -> new MVA(model, "method", "exact").getAvgTable());
show("NC", () -> new NC(model, "method", "exact").getAvgTable());
show("MAM", () -> new MAM(model).getAvgTable());
show("LDES", () -> new LDES(model, "seed", 23000, "samples", 5000).getAvgTable());

pauseForUser();

```

Running this edition prints

```

CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.

```

```

Station JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Delay   Class1         0.93078    0.93078    0.66667    0.39683    1.3962    1.3962
Delay   Class2         0.2057     0.2057    0.21667    0.087698   0.94939   0.94939
Queue1  Class1         0.077835   0.026527  0.55749    0.033184   0.13962   0.13962
Queue1  Class2         2.7857     0.94939   2.9342     1.1876     0.94939   0.94939
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Delay   Class1         0.93008    0.93008    0.66993    0.39877    1.4084    1.4078
Delay   Class2         0.20202    0.20202    0.21432    0.086747   0.9506    0.95064
Queue1  Class1         0.074672   0.018632  0.54991    0.032733   0.13725   0.13724
Queue1  Class2         2.8076     0.95301   2.9302     1.186      0.95064   0.95063
SSA analysis [method: default/nrm; type: approximate, randomized; lang: java; env: python] completed in <t>
>s. Iterations: 5000.
Station JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Delay   Class1         1.0033     1.0033    0.6656     0.39619    1.4262    1.5073

... (29 captured-output lines omitted; rerun the source example for the full output)

LDES analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 5000.
Station JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Delay   Class1         0.97939    0.99866   0.65869    0.39208    1.4591    1.498
Delay   Class2         0.21006    0.21195   0.21289    0.086169   1.0217    0.97823
Queue1  Class1         0.085317   0.030225  0.60334    0.035913   0.1498    0.15358
Queue1  Class2         2.7252     0.9423    2.7906     1.1295     0.97823   0.97899

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 8 solvers the script runs (CTMC, JMT, SSA, FLD, MVA, NC, MAM, LDES), which is the point of the example: the same model read by methods of different cost and different exactness.

2.2.15 cqn_twoqueues_multi

Two FCFS queues and a Delay with several closed classes.

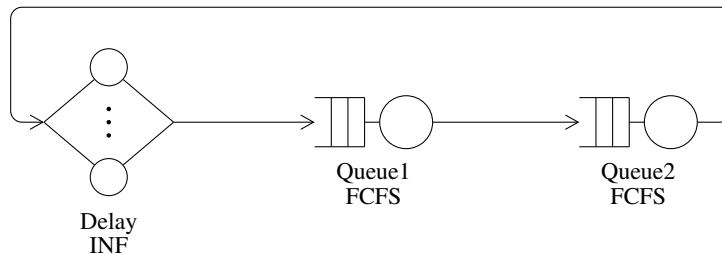


Figure 2.28: Topology of `cqn_twoqueues_multi`: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	2: Class1 (closed, $N = 10$), Class2 (closed, $N = 10$)
Scheduling	Delay: INF; Queue1: FCFS; Queue2: FCFS
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(0.6667); Class2: Exp(0.6667) Queue2 – Class1: Exp(0.3333); Class2: Exp(0.3333)
Solvers	MVA

Table 2.28: Features of `cqn_twoqueues_multi`.

The example is built and solved as follows.

```

Network model = ClosedModel.cqn_twoqueues_multi();
MVA solver = new MVA(model);

solver.getAvgTable().print();

pauseForUser();

```

Running this edition prints

```
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
Station JobClass      QLen      Util  RespT  ResidT      ArvR      Tput
Delay   Class1      0.16667  0.16667      1      1  0.16667  0.16667
Delay   Class2      0.16667  0.16667      1      1  0.16667  0.16667
Queue1  Class1      0.49999  0.25      3      3  0.16667  0.16667
Queue1  Class2      0.49999  0.25      3      3  0.16667  0.16667
Queue2  Class1      9.33333  0.5      56     56  0.16667  0.16667
Queue2  Class2      9.33333  0.5      56     56  0.16667  0.16667

[Running in non-interactive mode, continuing...]
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

2.3 Mixed networks

A mixed network carries open and closed classes at the same stations. The open classes contribute a fixed arrival rate, the closed ones a fixed population, and the two compete for the same servers, so neither can be solved in isolation.

The scripts pair a single-server and a multiserver station under FCFS and under PS. They are deliberately awkward instances: the load-dependent rates they induce are sparse, and they are the models on which the approximate solvers disagree first.

2.3.1 mqn_basic

Open and closed classes at the same stations, the smallest mixed model.

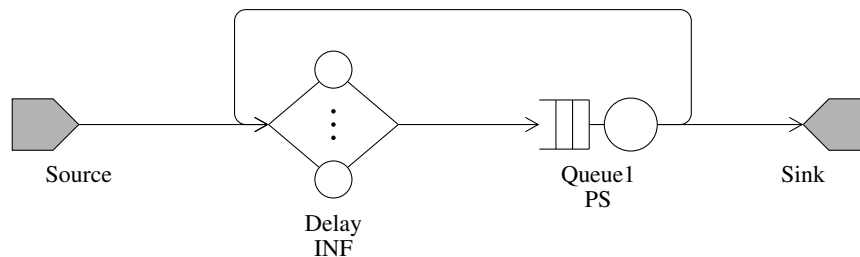


Figure 2.29: Topology of mqn_basic: an open network over 2 queueing stations.

Nodes	4: Source, Queue, Delay, Sink
Classes	2: ClosedClass (closed, $N = 2$), OpenClass (open)
Scheduling	Delay: INF; Queue1: PS
Arrivals	Source – ClosedClass: Disabled; OpenClass: Exp(0.1)
Service	Delay – ClosedClass: Erlang(3,2); OpenClass: HyperExp(0.5,0.5;3,10) Queue1 – ClosedClass: HyperExp(0.1,0.9;1,10); OpenClass: Exp(1)
Features	class switching on 1 link
Solvers	CTMC, JMT, SSA, MVA, LDES

Table 2.29: Features of mqn_basic.

The example is built and solved as follows.

```
Network model = MixedModel.mqn_basic();

// Create multiple solvers for comparison as in the notebook
CTMC solverCTMC = new CTMC(model, "cutoff", 3);
JMT solverJMT = new JMT(model, "seed", 23000, "verbose", true, "keep", false);
```

```

SSA solverSSA = new SSA(model, "seed", 23000, "verbose", false, "samples", 10000);
FLD solverFluid = new FLD(model);
MVA solverMVA = new MVA(model);
NC solverNC = new NC(model);
MAM solverMAM = new MAM(model);

// Solve and display results for each solver
Object[] solvers = {solverCTMC, solverJMT, solverSSA, solverFluid, solverMVA, solverNC,
    solverMAM};

for (Object solverObj : solvers) {
    try {
        if (solverObj instanceof CTMC) {
            CTMC solver = (CTMC) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof JMT) {
            JMT solver = (JMT) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof SSA) {
            SSA solver = (SSA) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof FLD) {

... (11 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

        }
    } catch (Exception e) {
        System.out.println("Error with solver: " + e.getMessage());
    }
}
pauseForUser();

```

Running this edition prints

```

WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 3 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Delay   ClosedClass    1.4362     1.4362     0.66667     0.66667     2.1543     2.1543
Delay   OpenClass       0.021607   0.021607   0.21667     0.21667     0.099723   0.099723
Queue1  ClosedClass    0.5638     0.40932    0.26171     0.26171     2.1543     2.1543
Queue1  OpenClass       0.17223    0.099723   1.7271     1.7271     0.099723   0.099723
Source  OpenClass        0          0          0          0          0          0.099723
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Delay   ClosedClass    1.4528     1.4528     0.66644     0.66644     2.168      2.1536
Delay   OpenClass       0.021961   0.021961   0.21581     0.21581     0.10001    0.10001
Queue1  ClosedClass    0.54717    0.40216    0.25983     0.25983     2.1536     2.1602
Queue1  OpenClass       0.16391    0.095781   1.711       1.711       0.10001    0.10001
Source  OpenClass        0          0          0          0          0          0.10001

... (23 captured-output lines omitted; rerun the source example for the full output)

Station JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Delay   ClosedClass    1.436     1.436     0.66667     0.66667     2.154     2.154
Delay   OpenClass       0.021667   0.021667   0.21667     0.21667     0.1       0.1
Queue1  ClosedClass    0.56402    0.40925    0.26185     0.26185     2.154     2.154
Queue1  OpenClass       0.17379    0.1       1.7379     1.7379     0.1       0.1
Source  OpenClass        0          0          0          0          0          0.1

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 6 solvers the script runs (CTMC, JMT, FLD, MVA, NC, MAM), which is the point of the example: the same model read by methods of different cost and different exactness.

2.3.2 mqn_multichain_cs

A mixed network whose classes are grouped into several chains by class switching.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository

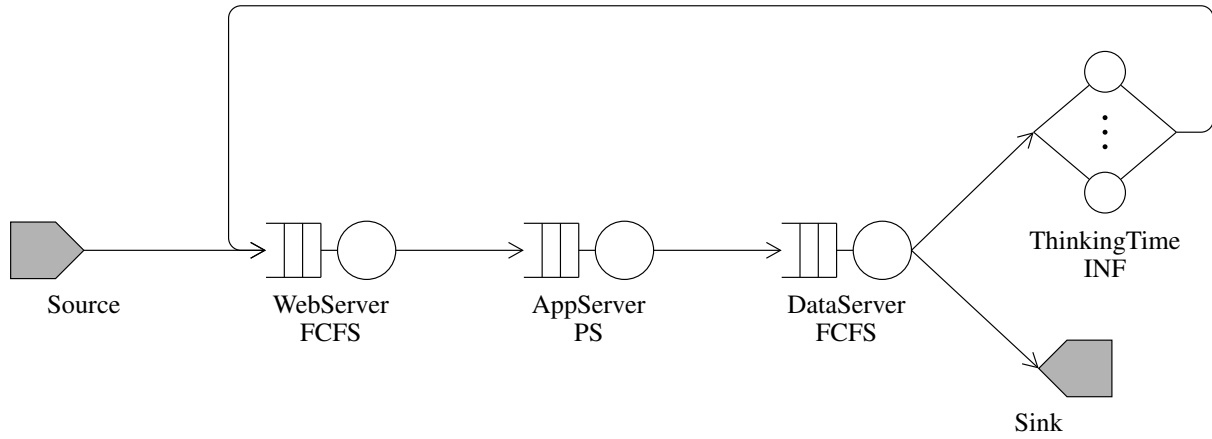


Figure 2.30: Topology of `mqn_multichain_cs`: an open network over 4 queueing stations.

Nodes	6: Source, Queue $\times$ 3, Delay, Sink
Classes	5: InteractiveA (closed, $N = 3$), InteractiveB (closed, $N = 2$), BatchA (open), BatchB (open), ExternalLoad (open)
Scheduling	ThinkingTime: INF; WebServer: FCFS; AppServer: PS; DataServer: FCFS
Arrivals	Source – InteractiveA: Disabled; InteractiveB: Disabled; BatchA: Exp(0.1); BatchB: Exp(0.05); ExternalLoad: Exp(0.05)
Service	ThinkingTime – InteractiveA: Exp(0.6667); InteractiveB: Exp(0.5); BatchA: Disabled; BatchB: Disabled; ExternalLoad: Disabled WebServer – InteractiveA: Exp(2.5); InteractiveB: Exp(2); BatchA: Exp(3.333); BatchB: Exp(2.857); ExternalLoad: Exp(5) AppServer – InteractiveA: Exp(1.25); InteractiveB: Exp(1); BatchA: Exp(1.667); BatchB: Exp(1.429); ExternalLoad: Exp(2) DataServer – InteractiveA: Exp(2); InteractiveB: Exp(1.667); BatchA: Exp(0.8333); BatchB: Exp(0.6667); ExternalLoad: Exp(1.25)
Features	drop rule {'BatchA': 'waitingQueue', 'BatchB': 'waitingQueue', 'ExternalLoad': 'waitingQueue'} at ThinkingTime; class switching on 1 link
Solvers	MVA, SSA

Table 2.30: Features of `mqn_multichain_cs`.

for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.3.3 mqn_multiserver_fcfs

The multiserver FCFS version of the same model.

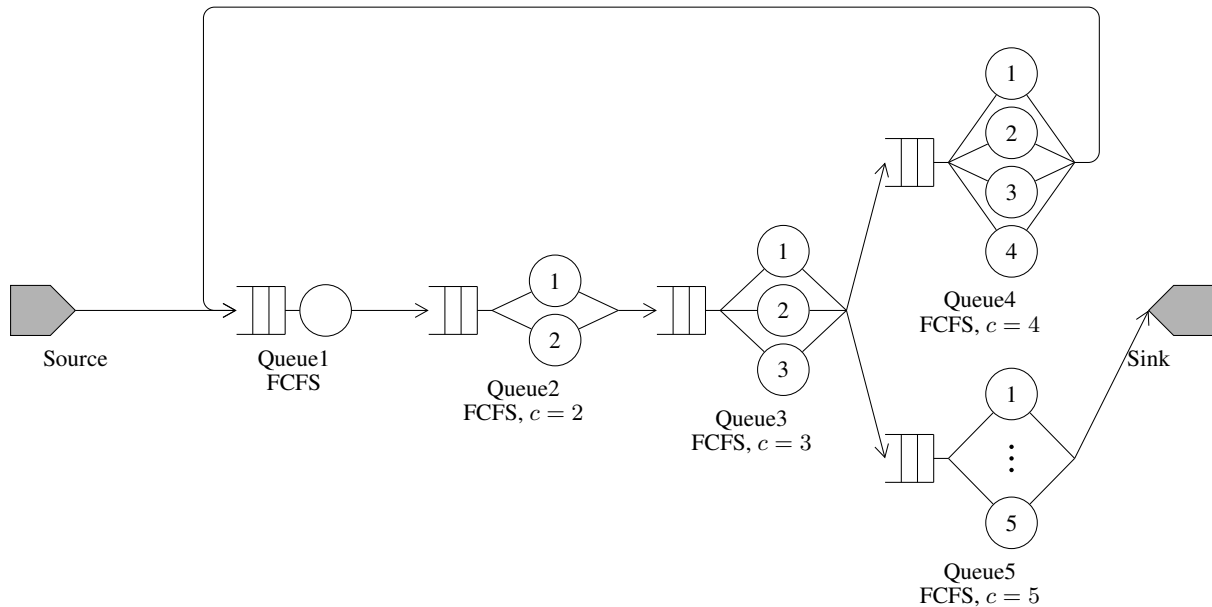


Figure 2.31: Topology of mqn_multiserver_fcfs: an open network over 5 queueing stations.

Nodes	7: Source, Queue $\times$ 5, Sink
Classes	2: ClosedClass (closed, $N = 3$), OpenClass (open)
Scheduling	Queue1: FCFS; Queue2: FCFS ($c = 2$); Queue3: FCFS ($c = 3$); Queue4: FCFS ($c = 4$); Queue5: FCFS ($c = 5$)
Arrivals	Source – ClosedClass: Disabled; OpenClass: Exp(0.3)
Service	Queue1 – ClosedClass: Exp(1); OpenClass: Exp(1) Queue2 – ClosedClass: Exp(2); OpenClass: Exp(1.414) Queue3 – ClosedClass: Exp(3); OpenClass: Exp(1.732) Queue4 – ClosedClass: Exp(4); OpenClass: Exp(2) Queue5 – ClosedClass: Exp(5); OpenClass: Exp(2.236)
Features	class switching on 3 links
Solvers	CTMC, JMT, SSA, MVA, LDES

Table 2.31: Features of mqn_multiserver_fcfs.

The example is built and solved as follows.

```
Network model = MixedModel.mqn_multiserver_fcfs();

// Create solvers with appropriate configurations as in notebook
try {
    CTMC solverCTMC = new CTMC(model, "cutoff", 3);
    solverCTMC.getAvgTable().print();
} catch (Exception e) {
    System.out.println("CTMC solver not available: " + e.getMessage());
}

try {
    JMT solverJMT = new JMT(model, "samples", 100000, "seed", 23000);
    solverJMT.getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT solver not available: " + e.getMessage());
}
```

```

}

try {
    SSA solverSSA = new SSA(model, "cutoff", 3, "seed", 23000);
    solverSSA.getAvgTable().print();
} catch (Exception e) {
    System.out.println("SSA solver not available: " + e.getMessage());
}

try {
    MVA solverMVA = new MVA(model);

    ... (6 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

    MAM solverMAM = new MAM(model, "cutoff", 3, "seed", 23000, "keep", false, "verbose", true);
    solverMAM.getAvgTable().print();
} catch (Exception e) {
    System.out.println("MAM solver not available: " + e.getMessage());
}

pauseForUser();

```

Running this edition prints

```

WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 3 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Queue1 ClosedClass 2.1969 0.72 3.0513 3.0513 0.72 0.72
Queue1 OpenClass 0.95338 0.24763 3.8501 3.8501 0.24763 0.24763
Queue2 ClosedClass 0.38259 0.18 0.53137 0.53137 0.72 0.72
Queue2 OpenClass 0.1841 0.087549 0.74346 0.74346 0.24763 0.24763
Queue3 ClosedClass 0.2405 0.08 0.33402 0.33402 0.72 0.72
Queue3 OpenClass 0.14315 0.047656 0.57809 0.57809 0.24763 0.24763
Queue4 ClosedClass 0.18 0.045 0.25 0.25 0.72 0.72
Queue5 OpenClass 0.11074 0.022148 0.44721 0.44721 0.24763 0.24763
Source OpenClass 0 0 0 0 0 0.24763
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Queue1 ClosedClass 2.245 0.6757 3.3433 3.3433 0.67091 0.67092

... (35 captured-output lines omitted; rerun the source example for the full output)

Queue2 OpenClass 0.22745 0.10607 0.75817 0.75817 0.3 0.3
Queue3 ClosedClass 0.23504 0.078197 0.33397 0.33397 0.70378 0.70378
Queue3 OpenClass 0.1735 0.057735 0.57834 0.57834 0.3 0.3
Queue4 ClosedClass 0.17594 0.043986 0.25 0.25 0.70378 0.70378
Queue5 OpenClass 0.13416 0.026833 0.44721 0.44721 0.3 0.3
Source OpenClass 0 0 0 0 0 0.3

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 5 solvers the script runs (CTMC, JMT, SSA, MVA, MAM), which is the point of the example: the same model read by methods of different cost and different exactness.

2.3.4 mqn_multiserver_ps

The multiserver PS version.

The example is built and solved as follows.

```

Network model = MixedModel.mqn_multiserver_ps();

// Create solvers with appropriate configurations as in notebook
try {
    CTMC solverCTMC = new CTMC(model, "cutoff", 3);
    solverCTMC.getAvgTable().print();
} catch (Exception e) {
    System.out.println("CTMC solver not available: " + e.getMessage());
}

```

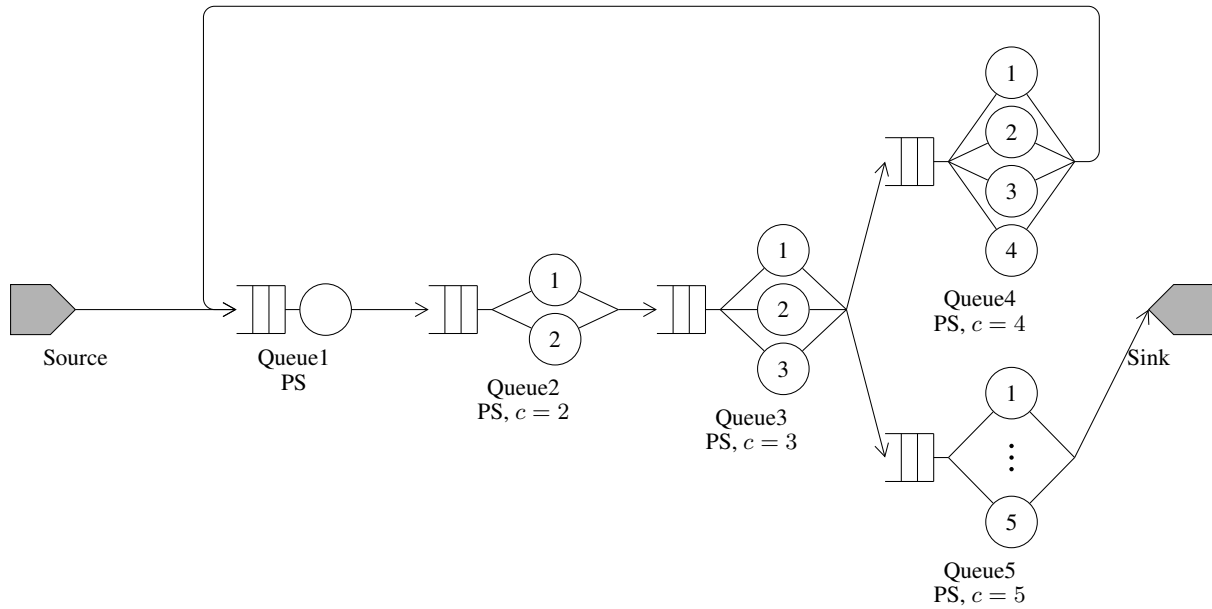


Figure 2.32: Topology of `mqn_multiserver_ps`: an open network over 5 queueing stations.

Nodes	7: Source, Queue $\times$ 5, Sink
Classes	2: ClosedClass (closed, $N = 3$), OpenClass (open)
Scheduling	Queue1: PS; Queue2: PS ($c = 2$); Queue3: PS ($c = 3$); Queue4: PS ($c = 4$); Queue5: PS ($c = 5$)
Arrivals	Source – ClosedClass: Disabled; OpenClass: Exp(0.3)
Service	Queue1 – ClosedClass: Exp(1); OpenClass: Exp(1) Queue2 – ClosedClass: Exp(2); OpenClass: Exp(1.414) Queue3 – ClosedClass: Exp(3); OpenClass: Exp(1.732) Queue4 – ClosedClass: Exp(4); OpenClass: Exp(2) Queue5 – ClosedClass: Exp(5); OpenClass: Exp(2.236)
Features	class switching on 3 links
Solvers	CTMC, JMT, MVA, LDES

Table 2.32: Features of `mqn_multiserver_ps`.

```

try {
    JMT solverJMT = new JMT(model, "samples", 100000, "seed", 23000);
    solverJMT.getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT solver not available: " + e.getMessage());
}

try {
    MVA solverMVA = new MVA(model, "method", "exact");
    solverMVA.getAvgTable().print();
} catch (Exception e) {
    System.out.println("MVA solver not available: " + e.getMessage());
}

try {
    MAM solverMAM = new MAM(model, "seed", 23000, "keep", false, "verbose", true);
    solverMAM.getAvgTable().print();
} catch (Exception e) {
    System.out.println("MAM solver not available: " + e.getMessage());
}

pauseForUser();

```

Running this edition prints

```

WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 3 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station  JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Queue1   ClosedClass    2.1993    0.72064    3.0519    3.0519    0.72064    0.72064
Queue1   OpenClass       0.9534    0.24747    3.8526    3.8526    0.24747    0.24747
Queue2   ClosedClass    0.37998    0.18016    0.52728    0.52728    0.72064    0.72064
Queue2   OpenClass     0.18649    0.087494    0.75357    0.75357    0.24747    0.24747
Queue3   ClosedClass    0.24057    0.080071    0.33384    0.33384    0.72064    0.72064
Queue3   OpenClass     0.14316    0.047626    0.57849    0.57849    0.24747    0.24747
Queue4   ClosedClass    0.18016    0.04504    0.25       0.25       0.72064    0.72064
Queue5   OpenClass     0.11067    0.022134    0.44721    0.44721    0.24747    0.24747
Source   OpenClass       0         0         0         0         0         0.24747
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station  JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Queue1   ClosedClass    2.2615    0.67284    3.3567    3.3567    0.6722    0.67218

... (24 captured-output lines omitted; rerun the source example for the full output)

Queue2   OpenClass     0.22824    0.10607    0.76081    0.76081    0.3       0.3
Queue3   ClosedClass    0.22475    0.074774    0.33396    0.33396    0.67297    0.67297
Queue3   OpenClass     0.17368    0.057735    0.57894    0.57894    0.3       0.3
Queue4   ClosedClass    0.16824    0.04206    0.25       0.25       0.67297    0.67297
Queue5   OpenClass     0.13416    0.026833    0.44721    0.44721    0.3       0.3
Source   OpenClass       0         0         0         0         0         0.3

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 4 solvers the script runs (CTMC, JMT, MVA, MAM), which is the point of the example: the same model read by methods of different cost and different exactness.

2.3.5 mqn_singleserver_fcfs

A single-server FCFS station under a mixed workload, a sparse load-dependent multiclass instance that stresses the solvers.

The example is built and solved as follows.

```

Network model = MixedModel.mqn_singleserver_fcfs();

// Create solvers as in notebook - CTMC/SSA omitted due to high population
try {
    JMT solverJMT = new JMT(model, "cutoff", 3, "keep", false, "verbose", true, "seed",
23000, "samples", 20000);
    solverJMT.getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT solver not available: " + e.getMessage());
}

```

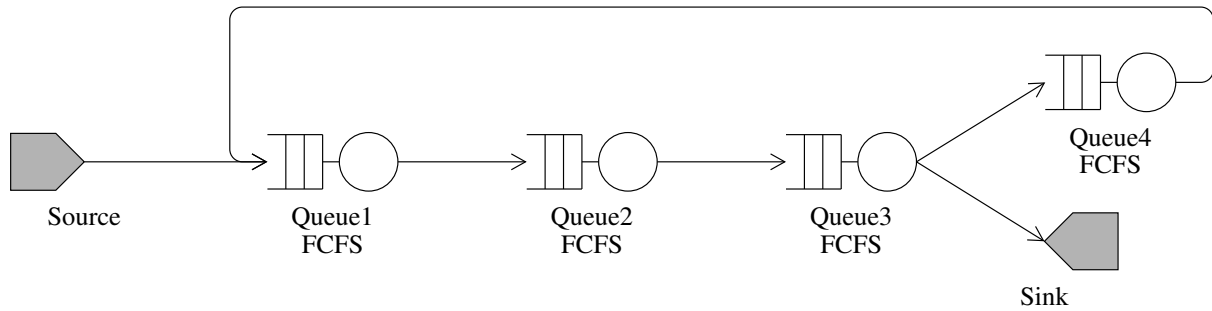


Figure 2.33: Topology of mqn_singleserver_fcfs: an open network over 4 queueing stations.

Nodes	6: Source, Queue $\times$ 4, Sink
Classes	2: ClosedClass (closed, $N = 100$), OpenClass (open)
Scheduling	Queue1: FCFS; Queue2: FCFS; Queue3: FCFS; Queue4: FCFS
Arrivals	Source – ClosedClass: Disabled; OpenClass: APH
Service	Queue1 – ClosedClass: Exp(1); OpenClass: Exp(1) Queue2 – ClosedClass: Exp(2); OpenClass: Exp(1.414) Queue3 – ClosedClass: Exp(3); OpenClass: Exp(1.732) Queue4 – ClosedClass: Exp(4); OpenClass: Exp(2)
Features	class switching on 2 links
Solvers	JMT, MVA, NC, LDES

Table 2.33: Features of mqn_singleserver_fcfs.

```

}

try {
    MVA solverMVA = new MVA(model, "method", "lin", "keep", false, "verbose", true, "seed",
        23000);
    solverMVA.getAvgTable().print();
} catch (Exception e) {
    System.out.println("MVA solver not available: " + e.getMessage());
}

try {
    NC solverNC = new NC(model, "keep", false, "verbose", true, "seed", 23000);
    solverNC.getAvgTable().print();
} catch (Exception e) {
    System.out.println("NC solver not available: " + e.getMessage());
}

try {
    MAM solverMAM = new MAM(model, "keep", false, "verbose", true, "seed", 23000);
    solverMAM.getAvgTable().print();
} catch (Exception e) {
    System.out.println("MAM solver not available: " + e.getMessage());
}

pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass      QLen    Util    RespT    ResidT    ArvR    Tput
Queue1  ClosedClass    98.44   0.68116  143.75   143.75   0.69274  0.68865
Queue1  OpenClass       56.624  0.29771  187.54   187.54   0.30425  0.30471
Queue2  ClosedClass     0.87067  0.3381  1.2547   1.2547   0.68865  0.68863
Queue2  OpenClass       0.53673  0.19108  1.7508   1.7508   0.30471  0.30472
Queue3  ClosedClass     0.43728  0.20633  0.62634  0.62634  0.68863  0.69273
Queue3  OpenClass       0.29725  0.18077  0.96959  0.96959  0.30472  0.29542
Queue4  ClosedClass     0.21893  0.1596  0.31419  0.31419  0.69273  0.69274
Source  OpenClass        0        0        0        0        0.30425
MVA analysis [method: lin; type: approximate, deterministic; lang: java; env: python] completed in <t>s.
Iterations: 232.
Station JobClass      QLen    Util    RespT    ResidT    ArvR    Tput

```

```

Queue1  ClosedClass  98.533  0.66435  148.31  148.31  0.66435  0.66435
Queue1  OpenClass   49.766  0.33333  149.3   149.3   0.33333  0.33333
Queue2  ClosedClass  0.83883  0.33218  1.2626  1.2626  0.66435  0.66435

```

... (19 captured-output lines omitted; rerun the source example for the full output)

```

Queue2  ClosedClass  0.99028  0.30339  1.6321  1.6321  0.60677  0.60677
Queue2  OpenClass   0.90071  0.2357  2.7021  2.7021  0.33333  0.33333
Queue3  ClosedClass  0.43798  0.20226  0.72182  0.72182  0.60677  0.60677
Queue3  OpenClass    0.42  0.19245  1.26  1.26  0.33333  0.33333
Queue4  ClosedClass  0.17882  0.15169  0.29471  0.29471  0.60677  0.60677
Source  OpenClass    0 0 0 0 0 0.33333

```

[Running in non-interactive mode, continuing...]

The rows repeat for each of the 4 solvers the script runs (JMT, MVA, NC, MAM), which is the point of the example: the same model read by methods of different cost and different exactness.

2.3.6 mqn_singleserver_ps

The processor sharing counterpart.

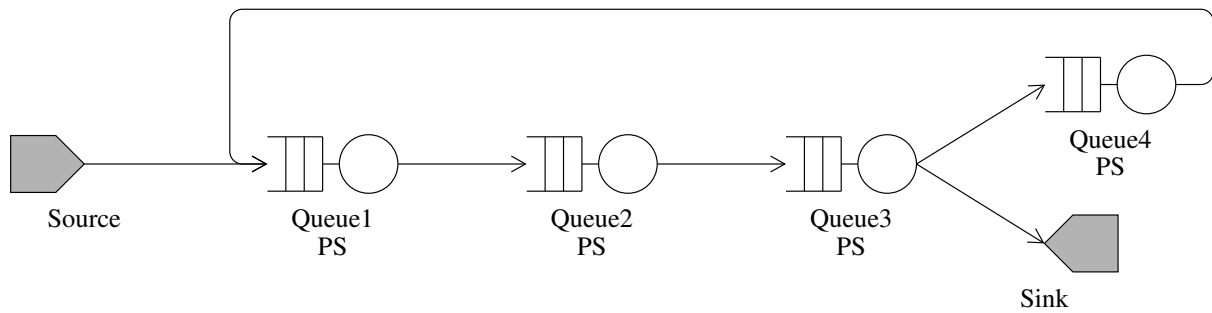


Figure 2.34: Topology of mqn_singleserver_ps: an open network over 4 queueing stations.

Nodes	6: Source, Queue $\times$ 4, Sink
Classes	2: ClosedClass (closed, $N = 100$), OpenClass (open)
Scheduling	Queue1: PS; Queue2: PS; Queue3: PS; Queue4: PS
Arrivals	Source – ClosedClass: Disabled; OpenClass: Exp(0.3)
Service	Queue1 – ClosedClass: Exp(1); OpenClass: Exp(1) Queue2 – ClosedClass: Exp(2); OpenClass: Exp(1.414) Queue3 – ClosedClass: Exp(3); OpenClass: Exp(1.732) Queue4 – ClosedClass: Exp(4); OpenClass: Exp(2)
Features	class switching on 2 links
Solvers	JMT, SSA, FLD, MVA, NC, LDES

Table 2.34: Features of mqn_singleserver_ps.

The example is built and solved as follows.

```

Network model = MixedModel.mqn_singleserver_ps();

// Create solvers as in notebook - comprehensive comparison
try {
    JMT solverJMT = new JMT(model, "cutoff", 3, "keep", false, "verbose", true, "seed",
        23000, "samples", 100000);
    solverJMT.getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT solver not available: " + e.getMessage());
}

try {
    SSA solverSSA = new SSA(model, "seed", 23000, "verbose", false, "samples", 100000);
    solverSSA.getAvgTable().print();
}

```

```

} catch (Exception e) {
    System.out.println("SSA solver not available: " + e.getMessage());
}

try {
    FLD solverFluid = new FLD(model, "cutoff", 3, "keep", false, "verbose", true, "seed",
        23000, "samples", 100000);
    solverFluid.getAvgTable().print();
} catch (Exception e) {
    System.out.println("Fluid solver not available: " + e.getMessage());
}

try {
    MVA solverMVA = new MVA(model, "method", "lin");

    ... (19 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

    LDES solverLDES = new LDES(model, "cutoff", 3, "keep", false, "verbose", true, "seed",
        23000, "samples", 100000);
    solverLDES.getAvgTable().print();
} catch (Exception e) {
    System.out.println("LDES solver not available: " + e.getMessage());
}

pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Queue1 ClosedClass 98.568 0.69772 141.74 141.74 0.70515 0.69804
Queue1 OpenClass 43.069 0.3019 143.85 143.85 0.30007 0.29886
Queue2 ClosedClass 0.82325 0.35985 1.1341 1.1341 0.69804 0.69752
Queue2 OpenClass 0.48659 0.20797 1.618 1.618 0.29886 0.29886
Queue3 ClosedClass 0.39465 0.23505 0.55513 0.55513 0.69752 0.69751
Queue3 OpenClass 0.28616 0.1685 0.96528 0.96528 0.29886 0.29905
Queue4 ClosedClass 0.21695 0.17845 0.30292 0.30292 0.69751 0.69752
Source OpenClass 0 0 0 0 0 0.30007
Fluid analysis [method: dae; type: approximate, deterministic; lang: java; env: python] completed in <t>s.
Fluid analysis [method: default/dae; type: approximate, deterministic; lang: java; env: python] completed
in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Queue1 ClosedClass 98.924 0.7 141.32 141.32 0.7 0.7
Queue1 OpenClass 42.396 0.3 141.32 141.32 0.3 0.3

... (41 captured-output lines omitted; rerun the source example for the full output)

Queue2 ClosedClass 0.82754 0.35638 1.1601 1.1601 0.71161 0.71161
Queue2 OpenClass 0.48802 0.20879 1.6363 1.6363 0.2979 0.2979
Queue3 ClosedClass 0.39489 0.23382 0.55368 0.55368 0.71161 0.71161
Queue3 OpenClass 0.28607 0.17041 0.95876 0.95876 0.2979 0.2979
Queue4 ClosedClass 0.217 0.17767 0.30493 0.30493 0.71161 0.71165
Source OpenClass 0 0 0 0 0 0.3

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 6 solvers the script runs (JMT, FLD, MVA, NC, MAM, LDES), which is the point of the example: the same model read by methods of different cost and different exactness.

2.4 Class switching

A job may change class as it moves. Switching is expressed either by off-diagonal blocks of the routing matrix, which give the probability of leaving node i as class r and arriving at node j as class s , or by a dedicated `ClassSwitch` node that performs the change in place.

Switching is what makes classes and chains differ: the classes that can reach one another form a chain, and it is the chain, not the class, that carries a conserved population. The diamond examples build a switching chain that is reducible on purpose, so that transient classes appear.

2.4.1 cs_implicit

A ClassSwitch node declared explicitly, against the implicit switching that a routing matrix with off-diagonal class blocks already expresses.

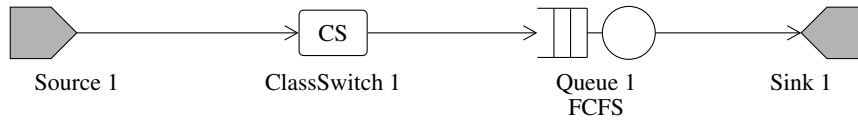


Figure 2.35: Topology of `cs_implicit`: an open network over 1 queueing station with a class-switch node.

Nodes	4: Source, Queue, ClassSwitch, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue 1: FCFS
Arrivals	Source 1 – Class1: Exp(0.1); Class2: Exp(0.5)
Service	Queue 1 – Class1: Exp(1); Class2: Exp(1)
Solvers	MVA

Table 2.35: Features of `cs_implicit`.

The example is built and solved as follows.

```

public static void cs_implicit() throws Exception {
    Network model = ClassSwitchingModel.cs_implicit();
    MVA solver = new MVA(model);

    solver.getAvgChainTable().print();
    pauseForUser();
}

```

Running this edition prints

```

MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 62.
Station Chain JobClasses QLen Util RespT ResidT ArvR Tput
Source 1 Chain1 (Class1 Class2) 0 0 0 0 0 0.6
Queue 1 Chain1 (Class1 Class2) 1.5 0.6 2.5 2.5 0.6 0.6
[Running in non-interactive mode, continuing...]

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

2.4.2 cs_multi_diamond

The same pattern with several branches.

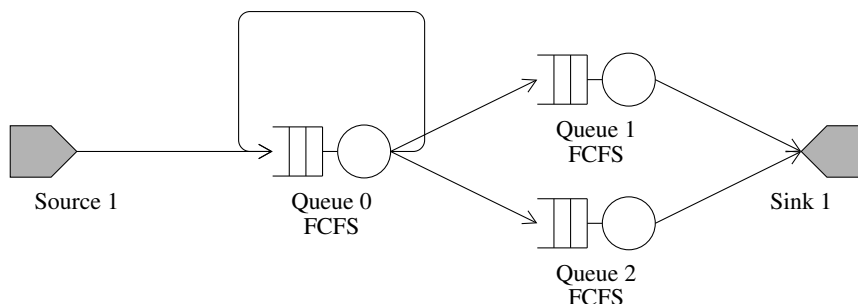


Figure 2.36: Topology of `cs_multi_diamond`: an open network over 3 queueing stations.

Nodes	5: Source, Queue $\times$ 3, Sink
Classes	3: Class1 (open), Class2 (open), Class3 (open)
Scheduling	Queue 0: FCFS; Queue 1: FCFS; Queue 2: FCFS
Arrivals	Source 1 – Class1: Exp(1); Class2: Disabled; Class3: Disabled
Service	Queue 0 – Class1: Exp(10); Class2: Disabled; Class3: Disabled Queue 1 – Class1: Disabled; Class2: Exp(20); Class3: Disabled Queue 2 – Class1: Disabled; Class2: Disabled; Class3: Exp(30)
Features	drop rule {'Class2': 'waitingQueue', 'Class3': 'waitingQueue'} at Queue 0; drop rule {'Class1': 'waitingQueue', 'Class3': 'waitingQueue'} at Queue 1; drop rule {'Class1': 'waitingQueue', 'Class2': 'waitingQueue'} at Queue 2; class switching on 1 link
Solvers	MVA

Table 2.36: Features of `cs_multi_diamond`.

The example is built and solved as follows.

```
public static void cs_multi_diamond() throws Exception {
    Network model = ClassSwitchingModel.cs_multi_diamond();
    MVA solver = new MVA(model);

    solver.getAvgChainTable().print();
    pauseForUser();
}
```

Running this edition prints

```
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 23.
Station Chain JobClasses QLen Util RespT ResidT ArvR Tput
Source 1 Chain1 (Class1 Class2 Class3) 0 0 0 0 0 1
Queue 0 Chain1 (Class1 Class2 Class3) 0.14286 0.125 0.11429 0.14286 1.25 1.25
Queue 1 Chain1 (Class1 Class2 Class3) 0.019108 0.01875 0.050955 0.019108 0.375 0.375
Queue 2 Chain1 (Class1 Class2 Class3) 0.021277 0.020833 0.034043 0.021277 0.625 0.625
[Running in non-interactive mode, continuing...]
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

2.4.3 `cs_single_diamond`

Class switching driven by a reducible Markov chain over the classes, the diamond pattern with one branch.

Nodes	3: Delay $\times$ 3
Classes	3: Class1 (closed, $N = 1$), Class2 (closed, $N = 0$), Class3 (closed, $N = 0$)
Scheduling	Queue 0: INF; Queue 1: INF; Queue 2: INF
Service	Queue 0 – Class1: Exp(1); Class2: Disabled; Class3: Disabled Queue 1 – Class1: Disabled; Class2: Exp(0.5); Class3: Disabled Queue 2 – Class1: Disabled; Class2: Disabled; Class3: Exp(0.3333)
Features	drop rule {'Class2': 'waitingQueue', 'Class3': 'waitingQueue'} at Queue 0; drop rule {'Class1': 'waitingQueue', 'Class3': 'waitingQueue'} at Queue 1; drop rule {'Class1': 'waitingQueue', 'Class2': 'waitingQueue'} at Queue 2
Solvers	MVA

Table 2.37: Features of `cs_single_diamond`.

The example is built and solved as follows.

```
public static void cs_single_diamond() throws Exception {
    Network model = ClassSwitchingModel.cs_single_diamond();
    MVA solver = new MVA(model);
}
```

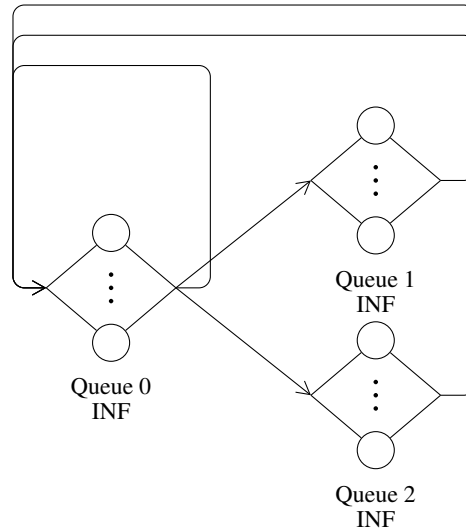


Figure 2.37: Topology of `cs_single_diamond`: a closed network over 3 queueing stations.

```
solver.getAvgChainTable().print();
pauseForUser();
```

Running this edition prints

```
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
Station Chain JobClasses QLen Util RespT ResidT ArvR Tput
Queue 0 Chain1 (Class1 Class2 Class3) 0.32258 0.32258 1 1 0.32258 0.32258
Queue 1 Chain1 (Class1 Class2 Class3) 0.19355 0.19355 2 0.6 0.096774 0.096774
Queue 2 Chain1 (Class1 Class2 Class3) 0.48387 0.48387 3 1.5 0.16129 0.16129

[Running in non-interactive mode, continuing...]
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

2.4.4 `cs_transient_class`

A class that is transient in the switching chain, so it is visited but never returned to.

Nodes	3: Delay $\times$ 3
Classes	3: Class1 (closed, $N = 1$), Class2 (closed, $N = 0$), Class3 (closed, $N = 0$)
Scheduling	Queue 0: INF; Queue 1: INF; Queue 2: INF
Service	Queue 0 – Class1: Exp(1); Class2: Exp(1); Class3: Exp(1) Queue 1 – Class1: Disabled; Class2: Exp(1); Class3: Disabled Queue 2 – Class1: Disabled; Class2: Disabled; Class3: Exp(1)
Features	drop rule {'Class1': 'waitingQueue', 'Class3': 'waitingQueue'} at Queue 1; drop rule {'Class1': 'waitingQueue', 'Class2': 'waitingQueue'} at Queue 2
Solvers	MVA

Table 2.38: Features of `cs_transient_class`.

The example is built and solved as follows.

```
public static void cs_transient_class() throws Exception {
    Network model = ClassSwitchingModel.cs_transient_class();
    model.printRoutingMatrix();

    jline.lang.NetworkStruct sn = model.getStruct(true);
    System.out.println("\n=== rt (station routing matrix) ===");
    sn.rt.print();
}
```

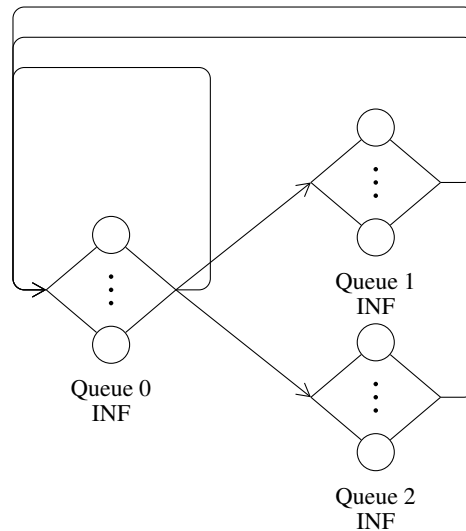


Figure 2.38: Topology of `cs_transient_class`: a closed network over 3 queueing stations.

```
// Debug SCC
int K = sn.nclasses;
int M = sn.nstateful;
jline.util.matrix.Matrix inchain = sn.inchain.get(0);
java.util.List<Integer> cols = new java.util.ArrayList<>();
for (int ist = 0; ist < M; ist++) {
    for (int ik = 0; ik < inchain.length(); ik++) {
        cols.add(ist * K + (int) inchain.get(ik));
    }
}
jline.util.matrix.Matrix Pchain = new jline.util.matrix.Matrix(cols.size(), cols.size());
for (int row = 0; row < cols.size(); row++) {
    for (int col = 0; col < cols.size(); col++) {
        Pchain.set(row, col, sn.rt.get(cols.get(row), cols.get(col)));
    }
}
jline.util.graph.DirectedGraph graph = new jline.util.graph.DirectedGraph(Pchain);
jline.util.graph.DirectedGraph.SCCResult sccResult = graph.stronglyconncomp();

... (15 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

System.out.println("\n=== Per-Class Table ===");
solver.getAvgTable().print();
System.out.println("\n=== Chain Table ===");
solver.getAvgChainTable().print();
pauseForUser();
```

Running this edition prints

```
Queue 0 [Class1] => Queue 0 [Class1] : Pr=0.200000
Queue 0 [Class1] => CS_Queue 0_to_Queue 1 [Class1] : Pr=0.300000
Queue 0 [Class1] => CS_Queue 0_to_Queue 2 [Class1] : Pr=0.500000
Queue 0 [Class2] => CS_Queue 0_to_Queue 1 [Class2] : Pr=1.000000
Queue 0 [Class3] => CS_Queue 0_to_Queue 2 [Class3] : Pr=1.000000
Queue 1 [Class2] => Queue 0 [Class2] : Pr=1.000000
Queue 2 [Class3] => Queue 0 [Class3] : Pr=1.000000
CS_Queue 0_to_Queue 1 [Class2] => Queue 1 [Class2] : Pr=1.000000
CS_Queue 0_to_Queue 2 [Class3] => Queue 2 [Class3] : Pr=1.000000

=== rt (station routing matrix) ===
[0.2  0  0  0  0.3  0  0  0  0.5
 0  0  0  0  1.0  0  0  0  0
 0  0  0  0  0  0  0  0  1.0
 1.0  0  0  0  0  0  0  0  0
 0  1.0  0  0  0  0  0  0  0]
```

```
... (23 captured-output lines omitted; rerun the source example for the full output)
```

```
=== Chain Table ===
```

Station	Chain	JobClasses	QLen	Util	RespT	ResidT	ArvR	Tput
Queue 0	Chain1	(Class1 Class2 Class3)	0.5	0.5	1	1	0.5	0.5
Queue 1	Chain1	(Class1 Class2 Class3)	0.1875	0.1875	1	0.375	0.1875	0.1875
Queue 2	Chain1	(Class1 Class2 Class3)	0.3125	0.3125	1	0.625	0.3125	0.3125

```
[Running in non-interactive mode, continuing...]
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

2.5 Priority models

Priorities are declared on the class and enforced by the discipline of the station. Head-of-line (HOL) service is non-preemptive: a low-priority job in service completes, and only the waiting order changes. The preemptive variants interrupt it instead, and the priority processor sharing family divides the capacity in proportion to the priority weights.

Priority breaks product form, so these models are the province of the exact CTMC solver, of simulation, and of the approximations that the MVA solver offers for HOL.

2.5.1 prio_hol_closed

The closed counterpart.

Nodes	6: Queue $\times$ 4, Delay $\times$ 2
Classes	3: Class1 (closed, $N = 18$), Class2 (closed, $N = 18$), Class3 (closed, $N = 18$)
Scheduling	SlowDelay: INF; FCFSQueue: FCFS; SIROQueue: SIRO; PSQueue: PS; HOLQueue: HOL; FastDelay: INF
Service	SlowDelay – Class1: Exp(0.1); Class2: Exp(0.1); Class3: Exp(0.1) FCFSQueue – Class1: Exp(3.333); Class2: Exp(2); Class3: Exp(1.667) SIROQueue – Class1: Exp(0.9091); Class2: Exp(0.7692); Class3: Exp(0.6667) PSQueue – Class1: Exp(1); Class2: Exp(0.9091); Class3: Exp(0.5263) HOLQueue – Class1: Exp(0.4); Class2: Exp(0.5263); Class3: Exp(0.2326) FastDelay – Class1: Exp(1); Class2: Exp(1); Class3: Exp(1)
Solvers	MVA, JMT, SSA, LDES

Table 2.39: Features of prio_hol_closed.

The example is built and solved as follows.

```
Network model = PrioModel.prio_hol_closed();

// Create solvers as in MATLAB version (lines 82, 84)
MVA solverMVA = new MVA(model, "seed", 23000, "cutoff", 1, "samples", 10000);
JMT solverJMT = new JMT(model, "seed", 23000, "cutoff", 1, "samples", 10000);

solverMVA.getAvgTable().print();
solverJMT.getAvgTable().print();

pauseForUser();
```

Running this edition prints

```
Priority: highest=Class1,Class3, lowest=Class2
Priority: highest=Class1,Class3, lowest=Class2
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 102.
Station  JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
SlowDelay Class1      1.4877     1.4877      10         10        0.14877     0.14877
SlowDelay Class2      0.0078005  0.0078005   10         10        0.00078005  0.00078005
```

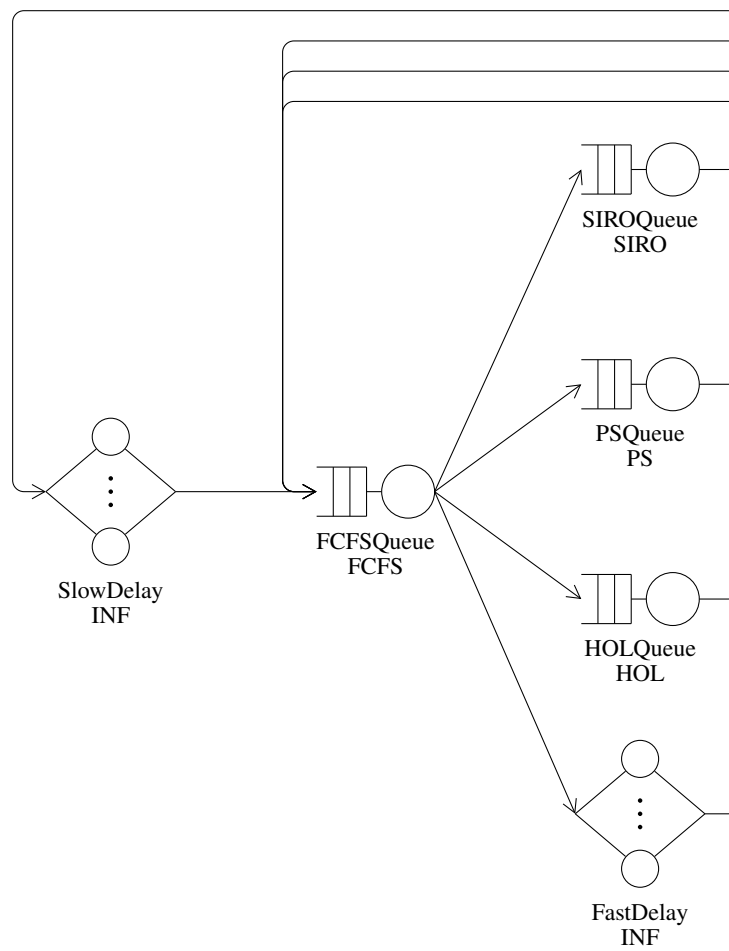


Figure 2.39: Topology of `prio_hol_closed`: a closed network over 6 queueing stations.

```

SlowDelay Class3      1.447      1.447      10      10      0.1447      0.1447
FCFSQueue Class1      0.49316    0.17853    0.8287    3.3148    0.59509    0.59509
FCFSQueue Class2      0.0032351  0.0015601  1.0368    4.1473    0.0031202  0.0031202
FCFSQueue Class3      0.6456     0.34728    1.1154    4.4616    0.5788     0.5788
SIROQueue Class1      0.28096    0.16365    1.8885    1.8885    0.14877    0.14877
SIROQueue Class2      0.0016424  0.0010141  2.1056    2.1056    0.00078005 0.00078005
SIROQueue Class3      0.32966    0.21705    2.2782    2.2782    0.1447     0.1447
PSQueue Class1        0.25406    0.14877    1.7077    1.7077    0.14877    0.14877
PSQueue Class2        0.0014773  0.00085805 1.8939    1.8939    0.00078005 0.00078005
PSQueue Class3        0.46625    0.27493    3.2222    3.2222    0.1447     0.1447

... (20 captured-output lines omitted; rerun the source example for the full output)

PSQueue Class3      0.42896    0.28248    3.1486    3.1486    0.14179    0.14205
HOLQueue Class1     15.239     0.3766     103.3     103.3     0.14672    0.14744
HOLQueue Class2        0          0          0          0          0.17254     0
HOLQueue Class3     14.951     0.6199     103.84    103.84    0.1435     0.14281
FastDelay Class1     0.14942    0.14942    0.99582    0.99582    0.14871    0.14872
FastDelay Class3     0.1443     0.1443     0.9783     0.9783    0.14411    0.14394

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (MVA, JMT), which is the point of the example: the same model read by methods of different cost and different exactness.

2.5.2 prio_hol_open

Head-of-line non-preemptive priority in an open network, with PS, SIRO and FCFS stations alongside.

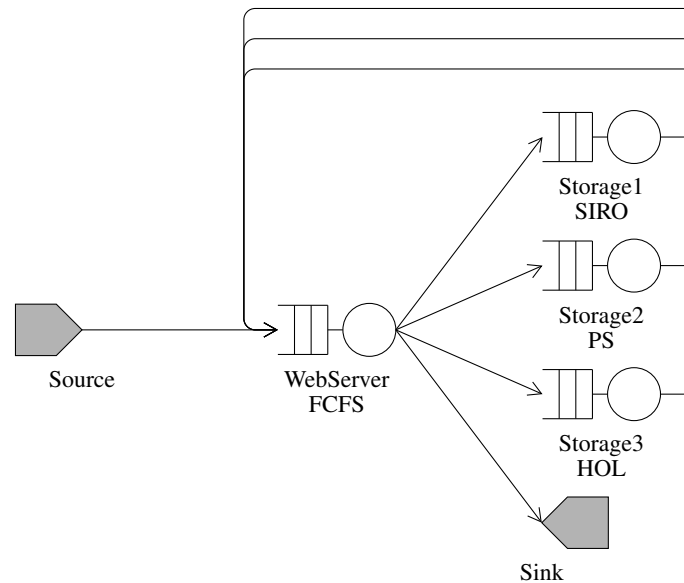


Figure 2.40: Topology of prio_hol_open: an open network over 4 queueing stations.

Nodes	6: Source, Queue × 4, Sink
Classes	3: Class1 (open), Class2 (open), Class3 (open)
Scheduling	WebServer: FCFS; Storage1: SIRO; Storage2: PS; Storage3: HOL
Arrivals	Source – Class1: Exp(0.1); Class2: Exp(0.1); Class3: Exp(0.1)
Service	WebServer – Class1: Exp(3.333); Class2: Exp(2); Class3: Exp(1.667) Storage1 – Class1: Exp(0.9091); Class2: Exp(0.7692); Class3: Exp(0.6667) Storage2 – Class1: Exp(0.5); Class2: Exp(0.4762); Class3: Exp(0.5263) Storage3 – Class1: Exp(0.4); Class2: Exp(0.5263); Class3: Exp(0.2326)
Solvers	CTMC, MVA, JMT, SSA

Table 2.40: Features of prio_hol_open.

The example is built and solved as follows.

```
Network model = PrioModel.prio_hol_open();

// Create solvers as in MATLAB version (lines 69, 71, 73, 74)
CTMC solverCTMC = new CTMC(model, "seed", 23000, "cutoff", 1, "samples", 10000);
MVA solverMVA = new MVA(model, "seed", 23000, "cutoff", 1, "samples", 10000);
JMT solverJMT = new JMT(model, "seed", 23000, "cutoff", 1, "samples", 10000);
SSA solverSSA = new SSA(model, "seed", 23000, "cutoff", 1, "samples", 10000);

solverCTMC.getAvgTable().print();
solverMVA.getAvgTable().print();
solverJMT.getAvgTable().print();
solverSSA.getAvgTable().print();

pauseForUser();
```

Running this edition prints

```
Priority: highest=Class1,Class3, lowest=Class2
Priority: highest=Class1,Class3, lowest=Class2
Priority: highest=Class1,Class3, lowest=Class2
WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 1 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station  JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Source    Class1          0          0          0          0          0  0.052281
Source    Class2          0          0          0          0          0  0.050449
Source    Class3          0          0          0          0          0  0.046082
WebServer Class1    0.090117  0.062737  0.43092  1.7237  0.20912  0.20912
WebServer Class2    0.12132  0.1009  0.60119  2.4048  0.20179  0.20179
WebServer Class3    0.12559  0.1106  0.68136  2.7255  0.18433  0.18433
Storage1  Class1    0.06951  0.057509  1.3295  1.3295  0.052281  0.052281
Storage1  Class2    0.075291  0.065583  1.4924  1.4924  0.050449  0.050449
Storage1  Class3    0.076531  0.069123  1.6607  1.6607  0.046082  0.046082
Storage2  Class1    0.12865  0.10456  2.4607  2.4607  0.052281  0.052281

... (51 captured-output lines omitted; rerun the source example for the full output)

Storage2  Class1    0.51203  0.20712  4.9442  4.9442  0.099783  0.10356
Storage2  Class2    0.66933  0.25871  5.4331  5.4331  0.10761  0.1232
Storage2  Class3    0.42879  0.18429  4.4208  4.4208  0.10307  0.096992
Storage3  Class1    1.2392  0.26539  11.673  11.673  0.099783  0.10616
Storage3  Class2    8.5617  0.21107  77.071  77.071  0.10761  0.11109
Storage3  Class3    1.2076  0.38531  13.477  13.477  0.10307  0.089607

[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 4 solvers the script runs (CTMC, MVA, JMT, SSA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.5.3 prio_identical

Priority classes with identical parameters, the sanity check that priority alone does not move the aggregate metrics.

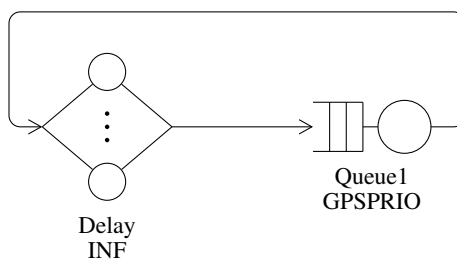


Figure 2.41: Topology of prio_identical: a closed network over 2 queueing stations.

The example is built and solved as follows.

Nodes	2: Queue, Delay
Classes	4: Class1 (closed, $N = 6$), Class2 (closed, $N = 4$), Class3 (closed, $N = 4$), Class4 (closed, $N = 1$)
Scheduling	Delay: INF; Queue1: GPSPRIO
Service	Delay – Class1: Erlang(3,2); Class2: Exp(1); Class3: Exp(1); Class4: Exp(2) Queue1 – Class1: Exp(30); Class2: Exp(2); Class3: Exp(12); Class4: Exp(1)
Solvers	CTMC, JMT

Table 2.41: Features of prio_identical.

```

Network model = PrioModel.prio_identical();

// Create solvers as in MATLAB version (lines 35, 36)
CTMC solverCTMC = new CTMC(model);
JMT solverJMT = new JMT(model, "seed", 23000, "verbose", true, "samples", 10000,
    "keep", true);

solverCTMC.getAvgTable().print();
solverJMT.getAvgTable().print();

pauseForUser();

```

Running this edition prints

```

Priority: highest=Class1, lowest=Class4
Priority: highest=Class1, lowest=Class4
Priority: highest=Class1, lowest=Class4
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 5.6372 5.6372 0.66667 0.66667 8.4558 8.4558
Delay Class2 0.91384 0.91384 1 1 0.91384 0.91384
Delay Class3 3.0493 3.0493 1 1 3.0493 3.0493
Delay Class4 0.0031954 0.0031954 0.5 0.5 0.0063909 0.0063909
Queue1 Class1 0.3628 0.28186 0.042905 0.042905 8.4558 8.4558
Queue1 Class2 3.0862 0.45692 3.3772 3.3772 0.91384 0.91384
Queue1 Class3 0.9507 0.25411 0.31177 0.31177 3.0493 3.0493
Queue1 Class4 0.9968 0.0063909 155.97 155.97 0.0063909 0.0063909
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 5.6487 5.6487 0.66214 0.66214 8.5567 8.5533
Delay Class2 0.89602 0.89602 0.99227 0.99227 0.89476 0.90847
Delay Class3 3.0308 3.0308 0.97343 0.97343 3.1087 3.0983
Delay Class4 0.0031932 0.0031932 0.50558 0.50558 0.0065312 0.0065312
Queue1 Class1 0.35125 0.28302 0.040881 0.040881 8.5533 8.5531
Queue1 Class2 3.104 0.42925 3.417 3.417 0.90847 0.90646
Queue1 Class3 0.96924 0.27143 0.30885 0.30885 3.0983 3.0462
Queue1 Class4 0.99678 0.0065642 152.63 152.63 0.0065312 0.0065303

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (CTMC, JMT), which is the point of the example: the same model read by methods of different cost and different exactness.

2.5.4 prio_psprio

Priority processor sharing, where the share of a class is set by its priority.

The example is built and solved as follows.

```

Network model = PrioModel.prio_psprio();

// Create solvers as in MATLAB version (lines 27, 28, 29)
CTMC solverCTMC = new CTMC(model);
JMT solverJMT = new JMT(model, "seed", 23000, "verbose", true, "samples", 5000);
SSA solverSSA = new SSA(model, "seed", 23000, "verbose", true, "samples", 5000);

solverCTMC.getAvgTable().print();

```

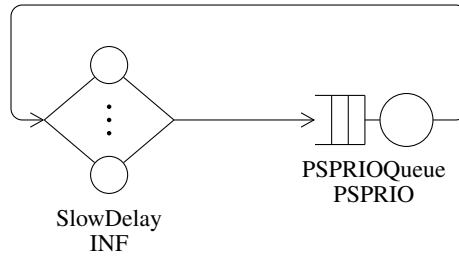



Figure 2.42: Topology of prio_psprio: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 2$), Class2 (closed, $N = 2$)
Scheduling	SlowDelay: INF; PSPRIOQueue: PSPRIO
Service	SlowDelay – Class1: Erlang(3,2); Class2: HyperExp(0.5,0.5;3,10) PSPRIOQueue – Class1: HyperExp(0.1,0.9;1,10); Class2: Exp(1)
Solvers	CTMC, JMT, SSA

Table 2.42: Features of prio_psprio.

```

solverJMT.getAvgTable().print();
solverSSA.getAvgTable().print();

pauseForUser();

```

Running this edition prints

```

Priority: highest=Class1, lowest=Class2
Priority: highest=Class1, lowest=Class2
Priority: highest=Class1, lowest=Class2
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station  JobClass  QLen    Util    RespT    ResidT    ArvR    Tput
SlowDelay  Class1    1.4834   1.4834   0.66667   0.66667   2.2252   2.2252
SlowDelay  Class2    0.12349  0.12349  0.21667   0.21667   0.56996  0.56996
PSPRIOQueue Class1    0.51655  0.42278  0.23214   0.23214   2.2252   2.2252
PSPRIOQueue Class2    1.8765   0.56996  3.2923    3.2923    0.56996  0.56996
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station  JobClass  QLen    Util    RespT    ResidT    ArvR    Tput
SlowDelay  Class1    1.5161   1.5161   0.66563   0.66563   2.2873   2.2388
SlowDelay  Class2    0.12717  0.12717  0.20933   0.20933   0.58227  0.58227
PSPRIOQueue Class1    0.48386  0.40341  0.21719   0.21719   2.2388   2.2385
PSPRIOQueue Class2    1.87     0.59305  3.179     3.179     0.58227  0.57109
SSA analysis [method: default/nrm; type: approximate, randomized; lang: java; env: python] completed in <t>
>s. Iterations: 5000.
Station  JobClass  QLen    Util    RespT    ResidT    ArvR    Tput
SlowDelay  Class1    1.4506   1.4506   0.65585   0.65585   2.1514   2.2118
SlowDelay  Class2    0.13061  0.13061  0.21605   0.21605   0.55583  0.60456
PSPRIOQueue Class1    0.54936  0.43671  0.25535   0.25535   2.2118   2.1514
PSPRIOQueue Class2    1.8694   0.55583  3.3632    3.3632    0.60456  0.55583

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 3 solvers the script runs (CTMC, JMT, SSA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6 Fork-join networks

A `Fork` node splits an arriving job into sibling tasks, one per outgoing branch, and a `Join` node recombines them. The response time of the parent job is the maximum over its siblings, which is why fork-join has no product-form solution and why the join is where the modeling subtleties live: how many siblings are required (the quorum), whether the join is present at all, and what happens when branches share a station.

The family is the largest in the basic set because each of those questions has its own script. Nesting a fork inside a branch, three or more branches, asymmetric branch demands and class switching on the siblings are all covered separately, so a failure can be localized.

2.6.1 fj_asymm

Asymmetric branches, whose response time is set by the slowest one.

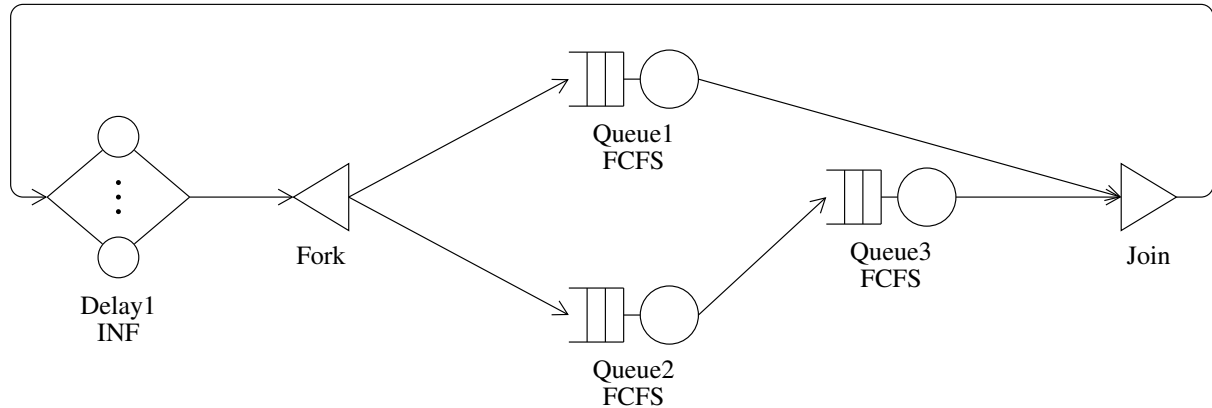


Figure 2.43: Topology of fj_asymm: a closed network over 4 queueing stations with a fork-join block.

Nodes	6: Queue $\times$ 3, Delay, Fork, Join
Classes	1: class1 (closed, $N = 10$)
Scheduling	Delay1: INF; Queue1: FCFS; Queue2: FCFS; Queue3: FCFS
Service	Delay1 – class1: Exp(0.5) Queue1 – class1: Exp(1) Queue2 – class1: Exp(2) Queue3 – class1: Exp(1)
Solvers	JMT, MVA

Table 2.43: Features of fj_asymm.

The example is built and solved as follows.

```

Network model = ForkJoinModel.fj_asymm();

// Solve with multiple methods as in notebook
JMT solverJMT = new JMT(model, "seed", 23000);
MVA solverMVA = new MVA(model);

Object[] solvers = {solverJMT, solverMVA};

for (int i = 0; i < solvers.length; i++) {
    try {
        if (solvers[i] instanceof JMT) {
            JMT solver = (JMT) solvers[i];
            solver.getAvgTable().print();
        } else if (solvers[i] instanceof MVA) {
            MVA solver = (MVA) solvers[i];
            solver.getAvgTable().print();
        }
    } catch (Exception e) {
        System.out.println("Error: " + e.getMessage());
    }
}

pauseForUser();
  
```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 class1 1.8108 1.8108 1.9614 1.9614 0.94914 0.94982
Queue1 class1 6.1513 0.95281 6.5892 6.5892 0.94982 0.94864
Queue2 class1 0.82403 0.47064 0.8557 0.8557 0.94982 0.95192
Queue3 class1 5.1269 0.93083 5.4049 5.4049 0.95192 0.94412
Join class1 4.293 0 2.2396 2.2396 1.8954 0.94685
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 673.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 class1 1.7139 1.7139 2 2 0.85693 0.85693
Queue1 class1 5.2237 0.85693 6.0958 6.0958 0.85693 0.85693
Queue2 class1 0.73602 0.42846 0.85891 0.85891 0.85693 0.85693
Queue3 class1 5.2237 0.85693 6.0958 6.0958 0.85693 0.85693
Join class1 5.6159 0 3.2768 3.2768 1.7139 0.85693

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.2 fj_basic_closed

The closed counterpart, on PS branches.

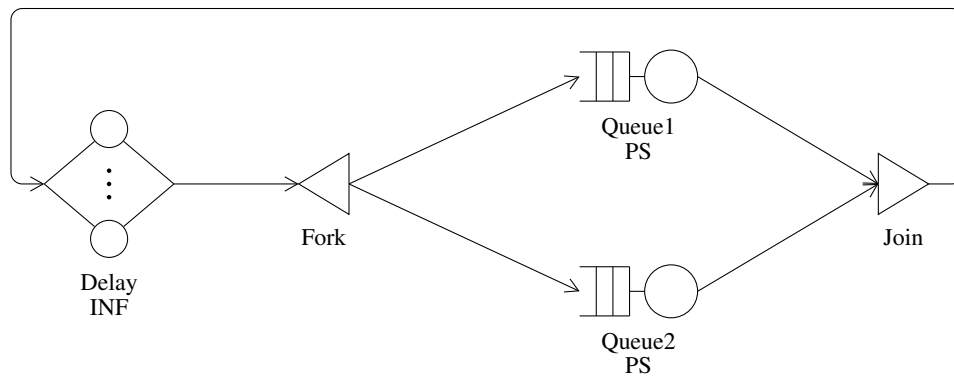


Figure 2.44: Topology of `fj_basic_closed`: a closed network over 3 queueing stations with a fork-join block.

Nodes	5: Queue $\times$ 2, Delay, Fork, Join
Classes	1: class1 (closed, $N = 5$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS
Service	Delay – class1: Exp(1) Queue1 – class1: Exp(1) Queue2 – class1: Exp(1)
Solvers	JMT, MVA

Table 2.44: Features of `fj_basic_closed`.

The example is built and solved as follows.

```

Network model = ForkJoinModel.fj_basic_closed();

// Solve with multiple methods as in notebook
JMT solverJMT = new JMT(model, "seed", 23000);
MVA solverMVA = new MVA(model, "fork_join", "ht", "method", "amva");

Object[] solvers = {solverJMT, solverMVA};

for (int i = 0; i < solvers.length; i++) {
    try {
        if (solvers[i] instanceof JMT) {

```

```

        JMT solver = (JMT) solvers[i];
        solver.getAvgTable().print();
    } else if (solvers[i] instanceof MVA) {
        MVA solver = (MVA) solvers[i];
        solver.getAvgTable().print();
    }
} catch (Exception e) {
    System.out.println("Error: " + e.getMessage());
}
}
pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass   QLen    Util    RespT    ResidT    ArvR    Tput
Delay   class1     0.86625  0.86625  0.99611  0.99611  0.89442  0.89768
Queue1  class1     2.6475  0.88812  3.0945  3.0945  0.89768  0.89761
Queue2  class1     2.5863  0.89336  2.815   2.815   0.89768  0.89757
Join    class1     2.9633   0      1.685   1.685   1.724   0.88799
MVA analysis [method: egflin; type: approximate, deterministic; lang: java; env: python] completed in <t>s
. Iterations: 489.
Station JobClass   QLen    Util    RespT    ResidT    ArvR    Tput
Delay   class1     0.77057  0.77057  1        1        0.77057  0.77057
Queue1  class1     2.8911  0.77057  3.7519  3.7519  0.77057  0.77057
Queue2  class1     2.8911  0.77057  3.7519  3.7519  0.77057  0.77057
Join    class1     2.8911   0      1.876   1.876   1.5411  0.77057
[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.3 fj_basic_nesting

A fork inside a branch of another fork, with open and closed classes.

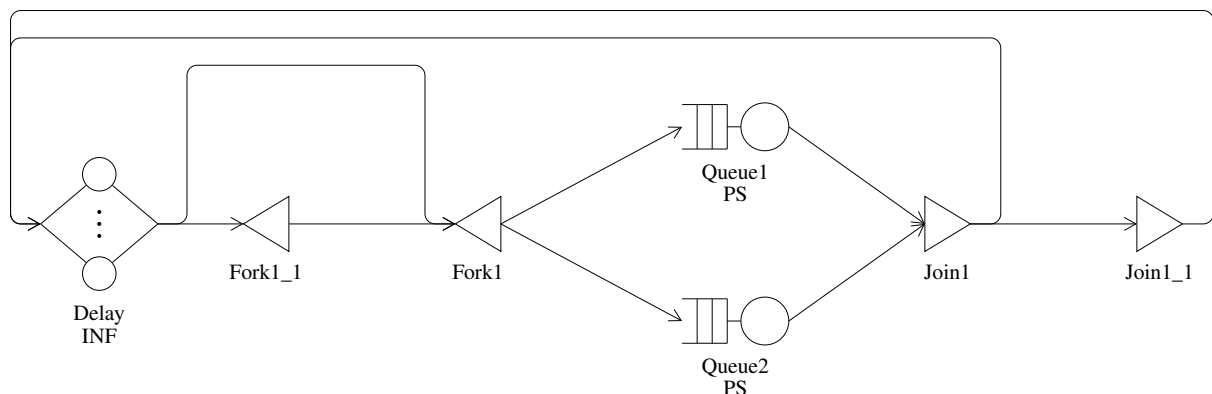


Figure 2.45: Topology of fj_basic_nesting: a closed network over 3 queueing stations with a fork-join block.

The example is built and solved as follows.

```

Network model = ForkJoinModel.fj_basic_nesting();
JMT solver = new JMT(model, "seed", 23000);
solver.getAvgTable().print();
MVA solverMVA = new MVA(model);
solverMVA.getAvgTable().print();
pauseForUser();

```

Running this edition prints

Nodes	7: Queue $\times$ 2, Delay, Fork $\times$ 2, Join $\times$ 2
Classes	2: class1 (closed, $N = 5$), class2 (closed, $N = 2$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS
Service	Delay – class1: Exp(0.25); class2: Exp(0.25) Queue1 – class1: Exp(1); class2: Exp(2) Queue2 – class1: Exp(0.75); class2: Exp(2)
Features	class switching on 2 links
Solvers	JMT, MVA

Table 2.45: Features of fj_basic_nesting.

WARNING: [setTasksPerLink] The setTasksPerLink feature is experimental and results may be inaccurate for analytical solvers.

JMT Model: <tmp>

JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.

Station	JobClass	QLen	Util	RespT	ResidT	ArvR	Tput
Delay	class1	2.0368	2.0368	4.0283	4.0283	0.50637	0.50487
Delay	class2	0.98642	0.98642	3.9138	3.9138	0.25427	0.2541
Join1	class1	2.2026	0	2.1777	2.1777	0.99731	0.50798
Join1	class2	1.0479	0	1.0011	1.0011	1.0445	0.51183
Join1_1	class2	0.54106	0	1.0555	1.0555	0.51183	0.2541
Queue1	class1	1.3447	0.49509	2.6485	2.6485	0.50487	0.5051
Queue1	class2	0.82969	0.26813	1.6297	1.6297	0.51178	0.51204
Queue2	class1	2.3299	0.6631	4.6827	4.6827	0.50487	0.50447
Queue2	class2	1.1149	0.25869	2.1947	2.1947	0.51178	0.51352

MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed in <t>s. Iterations: 1558.

Station	JobClass	QLen	Util	RespT	ResidT	ArvR	Tput
Delay	class1	1.762	1.762	4	4	0.44051	0.44051
Delay	class2	0.93501	0.93501	4	4	0.23375	0.23375
Join1	class1	2.4047	0	2.7294	2.7294	0.88102	0.44051
Join1	class2	0.99834	0	0.85419	0.85419	0.93501	0.4675
Join1_1	class2	0.71667	0	1.533	1.533	0.4675	0.23375
Queue1	class1	1.2807	0.44051	2.9074	2.9074	0.44051	0.44051
Queue1	class2	0.68934	0.23375	1.4745	1.4745	0.23375	0.4675
Queue2	class1	2.9009	0.58734	6.5853	6.5853	0.44051	0.44051
Queue2	class2	1.179	0.23375	2.5219	2.5219	0.23375	0.4675

[Running in non-interactive mode, continuing...]

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.4 fj_basic_open

One fork, two parallel FCFS branches and a join, in an open network.

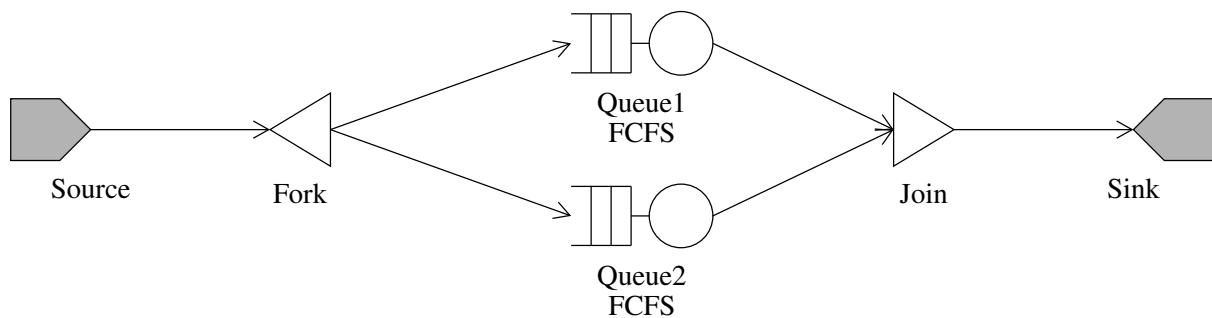


Figure 2.46: Topology of fj_basic_open: an open network over 2 queueing stations with a fork-join block.

The example is built and solved as follows.

```
Network model = ForkJoinModel.fj_basic_open();
```

Nodes	6: Source, Queue $\times$ 2, Fork, Join, Sink
Classes	1: class1 (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	Source – class1: Exp(0.05)
Service	Queue1 – class1: Exp(1) Queue2 – class1: Exp(2)
Solvers	JMT, MVA, LDES, MAM

Table 2.46: Features of fj_basic_open.

```
// The solvers the reference runs, in its order. MAM is pinned to the
// source-decomposition MMAP analyzer: the default analyzer answers a
// different question on a forked open network.
try {
    new JMT(model, "seed", 23000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT failed: " + e.getMessage());
}
try {
    new MVA(model).getAvgTable().print();
} catch (Exception e) {
    System.out.println("MVA failed: " + e.getMessage());
}
try {
    new LDES(model, "seed", 23000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("LDES failed: " + e.getMessage());
}
try {
    new MAM(model, "method", "dec.source.mmap").getAvgTable().print();
} catch (Exception e) {
    System.out.println("MAM failed: " + e.getMessage());
}
pauseForUser();
```

Running this edition prints

```
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Source   class1         0         0         0         0         0  0.050494
Queue1   class1      0.051923  0.049524  1.0478    1.0478    0.050494  0.050812
Queue2   class1      0.026691  0.026329  0.50973   0.50973   0.050494  0.050896
Join     class1      0.044462         0  0.43237   0.43237   0.10395   0.050857
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 396.
Station JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Source   class1         0         0         0         0         0  0.05
Queue1   class1      0.052631  0.05     1.0526    1.0526    0.05     0.05
Queue2   class1      0.025641  0.025    0.51282   0.51282   0.05     0.05
Join     class1      0.043789         0  0.43789   0.43789   0.1       0.05
LDES events:  4000    8000   12000   16000   20000   24000   28000   32000   36000   40000   44000   48000   52000
               56000   60000   64000   68000   72000   76000   80000   84000   88000   92000   96000  100000  104000  108000
               112000  116000  120000  124000  128000  132000  136000  140000  144000  148000  152000  156000  160000  164000
               168000  172000  176000  180000  184000  188000  192000  196000  200000  200000
LDES analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 200000.
Station JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
... (4 captured-output lines omitted; rerun the source example for the full output)
MAM analysis [method: dec.source.mmap; type: approximate, deterministic; lang: java; env: python]
completed in <t>s. Iterations: 3.
Station JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Source   class1         0         0         0         0         0  0.05
Queue1   class1      0.052632  0.05     1.0526    1.0526    0.05     0.05
Queue2   class1      0.025641  0.025    0.51282   0.51282   0.05     0.05
Join     class1      0.04379         0  0.4379    0.4379    0.1       0.05

[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 4 solvers the script runs (JMT, MVA, LDES, MAM), which is the point

of the example: the same model read by methods of different cost and different exactness.

2.6.5 fj_complex_serial

Several fork-join blocks in series inside a larger network.

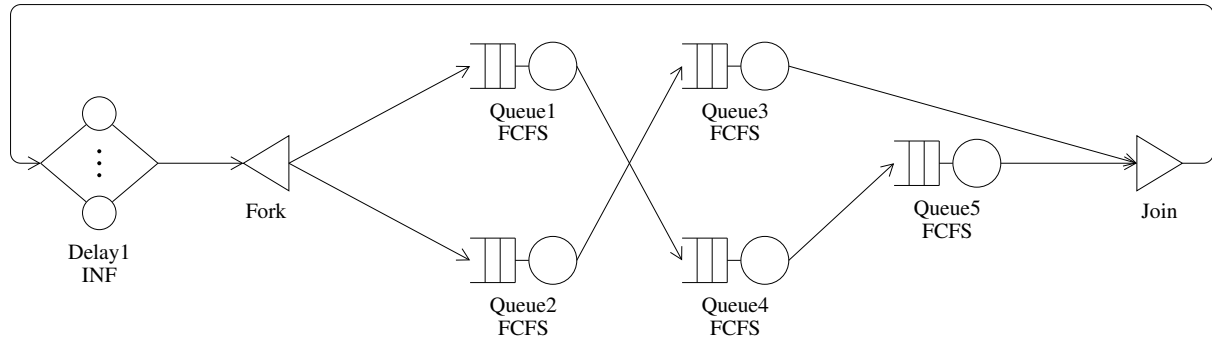


Figure 2.47: Topology of fj_complex_serial: a closed network over 6 queueing stations with a fork-join block.

Nodes	8: Queue × 5, Delay, Fork, Join
Classes	1: class1 (closed, $N = 10$)
Scheduling	Delay1: INF; Queue1: FCFS; Queue2: FCFS; Queue3: FCFS; Queue4: FCFS; Queue5: FCFS
Service	Delay1 – class1: Exp(0.5) Queue1 – class1: Exp(1) Queue2 – class1: Exp(2) Queue3 – class1: Exp(1) Queue4 – class1: Exp(3) Queue5 – class1: Exp(0.8)
Solvers	JMT, MVA

Table 2.47: Features of fj_complex_serial.

The example is built and solved as follows.

```

Network model = ForkJoinModel.fj_complex_serial();
JMT solverJMT = new JMT(model, "seed", 23000);
MVA solverMVA = new MVA(model);

Object[] solvers = {solverJMT, solverMVA};

for (Object solverObj : solvers) {
    try {
        if (solverObj instanceof JMT) {
            JMT solver = (JMT) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof MVA) {
            MVA solver = (MVA) solverObj;
            solver.getAvgTable().print();
        }
    } catch (Exception e) {
        System.out.println("Error with solver: " + e.getMessage());
    }
}
pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 class1 1.5109 1.5109 1.9611 1.9611 0.76579 0.75823
Queue1 class1 2.5753 0.76218 3.3293 3.3293 0.75823 0.75929
Queue2 class1 0.5914 0.38738 0.79434 0.79434 0.75823 0.76578
Queue3 class1 2.7793 0.76354 3.4472 3.4472 0.76578 0.75992
Queue4 class1 0.32517 0.24092 0.41843 0.41843 0.75929 0.75933

```

```

Queue5  class1  5.3388  0.95413  7.2633  7.2633  0.75933  0.76579
Join    class1  5.3359  0        3.4698  3.4698  1.5125  0.75865
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 1201.
Station JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
Delay1  class1    1.3867  1.3867  2      2      0.69335  0.69335
Queue1  class1    2.142   0.69335  3.0894  3.0894  0.69335  0.69335
Queue2  class1    0.52374  0.34667  0.75538  0.75538  0.69335  0.69335
Queue3  class1    2.142   0.69335  3.0894  3.0894  0.69335  0.69335
Queue4  class1    0.29836  0.23112  0.43031  0.43031  0.69335  0.69335
Queue5  class1    5.6088  0.86669  8.0894  8.0894  0.69335  0.69335
Join    class1    6.7098  0      4.8387  4.8387  1.3867  0.69335

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.6 fj_cs_multi_visits

Class switching with several visits per branch.

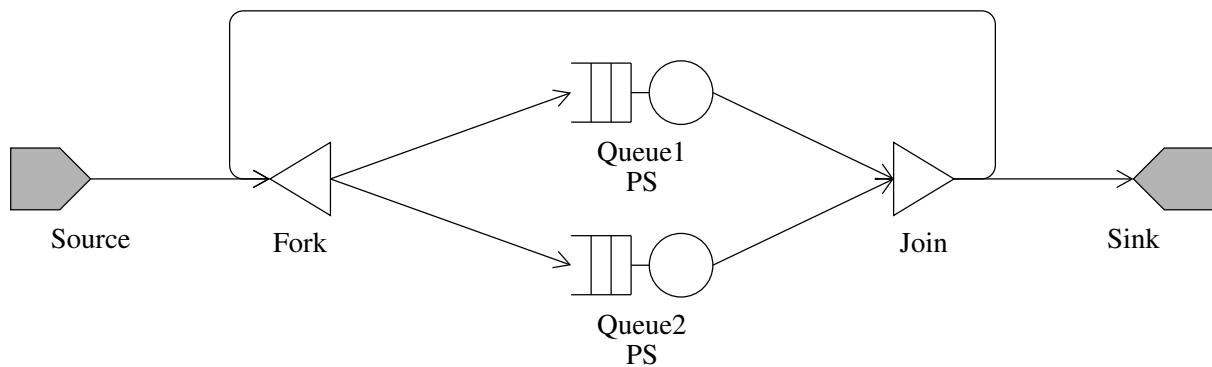


Figure 2.48: Topology of `fj_cs_multi_visits`: an open network over 2 queueing stations with a fork-join block.

Nodes	6: Source, Queue × 2, Fork, Join, Sink
Classes	2: class1 (open), class2 (open)
Scheduling	Queue1: PS; Queue2: PS
Arrivals	Source – class1: Exp(0.1); class2: Disabled
Service	Queue1 – class1: Exp(1); class2: Exp(1) Queue2 – class1: Exp(1); class2: Exp(1)
Features	class switching on 1 link
Solvers	MVA, JMT

Table 2.48: Features of `fj_cs_multi_visits`.

The example is built and solved as follows.

```

Network model = ForkJoinModel.fj_cs_multi_visits();
JMT solverJMT = new JMT(model, "seed", 23000);
MVA solverMVA = new MVA(model);

Object[] solvers = {solverJMT, solverMVA};

for (Object solverObj : solvers) {
    try {
        if (solverObj instanceof JMT) {
            JMT solver = (JMT) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof MVA) {
            MVA solver = (MVA) solverObj;

```



```

        solver.getAvgTable().print();
    }
} catch (Exception e) {
    System.out.println("Error with solver: " + e.getMessage());
}
}
pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Source   class1     0      0      0      0      0      0.099935
Queue1   class1   0.12819 0.09701 1.2308 1.2308 0.099935 0.099937
Queue1   class2   0.1269 0.1011 1.248 1.248 0.099931 0.099928
Queue2   class1   0.12642 0.10181 1.244 1.244 0.099935 0.099931
Queue2   class2   0.1243 0.1007 1.2318 1.2318 0.099931 0.099981
Join     class1   0.13493 0      0.66288 0.66288 0.19881 0.099931
Join     class2   0.12888 0      0.6401 0.6401 0.19882 0.099924
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 544.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Source   class1     0      0      0      0      0      0.1
Queue1   class1   0.12474 0.1 1.2474 1.2474 0.1 0.1
Queue1   class2   0.12474 0.1 1.2474 1.2474 0.1 0.1
Queue2   class1   0.12474 0.1 1.2474 1.2474 0.1 0.1
Queue2   class2   0.12474 0.1 1.2474 1.2474 0.1 0.1
Join     class1   0.12474 0 0.62368 0.62368 0.2 0.1
Join     class2   0.12474 0 0.62368 0.62368 0.2 0.1
[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.7 fj_cs_postfork

Class switching after the fork, on the sibling tasks.

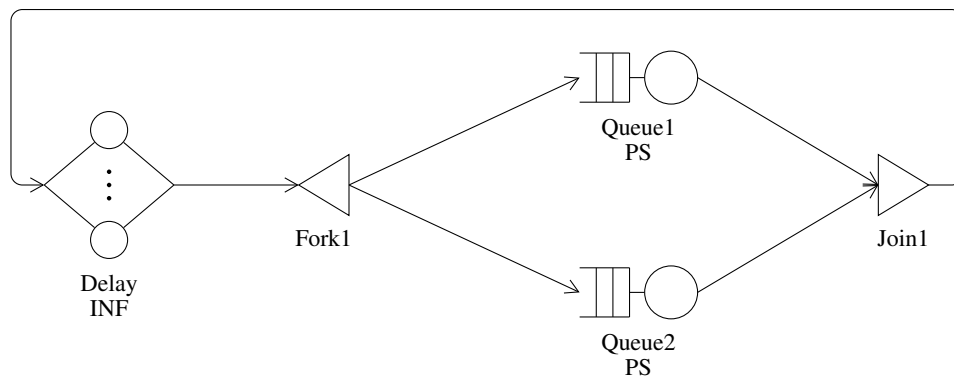


Figure 2.49: Topology of `fj_cs_postfork`: a closed network over 3 queueing stations with a fork-join block.

The example is built and solved as follows.

```

Network model = ForkJoinModel.fj_cs_postfork();
JMT solverJMT = new JMT(model, "seed", 23000);
MVA solverMVA = new MVA(model);

Object[] solvers = {solverJMT, solverMVA};

for (Object solverObj : solvers) {
    try {

```

Nodes	5: Queue $\times$ 2, Delay, Fork, Join
Classes	2: class1 (closed, $N = 1$), class2 (closed, $N = 1$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS
Service	Delay – class1: Exp(0.25); class2: Exp(0.25) Queue1 – class1: Exp(2); class2: Exp(2) Queue2 – class1: Exp(2); class2: Exp(2)
Features	class switching on 3 links
Solvers	MVA, JMT

Table 2.49: Features of `fj_cs_postfork`.

```

if (solverObj instanceof JMT) {
    JMT solver = (JMT) solverObj;
    solver.getAvgTable().print();
} else if (solverObj instanceof MVA) {
    MVA solver = (MVA) solverObj;
    solver.getAvgTable().print();
}
} catch (Exception e) {
    System.out.println("Error with solver: " + e.getMessage());
}
}
pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay   class1   0.83919 0.83919 4.0008  4.0008  0.20697 0.20977
Delay   class2   0.82894 0.82894 3.8875  0      0.20948 0.21192
Join1   class1   0.21374 0      0.2649  0.2649  0.8241  0.20774
Join1   class2   0      0      0      0      0      0.21192
Queue1  class1   0.21284 0.19531 0.5414  0.5414  0.41471 0.41463
Queue2  class1   0.22108 0.19926 0.54182 0.54182 0.41471 0.4147
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 546.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay   class1   1.6364 1.6364 4      4      0.40911 0.40911
Join1   class1   0.24644 0      0.30119 0.30119 0.81822 0.40911
Queue1  class1   0.24644 0.20455 0.60238 0.60238 0.40911 0.40911
Queue2  class1   0.24644 0.20455 0.60238 0.60238 0.40911 0.40911

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.8 `fj_cs_prefork`

Class switching before the fork.

Nodes	6: Queue $\times$ 2, Delay $\times$ 2, Fork, Join
Classes	2: class1 (closed, $N = 10$), class2 (closed, $N = 10$)
Scheduling	Delay1: INF; Delay2: INF; Queue1: PS; Queue2: PS
Service	Delay1 – class1: Exp(0.5); class2: Exp(0.5) Delay2 – class1: Disabled; class2: Exp(2) Queue1 – class1: Exp(1); class2: Disabled Queue2 – class1: Exp(1); class2: Disabled
Features	drop rule {'class1': 'waitingQueue'} at Delay2; drop rule {'class2': 'waitingQueue'} at Queue1; drop rule {'class2': 'waitingQueue'} at Queue2; class switching on 5 links
Solvers	JMT, MVA

Table 2.50: Features of `fj_cs_prefork`.

The example is built and solved as follows.

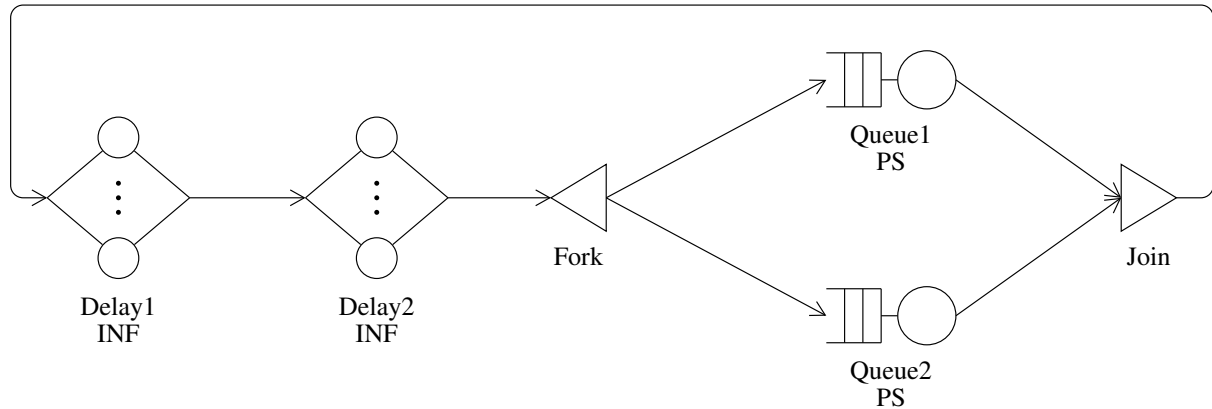


Figure 2.50: Topology of `fj_cs_prefork`: a closed network over 4 queueing stations with a fork-join block.

```
Network model = ForkJoinModel.fj_cs_prefork();
JMT solverJMT = new JMT(model, "seed", 23000);
MVA solverMVA = new MVA(model);

Object[] solvers = {solverJMT, solverMVA};

for (Object solverObj : solvers) {
    try {
        if (solverObj instanceof JMT) {
            JMT solver = (JMT) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof MVA) {
            MVA solver = (MVA) solverObj;
            solver.getAvgTable().print();
        }
    } catch (Exception e) {
        System.out.println("Error with solver: " + e.getMessage());
    }
}
pauseForUser();
```

Running this edition prints

```
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 class1 1.9083 1.9083 2.0226 2.0226 0.9777 0.97739
Delay2 class2 0.48212 0.48212 0.49535 0.49535 0.97739 0.97737
Queue1 class1 12.772 0.97594 12.462 12.462 0.97737 0.97823
Queue2 class1 8.7839 0.94243 8.9988 8.9988 0.97737 0.97343
Join class1 13.632 0 7.0837 7.0837 1.92 0.9777
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 887.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 class1 1.8659 1.8659 2 2 0.93293 0.93293
Delay2 class2 0.46646 0.46646 0.5 0.5 0.93293 0.93293
Queue1 class1 11.87 0.93293 12.723 12.723 0.93293 0.93293
Queue2 class1 11.87 0.93293 12.723 12.723 0.93293 0.93293
Join class1 11.87 0 6.3615 6.3615 1.8659 0.93293

[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.9 fj_deep_nesting

Several nesting levels, the stress case for the join bookkeeping.

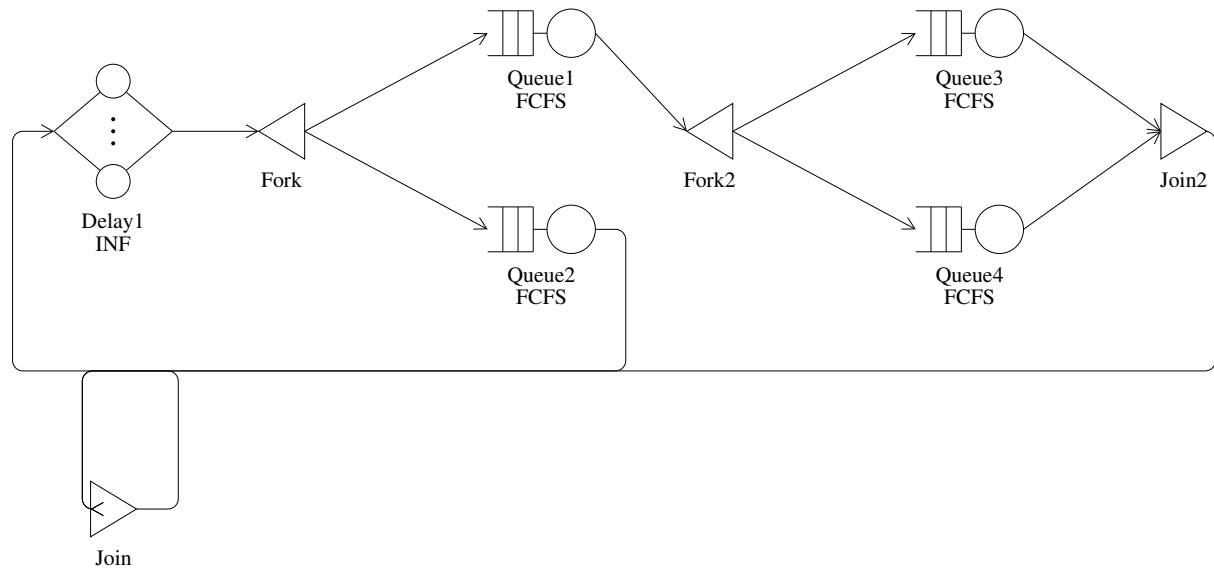


Figure 2.51: Topology of `fj_deep_nesting`: a closed network over 5 queueing stations with a fork-join block.

Nodes	9: Queue $\times$ 4, Delay, Fork $\times$ 2, Join $\times$ 2
Classes	1: class1 (closed, $N = 1$)
Scheduling	Delay1: INF; Queue1: FCFS; Queue2: FCFS; Queue3: FCFS; Queue4: FCFS
Service	Delay1 – class1: Exp(0.5) Queue1 – class1: Exp(1) Queue2 – class1: Exp(1) Queue3 – class1: Exp(2) Queue4 – class1: Exp(2)
Solvers	JMT, MVA

Table 2.51: Features of `fj_deep_nesting`.

The example is built and solved as follows.

```
Network model = ForkJoinModel.fj_deep_nesting();
JMT solverJMT = new JMT(model, "seed", 23000);
MVA solverMVA = new MVA(model);

Object[] solvers = {solverJMT, solverMVA};

for (Object solverObj : solvers) {
    try {
        if (solverObj instanceof JMT) {
            JMT solver = (JMT) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof MVA) {
            MVA solver = (MVA) solverObj;
            solver.getAvgTable().print();
        }
    } catch (Exception e) {
        System.out.println("Error with solver: " + e.getMessage());
    }
}
pauseForUser();
```

Running this edition prints

```
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 class1 0.50301 0.50301 1.9945 1.9945 0.2517 0.24907
Queue1 class1 0.2479 0.2479 1.0192 1.0192 0.24907 0.24906
Queue2 class1 0.24341 0.24341 0.95781 0.95781 0.24907 0.25171
Join class1 0.30974 0 0.62676 0.62676 0.50588 0.24906
Queue3 class1 0.12468 0.12468 0.50511 0.50511 0.24906 0.24844
Queue4 class1 0.12783 0.12783 0.50431 0.50431 0.24906 0.24907
Join2 class1 0.12602 0 0.24695 0.24695 0.50768 0.24845
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 549.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 class1 0.45192 0.45192 2 2 0.22596 0.22596
Queue1 class1 0.27306 0.22596 1.2084 1.2084 0.22596 0.22596
Queue2 class1 0.27306 0.22596 1.2084 1.2084 0.22596 0.22596
Join class1 0.39228 0 0.86802 0.86802 0.45192 0.22596
Queue3 class1 0.12639 0.11298 0.55933 0.55933 0.22596 0.22596
Queue4 class1 0.12639 0.11298 0.55933 0.55933 0.22596 0.22596
Join2 class1 0.12639 0 0.22373 0.22373 0.45192 0.22596
[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.10 fj_delays

Delay stations inside the branches.

Nodes	6: Queue $\times$ 2, Delay $\times$ 2, Fork, Join
Classes	1: class1 (closed, $N = 10$)
Scheduling	Delay1: INF; Delay2: INF; Queue1: FCFS; Queue2: FCFS
Service	Delay1 – class1: Exp(0.5) Delay2 – class1: Exp(2) Queue1 – class1: Exp(1) Queue2 – class1: Exp(1)
Solvers	JMT, MVA

Table 2.52: Features of fj_delays.

The example is built and solved as follows.

```
Network model = ForkJoinModel.fj_delays();
JMT solverJMT = new JMT(model, "seed", 23000);
```

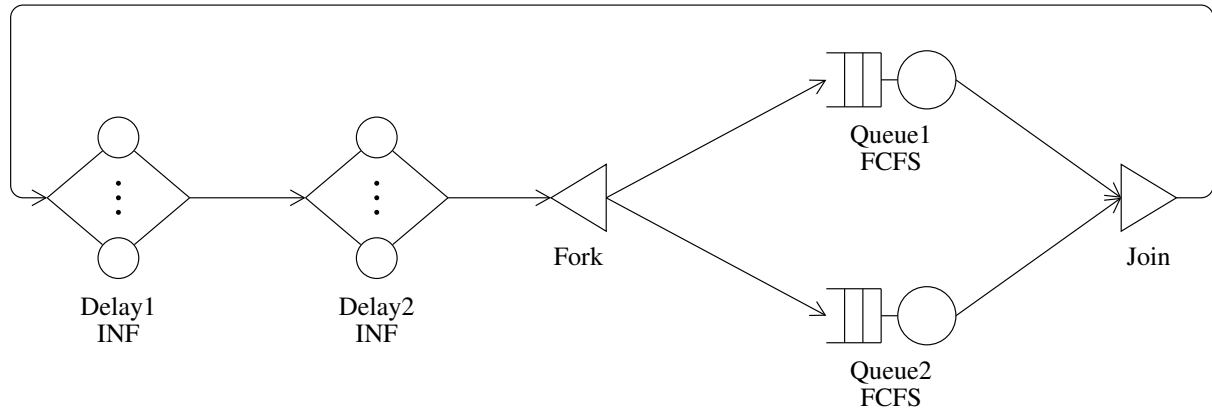


Figure 2.52: Topology of `fj_delays`: a closed network over 4 queueing stations with a fork-join block.

```

MVA solverMVA = new MVA(model);

Object[] solvers = {solverJMT, solverMVA};

for (Object solverObj : solvers) {
    try {
        if (solverObj instanceof JMT) {
            JMT solver = (JMT) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof MVA) {
            MVA solver = (MVA) solverObj;
            solver.getAvgTable().print();
        }
    } catch (Exception e) {
        System.out.println("Error with solver: " + e.getMessage());
    }
}

pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay1  class1      1.9728  1.9728   2.035   2.035   0.96734 0.95415
Delay2  class1      0.48219 0.48219 0.49433 0.49433 0.95415 0.95412
Queue1  class1      6.0429  0.95704 6.081   6.081   0.95412 0.96494
Queue2  class1      5.1463  0.94515 5.7192  5.7192  0.95412 0.96616
Join    class1      3.9749  0       2.0321  2.0321  1.9436 0.96744
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 628.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay1  class1      1.7184  1.7184   2       2       0.85919 0.85919
Delay2  class1      0.4296  0.4296   0.5     0.5     0.85919 0.85919
Queue1  class1      5.3091  0.85919 6.1792  6.1792  0.85919 0.85919
Queue2  class1      5.3091  0.85919 6.1792  6.1792  0.85919 0.85919
Join    class1      5.3091  0       3.0896  3.0896  1.7184 0.85919

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.11 `fj_mixed_openclosed`

Open and closed classes through one fork-join block, with a class disabled on a branch. The example is built and solved as follows.

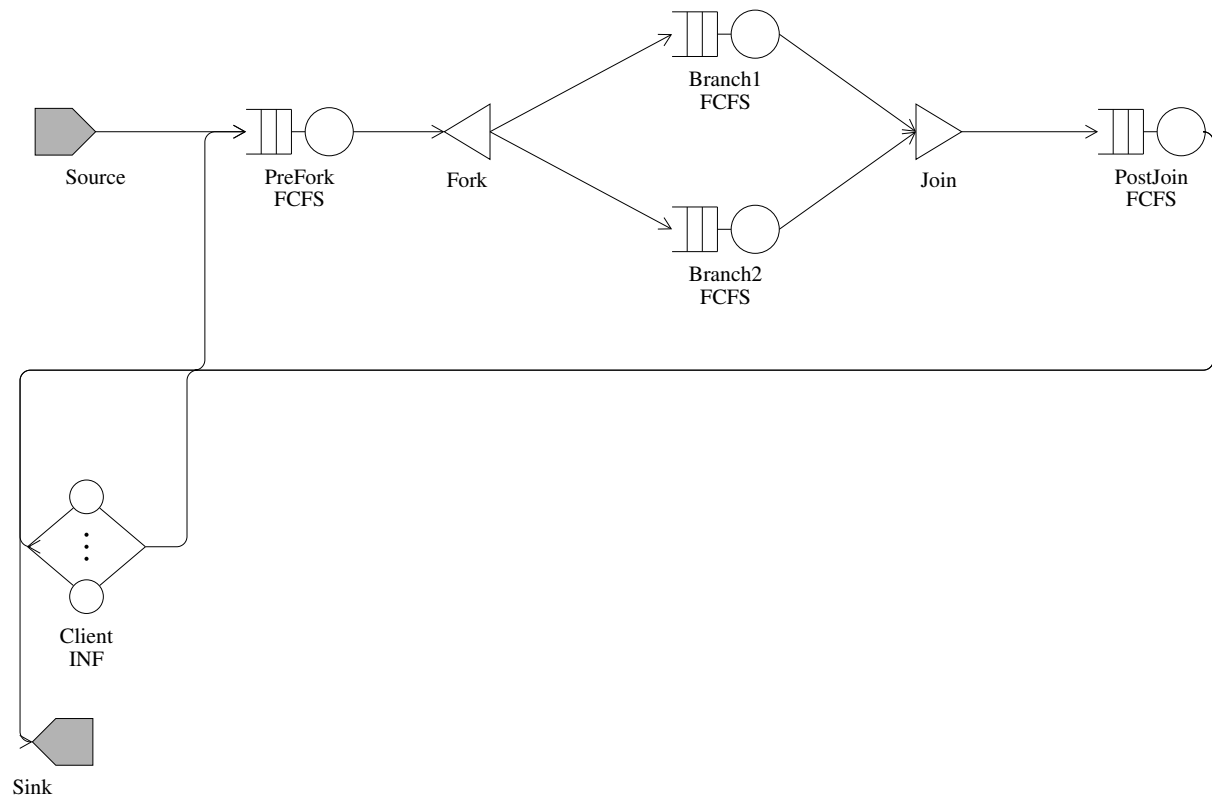


Figure 2.53: Topology of `fj_mixed_openclosed`: an open network over 5 queueing stations with a fork-join block.

Nodes	9: Source, Queue $\times$ 4, Delay, Fork, Join, Sink
Classes	2: Open (open), Closed (closed, $N = 1$)
Scheduling	Client: INF; PreFork: FCFS; Branch1: FCFS; Branch2: FCFS; PostJoin: FCFS
Arrivals	Source – Open: Exp(0.1); Closed: Disabled
Service	Client – Open: Disabled; Closed: Exp(1) PreFork – Open: Exp(5); Closed: Exp(5) Branch1 – Open: Exp(3.333); Closed: Exp(3.333) Branch2 – Open: Exp(2.5); Closed: Exp(2.5) PostJoin – Open: Exp(10); Closed: Exp(10)
Features	class switching on 2 links
Solvers	MVA, JMT

Table 2.53: Features of `fj_mixed_openclosed`.

```

Network model = new Network("ForkJoinOpenClosed");

Source source = new Source(model, "Source");
Delay client = new Delay(model, "Client");
Queue prefork = new Queue(model, "PreFork", SchedStrategy.FCFS);
Queue branch1 = new Queue(model, "Branch1", SchedStrategy.FCFS);
Queue branch2 = new Queue(model, "Branch2", SchedStrategy.FCFS);
Queue postjoin = new Queue(model, "PostJoin", SchedStrategy.FCFS);
Fork fork = new Fork(model, "Fork");
Join join = new Join(model, "Join", fork);
Sink sink = new Sink(model, "Sink");

OpenClass oclass = new OpenClass(model, "Open");
ClosedClass cclass = new ClosedClass(model, "Closed", 1, client);

source.setArrival(oclass, new Exp(0.1));
client.setService(cclass, Exp.fitMean(1.0));
client.setService(oclass, Disabled.getInstance());
prefork.setService(oclass, Exp.fitMean(0.2));
prefork.setService(cclass, Exp.fitMean(0.2));
branch1.setService(oclass, Exp.fitMean(0.3));
branch1.setService(cclass, Exp.fitMean(0.3));
branch2.setService(oclass, Exp.fitMean(0.4));
branch2.setService(cclass, Exp.fitMean(0.4));
postjoin.setService(oclass, Exp.fitMean(0.1));
postjoin.setService(cclass, Exp.fitMean(0.1));

... (18 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

P.set(cclass, cclass, join, postjoin, 1);
P.set(cclass, cclass, postjoin, client, 1);

model.link(P);

return model;

```

Running this edition prints

```
jline.lang.Network@621be5d1
```

2.6.12 fj_nojoin

A fork with no join, so the siblings leave through the sink independently.

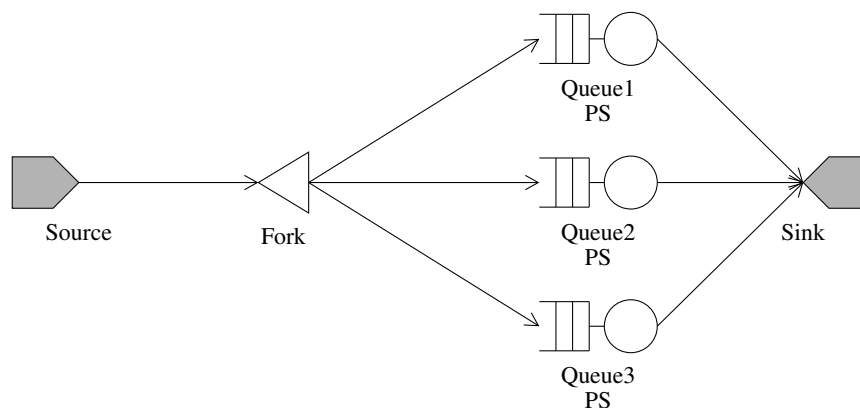


Figure 2.54: Topology of fj_nojoin: an open network over 3 queueing stations with a fork-join block.

The example is built and solved as follows.

```
Network model = ForkJoinModel.fj_nojoin();
```


Nodes	6: Source, Queue $\times$ 3, Fork, Sink
Classes	1: class1 (open)
Scheduling	Queue1: PS; Queue2: PS; Queue3: PS
Arrivals	Source – class1: Exp(0.5)
Service	Queue1 – class1: Exp(1) Queue2 – class1: Exp(2) Queue3 – class1: Exp(3)
Solvers	MVA, JMT

Table 2.54: Features of fj_nojoin.

```

JMT solverJMT = new JMT(model, "seed", 23000);
MVA solverMVA = new MVA(model);

Object[] solvers = {solverJMT, solverMVA};

for (Object solverObj : solvers) {
    try {
        if (solverObj instanceof JMT) {
            JMT solver = (JMT) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof MVA) {
            MVA solver = (MVA) solverObj;
            solver.getAvgTable().print();
        }
    } catch (Exception e) {
        System.out.println("Error with solver: " + e.getMessage());
    }
}
pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT Command output: JSIMEngine - Warning: A class1 task generated by a fork or scaler reached Sink before
merging at a join.

JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass   QLen    Util    RespT    ResidT    ArvR    Tput
Source   class1      0        0        0        0        0    0.5062
Queue1   class1      0.97824  0.49952  1.9229   1.9229   0.5062 0.50103
Queue2   class1      0.32722  0.24452  0.6673   0.6673   0.5062 0.50398
Queue3   class1      0.2038   0.16807  0.41145  0.41145  0.5062 0.50621
WARNING: [mmt] No join node found for fork node 4.
WARNING: [sort_forks] No join node found for fork node 4 in nested forks processing.
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 1029.
Station JobClass   QLen    Util    RespT    ResidT    ArvR    Tput
Source   class1      0        0        0        0        0    0.5
Queue1   class1      1        0.5      2        2        0.5    0.5
Queue2   class1      0.33314  0.25     0.66629  0.66629  0.5    0.5
Queue3   class1      0.19913  0.16667  0.39826  0.39826  0.5    0.5

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.13 fj_route_overlap

Branches that share a station, so their visits overlap.

The example is built and solved as follows.

```

Network model = ForkJoinModel.fj_route_overlap();
JMT solverJMT = new JMT(model, "seed", 23000);
MVA solverMVA = new MVA(model);

Object[] solvers = {solverJMT, solverMVA};

```

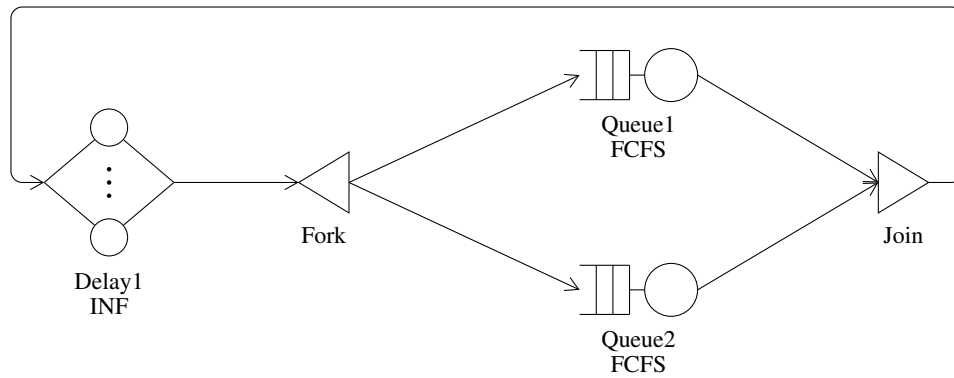


Figure 2.55: Topology of `fj_route_overlap`: a closed network over 3 queueing stations with a fork-join block.

Nodes	5: Queue $\times$ 2, Delay, Fork, Join
Classes	2: class1 (closed, $N = 10$), class2 (closed, $N = 10$)
Scheduling	Delay1: INF; Queue1: FCFS; Queue2: FCFS
Service	Delay1 – class1: Exp(0.5); class2: Exp(0.2) Queue1 – class1: Exp(1); class2: Exp(1) Queue2 – class1: Exp(2); class2: Exp(2)
Solvers	JMT, MVA

Table 2.55: Features of `fj_route_overlap`.

```

for (Object solverObj : solvers) {
    try {
        if (solverObj instanceof JMT) {
            JMT solver = (JMT) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof MVA) {
            MVA solver = (MVA) solverObj;
            solver.getAvgTable().print();
        }
    } catch (Exception e) {
        System.out.println("Error with solver: " + e.getMessage());
    }
}
pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 class1 1.0434 1.0434 1.9568 1.9568 0.53561 0.53361
Delay1 class2 2.2482 2.2482 4.9237 4.9237 0.46316 0.46313
Queue1 class1 8.9566 0.53833 16.567 16.567 0.53361 0.53561
Queue1 class2 7.7518 0.45808 16.743 16.743 0.46313 0.4626
Queue2 class1 0.52817 0.27261 0.99797 0.99797 0.53361 0.5331
Queue2 class2 0.47122 0.22608 1.0143 1.0143 0.46313 0.4629
Join class1 8.4186 0 7.8315 7.8315 1.0916 0.53561
Join class2 7.2339 0 7.7245 7.7245 0.91758 0.4626
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 3519.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 class1 1.0241 1.0241 2 2 0.51206 0.51206
Delay1 class2 2.215 2.215 5 5 0.443 0.443
Queue1 class1 9.0096 0.51206 17.595 17.595 0.51206 0.51206
Queue1 class2 7.8075 0.443 17.624 17.624 0.443 0.443
Queue2 class1 0.48432 0.25603 0.94582 0.94582 0.51206 0.51206
Queue2 class2 0.41936 0.2215 0.94663 0.94663 0.443 0.443
Join class1 8.5747 0 8.3727 8.3727 1.0241 0.51206
Join class2 7.4309 0 8.387 8.387 0.886 0.443

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.14 fj_serialfjs_closed

The closed version.

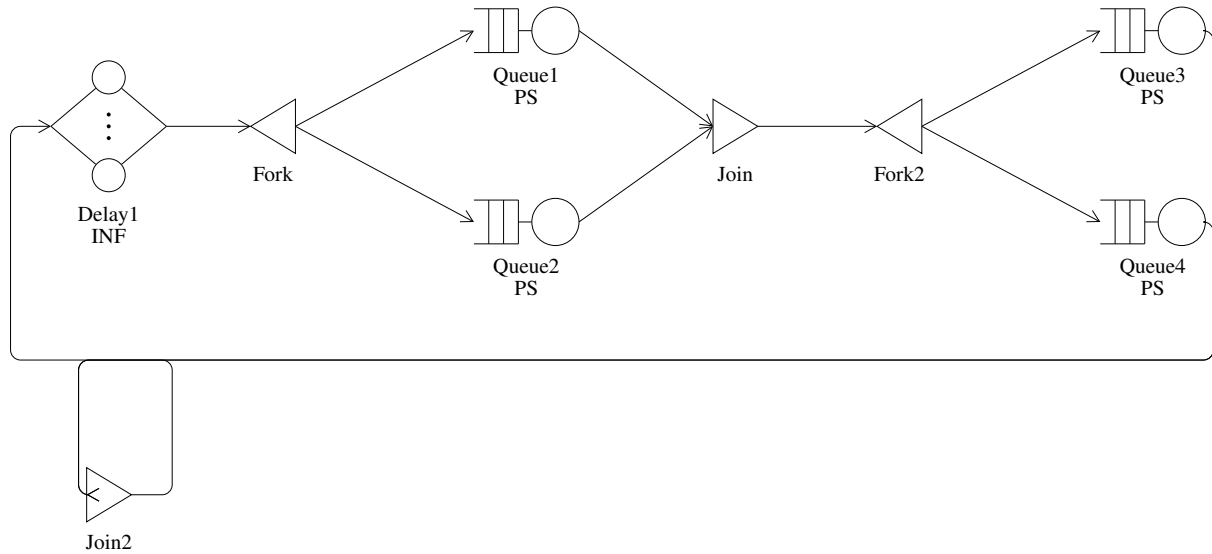


Figure 2.56: Topology of fj_serialfjs_closed: a closed network over 5 queueing stations with a fork-join block.

Nodes	9: Queue $\times$ 4, Delay, Fork $\times$ 2, Join $\times$ 2
Classes	1: class1 (closed, $N = 10$)
Scheduling	Delay1: INF; Queue1: PS; Queue2: PS; Queue3: PS; Queue4: PS
Service	Delay1 – class1: Exp(0.5) Queue1 – class1: Exp(1) Queue2 – class1: Exp(1) Queue3 – class1: Exp(1) Queue4 – class1: Exp(1)
Solvers	JMT, MVA

Table 2.56: Features of fj_serialfjs_closed.

The example is built and solved as follows.

```
Network model = ForkJoinModel.fj_serialfjs_closed();
JMT solverJMT = new JMT(model, "seed", 23000);
MVA solverMVA = new MVA(model);

Object[] solvers = {solverJMT, solverMVA};

for (Object solverObj : solvers) {
    try {
        if (solverObj instanceof JMT) {
            JMT solver = (JMT) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof MVA) {
            MVA solver = (MVA) solverObj;
            solver.getAvgTable().print();
        }
    } catch (Exception e) {
        System.out.println("Error with solver: " + e.getMessage());
    }
}
pauseForUser();
```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 class1 1.5478 1.5478 2.0184 2.0184 0.80819 0.81366
Queue1 class1 2.6984 0.81744 3.4341 3.4341 0.81366 0.80809
Queue2 class1 2.8035 0.80343 3.4227 3.4227 0.81366 0.81267
Join class1 3.1856 0 1.9318 1.9318 1.588 0.80888
Queue3 class1 2.5598 0.80584 3.1944 3.1944 0.80888 0.80815
Queue4 class1 2.6804 0.80079 3.3994 3.3994 0.80888 0.80952
Join2 class1 2.8828 0 1.7538 1.7538 1.6209 0.808
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 418.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 class1 1.5101 1.5101 2 2 0.75503 0.75503
Queue1 class1 2.8654 0.75503 3.7951 3.7951 0.75503 0.75503
Queue2 class1 2.8654 0.75503 3.7951 3.7951 0.75503 0.75503
Join class1 2.8654 0 1.8975 1.8975 1.5101 0.75503
Queue3 class1 2.8654 0.75503 3.7951 3.7951 0.75503 0.75503
Queue4 class1 2.8654 0.75503 3.7951 3.7951 0.75503 0.75503
Join2 class1 2.8654 0 1.8975 1.8975 1.5101 0.75503

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.15 fj_serialfjs_open

A chain of fork-join blocks, open.

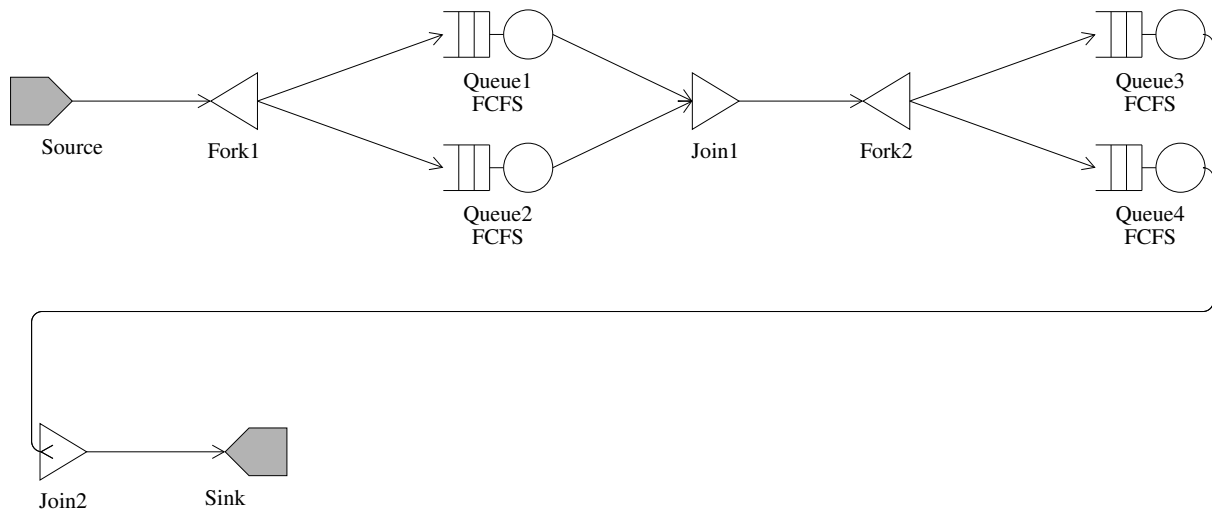


Figure 2.57: Topology of `fj_serialfjs_open`: an open network over 4 queueing stations with a fork-join block.

Nodes	10: Source, Queue $\times$ 4, Fork $\times$ 2, Join $\times$ 2, Sink
Classes	1: class1 (open)
Scheduling	Queue1: FCFS; Queue2: FCFS; Queue3: FCFS; Queue4: FCFS
Arrivals	Source – class1: Exp(0.4)
Service	Queue1 – class1: Exp(1) Queue2 – class1: Exp(1) Queue3 – class1: Exp(1) Queue4 – class1: Exp(1)
Solvers	JMT, MVA

Table 2.57: Features of `fj_serialfjs_open`.

The example is built and solved as follows.

```

Network model = ForkJoinModel.fj_serialfjs_open();
JMT solverJMT = new JMT(model, "seed", 23000);
MVA solverMVA = new MVA(model);

Object[] solvers = {solverJMT, solverMVA};

for (Object solverObj : solvers) {
    try {
        if (solverObj instanceof JMT) {
            JMT solver = (JMT) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof MVA) {
            MVA solver = (MVA) solverObj;
            solver.getAvgTable().print();
        }
    } catch (Exception e) {
        System.out.println("Error with solver: " + e.getMessage());
    }
}
pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass    QLen    Util    RespT    ResidT    ArvR    Tput
Source  class1      0        0        0        0        0    0.40474
Queue1  class1    0.68628  0.40101  1.6675   1.6675   0.40474  0.4092
Queue2  class1    0.69566  0.40012  1.7616   1.7616   0.40474  0.40471
Join1   class1    0.59155  0        0.76312  0.76312  0.83292  0.40867
Queue3  class1    0.68976  0.41292  1.6883   1.6883   0.40867  0.40872
Queue4  class1    0.6731  0.41363  1.6489   1.6489   0.40867  0.41236
Join2   class1    0.61305  0        0.7252   0.7252   0.83343  0.41225
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 809.
Station JobClass    QLen    Util    RespT    ResidT    ArvR    Tput
Source  class1      0        0        0        0        0    0.4
Queue1  class1    0.66666  0.4    1.6667   1.6667   0.4    0.4
Queue2  class1    0.66666  0.4    1.6667   1.6667   0.4    0.4
Join1   class1    0.66666  0    0.83333  0.83333  0.8    0.4
Queue3  class1    0.66666  0.4    1.6667   1.6667   0.4    0.4
Queue4  class1    0.66666  0.4    1.6667   1.6667   0.4    0.4
Join2   class1    0.66666  0    0.83333  0.83333  0.8    0.4
[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.16 fj_threebranches

Three parallel branches rather than two.

Nodes	6: Queue × 3, Delay, Fork, Join
Classes	2: class1 (closed, $N = 10$), class2 (closed, $N = 10$)
Scheduling	Delay1: INF; Queue1: PS; Queue2: PS; Queue3: PS
Service	Delay1 – class1: Exp(0.5); class2: Exp(0.8) Queue1 – class1: Exp(1.5); class2: Exp(2.8) Queue2 – class1: Exp(1.1); class2: Exp(3) Queue3 – class1: Exp(2.5); class2: Exp(1)
Solvers	MVA

Table 2.58: Features of fj_threebranches.

The example is built and solved as follows.

```

Network model = ForkJoinModel.fj_threebranches();
JMT solverJMT = new JMT(model, "seed", 23000);
MVA solverMVA = new MVA(model);

```

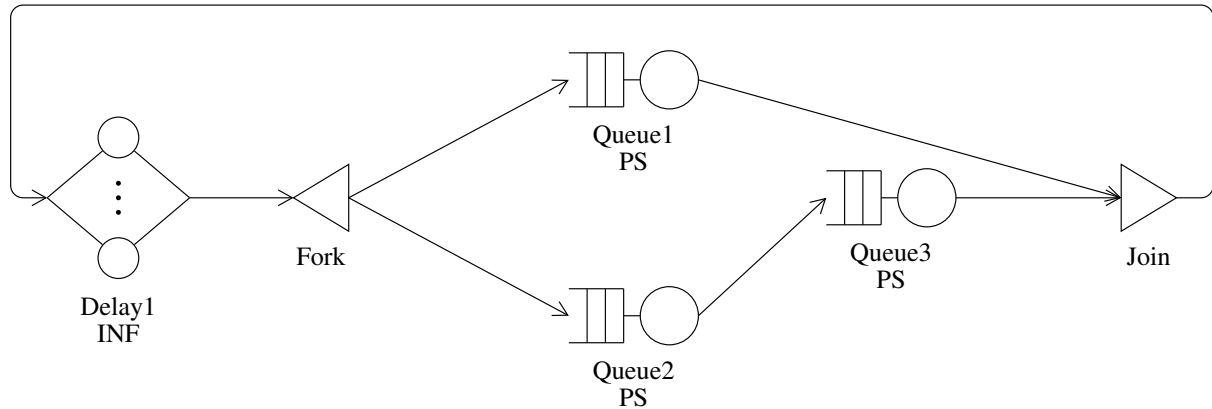


Figure 2.58: Topology of fj_threebranches: a closed network over 4 queueing stations with a fork-join block.

```

Object[] solvers = {solverJMT, solverMVA};

for (Object solverObj : solvers) {
    try {
        if (solverObj instanceof JMT) {
            JMT solver = (JMT) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof MVA) {
            MVA solver = (MVA) solverObj;
            solver.getAvgTable().print();
        }
    } catch (Exception e) {
        System.out.println("Error with solver: " + e.getMessage());
    }
}
pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 class1 1.5598 1.5598 1.9988 1.9988 0.79323 0.79186
Delay1 class2 0.86162 0.86162 1.2766 1.2766 0.67451 0.67426
Queue1 class1 1.8434 0.53176 2.288 2.288 0.79186 0.79224
Queue1 class2 0.94275 0.23676 1.3347 1.3347 0.67426 0.6734
Queue2 class1 4.1168 0.72262 5.2241 5.2241 0.79186 0.79179
Queue2 class2 1.5395 0.22418 2.2137 2.2137 0.67426 0.67346
Queue3 class1 4.011 0.328 5.052 5.052 0.79179 0.79102
Queue3 class2 7.526 0.67036 11.084 11.084 0.67346 0.67391
Join class1 6.8098 0 4.1768 4.1768 1.5954 0.79323
Join class2 8.2743 0 5.9072 5.9072 1.3957 0.67391
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 3224.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay1 class1 1.4591 1.4591 2 2 0.72955 0.72955

... (3 captured-output lines omitted; rerun the source example for the full output)

Queue2 class1 4.8036 0.66323 6.5844 6.5844 0.72955 0.72955
Queue2 class2 1.5573 0.21268 2.4407 2.4407 0.63804 0.63804
Queue3 class1 3.529 0.29182 4.8372 4.8372 0.72955 0.72955
Queue3 class2 7.6515 0.63804 11.992 11.992 0.63804 0.63804
Join class1 7.2315 0 4.9561 4.9561 1.4591 0.72955
Join class2 8.555 0 6.7041 6.7041 1.2761 0.63804

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.17 fj_tiny_closed

The smallest closed fork-join model, small enough for an exact CTMC solve.

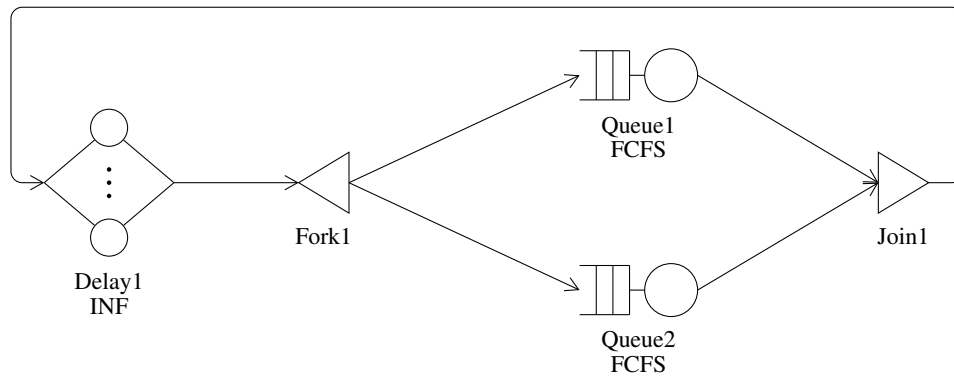


Figure 2.59: Topology of `fj_tiny_closed`: a closed network over 3 queueing stations with a fork-join block.

Nodes	5: Queue $\times$ 2, Delay, Fork, Join
Classes	1: Class1 (closed, $N = 1$)
Scheduling	Delay1: INF; Queue1: FCFS; Queue2: FCFS
Service	Delay1 – Class1: Exp(1) Queue1 – Class1: Exp(2) Queue2 – Class1: Exp(3)
Solvers	CTMC, SSA, JMT

Table 2.59: Features of `fj_tiny_closed`.

The example is built and solved as follows.

```
Network model = ForkJoinModel.fj_tiny_closed();
new CTMC(model, "exact").getAvgTable().print();
new SSA(ForkJoinModel.fj_tiny_closed(), "seed", 23000, "samples", 5e4)
    .getAvgTable().print();
new JMT(ForkJoinModel.fj_tiny_closed(), "seed", 23000).getAvgTable().print();

// closed-form check (N=1): X = 1 / (1 + 1/2 + 1/3 - 1/5) = 0.612244...
System.out.printf("%nclosed form X = %.6f%n", 1.0 / (1 + 1.0 / 2 + 1.0 / 3 - 1.0 / 5));
pauseForUser();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.6.18 fj_twoclasses_forked

Two classes forked independently through the same block.

Nodes	6: Source, Queue $\times$ 2, Fork, Join, Sink
Classes	2: class1 (open), class2 (open)
Scheduling	Queue1: PS; Queue2: PS
Arrivals	Source – class1: Exp(0.25); class2: Exp(0.25)
Service	Queue1 – class1: Exp(1); class2: Immediate Queue2 – class1: Exp(0.75); class2: Exp(2)
Solvers	JMT, MVA

Table 2.60: Features of `fj_twoclasses_forked`.

The example is built and solved as follows.

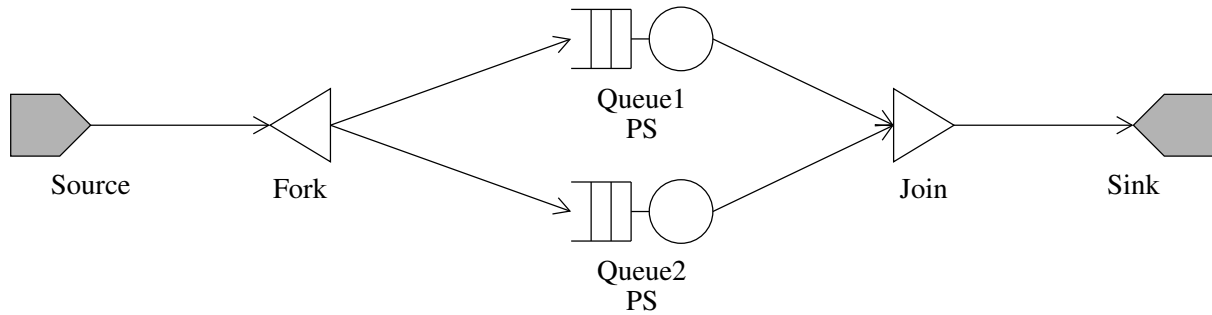


Figure 2.60: Topology of `fj_twoclasses_forked`: an open network over 2 queueing stations with a fork-join block.

```

Network model = ForkJoinModel.fj_twoclasses_forked();
JMT solverJMT = new JMT(model, "seed", 23000);
MVA solverMVA = new MVA(model);

Object[] solvers = {solverJMT, solverMVA};

for (Object solverObj : solvers) {
    try {
        if (solverObj instanceof JMT) {
            JMT solver = (JMT) solverObj;
            solver.getAvgTable().print();
        } else if (solverObj instanceof MVA) {
            MVA solver = (MVA) solverObj;
            solver.getAvgTable().print();
        }
    } catch (Exception e) {
        System.out.println("Error with solver: " + e.getMessage());
    }
}
pauseForUser();

```

Running this edition prints

```

WARNING: [setTasksPerLink] The setTasksPerLink feature is experimental and results may be inaccurate for
analytical solvers.
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Source class1 0 0 0 0 0 0.25128
Source class2 0 0 0 0 0 0.25184
Queue1 class1 1.4252 0.49231 2.9402 2.9402 0.50253 0.49004
Queue1 class2 0 0 0 0 0.50531 0.50531
Queue2 class1 8.7958 0.65567 17.406 17.406 0.50253 0.50863
Queue2 class2 3.4512 0.25836 6.7395 6.7395 0.50531 0.50463
Join class1 14.468 0 14.673 14.673 0.9868 0.24763
Join class2 6.754 0 6.305 6.305 1.028 0.25154
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 7220.
Station JobClass QLen Util RespT ResidT ArvR Tput
Source class1 0 0 0 0 0 0.25
Source class2 0 0 0 0 0 0.25
Queue1 class1 1 0.5 2 2 0.25 0.5
Queue1 class2 0 0 2e-08 0 0.25 0.5
Queue2 class1 8 0.66667 16 16 0.25 0.5
Queue2 class2 3 0.25 6 6 0.25 0.5
Join class1 15.082 0 15.082 15.082 1 0.25
Join class2 6 0 6 6 1 0.25
[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.6.19 fj_variable_fanout

A forking level that is not one fixed number: a per-link vector, a random degree, or a branch that fires only part of the time.

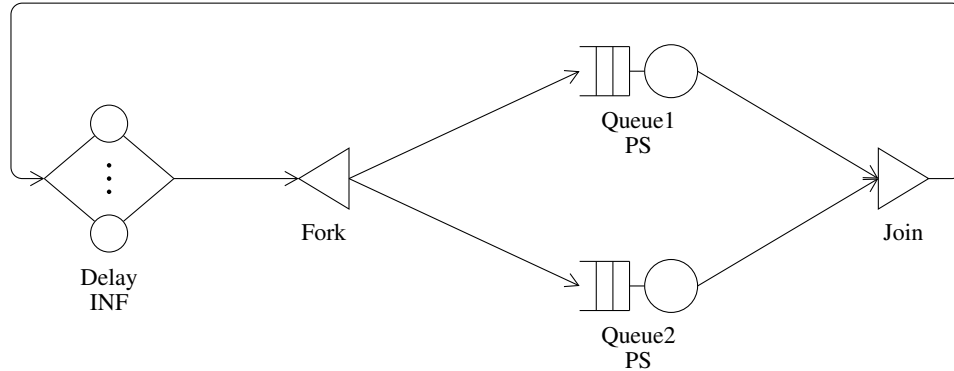


Figure 2.61: Topology of fj_variable_fanout: a closed network over 3 queueing stations with a fork-join block.

Nodes	5: Queue $\times$ 2, Delay, Fork, Join
Classes	1: class1 (closed, $N = 4$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS
Service	Delay – class1: Exp(1) Queue1 – class1: Exp(2) Queue2 – class1: Exp(2)
Solvers	JMT

Table 2.61: Features of fj_variable_fanout.

The example is built and solved as follows.

```
Network model = new Network("model");
Delay delay = new Delay(model, "Delay");
Queue queue1 = new Queue(model, "Queue1", SchedStrategy.PS);
Queue queue2 = new Queue(model, "Queue2", SchedStrategy.PS);
Fork fork = new Fork(model, "Fork");
Join join = new Join(model, "Join", fork);

ClosedClass jobclass1 = new ClosedClass(model, "class1", 4, delay);

delay.setService(jobclass1, new Exp(1.0));
queue1.setService(jobclass1, new Exp(2.0));
queue2.setService(jobclass1, new Exp(2.0));

if ("vector".equals(mode)) {
    fork.setTasksPerLink(jobclass1, 3.0);
} else if ("random".equals(mode)) {
    Matrix pmf = new Matrix(1, 2);
    pmf.set(0, 0, 0.5);
    pmf.set(0, 1, 0.5);
    Matrix pts = new Matrix(1, 2);
    pts.set(0, 0, 1.0);
    pts.set(0, 1, 3.0);
    fork.setTasksPerLinkDistribution(jobclass1, new DiscreteSampler(pmf, pts));
} else if ("prob".equals(mode)) {
    // an uncertain branch needs a Join that does not wait for it
    join.setStrategy(jobclass1, JoinStrategy.PARTIAL);

    ... (8 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

routingMatrix.set(jobclass1, jobclass1, queue1, join, 1.0);
routingMatrix.set(jobclass1, jobclass1, queue2, join, 1.0);
routingMatrix.set(jobclass1, jobclass1, join, delay, 1.0);
```

```
model.link(routingMatrix);
return model;
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

2.7 Layered queueing networks

A layered queueing network (LQN) models software resources explicitly. A `Processor` hosts `Tasks`, each task exposes `Entry` points, and each entry runs an `Activity` graph that consumes host demand and makes synchronous or asynchronous calls to the entries of other tasks. A synchronous call blocks the caller until the reply arrives, which is what creates the layers: a task is simultaneously a server to its callers and a client of the tasks it calls.

The models are solved by `SolverLN`, which decomposes the layers into queueing submodels and iterates to a fixed point, and by the external `lqns` and `lqsim` tools. The same model is also readable from, and writable to, the `.lqn` XML interchange format.

2.7.1 lqn_basic

One reference task calling one server task, the smallest layered model.

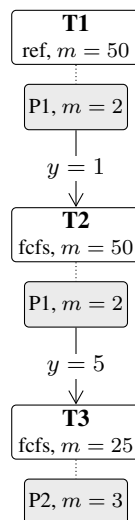


Figure 2.62: Call graph of `lqn_basic`: 3 tasks over 2 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	2: P1 (ps, $m = 2$), P2 (ps, $m = 3$)
Tasks	3: T1 (ref, $m = 50$), T2 (fcfs, $m = 50$), T3 (fcfs, $m = 25$)
Entries	3: E1, E2, E3
Activities	3, host demands AS1: 0.1, AS2: 0.05, AS3: 0.02
Calls	2 (2 synchronous, 0 asynchronous)
Think times	T1: 2.0

Table 2.62: Features of `lqn_basic`.

The example is built and solved as follows.

```

public static void lqn_basic() throws Exception {
    LayeredNetwork model = LayeredModel.lqn_basic();

    // The reference solves this with the layered solver on its default layers.
    new LN(model).getAvgTable().print();

    pauseForUser();
}

```

Running this edition prints

```

Iter 1. Analyze time: 0.302s. Update time: 0.021s. Runtime: 0.304s.
Iter 2. Analyze time: 0.186s. Update time: 0.006s. Runtime: 0.516s.
Iter 3. Analyze time: 0.174s. Update time: 0.017s. Runtime: 0.696s.
Iter 4. Analyze time: 0.142s. Update time: 0.011s. Runtime: 0.860s.
Iter 5. Analyze time: 0.173s. Update time: 0.022s. Runtime: 1.058s.
Iter 6. Analyze time: 0.158s. Update time: 0.028s. Runtime: 1.245s.
Iter 7. Analyze time: 0.138s. Update time: 0.004s. Runtime: 1.412s.
Iter 8. Analyze time: 0.129s. Update time: 0.011s. Runtime: 1.549s.
Iter 9. Analyze time: 0.073s. Update time: 0.007s. Runtime: 1.634s.
Iter 10. Analyze time: 0.127s. Update time: 0.005s. Runtime: 1.773s.
Iter 11. Analyze time: 0.117s. Update time: 0.017s. Runtime: 1.897s.
Iter 12. Analyze time: 0.180s. Update time: 0.014s. Runtime: 2.099s.
Iter 13. Analyze time: 0.136s. Update time: 0.006s. Runtime: 2.250s.
Iter 14. Analyze time: 0.216s. Update time: 0.012s. Runtime: 2.474s.
Iter 15. Analyze time: 0.149s. Update time: 0.007s. Runtime: 2.636s.
Iter 16. Analyze time: 0.117s. Update time: 0.003s. Runtime: 2.770s.

... (43 captured-output lines omitted; rerun the source example for the full output)

E1   Entry      23.4  0.66474  1.7609      NaN   NaN  13.3
E2   Entry      8.7259  0.33237  0.65634      NaN   NaN  13.3
E3   Entry      1.3295  0.44316  0.02         NaN   NaN  66.5
AS1  Activity    23.4  0.66474  1.7609  1.0848      NaN  13.3
AS2  Activity    8.7259  0.33237  0.65634  0.55607     NaN  13.3
AS3  Activity    1.3295  0.44316  0.02       0.02       NaN  66.5

[Running in non-interactive mode, continuing...]

```

2.7.2 lqn_bpmn

A BPMN process translated into a layered model with forks, joins and immediate activities.

Processors	7: R1_Processor (fcfs, $m = 100$), R2_Processor (inf, $m = 1$), R3_Processor (fcfs, $m = 2$), R1A_Processor (fcfs, $m = 7$), R1B_Processor (fcfs, $m = 3$), R2A_Processor (fcfs, $m = 4$), R2B_Processor (fcfs, $m = 5$)
Tasks	7: R1_Task (ref, $m = 100$), R2_Task (inf, $m = 1$), R3_Task (fcfs, $m = 2$), R1A_Task (fcfs, $m = 7$), R1B_Task (fcfs, $m = 3$), R2A_Task (fcfs, $m = 4$), R2B_Task (fcfs, $m = 5$)
Entries	16: R1_Ref_Entry, R2_Synch_A2_Entry, R2_Synch_A5_Entry, R3_Synch_A9_Entry, R1A_Synch_A1_Entry, R1A_Synch_A2_Entry, R1A_Synch_A3_Entry, R1B_Synch_A4_Entry, R1B_Synch_A5_Entry, R1B_Synch_A6_Entry, R2A_Synch_A7_Entry, R2A_Synch_A8_Entry, R2A_Synch_A11_Entry, R2B_Synch_A9_Entry, R2B_Synch_A10_Entry, R2B_Synch_A12_Entry
Activities	29, host demands A1_Empty: 0.0, A2_Empty: 0.0, A5_Empty: 0.0, A6_Empty: 0.0, A3_Empty: 0.0, A4_Empty: 0.0, E4_Empty: 0.0, A7_Empty: 0.0, ...
Calls	15 (15 synchronous, 0 asynchronous)
Think times	R1_Task: 20.0

Table 2.63: Features of lqn_bpmn.

The example is built and solved as follows.

```

LayeredNetwork model = LayeredModel.lqn_bpmn();

LQNS solver = new LQNS(model);
AvgTable avgTable = solver.getLNAvgTable();
avgTable.print();

```

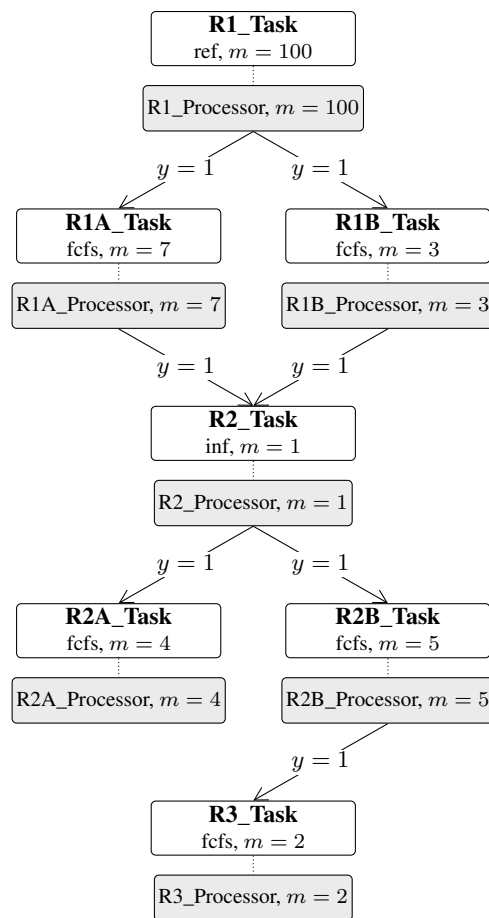


Figure 2.63: Call graph of `lqn_bpmn`: 7 tasks over 7 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

```
pauseForUser();
```

Running this edition prints

Node	NodeType	QLen	Util	RespT	ResidT	ArvR	Tput
R1_Processor	Processor	NaN	0	NaN	NaN	NaN	NaN
R2_Processor	Processor	NaN	0	NaN	NaN	NaN	NaN
R3_Processor	Processor	NaN	1.1895	NaN	NaN	NaN	NaN
R1A_Processor	Processor	NaN	2.0111	NaN	NaN	NaN	NaN
R1B_Processor	Processor	NaN	1.8962	NaN	NaN	NaN	NaN
R2A_Processor	Processor	NaN	1.3324	NaN	NaN	NaN	NaN
R2B_Processor	Processor	NaN	1.7575	NaN	NaN	NaN	NaN
R1_Task	RefTask	98.06	0	NaN	NaN	NaN	0.14365
R2_Task	Task	6.173	0	NaN	NaN	NaN	0.27712
R3_Task	Task	1.2198	1.1895	NaN	NaN	NaN	0.11895
R1A_Task	Task	7.1285	2.0111	NaN	NaN	NaN	0.37349
R1B_Task	Task	3.1688	1.8962	NaN	NaN	NaN	0.34476
R2A_Task	Task	1.3332	1.3324	NaN	NaN	NaN	0.21794
R2B_Task	Task	3.0992	1.7575	NaN	NaN	NaN	0.31185
R1_Ref_Entry	Entry	98.06	0	682.64	NaN	NaN	0.14365

... (38 captured-output lines omitted; rerun the source example for the full output)

A8	Activity	0.64744	0.64716	8.0035	NaN	NaN	0.080894
A11	Activity	0.27434	0.2741	4.0035	NaN	NaN	0.068525
A9	Activity	0.49165	0.49153	4.0009	NaN	NaN	0.12288
A10	Activity	0.73741	0.7373	6.0009	NaN	NaN	0.12288
A12	Activity	0.5287	0.52864	8.0009	NaN	NaN	0.06608
A9_Res_Empty	Activity	1.3414	0	10.916	NaN	NaN	0.12288

[Running in non-interactive mode, continuing...]

2.7.3 lqn_bpmn_trace

The BPMN translation with the activity trace printed.

Processors	7: R1_Processor (fcfs, $m = 100$), R2_Processor (inf, $m = 1$), R3_Processor (fcfs, $m = 2$), R1A_Processor (fcfs, $m = 7$), R1B_Processor (fcfs, $m = 3$), R2A_Processor (fcfs, $m = 4$), R2B_Processor (fcfs, $m = 5$)
Tasks	7: R1_Task (ref, $m = 100$), R2_Task (inf, $m = 1$), R3_Task (fcfs, $m = 2$), R1A_Task (fcfs, $m = 7$), R1B_Task (fcfs, $m = 3$), R2A_Task (fcfs, $m = 4$), R2B_Task (fcfs, $m = 5$)
Entries	16: R1_Ref_Entry, R2_Synch_A2_Entry, R2_Synch_A5_Entry, R3_Synch_A9_Entry, R1A_Synch_A1_Entry, R1A_Synch_A2_Entry, R1A_Synch_A3_Entry, R1B_Synch_A4_Entry, R1B_Synch_A5_Entry, R1B_Synch_A6_Entry, R2A_Synch_A7_Entry, R2A_Synch_A8_Entry, R2A_Synch_A11_Entry, R2B_Synch_A9_Entry, R2B_Synch_A10_Entry, R2B_Synch_A12_Entry
Activities	29, host demands A1_Empty: 0.0, A2_Empty: 0.0, A5_Empty: 0.0, A6_Empty: 0.0, A3_Empty: 0.0, A4_Empty: 0.0, E4_Empty: 0.0, A7_Empty: 0.0, ...
Calls	15 (15 synchronous, 0 asynchronous)
Think times	R1_Task: 20.0

Table 2.64: Features of lqn_bpmn_trace.

The example is built and solved as follows.

```
LayeredNetwork model = LayeredModel.lqn_bpmn_trace();

SolverOptions lnoptions = LN.defaultOptions();
lnoptions.verbose = VerboseLevel.SILENT;
lnoptions.iter_max = 10;
SolverOptions options = MVA.defaultOptions();
options.verbose = VerboseLevel.SILENT;
LN solverLN = new LN(model, m -> new MVA(m, options), lnoptions);

System.out.println("=== Running LN solver with tracing ===");
solverLN.init();

Map<Integer, Map<Integer, SolverResult>> results =
```

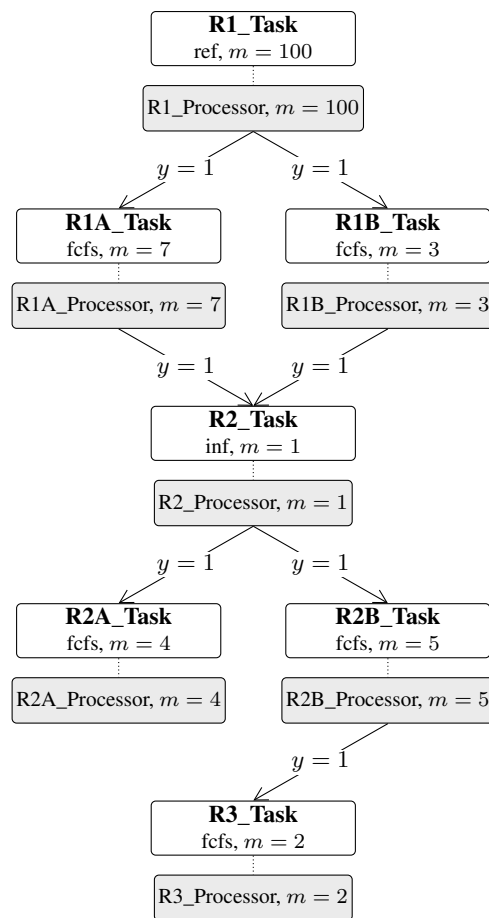


Figure 2.64: Call graph of `lqn_bpmn_trace`: 7 tasks over 7 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

```

    new HashMap<Integer, Map<Integer, SolverResult>>();
for (int it = 1; it <= 5; it++) {
    System.out.println("=== Iteration " + it + " ===");
    solverLN.pre(it);
    Map<Integer, SolverResult> perLayer = new HashMap<Integer, SolverResult>();
    // the layer index is 0-based here and 1-based in the reference, and
    // post() reads the map back by that index, so only the printed
    // number is shifted
    for (int e = 0; e < solverLN.nlayers; e++) {
        SolverResult res = solverLN.analyze(it, e);
        if (res == null) {
            continue;
        }

        ... (25 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

        int idx = nodeNames.indexOf(taskName);
        if (idx >= 0) {
            System.out.printf("%s: Tput=%.6f%n", taskName, tput.get(idx));
        }
    }
}
pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.7.4 lqn_flatph

Method `flat.ph`: the squashed layering carrying the same composed phase-type encoding, one sub-model holding every processor and called task.

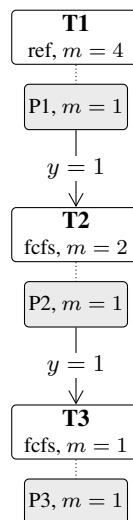


Figure 2.65: Call graph of `lqn_flatph`: 3 tasks over 3 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	3: P1 (ps, $m = 1$), P2 (ps, $m = 1$), P3 (ps, $m = 1$)
Tasks	3: T1 (ref, $m = 4$), T2 (fcfs, $m = 2$), T3 (fcfs, $m = 1$)
Entries	3: E1, E2, E3
Activities	3, host demands A1: 0.2, A2: 0.2, A3: 0.2
Calls	2 (2 synchronous, 0 asynchronous)
Think times	T1: 1.0

Table 2.65: Features of `lqn_flatph`.

The example is built and solved as follows.

```
LayeredNetwork model = LayeredModel.lqn_flatph();

String[] methods = {"srvn.cs", "srvn.ph", "flat.cs", "flat.ph"};
for (String method : methods) {
    SolverOptions options = LN.defaultOptions();
    options.method = method;
    options.verbose = VerboseLevel.SILENT;
    LN solver = new LN(model, options);
    AvgTable avgTable = solver.getAvgTable();
    System.out.printf("%n--- method=%s (built as %s, %d submodel(s))%n",
        method, solver.lnmethod, solver.getEnsemble().size());
    avgTable.setOptions(printing());
    avgTable.print();
}
pauseForUser();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.7.5 lqn_fork_open_arrival

An AND-fork under an open arrival stream.

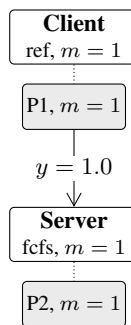


Figure 2.66: Call graph of `lqn_fork_open_arrival`: 2 tasks over 2 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	2: P1 (inf, $m = 1$), P2 (inf, $m = 1$)
Tasks	2: Client (ref, $m = 1$), Server (fcfs, $m = 1$)
Entries	3: CE, SE, OE
Activities	9, host demands CA: 0.5, RA1: 0.2, RA2: 0.3, RA3: 0.4, RA4: 0.1, OA1: 0.2, OA2: 0.3, OA3: 0.4, ...
Calls	1 (1 synchronous, 0 asynchronous)
Think times	Client: 1.0

Table 2.66: Features of `lqn_fork_open_arrival`.

The example is built and solved as follows.

```
LayeredNetwork model = new LayeredNetwork("lqnForkOpenArrival");

Processor P1 = new Processor(model, "P1", Integer.MAX_VALUE, SchedStrategy.INF);
Processor P2 = new Processor(model, "P2", Integer.MAX_VALUE, SchedStrategy.INF);

Task T1 = new Task(model, "Client", 1, SchedStrategy.REF).on(P1);
T1.setThinkTime(Exp.fitMean(1.0));
Task T2 = new Task(model, "Server", 1, SchedStrategy.FCFS).on(P2);
T2.setThinkTime(Immediate.getInstance());
```



```

Entry CE = new Entry(model, "CE").on(T1);
Entry SE = new Entry(model, "SE").on(T2);
Entry OE = new Entry(model, "OE").on(T2);
OE.setArrival(new Exp(0.1));

new Activity(model, "CA", Exp.fitMean(0.5)).on(T1).boundTo(CE).synchCall(SE, 1);

new Activity(model, "RA1", Exp.fitMean(0.2)).on(T2).boundTo(SE);
new Activity(model, "RA2", Exp.fitMean(0.3)).on(T2);
new Activity(model, "RA3", Exp.fitMean(0.4)).on(T2);
new Activity(model, "RA4", Exp.fitMean(0.1)).on(T2).repliesTo(SE);

new Activity(model, "OA1", Exp.fitMean(0.2)).on(T2).boundTo(OE);
new Activity(model, "OA2", Exp.fitMean(0.3)).on(T2);
new Activity(model, "OA3", Exp.fitMean(0.4)).on(T2);
new Activity(model, "OA4", Exp.fitMean(0.1)).on(T2);

... (8 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

openBranches.add("OA2");
openBranches.add("OA3");
T2.addPrecedence(ActivityPrecedence.AndFork("OA1", openBranches));
T2.addPrecedence(ActivityPrecedence.AndJoin(openBranches, "OA4"));

return model;

```

Running this edition prints

```
jline.lang.layered.LayeredNetwork@533ddba
```

2.7.6 lqn_init

A layered model whose SolverLN run is warm-started from the LQNS solution, and the elapsed time that initialization saves.

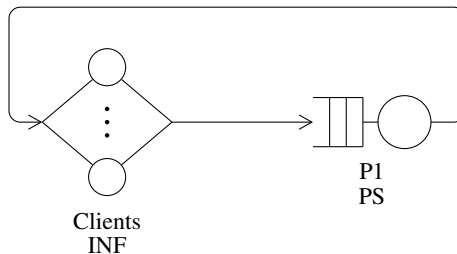


Figure 2.67: Topology of lqn_init: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	1: T1 (closed, $N = 10$)
Scheduling	Clients: INF; P1: PS
Service	Clients – T1: Exp(0.008971) P1 – T1: APH

Table 2.67: Features of lqn_init.

The example is built and solved as follows.

```

System.out.println(
    "This example illustrates the initialization of LN using the output of LQNS.");
LayeredNetwork model = LayeredModel.lqn_serial();

// LQNS, whose solution is what an LN run can be initialized from
if (LQNS.isAvailable()) {

```

```

    SolverOptions options = LQNS.defaultOptions();
    options.keep = true; // keep the intermediate XML files of the translation
    AvgTable lqnsTable = new LQNS(model, options).getLNAvgTable();
    System.out.println("\nLQNS Results:");
    lqnsTable.setOptions(printing());
    lqnsTable.print();
} else {
    System.out.println("\nLQNS solver not available - skipping.");
}

// LN without initialization, for the elapsed time the initialization saves
System.out.println("\nSolve with LN without initialization:");
long t0 = System.nanoTime();
AvgTable lnTable = new LN(model, m -> new MVA(m)).getAvgTable();
double elapsed = (System.nanoTime() - t0) / 1e9;
lnTable.setOptions(printing());
lnTable.print();
System.out.printf("Time elapsed: %.3fs\n", elapsed);

// CDF of response times, taken on the layer the calls terminate in

    ... (14 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

    }
} else {
    System.out.println("Model ensemble has fewer than three layers.");
}
pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.7.7 lqn_jsq

Join-the-shortest-queue dispatching between replicated tasks.

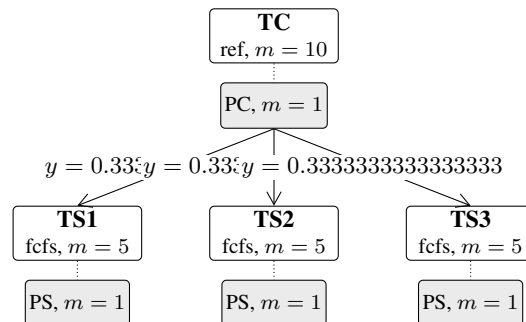


Figure 2.68: Call graph of lqn_jsq: 4 tasks over 2 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	2: PC (inf, $m = 1$), PS (ps, $m = 1$)
Tasks	4: TC (ref, $m = 10$), TS1 (fcfs, $m = 5$), TS2 (fcfs, $m = 5$), TS3 (fcfs, $m = 5$)
Entries	4: EC, ES1, ES2, ES3
Activities	4, host demands AC: 0.5, AS1: 1.0, AS2: 1.0, AS3: 1.0
Calls	3 (3 synchronous, 0 asynchronous)
Think times	TC: 5.0

Table 2.68: Features of lqn_jsq.

The example is built and solved as follows.

```

LayeredNetwork model = LayeredModel.lqn_jsq();

SolverOptions options = LN.defaultOptions();
options.config.layering = "flat";
options.verbose = VerboseLevel.SILENT;
AvgTable avgTable = new LN(model, m -> new SSA(m, "verbose", false), options,
    SolverType.SSA).getAvgTable();
avgTable.setOptions(printing());
avgTable.print();
pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.7.8 lqn_moment3

The third moment of the entry service time, and how it propagates up the layers.

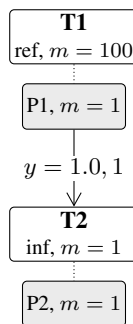


Figure 2.69: Call graph of `lqn_moment3`: 2 tasks over 2 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	2: P1 (ps, $m = 1$), P2 (ps, $m = 1$)
Tasks	2: T1 (ref, $m = 100$), T2 (inf, $m = 1$)
Entries	3: E1, E2, E3
Activities	5, host demands A1: 1.0, A20: 1.0, A21: 1.0, A22: 1.0, A3: 1.0
Calls	2 (2 synchronous, 0 asynchronous)
Think times	T1: 10.0
Solvers	FLD

Table 2.69: Features of `lqn_moment3`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.7.9 lqn_multi_solvers

One layered model solved by every layered solver, side by side.

The example is built and solved as follows.

```

LayeredNetwork model = LayeredModel.lqn_multi_solvers();

// LQNS solver as in MATLAB (lines 27-29)
SolverOptions lqnsOptions = LQNS.defaultOptions();
lqnsOptions.keep = true;

```

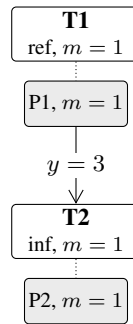


Figure 2.70: Call graph of `lqn_multi_solvers`: 2 tasks over 2 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	2: P1 (inf, $m = 1$), P2 (inf, $m = 1$)
Tasks	2: T1 (ref, $m = 1$), T2 (inf, $m = 1$)
Entries	2: E1, E2
Activities	2, host demands A1: 1.0, A2: 1.0
Calls	1 (1 synchronous, 0 asynchronous)
Think times	T1: 0.0001

Table 2.70: Features of `lqn_multi_solvers`.

```

lqnsOptions.verbose = VerboseLevel.STD;
LQNS lqnsSolver = new LQNS(model, lqnsOptions);
AvgTable avgTableLQNS = lqnsSolver.getLNAvgTable();
avgTableLQNS.print();

// LN with MVA solver in each layer (lines 37-39)
SolverOptions lnOptions = LN.defaultOptions();
lnOptions.verbose = VerboseLevel.SILENT;
lnOptions.seed = 2300;
SolverOptions mvaOptions = MVA.defaultOptions();
mvaOptions.verbose = VerboseLevel.SILENT;
LN solverLN_MVA = new LN(model, (subModel) -> new MVA(subModel, mvaOptions), lnOptions);
AvgTable avgTableLN_MVA = solverLN_MVA.getAvgTable();
avgTableLN_MVA.print();

// LN with NC solver in each layer (lines 47-49)
SolverOptions ncOptions = NC.defaultOptions();
ncOptions.verbose = VerboseLevel.SILENT;
LN solverLN_NC = new LN(model, (subModel) -> new NC(subModel, ncOptions), lnOptions);
AvgTable avgTableLN_NC = solverLN_NC.getAvgTable();
avgTableLN_NC.print();

pauseForUser();

```

Running this edition prints

Node	NodeType	QLen	Util	RespT	ResidT	ArvR	Tput
P1	Processor	NaN	0.24999	NaN	NaN	NaN	NaN
P2	Processor	NaN	0.74998	NaN	NaN	NaN	NaN
T1	RefTask	0.99998	0.24999	NaN	NaN	NaN	0.24999
T2	Task	0.74998	0.74998	NaN	NaN	NaN	0.74998
E1	Entry	0.99998	0.24999	4	NaN	NaN	0.24999
E2	Entry	0.74998	0.74998	1	NaN	NaN	0.74998
A1	Activity	0.99998	0.24999	4	NaN	NaN	0.24999
A2	Activity	0.74998	0.74998	1	NaN	NaN	0.74998

[Running in non-interactive mode, continuing...]

2.7.10 lqn_ofbiz

The Apache OFBiz case study, read from its .lqn file.

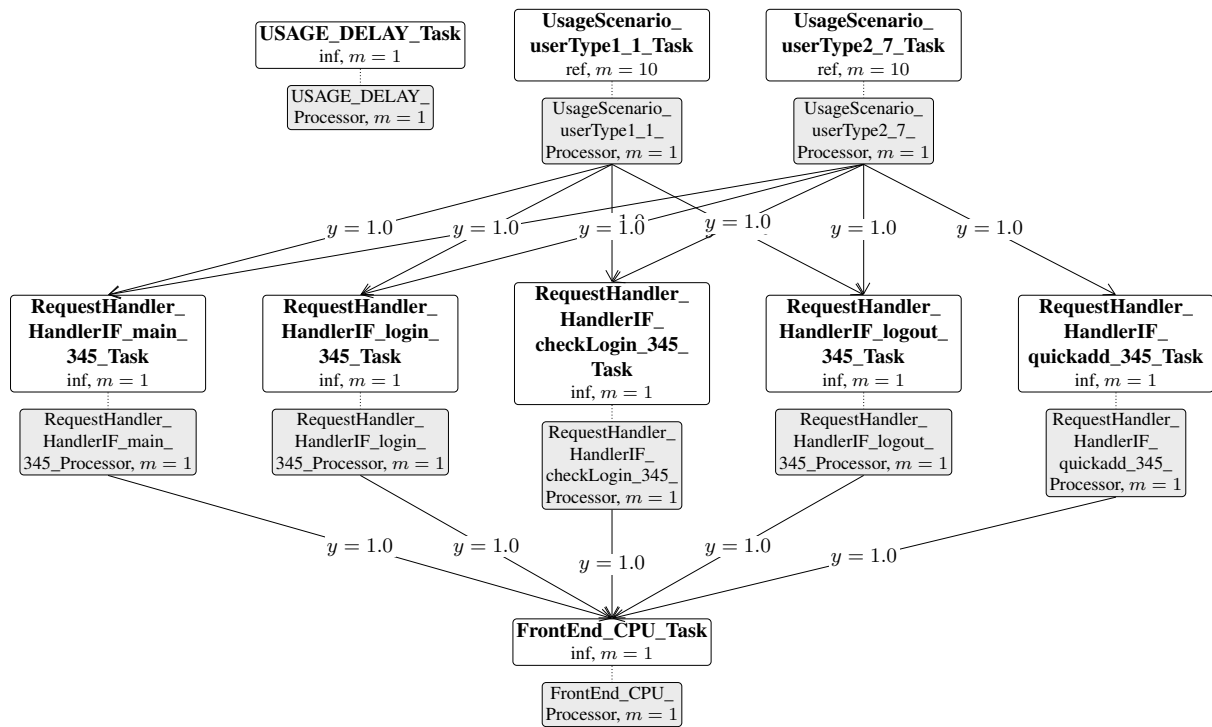


Figure 2.71: Call graph of lqn_ofbiz: 9 tasks over 9 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

The example is built and solved as follows.

```
LayeredNetwork model = LayeredModel.lqn_ofbiz();

try {
    new LQNS(model).getLNAvgTable().print();
} catch (Exception e) {
    System.out.println("LQNS failed: " + e.getMessage());
}

// NC layers: the golden keys this row LN(NC).
new LN(model, (subModel) -> new NC(subModel, "verbose", false)).getAvgTable().print();

pauseForUser();
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

2.7.11 lqn_open_arrival

An open arrival stream into an entry, instead of a closed reference task.

The example is built and solved as follows.

```
LayeredNetwork model = new LayeredNetwork("openArrivalLQN");

Processor P1 = new Processor(model, "P1", 1, SchedStrategy.PS);
Task T1 = new Task(model, "T1", 1, SchedStrategy.FCFS).on(P1);
T1.setThinkTime(Immediate.getInstance());
Entry E1 = new Entry(model, "E1").on(T1);
E1.setArrival(new Exp(0.2));
```

Processors	9: FrontEnd_CPU_Processor (ps, $m = 1$), USAGE_DELAY_Processor (inf, $m = 1$), UsageScenario_userType1_1_Processor (fcfs, $m = 1$), RequestHandler_HandlerIF_main_345_Processor (fcfs, $m = 1$), RequestHandler_HandlerIF_login_345_Processor (fcfs, $m = 1$), RequestHandler_HandlerIF_checkLogin_345_Processor (fcfs, $m = 1$), RequestHandler_HandlerIF_logout_345_Processor (fcfs, $m = 1$), UsageScenario_userType2_7_Processor (fcfs, $m = 1$), RequestHandler_HandlerIF_quickadd_345_Processor (fcfs, $m = 1$)
Tasks	9: FrontEnd_CPU_Task (inf, $m = 1$), USAGE_DELAY_Task (inf, $m = 1$), UsageScenario_userType1_1_Task (ref, $m = 10$), RequestHandler_HandlerIF_main_345_Task (inf, $m = 1$), RequestHandler_HandlerIF_login_345_Task (inf, $m = 1$), RequestHandler_HandlerIF_checkLogin_345_Task (inf, $m = 1$), RequestHandler_HandlerIF_logout_345_Task (inf, $m = 1$), UsageScenario_userType2_7_Task (ref, $m = 10$), RequestHandler_HandlerIF_quickadd_345_Task (inf, $m = 1$)
Entries	14: FrontEnd_CPU_Entry, InternalAction_main_Iu1-wMhoEeKON4DtRoKCMw_34_50_Entry, InternalAction_login_YgxRGw26EeSPwb7XgvxhWQ_34_50_Entry, InternalAction_CheckLogin_vHmZsMhoEeKON4DtRoKCMw_34_50_Entry, InternalAction_logout_j-2E4MhrEeKON4DtRoKCMw_34_50_Entry, InternalAction_addcart_5JEHQMhoEeKON4DtRoKCMw_34_50_Entry, USAGE_DELAY0_Entry, UsageScenario_userType1_1_Entry, RequestHandler_HandlerIF_main_345_Entry, RequestHandler_HandlerIF_login_345_Entry, RequestHandler_HandlerIF_checkLogin_345_Entry, RequestHandler_HandlerIF_logout_345_Entry, UsageScenario_userType2_7_Entry, RequestHandler_HandlerIF_quickadd_345_Entry
Activities	40, host demands FrontEnd_CPU_Activity: 0.0, InternalAction_main_Iu1-wMhoEeKON4DtRoKCMw_34_50_Activity: 0.01, InternalAction_login_YgxRGw26EeSPwb7XgvxhWQ_34_50_Activity: 0.01, InternalAction_CheckLogin_vHmZsMhoEeKON4DtRoKCMw_34_50_Activity: 0.01, InternalAction_logout_j-2E4MhrEeKON4DtRoKCMw_34_50_Activity: 0.01, InternalAction_addcart_5JEHQMhoEeKON4DtRoKCMw_34_50_Activity: 0.01, USAGE_DELAY0_Activity: 0.0, Start2: 0.0, ...
Calls	19 (19 synchronous, 0 asynchronous)
Think times	UsageScenario_userType1_1_Task: 10.0, UsageScenario_userType2_7_Task: 14.0

Table 2.71: Features of lqn_ofbiz.

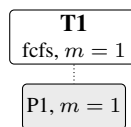


Figure 2.72: Call graph of lqn_open_arrival: 1 task over 1 processor, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	1: P1 (ps, $m = 1$)
Tasks	1: T1 (fcfs, $m = 1$)
Entries	1: E1
Activities	1, host demands A1: 1.6

Table 2.72: Features of lqn_open_arrival.

```
new Activity(model, "A1", Exp.fitMean(1.6)).on(T1).boundTo(E1).repliesTo(E1);

return model;
```

Running this edition prints

```
jline.lang.layered.LayeredNetwork@377dca04
```

2.7.12 lqn_paramident

Parameter identification on a layered model, fitting host demands to measurements.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows.

```
LayeredNetwork model = LayeredModel.lqn_paramident();

// what to estimate (paramSpec) and what is observed (obsSpec)
List<ParamSpec> paramSpec = Arrays.asList(
    ParamSpec.think("T1"), ParamSpec.hostDemand("AS3"));
List<ObsSpec> obsSpec = Arrays.asList(
    ObsSpec.respT("E1"), ObsSpec.util("P1"), ObsSpec.util("P2"));

// ground-truth parameter trajectory with a step change
int nsteps = 24;
Matrix aTrueSeq = new Matrix(2, nsteps);
for (int k = 0; k < nsteps; k++) {
    aTrueSeq.set(0, k, k >= 12 ? 1.0 : 0.5); // step: Z doubles at step 13
    aTrueSeq.set(1, k, (k >= 6 && k <= 17) ? 1.0 / 25 : 1.0 / 50); // S_d pulse
}

// synthesise the measurement matrix Z from the true model plus noise
SolverOptions solveropts = LN.defaultOptions();
solveropts.verbose = VerbosityLevel.SILENT;
int no = obsSpec.size();
Matrix Z = new Matrix(no, nsteps);
java.util.Random rng = new java.util.Random(12345);
for (int k = 0; k < nsteps; k++) {
    Matrix a = new Matrix(2, 1);
    a.set(0, 0, aTrueSeq.get(0, k));
    a.set(1, 0, aTrueSeq.get(1, k));

    ... (35 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

    info.ahat.get(0, nsteps - 1, aTrueSeq.get(0, nsteps - 1),
        info.ahat.get(1, nsteps - 1, aTrueSeq.get(1, nsteps - 1)));
System.out.printf(
    "RMS parameter tracking error Ea = %.4g, prediction error Er = %.4g%n",
    info.Ea, info.Er);
pauseForUser();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.7.13 lqn_rrrobin

Round-robin dispatching between replicated tasks.

The example is built and solved as follows.

```
LayeredNetwork model = LayeredModel.lqn_rrrobin();

SolverOptions options = LN.defaultOptions();
options.config.layering = "flat";
```

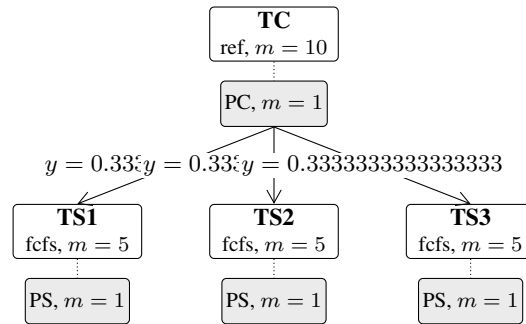


Figure 2.73: Call graph of `lqn_rrabin`: 4 tasks over 2 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	2: PC (inf, $m = 1$), PS (ps, $m = 1$)
Tasks	4: TC (ref, $m = 10$), TS1 (fcfs, $m = 5$), TS2 (fcfs, $m = 5$), TS3 (fcfs, $m = 5$)
Entries	4: EC, ES1, ES2, ES3
Activities	4, host demands AC: 0.5, AS1: 1.0, AS2: 1.0, AS3: 1.0
Calls	3 (3 synchronous, 0 asynchronous)
Think times	TC: 5.0

Table 2.73: Features of `lqn_rrabin`.

```

options.verbose = VerboseLevel.SILENT;
AvgTable avgTable = new LN(model, m -> new SSA(m, "verbose", false), options,
    SolverType.SSA).getAvgTable();
avgTable.setOptions(printing());
avgTable.print();
pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.7.14 `lqn_serial`

A serial call chain across three layers, with an immediate activity.

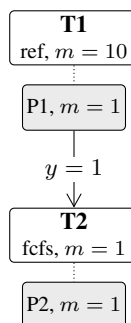


Figure 2.74: Call graph of `lqn_serial`: 2 tasks over 2 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

The example is built and solved as follows.

```

LayeredNetwork model = LayeredModel.lqn_serial();

// Create LQNS solver as in MATLAB (lines 32-34)
SolverOptions options = LQNS.defaultOptions();

```


Processors	2: P1 (ps, $m = 1$), P2 (ps, $m = 1$)
Tasks	2: T1 (ref, $m = 10$), T2 (fcfs, $m = 1$)
Entries	2: E1, E2
Activities	4, host demands AS1: 1.6, AS2: 0.0, AS3: 5.0, AS4: 1.0
Calls	1 (1 synchronous, 0 asynchronous)
Think times	T1: 100.0

Table 2.74: Features of `lqn_serial`.

```

options.keep = true;
LQNS solver = new LQNS(model, options);
AvgTable avgTable = solver.getLNAvgTable();
avgTable.print();

// Get raw avg tables as in MATLAB (lines 37-39)
AvgTable[] rawTables = solver.getRawAvgTables();
rawTables[0].print();
if (rawTables.length > 1) {
    rawTables[1].print();
}

// LN as in MATLAB (line 41)
LN solverLN = new LN(model, SolverType.MVA);
AvgTable avgTableLN = solverLN.getAvgTable();
avgTableLN.print();

pauseForUser();

```

Running this edition prints

Node	NodeType	QLen	Util	RespT	ResidT	ArvR	Tput
P1	Processor	NaN	0.14206	NaN	NaN	NaN	NaN
P2	Processor	NaN	0.53271	NaN	NaN	NaN	NaN
T1	RefTask	1.1215	0.14206	NaN	NaN	NaN	0.088786
T2	Task	0.53271	0.53271	NaN	NaN	NaN	0.088786
E1	Entry	1.1215	0.14206	12.631	NaN	NaN	0.088786
E2	Entry	0.53271	0.53271	6	NaN	NaN	0.088786
AS1	Activity	0.16272	0.14206	1.8327	NaN	NaN	0.088786
AS2	Activity	0.95874	0	10.798	NaN	NaN	0.088786
AS3	Activity	0.44393	0.44393	5	NaN	NaN	0.088786
AS4	Activity	0.088786	0.088786	1	NaN	NaN	0.088786

Node	NodeType	Util	P1Util	P2Util	P1SvcT	P2SvcT	Tput	ProcWait	ProcUtil
P1	Processor	0	0	0	0	0	0	0	0.14206
P2	Processor	0	0	0	0	0	0	0	0.53271
T1	Task	1.1215	1.1215	0	0	0	0.088786	0	0.14206
T2	Task	0.53271	0.53271	0	0	0	0.088786	0	0.53271

... (67 captured-output lines omitted; rerun the source example for the full output)

E1	Entry	1.1733	0.14123	13.3	NaN	NaN	0.088267		
E2	Entry	0.5296	0.5296	6	NaN	NaN	0.088267		
AS1	Activity	0.16143	0.14123	1.8289	1.8289	NaN	0.088267		
AS2	Activity	1.0119	0	11.464	0	NaN	0.088267		
AS3	Activity	0.44133	0.44133	5	5	NaN	0.088267		
AS4	Activity	0.088267	0.088267	1	1	NaN	0.088267		

[Running in non-interactive mode, continuing...]

2.7.15 `lqn_server_pools`

A layered model with heterogeneous server pools, compared with the compatibility representation that treats the servers as interchangeable.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows.

```

SolverOptions opt = LN.defaultOptions();
// a compatibility declaration is a station rate law, which only the
// class-switching layerings can carry
opt.method = "srvn.cs";

System.out.println("--- homogeneous pool on P1 ---");
new LN(LayeredModel.lqn_server_pools(false), opt).getAvgTable().print();

System.out.println("--- compatibility pool on P1 ---");
new LN(LayeredModel.lqn_server_pools(true), opt).getAvgTable().print();

pauseForUser();

```

Running this edition prints

```

--- homogeneous pool on P1 ---

Iter 1. Analyze time: 0.271s. Update time: 0.027s. Runtime: 0.272s.
Iter 2. Analyze time: 0.167s. Update time: 0.023s. Runtime: 0.467s.
Iter 3. Analyze time: 0.140s. Update time: 0.029s. Runtime: 0.631s.
Iter 4. Analyze time: 0.069s. Update time: 0.011s. Runtime: 0.735s.
Iter 5. Analyze time: 0.036s. Update time: 0.026s. Runtime: 0.783s.
Iter 6. Analyze time: 0.043s. Update time: 0.022s. Runtime: 0.852s.
Iter 7. Analyze time: 0.069s. Update time: 0.022s. Runtime: 0.944s.
Iter 8. Analyze time: 0.083s. Update time: 0.028s. Runtime: 1.055s.
Iter 9. Analyze time: 0.071s. Update time: 0.018s. Runtime: 1.159s.
Iter 10. Analyze time: 0.081s. Update time: 0.032s. Runtime: 1.270s.
Iter 11. Analyze time: 0.064s. Update time: 0.029s. Runtime: 1.366s.
Iter 12. Analyze time: 0.069s. Update time: 0.028s. Runtime: 1.467s.
Iter 13. Analyze time: 0.096s. Update time: 0.031s. Runtime: 1.595s.
Iter 14. Analyze time: 0.078s. Update time: 0.023s. Runtime: 1.722s.

... (112 captured-output lines omitted; rerun the source example for the full output)

E1  Entry      1.7208  0.63961  1.3452      NaN  NaN  1.2792
E2  Entry      0.28853  0.14212  0.22558      NaN  NaN  1.279
E3  Entry      0.4248  0.21311  0.33222      NaN  NaN  1.2787
A1  Activity   1.7225  0.63961  1.3465  0.78479  NaN  1.2792
A2  Activity   0.28853  0.14212  0.22558  0.22558  NaN  1.279
A3  Activity   0.4248  0.21311  0.33222  0.33222  NaN  1.2787

[Running in non-interactive mode, continuing...]

```

2.7.16 lqn_setup

Task setup and initialization activities before service.

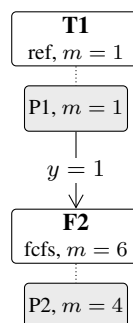


Figure 2.75: Call graph of `lqn_setup`: 2 tasks over 2 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

The example is built and solved as follows.

```

LayeredNetwork model = LayeredModel.lqn_setup();

// The reference drives MVA layers, which is what its golden holds.

```

Processors	2: P1 (inf, $m = 1$), P2 (fcfs, $m = 4$)
Tasks	2: T1 (ref, $m = 1$), F2 (fcfs, $m = 6$)
Entries	2: E1, E2
Activities	2, host demands A1: 1.0, A2: 0.3333333333333333
Calls	1 (1 synchronous, 0 asynchronous)

Table 2.75: Features of `lqn_setup`.

```
new LN(model, SolverType.MVA).getAvgTable().print();

pauseForUser();
```

Running this edition prints

```
pfqn_mvald: MVA-LD is numerically unstable on this model, LINE will force all probabilities to be non-
negative.

Iter 1. Analyze time: 0.110s. Update time: 0.017s. Runtime: 0.111s.
Iter 2. Analyze time: 0.023s. Update time: 0.018s. Runtime: 0.155s.
Iter 3. Analyze time: 0.025s. Update time: 0.013s. Runtime: 0.199s.
Iter 4. Analyze time: 0.026s. Update time: 0.025s. Runtime: 0.239s.
Iter 5. Analyze time: 0.019s. Update time: 0.010s. Runtime: 0.288s.
Iter 6. Analyze time: 0.018s. Update time: 0.013s. Runtime: 0.321s.
Iter 7. Analyze time: 0.025s. Update time: 0.014s. Runtime: 0.372s.
Iter 8. Analyze time: 0.034s. Update time: 0.015s. Runtime: 0.422s.
Iter 9. Analyze time: 0.012s. Update time: 0.008s. Runtime: 0.451s.
Iter 10. Analyze time: 0.007s. Update time: 0.029s. Runtime: 0.471s.
Iter 11. Analyze time: 0.016s. Update time: 0.012s. Runtime: 0.523s.
Iter 12. Analyze time: 0.013s. Update time: 0.024s. Runtime: 0.550s.
Iter 13. Analyze time: 0.016s. Update time: 0.020s. Runtime: 0.594s.
Iter 14. Analyze time: 0.009s. Update time: 0.011s. Runtime: 0.625s.

... (42 captured-output lines omitted; rerun the source example for the full output)

T1   RefTask      1      0.5      NaN      1      NaN      0.5
F2   Task         0.5  0.041667  NaN    0.33333  NaN      0.5
E1   Entry        1      0.5      2      NaN     NaN      0.5
E2   Entry        0.5  0.041667  1      NaN     NaN      0.5
A1   Activity     1      0.5      2      1      NaN      0.5
A2   Activity     0.16667 0.041667 0.33333 0.33333 NaN      0.5

[Running in non-interactive mode, continuing...]
```

2.7.17 `lqn_sockshop`

The Sock Shop microservice benchmark as a layered model.

Processors	7: P1 (inf, $m = 1$), P2_1 (ps, $m = 1$), P2_2 (ps, $m = 1$), P2_3 (ps, $m = 1$), P3_1 (ps, $m = 1$), P3_2 (ps, $m = 1$), P3_3 (ps, $m = 1$)
Tasks	7: T0 (ref, $m = 1000$), T1 (fcfs, $m = 24$), T2 (fcfs, $m = 21$), T6 (fcfs, $m = 100$), T3 (fcfs, $m = 139$), T4 (fcfs, $m = 16$), T5 (fcfs, $m = 151$)
Entries	12: E0, E1, E2, E3, E4, E11, E5, E6, E7, E8, E9, E10
Activities	24, host demands AS: 5e-08, AS0: 5e-08, AS1: 5e-08, AS2: 0.0022216, AH1: 5e-08, AH2: 0.0021319, AH3: 5e-08, AH4: 0.0037561, ...
Calls	14 (14 synchronous, 0 asynchronous)
Think times	T0: 7.0

Table 2.76: Features of `lqn_sockshop`.

The example is built and solved as follows.

```
LayeredNetwork model = LayeredModel.lqn_sockshop();

LN solverLN = new LN(model, SolverType.MVA);
AvgTable avgTable = solverLN.getAvgTable();
avgTable.print();
```

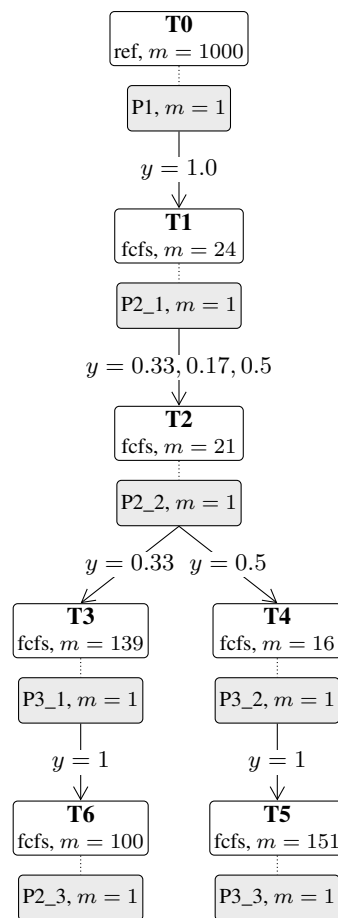


Figure 2.76: Call graph of `lqn_sockshop`: 7 tasks over 7 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

```
pauseForUser();
```

Running this edition prints

```
Iter 1. Analyze time: 0.262s. Update time: 0.113s. Runtime: 0.271s.
Iter 2. Analyze time: 0.154s. Update time: 0.067s. Runtime: 0.542s.
Iter 3. Analyze time: 0.137s. Update time: 0.074s. Runtime: 0.755s.
Iter 4. Analyze time: 0.268s. Update time: 0.074s. Runtime: 1.104s.
Iter 5. Analyze time: 0.323s. Update time: 0.039s. Runtime: 1.506s.
Iter 6. Analyze time: 0.149s. Update time: 0.069s. Runtime: 1.701s.
Iter 7. Analyze time: 0.253s. Update time: 0.072s. Runtime: 2.035s.
Iter 8. Analyze time: 0.280s. Update time: 0.042s. Runtime: 2.392s.
Iter 9. Analyze time: 0.063s. Update time: 0.030s. Runtime: 2.502s.
Iter 10. Analyze time: 0.080s. Update time: 0.042s. Runtime: 2.614s.
Iter 11. Analyze time: 0.082s. Update time: 0.030s. Runtime: 2.744s.
Iter 12. Analyze time: 0.080s. Update time: 0.023s. Runtime: 2.857s.
Iter 13. Analyze time: 0.075s. Update time: 0.027s. Runtime: 2.962s.
Iter 14. Analyze time: 0.050s. Update time: 0.030s. Runtime: 3.042s.
Iter 15. Analyze time: 0.043s. Update time: 0.029s. Runtime: 3.119s.
Iter 16. Analyze time: 0.060s. Update time: 0.046s. Runtime: 3.209s.

... (82 captured-output lines omitted; rerun the source example for the full output)

AS3 Activity 1.3068e-07 1.2107e-07 1.0793e-08 0 NaN 12.1
AS4 Activity 0.088229 0.042285 0.0072872 0.0018848 NaN 12.1
AS5 Activity 1.3068e-07 1.2107e-07 1.0793e-08 0 NaN 12.1
AS6 Activity 0.082004 0.036518 0.0067731 0.0016277 NaN 12.1
AH13 Activity 2.6262e-07 2.4214e-07 1.0846e-08 1.0846e-08 NaN 24.2
AH14 Activity 0.085177 0.078536 0.0035177 0.0035177 NaN 24.2

[Running in non-interactive mode, continuing...]
```

2.7.18 lqn_srvnph

Method `srvnph`: each entry activity graph composed into one phase-type service law, so an AND fork-join and a loop survive as a distribution rather than as routing.

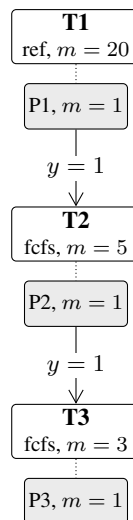


Figure 2.77: Call graph of `lqn_srvnph`: 3 tasks over 3 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

The example is built and solved as follows.

```
LayeredNetwork model = LayeredModel.lqn_srvnph();

SolverOptions lnoptions = LN.defaultOptions();
lnoptions.verbose = VerboseLevel.SILENT;
SolverOptions solveroptions = MVA.defaultOptions();
```

Processors	3: P1 (inf, $m = 1$), P2 (ps, $m = 1$), P3 (ps, $m = 1$)
Tasks	3: T1 (ref, $m = 20$), T2 (fcfs, $m = 5$), T3 (fcfs, $m = 3$)
Entries	4: E1, E2, E3, E4
Activities	10, host demands A1: 0.1, A20: 0.2, A21: 0.3, A22: 0.2, A23: 0.1, A30: 0.1, A31: 0.2, A32: 0.1, ...
Calls	3 (3 synchronous, 0 asynchronous)
Think times	T1: 1.0

Table 2.77: Features of `lqn_srvnph`.

```

solveroptions.verbose = VerboseLevel.SILENT;

long t0 = System.nanoTime();
LN defaultSolver = new LN(model, m -> new MVA(m, solveroptions), lnoptions);
AvgTable defaultTable = defaultSolver.getAvgTable();
double tDefault = (System.nanoTime() - t0) / 1e9;
System.out.printf("%nLN(default) Results [%3f s]:%n", tDefault);
defaultTable.setOptions(printing());
defaultTable.print();

SolverOptions lnoptions2 = LN.defaultOptions();
lnoptions2.verbose = VerboseLevel.SILENT;
// the method name; bare "srvnph" is unrecognised and falls back to srvn.cs
lnoptions2.method = "srvn.ph";
t0 = System.nanoTime();
LN phSolver = new LN(model, m -> new MVA(m, solveroptions), lnoptions2);
AvgTable phTable = phSolver.getAvgTable();
double tSrvnph = (System.nanoTime() - t0) / 1e9;
System.out.printf("%nLN(srvn.ph) Results [%3f s]:%n", tSrvnph);
phTable.setOptions(printing());
phTable.print();

... (7 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

nclassesSrvnph += layer.getNumberOfClasses();
}
System.out.printf(
    "%nLayer classes: default %d, srvn.ph %d. Runtime: %.3fs vs %.3fs (%.2fx).%n",
    nclassesDefault, nclassesSrvnph, tDefault, tSrvnph, tDefault / tSrvnph);
pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.7.19 `lqn_transient`

The transient response of a layered model, through `getTranAvg`.

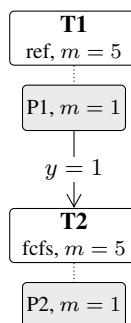


Figure 2.78: Call graph of `lqn_transient`: 2 tasks over 2 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	2: P1 (ps, $m = 1$), P2 (ps, $m = 1$)
Tasks	2: T1 (ref, $m = 5$), T2 (fcfs, $m = 5$)
Entries	2: E1, E2
Activities	2, host demands A1: 0.5, A2: 0.3333333333333333
Calls	1 (1 synchronous, 0 asynchronous)
Think times	T1: 1.0

Table 2.78: Features of `lqn_transient`.

The example is built and solved as follows.

```
LayeredNetwork model = LayeredModel.lqn_transient();

SolverOptions options = LN.defaultOptions();
options.verbose = VerboseLevel.SILENT;
LN solver = new LN(model, m -> new FLD(m, "verbose", false), options);
LNTranAvgResult tran = solver.getTranAvg();

int E = solver.nlayers;
System.out.printf("LN.getTranAvg returned transient traces for %d layers.%n", E);

// The block-diagonal offsets (r0,c0) advance by each layer's station and
// class counts, mirroring how getTranAvg stacks the per-layer blocks.
int r0 = 0;
int c0 = 0;
for (int e = 0; e < E; e++) {
    Network layer = solver.getEnsemble().get(e);
    int M = layer.getNumberOfStations();
    int K = layer.getNumberOfClasses();
    System.out.printf("%n--- Layer %d (%d stations, %d classes)%n", e + 1, M, K);
    for (int i = 0; i < M; i++) {
        for (int r = 0; r < K; r++) {
            Matrix trace = tran.QNt[r0 + i][c0 + r];
            Matrix t = tran.t[r0 + i][c0 + r];
            if (trace == null || trace.isEmpty() || trace.elementMax() <= 1e-6) {
                continue;
            }

            ... (5 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

        }
    }
    r0 += M;
    c0 += K;
}
pauseForUser();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.7.20 `lqn_twotasks`

Two tasks over separate processors, the two-layer archetype.

Processors	2: P1 (ps, $m = 1$), P2 (ps, $m = 1$)
Tasks	2: T1 (ref, $m = 100$), T2 (inf, $m = 1$)
Entries	3: E1, E2, E3
Activities	5, host demands A1: 1.0, A20: 1.0, A21: 1.0, A22: 1.0, A3: 1.0
Calls	2 (2 synchronous, 0 asynchronous)
Think times	T1: 10.0

Table 2.79: Features of `lqn_twotasks`.

The example is built and solved as follows.

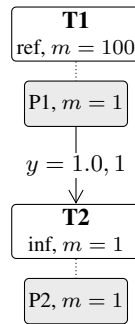


Figure 2.79: Call graph of `lqn_twotasks`: 2 tasks over 2 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

```

LayeredNetwork model = LayeredModel.lqn_twotasks();

try {
    new LQNS(model).getLNAvgTable().print();
} catch (Exception e) {
    System.out.println("LQNS failed: " + e.getMessage());
}

// NC LAYERS, NOT THE DEFAULT: MVA layers and NC layers are different fixed
// points, so the layer solver the reference pins is part of the golden.
new LN(model, (subModel) -> new NC(subModel, "verbose", false)).getAvgTable().print();

pauseForUser();

```

Running this edition prints

Node	NodeType	QLen	Util	RespT	ResidT	ArvR	Tput
P1	Processor	NaN	0.25	NaN	NaN	NaN	NaN
P2	Processor	NaN	1	NaN	NaN	NaN	NaN
T1	RefTask	97.5	0.25	NaN	NaN	NaN	0.25
T2	Task	97.168	1	NaN	NaN	NaN	0.5
E1	Entry	97.5	0.25	390	NaN	NaN	0.25
E2	Entry	72.876	0.75	291.5	NaN	NaN	0.25
E3	Entry	24.292	0.25	97.168	NaN	NaN	0.25
A1	Activity	97.5	0.25	390	NaN	NaN	0.25
A20	Activity	24.292	0.25	97.168	NaN	NaN	0.25
A21	Activity	24.292	0.25	97.168	NaN	NaN	0.25
A22	Activity	24.292	0.25	97.168	NaN	NaN	0.25
A3	Activity	24.292	0.25	97.168	NaN	NaN	0.25

Iter 1. Analyze time: 0.399s. Update time: 0.019s. Runtime: 0.400s.
 Iter 2. Analyze time: 0.124s. Update time: 0.020s. Runtime: 0.552s.

... (58 captured-output lines omitted; rerun the source example for the full output)

E3	Entry	24.3	0.25	97.2	NaN	NaN	0.25
A1	Activity	97.5	0.25	390	1.3275	NaN	0.25
A20	Activity	24.3	0.25	97.2	48.6	NaN	0.25
A21	Activity	24.3	0.25	97.2	48.6	NaN	0.25
A22	Activity	24.3	0.25	97.2	48.6	NaN	0.25
A3	Activity	24.3	0.25	97.2	48.6	NaN	0.25

[Running in non-interactive mode, continuing...]

2.7.21 lqn_workflows

Workflow patterns (sequence, AND-fork and join, loop) expressed as activity graphs. The example is built and solved as follows.

```

LayeredNetwork model = LayeredModel.lqn_workflows();

try {

```

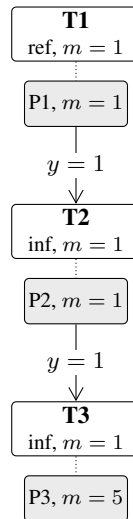



Figure 2.80: Call graph of `lqn_workflows`: 3 tasks over 3 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	3: P1 (inf, $m = 1$), P2 (inf, $m = 1$), P3 (ps, $m = 5$)
Tasks	3: T1 (ref, $m = 1$), T2 (inf, $m = 1$), T3 (inf, $m = 1$)
Entries	3: Entry, E2, E1
Activities	14, host demands A1: 1.0, A2: 2.0, A3: 3.0, B1: 0.1, B2: 0.2, B3: 0.3, B4: 0.4, B5: 0.5, ...
Calls	2 (2 synchronous, 0 asynchronous)
Think times	T1: 0.0

Table 2.80: Features of `lqn_workflows`.

```

new LQNS(model).getLNAvgTable().print();
} catch (Exception e) {
    System.out.println("LQNS failed: " + e.getMessage());
}
new LN(model).getAvgTable().print();
pauseForUser();

```

Running this edition prints

```

WARNING: [writeXML] Task T3 is not a reference task, so its think time (10.0000) is not written: the LQN
XML schema accepts think-time on reference tasks only.
Node  NodeType      QLen      Util      RespT      ResidT      ArvR      Tput
P1     Processor      NaN       0.79575   NaN        NaN         NaN       NaN
P2     Processor      NaN       0.16711   NaN        NaN         NaN       NaN
P3     Processor      NaN       0.072414  NaN        NaN         NaN       NaN
T1     RefTask         1         0.79575   NaN        NaN         NaN       0.079575
T2     Task            0.20425   0.16711   NaN        NaN         NaN       0.079575
T3     Task            0.072396  0.072414  NaN        NaN         NaN       0.079575
Entry  Entry           1         0.79575   12.567     NaN         NaN       0.079575
E2     Entry           0.20425   0.16711   2.5667     NaN         NaN       0.079575
E1     Entry           0.072396  0.072414  0.90978    NaN         NaN       0.079575
A1     Activity        0.079575  0.079575  1          NaN         NaN       0.079575
A2     Activity        0.47745   0.47745   2          NaN         NaN       0.23873
A3     Activity        0.44297   0.23873   5.5667     NaN         NaN       0.079575
B1     Activity        0.0079575 0.0079575 0.1         NaN         NaN       0.079575
B2     Activity        0.015915  0.015915  0.2         NaN         NaN       0.079575
... (81 captured-output lines omitted; rerun the source example for the full output)
B6     Activity        0.11726   0.046593  1.51        0.6         NaN       0.077655
C1     Activity        0.0078713 0.0078713 0.1          0.1         NaN       0.078713
C2     Activity        0.0047228 0.0047228 0.2          0.06        NaN       0.023614
C3     Activity        0.0070842 0.0070842 0.3          0.09        NaN       0.023614
C4     Activity        0.012594  0.012594  0.4          0.16        NaN       0.031485
C5     Activity        0.039357  0.039357  0.5          0.5         NaN       0.078713

```

[Running in non-interactive mode, continuing...]

2.8 Cache models

A `Cache` node holds a finite set of items under a replacement policy. Each request names an item drawn from a popularity distribution, typically Zipf; a hit is served at once, a miss must be fetched and inserted, which evicts an item. The metric of interest is the hit ratio, and everything else in the model follows from it.

The first group of scripts fixes the workload and varies the replacement policy (LRU, FIFO, RR, CLIMB, H-LRU, Q-LRU), so the hit ratios are directly comparable. The second group builds retrieval systems, in which the cache sits in front of a backing store and the miss path is itself a queueing network, and the delayed-hit effect appears: a request for an item already being fetched waits for that fetch rather than starting another.

2.8.1 `cache_compare_replc`

RR, FIFO and LRU on one item popularity. CTMC and MVA report all three hit ratios; NC does not support LRU and reports that cell as NaN rather than implying support.

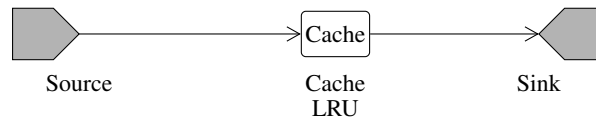


Figure 2.81: Topology of `cache_compare_replc`: an open network with a cache node.

Nodes	3: Source, Cache, Sink
Classes	3: InitClass (open), HitClass (open), MissClass (open)
Arrivals	Source – InitClass: Exp(2); HitClass: Disabled; MissClass: Disabled
Features	cache: LRU replacement, 5 items, capacity 2,1; class switching on 2 links

Table 2.81: Features of `cache_compare_replc`.

The example is built and solved as follows.

```

public static void cache_compare_replc() {
    System.out.println("=== Cache Compare Replacement Strategies ===");

    ReplacementStrategy[] replStrat = {ReplacementStrategy.RR, ReplacementStrategy.FIFO,
    ReplacementStrategy.LRU};

    for (int s = 0; s < replStrat.length; s++) {
        try {
            // Create a fresh model for each strategy (replacement strategy is set in
            constructor)
            Network model;
            if (replStrat[s] == ReplacementStrategy.RR) {
                model = CacheModel.cache_replc_rr();
            } else if (replStrat[s] == ReplacementStrategy.FIFO) {
                model = CacheModel.cache_replc_fifo();
            } else if (replStrat[s] == ReplacementStrategy.LRU) {
                model = CacheModel.cache_replc_lru();
            } else {
                model = CacheModel.cache_compare_replc();
            }
        }
    }
}
  
```

```

        jline.lang.nodes.Cache cacheNode = (jline.lang.nodes.Cache) model.getNodeByName("
Cache");

        // Skip CTMC solver as commented in the source notebook

        model.reset();
        MVA solver2 = new MVA(model, "seed", 1, "verbose", false);
        NetworkAvgNodeTable avgTable2 = solver2.getAvgNodeTable();

        ... (12 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

        System.out.printf("%s: MVA=%.8f, NC=%.8f\n", replStrat[s], mvaHitRatio,
ncHitRatio);

        } catch (Exception e) {
            System.out.println("Error testing " + replStrat[s] + ": " + e.getMessage());
        }
    }
}

```

The captured run completed successfully and produced no console output.

2.8.2 cache_itemsize_costcap

Per-item storage costs and a cap on the cost each list may hold, so an insertion that would breach the cap serves the request without changing the cache state.

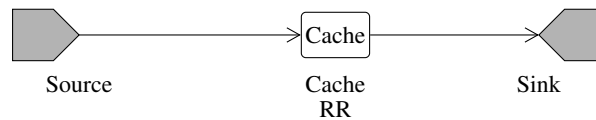


Figure 2.82: Topology of cache_itemsize_costcap: an open network with a cache node.

Nodes	3: Source, Cache, Sink
Classes	3: InitClass (open), HitClass (open), MissClass (open)
Arrivals	Source – InitClass: Exp(2); HitClass: Disabled; MissClass: Disabled
Features	cache: RR replacement, 6 items, capacity 1,1; class switching on 2 links
Solvers	NC

Table 2.82: Features of cache_itemsize_costcap.

The example is built and solved as follows.

```

Network model = new Network("model");

int n = 6; // number of items
Matrix m = new Matrix(1, 2); // two lists, one item each
m.set(0, 0, 1);
m.set(0, 1, 1);

Matrix sizes = new Matrix(1, n); // small and large items
double[] sizeValues = {1, 1, 1, 2, 2, 2};
for (int i = 0; i < n; i++) sizes.set(0, i, sizeValues[i]);
Matrix caps = new Matrix(1, 2); // list 2 admits small items only
caps.set(0, 0, 2);
caps.set(0, 1, 1);

Source source = new Source(model, "Source");
Cache cacheNode = new Cache(model, "Cache", n, m, ReplacementStrategy.RR);
Sink sink = new Sink(model, "Sink");

OpenClass jobClass = new OpenClass(model, "InitClass", 0);
OpenClass hitClass = new OpenClass(model, "HitClass", 0);
OpenClass missClass = new OpenClass(model, "MissClass", 0);

```

```

source.setArrival(jobClass, new Exp(2.0));
cacheNode.setRead(jobClass, new DiscreteSampler(new Matrix(1, n).fill(1.0 / n)));
cacheNode.setItemSizes(sizes);
cacheNode.setCostCaps(caps);

... (4 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

routingMatrix.set(jobClass, jobClass, source, cacheNode, 1.0);
routingMatrix.set(hitClass, hitClass, cacheNode, sink, 1.0);
routingMatrix.set(missClass, missClass, cacheNode, sink, 1.0);
model.link(routingMatrix);

return model;

```

Running this edition prints

```
jline.lang.Network@28f67ac7
```

2.8.3 cache_mmap_rr_env

A marked MAP request stream into a random-replacement cache inside a random environment.

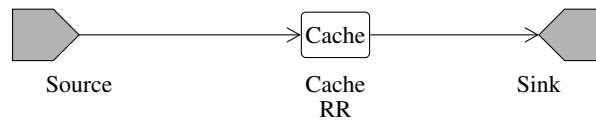


Figure 2.83: Topology of `cache_mmap_rr_env`: an open network with a cache node.

Nodes	3: Source, Cache, Sink
Classes	6: Read1 (open), Read2 (open), Hit1 (open), Miss1 (open), Hit2 (open), Miss2 (open)
Arrivals	Source – Read1: MMAP; Read2: MMAP; Hit1: Disabled; Miss1: Disabled; Hit2: Disabled; Miss2: Disabled
Features	cache: RR replacement, 4 items, capacity 2; class switching on 2 links
Solvers	LDES, CTMC, FLD

Table 2.83: Features of `cache_mmap_rr_env`.

The example is built and solved as follows.

```

jline.lang.processes.MarkedMAP mmap) {
Network model = new Network("MMAPCache");

int n = 4; // number of items
int m = 2; // cache capacity

Source source = new Source(model, "Source");
Cache cacheNode = new Cache(model, "Cache", n, m, ReplacementStrategy.RR);
Sink sink = new Sink(model, "Sink");

OpenClass rd1 = new OpenClass(model, "Read1", 0);
OpenClass rd2 = new OpenClass(model, "Read2", 0);
OpenClass hit1 = new OpenClass(model, "Hit1", 0);
OpenClass mis1 = new OpenClass(model, "Miss1", 0);
OpenClass hit2 = new OpenClass(model, "Hit2", 0);
OpenClass mis2 = new OpenClass(model, "Miss2", 0);

// Per-class item-popularity distributions (deliberately different)
Matrix p1 = new Matrix(1, n); // class 1 favors low-index items
Matrix p2 = new Matrix(1, n); // class 2 favors high-index items
double[] w1 = {8, 4, 2, 1};
for (int i = 0; i < n; i++) {

```

```

    p1.set(0, i, w1[i] / 15.0);
    p2.set(0, i, w1[n - 1 - i] / 15.0);
}
cacheNode.setRead(rd1, new DiscreteSampler(p1));

... (20 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

routingMatrix.set(mis1, mis1, cacheNode, sink, 1.0);
routingMatrix.set(hit2, hit2, cacheNode, sink, 1.0);
routingMatrix.set(mis2, mis2, cacheNode, sink, 1.0);
model.link(routingMatrix);

return model;

```

Running this edition prints

```
no no-arg static method cache_mmap_rr_env on jline.examples.java.basic.CacheModel
```

2.8.4 cache_replc_climb

The CLIMB policy, which promotes a hit by one list only.

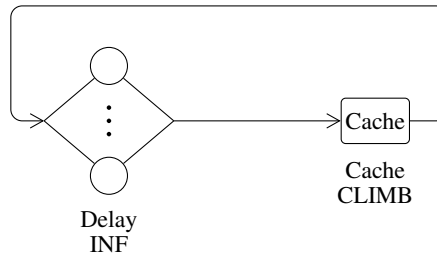


Figure 2.84: Topology of cache_replc_climb: a closed network over 1 queueing station with a cache node.

Nodes	2: Delay, Cache
Classes	3: JobClass (closed, $N = 1$), HitClass (closed, $N = 0$), MissClass (closed, $N = 0$)
Scheduling	Delay: INF
Service	Delay – JobClass: Exp(1); HitClass: Disabled; MissClass: Disabled
Features	drop rule {'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at Delay; cache: CLIMB replacement, 5 items, capacity 2; class switching on 2 links
Solvers	CTMC

Table 2.84: Features of cache_replc_climb.

The example is built and solved as follows.

```

Network model = new Network("model");

int n = 5; // Number of items
int m = 2; // Cache capacity

Delay delay = new Delay(model, "Delay");
Cache cacheNode = new Cache(model, "Cache", n, m, ReplacementStrategy.CLIMB);

ClosedClass jobClass = new ClosedClass(model, "JobClass", 1, delay, 0);
ClosedClass hitClass = new ClosedClass(model, "HitClass", 0, delay, 0);
ClosedClass missClass = new ClosedClass(model, "MissClass", 0, delay, 0);

delay.setService(jobClass, new Exp(1.0));
cacheNode.setRead(jobClass, new Zipf(1.2, n));
cacheNode.setHitClass(jobClass, hitClass);

```

```

cacheNode.setMissClass(jobClass, missClass);

RoutingMatrix routingMatrix = model.initRoutingMatrix();
routingMatrix.set(jobClass, jobClass, delay, cacheNode, 1.0);
routingMatrix.set(hitClass, jobClass, cacheNode, delay, 1.0);
routingMatrix.set(missClass, jobClass, cacheNode, delay, 1.0);
model.link(routingMatrix);

return model;

```

Running this edition prints

```
jline.lang.Network@2ff4f00f
```

2.8.5 cache_replc_fifo

First-in first-out replacement.

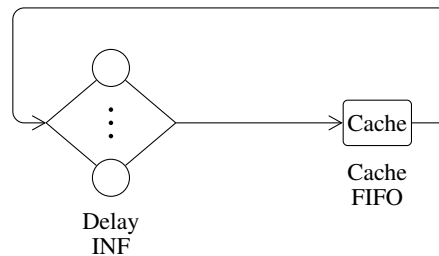


Figure 2.85: Topology of cache_replc_fifo: a closed network over 1 queueing station with a cache node.

Nodes	2: Delay, Cache
Classes	3: JobClass (closed, $N = 1$), HitClass (closed, $N = 0$), MissClass (closed, $N = 0$)
Scheduling	Delay: INF
Service	Delay – JobClass: Exp(1); HitClass: Disabled; MissClass: Disabled
Features	drop rule {'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at Delay; cache: FIFO replacement, 5 items, capacity 2; class switching on 2 links
Solvers	CTMC, SSA, MVA, FLD

Table 2.85: Features of cache_replc_fifo.

The example is built and solved as follows.

```

public static void cache_replc_fifo() {
    //System.out.println("=== Cache FIFO Example ===");
    Network model = CacheModel.cache_replc_fifo();

    try {
        CTMC solver1 = new CTMC(model, "keep", false, "seed", 1);
        NetworkAvgNodeTable avgTable1 = solver1.getAvgNodeTable();
        //System.out.println("--- CTMC Solver ---");
        avgTable1.print();

        model.reset();
        SSA solver2 = new SSA(model, "samples", 100000, "verbose", true, "method", "serial",
"seed", 23000);
        NetworkAvgNodeTable avgTable2 = solver2.getAvgNodeTable();
        //System.out.println("--- SSA Solver ---");
        avgTable2.print();

        model.reset();
        MVA solver3 = new MVA(model, "seed", 1);
    }
}

```

```

NetworkAvgNodeTable avgTable3 = solver3.getAvgNodeTable();
//System.out.println("--- MVA Solver ---");
avgTable3.print();

model.reset();
// The reference runs the RMF fluid limit here, which is what its golden
// holds; the default fluid closure answers a different question.
FLD solverFLD = new FLD(model, "method", "rmf");
solverFLD.getAvgNodeTable().print();

} catch (Exception e) {
    //System.out.println("Error in cache_replc_fifo: " + e.getMessage());
    e.printStackTrace();
}

```

Running this edition prints

```

CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Node      JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
Delay     JobClass  1     1     1       1       1     1
Cache     JobClass  0     0     0       0       1     0
Cache     HitClass  0     0     0       0       0     0.4
Cache     MissClass 0     0     0       0       0     0.6
CS_Cache_to_Delay JobClass 0     0     0       0       0     1
CS_Cache_to_Delay HitClass 0     0     0       0       0.4   0
CS_Cache_to_Delay MissClass 0     0     0       0       0.6   0
SSA analysis [method: serial; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 100001.
Node      JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
Delay     JobClass  1     1     1       1       1     1
Cache     JobClass  0     0     0       0       1     0
Cache     HitClass  0     0     0       0       0     0.39783
Cache     MissClass 0     0     0       0       0     0.60217
CS_Cache_to_Delay JobClass 0     0     0       0       0     1

... (12 captured-output lines omitted; rerun the source example for the full output)

Node      JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
Delay     JobClass  1     1     1       1       1     1
Cache     JobClass  0     0     0       0       1     0
Cache     HitClass  0     0     0       0       0     0.4
Cache     MissClass 0     0     0       0       0     0.6
CS_Cache_to_Delay JobClass 0     0     0       0       5.5511e-17 1
CS_Cache_to_Delay HitClass 0     0     0       0       0.4     0
CS_Cache_to_Delay MissClass 0     0     0       0       0.6     0

```

The rows repeat for each of the 4 solvers the script runs (CTMC, SSA, MVA, FLD), which is the point of the example: the same model read by methods of different cost and different exactness.

2.8.6 cache_replc_hlru

H-LRU, the multi-list generalization of LRU.

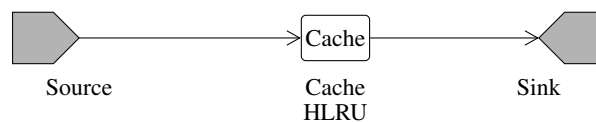


Figure 2.86: Topology of `cache_replc_hlru`: an open network with a cache node.

Nodes	3: Source, Cache, Sink
Classes	3: InitClass (open), HitClass (open), MissClass (open)
Arrivals	Source – InitClass: Exp(1); HitClass: Disabled; MissClass: Disabled
Features	cache: HLRU replacement, 6 items, capacity 2,1; class switching on 2 links
Solvers	CTMC, MVA, SSA

Table 2.86: Features of `cache_replc_hlru`.

The example is built and solved as follows.

```
Network model = new Network("model");

int n = 6; // Number of items
Matrix m = new Matrix(1, 2); // list 1 holds 2 items, list 2 holds 1
m.set(0, 0, 2);
m.set(0, 1, 1);

Source source = new Source(model, "Source");
Cache cacheNode = new Cache(model, "Cache", n, m, ReplacementStrategy.HLRU);
Sink sink = new Sink(model, "Sink");

OpenClass jobClass = new OpenClass(model, "InitClass", 0);
OpenClass hitClass = new OpenClass(model, "HitClass", 0);
OpenClass missClass = new OpenClass(model, "MissClass", 0);

source.setArrival(jobClass, new Exp(1.0));
cacheNode.setRead(jobClass, new Zipf(1.2, n));
cacheNode.setHitClass(jobClass, hitClass);
cacheNode.setMissClass(jobClass, missClass);

RoutingMatrix routingMatrix = model.initRoutingMatrix();
routingMatrix.set(jobClass, jobClass, source, cacheNode, 1.0);
routingMatrix.set(hitClass, hitClass, cacheNode, sink, 1.0);
routingMatrix.set(missClass, missClass, cacheNode, sink, 1.0);
model.link(routingMatrix);

return model;
```

Running this edition prints

```
jline.lang.Network@546a03af
```

2.8.7 cache_replc_lru

Least-recently-used replacement, with a closed class circulating between a Delay and the cache.

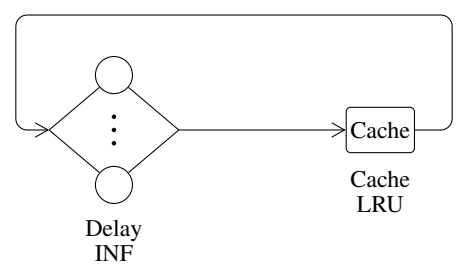


Figure 2.87: Topology of cache_replc_lru: a closed network over 1 queueing station with a cache node.

Nodes	2: Delay, Cache
Classes	3: JobClass (closed, $N = 1$), HitClass (closed, $N = 0$), MissClass (closed, $N = 0$)
Scheduling	Delay: INF
Service	Delay – JobClass: Exp(1); HitClass: Disabled; MissClass: Disabled
Features	drop rule { 'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue' } at Delay; cache: LRU replacement, 5 items, capacity 2; class switching on 2 links
Solvers	CTMC, SSA, MVA

Table 2.87: Features of cache_replc_lru.

The example is built and solved as follows.


```

Network model = CacheModel.cache_replc_lru();

try {
    new CTMC(model, "keep", false).getAvgNodeTable().print();
    model.reset();
} catch (Exception e) {
    System.out.println("CTMC failed: " + e.getMessage());
}

try {
    new SSA(model, "samples", 100000, "verbose", true, "method", "serial", "seed", 23000)
        .getAvgNodeTable().print();
    model.reset();
} catch (Exception e) {
    System.out.println("SSA failed: " + e.getMessage());
}

try {
    new MVA(model).getAvgNodeTable().print();
    model.reset();
} catch (Exception e) {
    System.out.println("MVA failed: " + e.getMessage());
}

```

Running this edition prints

```

CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Node      JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
Delay     JobClass   1     1     1       1     1     1
Cache     JobClass   0     0     0       0     1     0
Cache     HitClass   0     0     0       0     0     0.4
Cache     MissClass  0     0     0       0     0     0.6
CS_Cache_to_Delay JobClass  0     0     0       0     0     1
CS_Cache_to_Delay HitClass  0     0     0       0     0.4   0
CS_Cache_to_Delay MissClass 0     0     0       0     0.6   0
SSA analysis [method: serial; type: approximate, randomized; lang: java; env: python] completed in <t>s.
  Iterations: 100001.
Node      JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
Delay     JobClass   1     1     1       1     1     1
Cache     JobClass   0     0     0       0     1     0
Cache     HitClass   0     0     0       0     0     0.40033
Cache     MissClass  0     0     0       0     0     0.59967
CS_Cache_to_Delay JobClass  0     0     0       0     5.5511e-17 1
... (3 captured-output lines omitted; rerun the source example for the full output)
Node      JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
Delay     JobClass   1     1     1       1     1     1
Cache     JobClass   0     0     0       0     1     0
Cache     HitClass   0     0     0       0     0     0.4
Cache     MissClass  0     0     0       0     0     0.6
CS_Cache_to_Delay JobClass  0     0     0       0     0     1
CS_Cache_to_Delay HitClass  0     0     0       0     0.4   0
CS_Cache_to_Delay MissClass 0     0     0       0     0.6   0

```

The rows repeat for each of the 3 solvers the script runs (CTMC, SSA, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

2.8.8 cache_replc_qlru

Q-LRU, which promotes a hit with probability q .

The example is built and solved as follows.

```

Network model = new Network("model");

int n = 5; // Number of items
int m = 2; // Cache capacity

Delay delay = new Delay(model, "Delay");
Cache cacheNode = new Cache(model, "Cache", n, m, ReplacementStrategy.QLRU);
cacheNode.setAdmissionProb(0.5); // admit a missed item with probability q=0.5

```

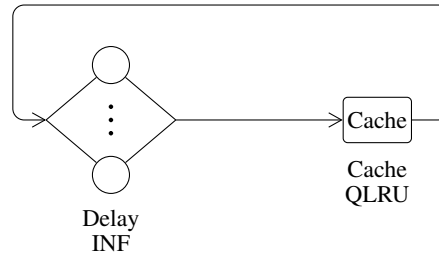


Figure 2.88: Topology of `cache_replc_qlru`: a closed network over 1 queueing station with a cache node.

Nodes	2: Delay, Cache
Classes	3: JobClass (closed, $N = 1$), HitClass (closed, $N = 0$), MissClass (closed, $N = 0$)
Scheduling	Delay: INF
Service	Delay – JobClass: Exp(1); HitClass: Disabled; MissClass: Disabled
Features	drop rule {'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at Delay; cache: QLRU replacement, 5 items, capacity 2; class switching on 2 links
Solvers	CTMC

Table 2.88: Features of `cache_replc_qlru`.

```

ClosedClass jobClass = new ClosedClass(model, "JobClass", 1, delay, 0);
ClosedClass hitClass = new ClosedClass(model, "HitClass", 0, delay, 0);
ClosedClass missClass = new ClosedClass(model, "MissClass", 0, delay, 0);

delay.setService(jobClass, new Exp(1.0));
cacheNode.setRead(jobClass, new Zipf(1.2, n));
cacheNode.setHitClass(jobClass, hitClass);
cacheNode.setMissClass(jobClass, missClass);

RoutingMatrix routingMatrix = model.initRoutingMatrix();
routingMatrix.set(jobClass, jobClass, delay, cacheNode, 1.0);
routingMatrix.set(hitClass, jobClass, cacheNode, delay, 1.0);
routingMatrix.set(missClass, jobClass, cacheNode, delay, 1.0);
model.link(routingMatrix);

return model;

```

Running this edition prints

```
jline.lang.Network@2ff4f00f
```

2.8.9 `cache_replc_routing`

Cache-aware routing: hits and misses leave the cache on different arcs, under RAND and round-robin routing.

Nodes	6: Source, Delay $\times 2$, Cache, Router, Sink
Classes	3: InitClass (open), HitClass (open), MissClass (open)
Scheduling	Delay1: INF; Delay2: INF
Arrivals	Source – InitClass: Exp(2)
Service	Delay1 – HitClass: Exp(10); MissClass: Exp(1) Delay2 – HitClass: Exp(20); MissClass: Exp(2)
Features	cache: FIFO replacement, 5 items, capacity 2; class switching on 6 links
Solvers	CTMC, SSA, MVA, NC, FLD

Table 2.89: Features of `cache_replc_routing`.

The example is built and solved as follows.

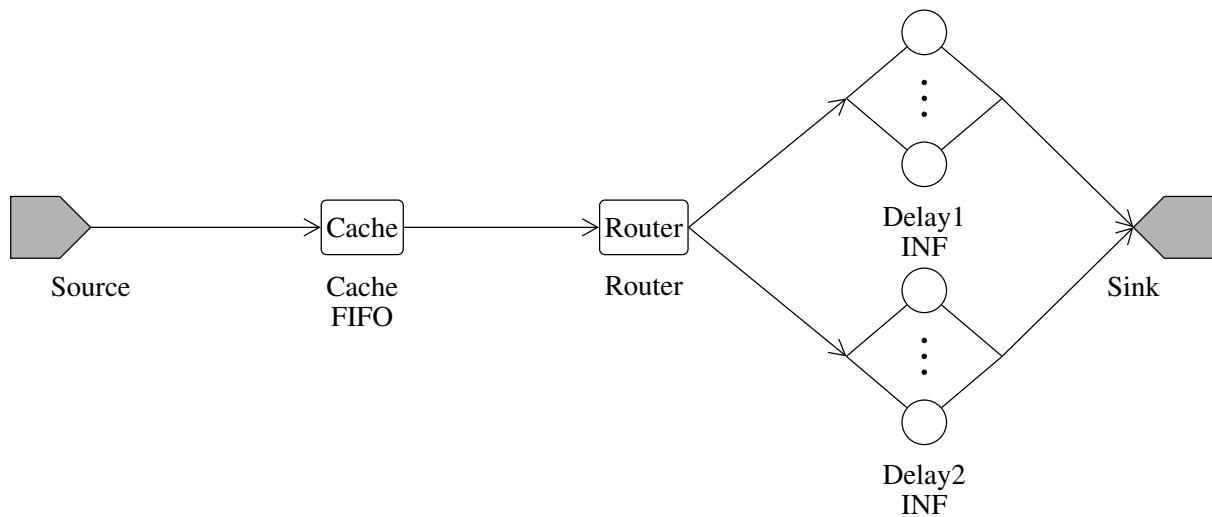


Figure 2.89: Topology of `cache_replc_routing`: an open network over 2 queueing stations with a cache node and a router.

```
public static void cache_replc_routing() {
    //System.out.println("=== Cache Routing Example ===");
    Network model = CacheModel.cache_replc_routing();

    try {
        CTMC solver1 = new CTMC(model, "keep", false, "cutoff", 1, "seed", 1);
        NetworkAvgNodeTable avgTable1 = solver1.getAvgNodeTable();
        //System.out.println("--- CTMC Solver ---");
        avgTable1.print();

        model.reset();
        SSA solver2 = new SSA(model, "samples", 10000, "verbose", true, "method", "serial", "seed", 1);
        NetworkAvgNodeTable avgTable2 = solver2.getAvgNodeTable();
        //System.out.println("--- SSA Solver ---");
        avgTable2.print();

        model.reset();
        MVA solver3 = new MVA(model, "seed", 1);
        NetworkAvgNodeTable avgTable3 = solver3.getAvgNodeTable();
        //System.out.println("--- MVA Solver ---");
        avgTable3.print();

        model.reset();
        NC solver4 = new NC(model, "seed", 1);
        NetworkAvgNodeTable avgTable4 = solver4.getAvgNodeTable();
        //System.out.println("--- NC Solver ---");

        ... (12 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

        //System.out.println("Miss Ratio: " + missRatio);

    } catch (Exception e) {
        //System.out.println("Error in cache_replc_routing: " + e.getMessage());
        e.printStackTrace();
    }
}
```

Running this edition prints

```
WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 1 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
WARNING: [solver_ctmc] CTMC: 40 vanishing state(s) have no immediate outgoing arc; the vanishing predicate
and the immediate-arc tagging disagree, so those rows keep their timed arcs.
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
```

Node	JobClass	QLen	Util	RespT	ResidT	ArvR	Tput
Source	InitClass	0	0	0	0	0	1.9808
Cache	InitClass	0	0	0	0	1.9808	0
Cache	HitClass	0	0	0	0	0	0.79232
Cache	MissClass	0	0	0	0	0	1.1885
Router	HitClass	0	0	0	0	0.79232	0.79232
Router	MissClass	0	0	0	0	1.1885	1.1885
Delay1	HitClass	0.038955	0.038955	0.1	0.02	0.39616	0.38955
Delay1	MissClass	0.42464	0.42464	1	0.3	0.59424	0.42464
Delay2	HitClass	0.020111	0.020111	0.05	0.01	0.39616	0.40223
Delay2	MissClass	0.29187	0.29187	0.5	0.15	0.59424	0.58373
Sink	HitClass	0	0	0	0	0.79232	0
Sink	MissClass	0	0	0	0	1.1885	0

... (48 captured-output lines omitted; rerun the source example for the full output)

Router	HitClass	0	0	0	0	0.8	0.8
Router	MissClass	0	0	0	0	1.2	1.2
Delay1	HitClass	0	0	0	0	0.4	0
Delay1	MissClass	0	0	0	0	0.6	0
Delay2	HitClass	0	0	0	0	0.4	0
Delay2	MissClass	0	0	0	0	0.6	0
Sink	HitClass	0	0	0	0	0.8	0
Sink	MissClass	0	0	0	0	1.2	0

The rows repeat for each of the 5 solvers the script runs (CTMC, SSA, MVA, NC, FLD), which is the point of the example: the same model read by methods of different cost and different exactness.

2.8.10 cache_replc_rr

Random replacement.

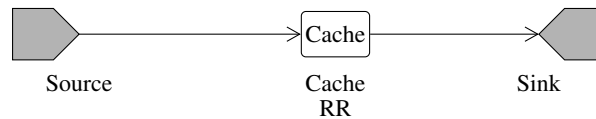


Figure 2.90: Topology of `cache_replc_rr`: an open network with a cache node.

Nodes	3: Source, Cache, Sink
Classes	3: InitClass (open), HitClass (open), MissClass (open)
Arrivals	Source – InitClass: Exp(2); HitClass: Disabled; MissClass: Disabled
Features	cache: RR replacement, 5 items, capacity 2; class switching on 2 links
Solvers	CTMC, SSA, MVA, NC, FLD

Table 2.90: Features of `cache_replc_rr`.

The example is built and solved as follows.

```

public static void cache_replc_rr() {
    //System.out.println("=== Cache RR Example ===");
    Network model = CacheModel.cache_replc_rr();

    try {
        CTMC solver1 = new CTMC(model, "keep", false, "cutoff", 1, "seed", 1);
        NetworkAvgNodeTable avgTable1 = solver1.getAvgNodeTable();
        //System.out.println("--- CTMC Solver ---");
        avgTable1.print();

        model.reset();
        SSA solver2 = new SSA(model, "samples", 10000, "verbose", true, "method", "serial", "seed", 1);
        NetworkAvgNodeTable avgTable2 = solver2.getAvgNodeTable();
        //System.out.println("--- SSA Solver ---");
        avgTable2.print();
    }
}

```

```

model.reset();
MVA solver3 = new MVA(model, "seed", 1);
NetworkAvgNodeTable avgTable3 = solver3.getAvgNodeTable();
//System.out.println("--- MVA Solver ---");
avgTable3.print();

model.reset();
NC solver4 = new NC(model, "seed", 1);
NetworkAvgNodeTable avgTable4 = solver4.getAvgNodeTable();
//System.out.println("--- NC Solver ---");

... (12 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

//System.out.println("Miss Ratio: " + missRatio);

} catch (Exception e) {
    //System.out.println("Error in cache_replc_rr: " + e.getMessage());
    e.printStackTrace();
}

```

Running this edition prints

```

WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 1 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Node   JobClass   QLen  Util  RespT  ResidT   ArvR   Tput
Source InitClass   0    0    0      0      0      2
Cache  InitClass   0    0    0      0      2      0
Cache  HitClass    0    0    0      0      0  1.1808
Cache  MissClass   0    0    0      0      0  0.81921
Sink   HitClass    0    0    0      0  1.1808      0
Sink   MissClass   0    0    0      0  0.81921      0
SSA analysis [method: serial; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 10001.
Node   JobClass   QLen  Util  RespT  ResidT   ArvR   Tput
Source InitClass   0    0    0      0      0      2
Cache  InitClass   0    0    0      0      2      0
Cache  HitClass    0    0    0      0      0  1.1633
Cache  MissClass   0    0    0      0      0  0.83669
Sink   HitClass    0    0    0      0  1.1633      0

... (17 captured-output lines omitted; rerun the source example for the full output)

Fluid analysis [method: rmf; type: approximate, deterministic; lang: java; env: python] completed in <t>s.
Node   JobClass   QLen  Util  RespT  ResidT   ArvR   Tput
Source InitClass   0    0    0      0      0      2
Cache  InitClass   0    0    0      0      2      0
Cache  HitClass    0    0    0      0      0  1.1246
Cache  MissClass   0    0    0      0      0  0.87542
Sink   HitClass    0    0    0      0  1.1246      0
Sink   MissClass   0    0    0      0  0.87542      0

```

The rows repeat for each of the 5 solvers the script runs (CTMC, SSA, MVA, NC, FLD), which is the point of the example: the same model read by methods of different cost and different exactness.

2.8.11 cache_rmf_transient

The transient hit ratio from the refined mean field approximation.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows.

```

int n = 10;           // number of items
int[] m = {3, 2};     // list capacities (2-list cache)
double alpha = 0.8;   // Zipf exponent

double[] p = new double[n];
double sum = 0.0;
for (int i = 0; i < n; i++) {

```

```

        p[i] = Math.pow(i + 1, -alpha);
        sum += p[i];
    }
    for (int i = 0; i < n; i++) p[i] /= sum;

    System.out.println("Cache parameters: n=" + n + ", m=[" + m[0] + ", " + m[1]
        + "], Zipf(" + alpha + ")");

    CacheRMF rmf = new CacheRMF(p, m);

    Object[] ss = rmf.meanFieldExpansionSteadyState(1);
    double[] pi = (double[]) ss[0];
    double[] V = (double[]) ss[1];
    double[] piRefined = new double[pi.length];
    for (int i = 0; i < pi.length; i++) piRefined[i] = pi[i] + V[i] / n;

    System.out.println("\nSteady-state results (refined mean field):");
    double totalHit = 0.0;
    for (int k = 1; k <= m.length; k++) {

        ... (25 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

        double hr = 0.0;
        for (int k = 1; k <= m.length; k++) hr += rmf.hitRate(xt, k);
        jline.io.LineResultRecorder.scalar("CACHE",
            String.format("t=%.3f", T[idx]), "Transient", hr);
        System.out.printf("  t=%.3f: hit_rate=%.6f\n", T[idx], hr);
    }
}

```

Running this edition prints

```

Cache parameters: n=10, m=[3, 2], Zipf(0.8)

Steady-state results (refined mean field):
Hit rate (list 1): 0.312854
Hit rate (list 2): 0.301606
Miss rate:      0.385540
Total hit prob: 0.614460
Total miss prob: 0.385540

Transient hit rates (refined, N=10):
t= 0.000: hit_rate=0.728003
t= 5.025: hit_rate=0.645750
t= 12.563: hit_rate=0.624411
t= 25.126: hit_rate=0.616758
t= 50.000: hit_rate=0.614587

```

2.8.12 cachenet_tandem

A tandem of two caches over one item set, where a miss at the first is a read of the same item at the second. Item identity is carried by a dedicated job class per item, so no class switching happens on the arc between the caches.

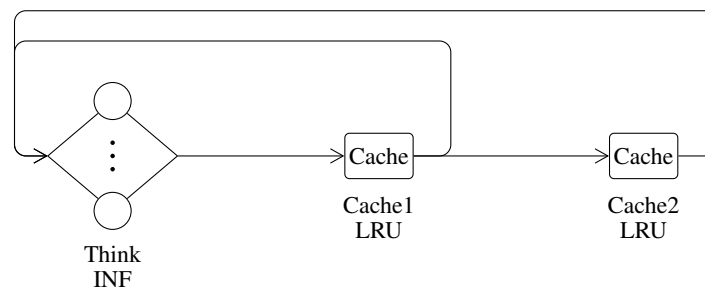


Figure 2.91: Topology of `cachenet_tandem`: a closed network over 1 queueing station with a cache node.

Nodes	3: Delay, Cache $\times$ 2
Classes	15: Read1 (closed, $N = 1$), Hit1_1 (closed, $N = 0$), Hit2_1 (closed, $N = 0$), Miss_1 (closed, $N = 0$), Read2 (closed, $N = 1$), Hit1_2 (closed, $N = 0$), Hit2_2 (closed, $N = 0$), Miss_2 (closed, $N = 0$), Read3 (closed, $N = 1$), Hit1_3 (closed, $N = 0$), Hit2_3 (closed, $N = 0$), Miss_3 (closed, $N = 0$), Cache2_item1 (closed, $N = 0$), Cache2_item2 (closed, $N = 0$), Cache2_item3 (closed, $N = 0$)
Scheduling Service	Think: INF Think – Read1: Exp(0.5); Hit1_1: Disabled; Hit2_1: Disabled; Miss_1: Disabled; Read2: Exp(0.3); Hit1_2: Disabled; Hit2_2: Disabled; Miss_2: Disabled; Read3: Exp(0.2); Hit1_3: Disabled; Hit2_3: Disabled; Miss_3: Disabled; Cache2_item1: Disabled; Cache2_item2: Disabled; Cache2_item3: Disabled
Features	drop rule {'Hit1_1': 'waitingQueue', 'Hit2_1': 'waitingQueue', 'Miss_1': 'waitingQueue', 'Hit1_2': 'waitingQueue', 'Hit2_2': 'waitingQueue', 'Miss_2': 'waitingQueue', 'Hit1_3': 'waitingQueue', 'Hit2_3': 'waitingQueue', 'Miss_3': 'waitingQueue', 'Cache2_item1': 'waitingQueue', 'Cache2_item2': 'waitingQueue', 'Cache2_item3': 'waitingQueue'} at Think; cache: LRU replacement, 3 items, capacity 1; cache: LRU replacement, 3 items, capacity 2; class switching on 3 links
Solvers	CTMC, SSA, LDES

Table 2.91: Features of `cachenet_tandem`.

The example is built and solved as follows.

```
new CTMC(CacheModel.cachenet_tandem()).getAvgCacheTable().print();
new SSA(CacheModel.cachenet_tandem(), "samples", 5e4, "seed", 1)
    .getAvgCacheTable().print();
new LDES(CacheModel.cachenet_tandem(), "samples", 2e5, "seed", 1)
    .getAvgCacheTable().print();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.8.13 retrieval_chain

A chain of caches, each backing the one before it.

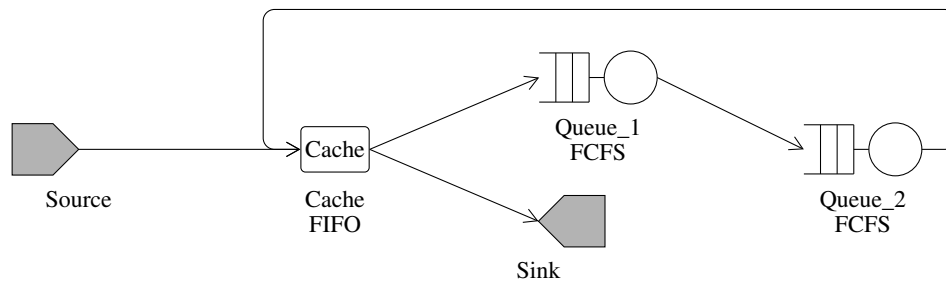


Figure 2.92: Topology of `retrieval_chain`: an open network over 2 queueing stations with a cache node.

The example is built and solved as follows.

```
Matrix serviceRates = new Matrix(new double[][] {
    {2.0, 2.0, 2.0}, // Queue_1 per-item rates
    {3.0, 3.0, 3.0} // Queue_2 per-item rates
});
Network model = CacheRetrievalSystemModel.chain_retrieval_system_model(
    new double[] {0.6, 0.3, 0.1}, serviceRates, 2, "[1]", SchedStrategy.FCFS);
solveAll(model);
```

Running this edition prints

Nodes	5: Source, Queue $\times$ 2, Cache, Sink
Classes	6: InitClass (open), HitClass (open), MissClass (open), InitClass_retrievalClass_1 (open), InitClass_retrievalClass_2 (open), InitClass_retrievalClass_3 (open)
Scheduling	Queue_1: FCFS; Queue_2: FCFS
Arrivals	Source – InitClass: Exp(1); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Disabled; InitClass_retrievalClass_2: Disabled; InitClass_retrievalClass_3: Disabled
Service	Queue_1 – InitClass: Exp(2); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Exp(2); InitClass_retrievalClass_2: Exp(2); InitClass_retrievalClass_3: Exp(2) Queue_2 – InitClass: Exp(3); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Exp(3); InitClass_retrievalClass_2: Exp(3); InitClass_retrievalClass_3: Exp(3)
Features	cache: FIFO replacement, 3 items, capacity 1; drop rule {'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at Queue_1; drop rule {'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at Queue_2; class switching on 4 links
Solvers	SSA, LDES, MVA, NC

Table 2.92: Features of retrieval_chain.

```

WARNING: [solver_ssa_analyzer_serial] Retrieval-system expected latency is not currently implemented;
reporting NaN.
SSA analysis [method: serial; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 100001.
Node  JobClass  List  ListCap  Items  HitProb  DelayedHitProb  MissProb  HitRate  DelayedHitRate
MissRate  ArvR  ResidT  ListCost
Cache  InitClass  0      1      3  0.43354      0.14244      0.42402  0.43354      0.14244
0.42402 0.56646      NaN      NaN
LDES events: 20000 40000 60000 80000 100000 120000 140000 160000 180000 200000 220000 240000 260000
280000 300000 320000 340000 360000 380000 400000 420000 440000 460000 480000 500000 520000 540000
560000 580000 600000 620000 640000 660000 680000 700000 720000 740000 760000 780000 800000 820000
840000 860000 880000 900000 920000 940000 960000 980000 1000000 1000000
LDES analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 1000000.
Node  JobClass  List  ListCap  Items  HitProb  DelayedHitProb  MissProb  HitRate  DelayedHitRate
MissRate  ArvR  ResidT  ListCost
Cache  InitClass  0      1      3  0.43802      0.14432      0.41766  0.43802      0.14432
0.41766 0.56198 0.83647      NaN
MVA analysis [method: default/fpi; type: approximate, deterministic; lang: java; env: python] completed in
<t>s.
Node  JobClass  List  ListCap  Items  HitProb  DelayedHitProb  MissProb  HitRate  DelayedHitRate
MissRate  ArvR  ResidT  ListCost
Cache  InitClass  0      1      3  0.40688      0.16346      0.42966  0.40688      0.16346
0.42966 0.59312 0.87138      NaN
NC analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>s
.
Node  JobClass  List  ListCap  Items  HitProb  DelayedHitProb  MissProb  HitRate  DelayedHitRate
MissRate  ArvR  ResidT  ListCost
Cache  InitClass  0      1      3  0.43729      0.14466      0.41805  0.43729      0.14466
0.41805 0.56271      NaN      NaN

```

The rows repeat for each of the 4 solvers the script runs (SSA, LDES, MVA, NC), which is the point of the example: the same model read by methods of different cost and different exactness.

2.8.14 retrieval_default

The same system with a PS backing store and the default retrieval parameters.

The example is built and solved as follows.

```

Network model = CacheRetrievalSystemModel.retrieval_default();
solveAll(model);

```

Running this edition prints

```

WARNING: [solver_ssa_analyzer_serial] Retrieval-system expected latency is not currently implemented;
reporting NaN.
SSA analysis [method: serial; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 100001.

```

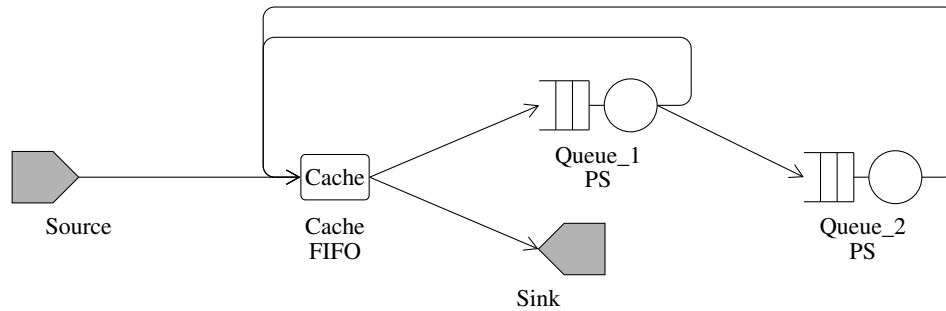



Figure 2.93: Topology of `retrieval_default`: an open network over 2 queueing stations with a cache node.

Nodes	5: Source, Queue $\times$ 2, Cache, Sink
Classes	6: InitClass (open), HitClass (open), MissClass (open), InitClass_retrievalClass_1 (open), InitClass_retrievalClass_2 (open), InitClass_retrievalClass_3 (open)
Scheduling	Queue_1: PS; Queue_2: PS
Arrivals	Source – InitClass: Exp(1); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Disabled; InitClass_retrievalClass_2: Disabled; InitClass_retrievalClass_3: Disabled
Service	Queue_1 – InitClass: Exp(2); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Exp(5); InitClass_retrievalClass_2: Exp(2); InitClass_retrievalClass_3: Exp(2) Queue_2 – InitClass: Exp(3); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Exp(3); InitClass_retrievalClass_2: Exp(3); InitClass_retrievalClass_3: Exp(3)
Features	cache: FIFO replacement, 3 items, capacity 1; drop rule {'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at Queue_1; drop rule {'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at Queue_2; class switching on 4 links
Solvers	SSA, LDES, MVA, NC

Table 2.93: Features of `retrieval_default`.

Node	JobClass	List	ListCap	Items	HitProb	DelayedHitProb	MissProb	HitRate	DelayedHitRate
	MissRate	ArvR	ResidT	ListCost					
Cache	InitClass	0	1	3	0.46222	0.076989	0.46079	0.46222	0.076989
	0.46079	0.53778	NaN	NaN					
LDES events: 20000 40000 60000 80000 100000 120000 140000 160000 180000 200000 220000 240000 260000 280000 300000 320000 340000 360000 380000 400000 420000 440000 460000 480000 500000 520000 540000 560000 580000 600000 620000 640000 660000 680000 700000 720000 740000 760000 780000 800000 820000 840000 860000 880000 900000 920000 940000 960000 980000 1000000 1000000									
LDES analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s. Iterations: 1000000.									
Node	JobClass	List	ListCap	Items	HitProb	DelayedHitProb	MissProb	HitRate	DelayedHitRate
	MissRate	ArvR	ResidT	ListCost					
Cache	InitClass	0	1	3	0.46348	0.078748	0.45777	0.46348	0.078748
	0.45777	0.53652	0.57635	NaN					
MVA analysis [method: default/fpi; type: approximate, deterministic; lang: java; env: python] completed in <t>s.									
Node	JobClass	List	ListCap	Items	HitProb	DelayedHitProb	MissProb	HitRate	DelayedHitRate
	MissRate	ArvR	ResidT	ListCost					
Cache	InitClass	0	1	3	0.42401	0.084524	0.49147	0.42401	0.084524
	0.49147	0.57599	0.54262	NaN					
NC analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.									
Node	JobClass	List	ListCap	Items	HitProb	DelayedHitProb	MissProb	HitRate	DelayedHitRate
	MissRate	ArvR	ResidT	ListCost					
Cache	InitClass	0	1	3	0.46383	0.078456	0.45771	0.46383	0.078456
	0.45771	0.53617	NaN	NaN					

The rows repeat for each of the 4 solvers the script runs (SSA, LDES, MVA, NC), which is the point of the example: the same model read by methods of different cost and different exactness.

2.8.15 retrieval_ps

The PS variant, with the delayed-hit behaviour made explicit.

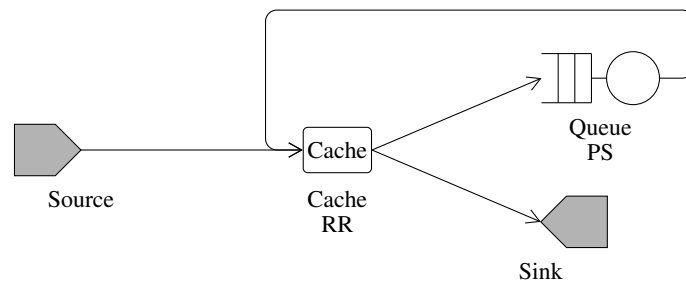


Figure 2.94: Topology of retrieval_ps: an open network over 1 queueing station with a cache node.

The example is built and solved as follows.

```

double[] accessProb = {49, 49, 49, 49, 7, 1, 1};
double total = 0.0;
for (double p : accessProb) total += p;
for (int i = 0; i < accessProb.length; i++) accessProb[i] /= total; // lambda =
(49,49,49,49,7,1,1)/205
Network model = CacheRetrievalSystemModel.simple_retrieval_system_model(
    accessProb,
    new double[] {1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0}, // mu_i = 1
    "[6]", SchedStrategy.PS, ReplacementStrategy.RR);
solveAll(model);

```

Running this edition prints

```

WARNING: [solver_ssa_analyzer_serial] Retrieval-system expected latency is not currently implemented;
reporting NaN.
SSA analysis [method: serial; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 100001.
Node JobClass List ListCap Items HitProb DelayedHitProb MissProb HitRate DelayedHitRate
MissRate ArvR ResidT ListCost

```

Nodes	4: Source, Queue, Cache, Sink
Classes	10: InitClass (open), HitClass (open), MissClass (open), InitClass_retrievalClass_1 (open), InitClass_retrievalClass_2 (open), InitClass_retrievalClass_3 (open), InitClass_retrievalClass_4 (open), InitClass_retrievalClass_5 (open), InitClass_retrievalClass_6 (open), InitClass_retrievalClass_7 (open)
Scheduling	Queue: PS
Arrivals	Source – InitClass: Exp(1); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Disabled; InitClass_retrievalClass_2: Disabled; InitClass_retrievalClass_3: Disabled; InitClass_retrievalClass_4: Disabled; InitClass_retrievalClass_5: Disabled; InitClass_retrievalClass_6: Disabled; InitClass_retrievalClass_7: Disabled
Service	Queue – InitClass: Exp(1); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Exp(1); InitClass_retrievalClass_2: Exp(1); InitClass_retrievalClass_3: Exp(1); InitClass_retrievalClass_4: Exp(1); InitClass_retrievalClass_5: Exp(1); InitClass_retrievalClass_6: Exp(1); InitClass_retrievalClass_7: Exp(1)
Features	cache: RR replacement, 7 items, capacity 6; drop rule {'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at Queue; class switching on 3 links
Solvers	SSA, LDES, MVA, NC

Table 2.94: Features of retrieval_ps.

Cache	InitClass	0	6	7	0.98494	0.0014294	0.013632	0.98494	0.0014294
		0.013632	0.015061	NaN	NaN				
LDES events: 20000 40000 60000 80000 100000 120000 140000 160000 180000 200000 220000 240000 260000 280000 300000 320000 340000 360000 380000 400000 420000 440000 460000 480000 500000 520000 540000 560000 580000 600000 620000 640000 660000 680000 700000 720000 740000 760000 780000 800000 820000 840000 860000 880000 900000 920000 940000 960000 980000 1000000 1000000									
LDES analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s. Iterations: 1000000.									
Node	JobClass	List	ListCap	Items	HitProb	DelayedHitProb	MissProb	HitRate	DelayedHitRate
	MissRate	ArvR	ResidT	ListCost					
Cache	InitClass	0	6	7	0.98298	0.002025	0.014991	0.98298	0.002025
		0.014991	0.017016	0.99256	NaN				
MVA analysis [method: default/fpi; type: approximate, deterministic; lang: java; env: python] completed in <t>s.									
Node	JobClass	List	ListCap	Items	HitProb	DelayedHitProb	MissProb	HitRate	DelayedHitRate
	MissRate	ArvR	ResidT	ListCost					
Cache	InitClass	0	6	7	0.97573	0.0034469	0.020826	0.97573	0.0034469
		0.020826	0.024273	1.0155	NaN				
NC analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.									
Node	JobClass	List	ListCap	Items	HitProb	DelayedHitProb	MissProb	HitRate	DelayedHitRate
	MissRate	ArvR	ResidT	ListCost					
Cache	InitClass	0	6	7	0.98272	0.0021597	0.015118	0.98272	0.0021597
		0.015118	0.017278	NaN	NaN				

The rows repeat for each of the 4 solvers the script runs (SSA, LDES, MVA, NC), which is the point of the example: the same model read by methods of different cost and different exactness.

2.8.16 retrieval_routing

Retrieval with routing that separates the hit and miss paths.

The example is built and solved as follows.

```
Matrix serviceRates = new Matrix(new double[][] {
    {2.0, 2.0, 2.0}, // IS queue
    {3.0, 3.0, 3.0}, // Queue 1
    {3.0, 3.0, 3.0}  // Queue 2
});
Network model = CacheRetrievalSystemModel.retrieval_system_with_probabilistic_routing(
    new double[] {0.6, 0.3, 0.1}, serviceRates, "[2]", SchedStrategy.FCFS);
solveAll(model);
```

Running this edition prints

```
WARNING: [solver_ssa_analyzer_serial] Retrieval-system expected latency is not currently implemented;
reporting NaN.
```

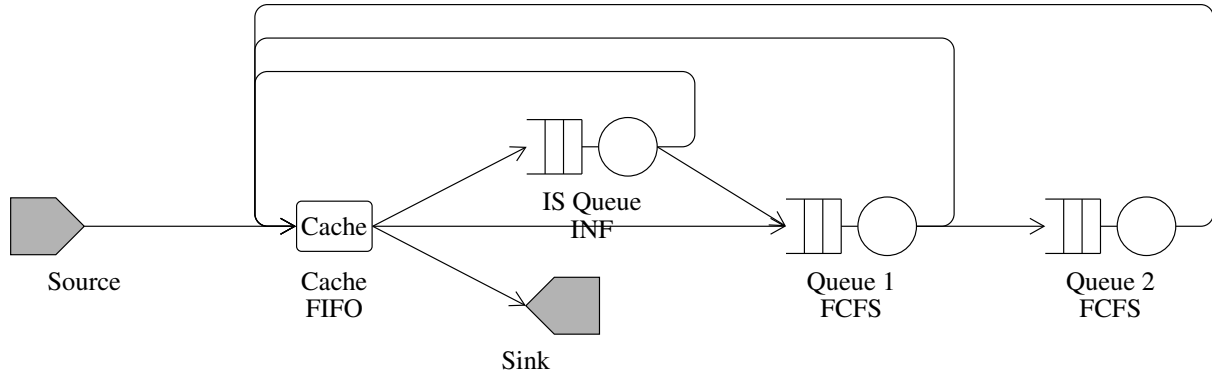


Figure 2.95: Topology of `retrieval_routing`: an open network over 3 queueing stations with a cache node.

Nodes	6: Source, Queue $\times$ 3, Cache, Sink
Classes	6: InitClass (open), HitClass (open), MissClass (open), InitClass_retrievalClass_1 (open), InitClass_retrievalClass_2 (open), InitClass_retrievalClass_3 (open)
Scheduling	IS Queue: INF; Queue 1: FCFS; Queue 2: FCFS
Arrivals	Source – InitClass: Exp(1); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Disabled; InitClass_retrievalClass_2: Disabled; InitClass_retrievalClass_3: Disabled
Service	IS Queue – InitClass: Exp(2); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Exp(2); InitClass_retrievalClass_2: Exp(2); InitClass_retrievalClass_3: Exp(2) Queue 1 – InitClass: Exp(3); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Exp(3); InitClass_retrievalClass_2: Exp(3); InitClass_retrievalClass_3: Exp(3) Queue 2 – InitClass: Exp(3); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Exp(3); InitClass_retrievalClass_2: Exp(3); InitClass_retrievalClass_3: Exp(3)
Features	cache: FIFO replacement, 3 items, capacity 2; drop rule {'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at IS Queue; drop rule {'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at Queue 1; drop rule {'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at Queue 2; class switching on 5 links
Solvers	SSA, LDES, MVA, NC

Table 2.95: Features of `retrieval_routing`.

```

SSA analysis [method: serial; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 100001.
Node   JobClass   List   ListCap   Items   HitProb   DelayedHitProb   MissProb   HitRate   DelayedHitRate
MissRate   ArvR   ResidT   ListCost
Cache   InitClass   0       2       3   0.78896       0.038947   0.17209   0.78896       0.038947
0.17209   0.21104   NaN     NaN
LDES events: 20000  40000  60000  80000  100000  120000  140000  160000  180000  200000  220000  240000  260000
280000  300000  320000  340000  360000  380000  400000  420000  440000  460000  480000  500000  520000  540000
560000  580000  600000  620000  640000  660000  680000  700000  720000  740000  760000  780000  800000  820000
840000  860000  880000  900000  920000  940000  960000  980000  1000000 1000000
LDES analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 1000000.
Node   JobClass   List   ListCap   Items   HitProb   DelayedHitProb   MissProb   HitRate   DelayedHitRate
MissRate   ArvR   ResidT   ListCost
Cache   InitClass   0       2       3   0.78284       0.039019   0.17814   0.78284       0.039019
0.17814   0.21716   0.61714   NaN
MVA analysis [method: default/fpi; type: approximate, deterministic; lang: java; env: python] completed in
<t>s.
Node   JobClass   List   ListCap   Items   HitProb   DelayedHitProb   MissProb   HitRate   DelayedHitRate
MissRate   ArvR   ResidT   ListCost
Cache   InitClass   0       2       3   0.74517       0.050821   0.20401   0.74517       0.050821
0.20401   0.25483   0.62901   NaN
NC analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>s
.
Node   JobClass   List   ListCap   Items   HitProb   DelayedHitProb   MissProb   HitRate   DelayedHitRate
MissRate   ArvR   ResidT   ListCost
Cache   InitClass   0       2       3   0.78352       0.038664   0.17781   0.78352       0.038664
0.17781   0.21648   NaN     NaN

```

The rows repeat for each of the 4 solvers the script runs (SSA, LDES, MVA, NC), which is the point of the example: the same model read by methods of different cost and different exactness.

2.8.17 retrieval_simple

A retrieval system: a cache in front of an infinite-server backing store, hits served at once and misses fetched.

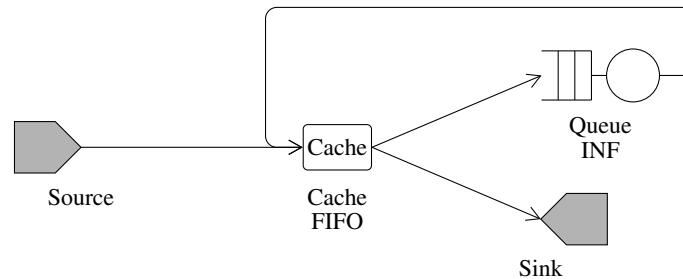


Figure 2.96: Topology of `retrieval_simple`: an open network over 1 queueing station with a cache node.

Nodes	4: Source, Queue, Cache, Sink
Classes	6: InitClass (open), HitClass (open), MissClass (open), InitClass_retrievalClass_1 (open), InitClass_retrievalClass_2 (open), InitClass_retrievalClass_3 (open)
Scheduling	Queue: INF
Arrivals	Source – InitClass: Exp(1); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Disabled; InitClass_retrievalClass_2: Disabled; InitClass_retrievalClass_3: Disabled
Service	Queue – InitClass: Exp(2); HitClass: Disabled; MissClass: Disabled; InitClass_retrievalClass_1: Exp(2); InitClass_retrievalClass_2: Exp(2); InitClass_retrievalClass_3: Exp(2)
Features	cache: FIFO replacement, 3 items, capacity 1; drop rule {'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at Queue; class switching on 3 links
Solvers	SSA, LDES, MVA, NC

Table 2.96: Features of `retrieval_simple`.

The example is built and solved as follows.

```

Network model = CacheRetrievalSystemModel.simple_retrieval_system_model(
    new double[] {0.6, 0.3, 0.1},
    new double[] {2.0, 2.0, 2.0},
    "[1]", SchedStrategy.INF);
solveCtmc(model);
solveAll(model);

```

Running this edition prints

```

WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 1 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
WARNING: [solver_ctmc_analyzer] Retrieval-system expected latency is not currently implemented; reporting
NaN.
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Node   JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Source InitClass   0     0     0       0       0       1
Cache  InitClass   0     0     0       0       1       0
Cache  HitClass    0     0     0       0       0  0.51077
Cache  MissClass    0     0     0       0       0  0.48923
Sink   HitClass    0     0     0       0  0.51077  0
Sink   MissClass    0     0     0       0  0.48923  0
CTMC hit=[ 0.510772584520104   NaN   NaN   NaN   NaN   NaN ]
miss=[ 0.489227415479896   NaN   NaN   NaN   NaN   NaN ]
expectedLatency=[ NaN   NaN   NaN   NaN   NaN   NaN ]

WARNING: [solver_ssa_analyzer_serial] Retrieval-system expected latency is not currently implemented;
reporting NaN.
SSA analysis [method: serial; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Iterations: 100001.

... (4 captured-output lines omitted; rerun the source example for the full output)

Node   JobClass   List   ListCap   Items   HitProb   DelayedHitProb   MissProb   HitRate   DelayedHitRate
MissRate   ArvR   ResidT   ListCost
Cache  InitClass   0       1       3   0.4446     0.091824   0.46358   0.4446     0.091824
0.46358 0.5554 0.4997   NaN
MVA analysis [method: default/fpi; type: approximate, deterministic; lang: java; env: python] completed in
<t>s.
Node   JobClass   List   ListCap   Items   HitProb   DelayedHitProb   MissProb   HitRate   DelayedHitRate
MissRate   ArvR   ResidT   ListCost
Cache  InitClass   0       1       3  0.41334     0.10048   0.48618   0.41334     0.10048
0.48618 0.58666 0.5     NaN
NC analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>s
.
Node   JobClass   List   ListCap   Items   HitProb   DelayedHitProb   MissProb   HitRate   DelayedHitRate
MissRate   ArvR   ResidT   ListCost
Cache  InitClass   0       1       3  0.44605     0.091157   0.4628    0.44605     0.091157
0.4628 0.55395   NaN     NaN

```

The rows repeat for each of the 5 solvers the script runs (CTMC, SSA, LDES, MVA, NC), which is the point of the example: the same model read by methods of different cost and different exactness.

2.9 Cluster models

A cluster is a dispatcher in front of a pool of servers. It is not a new node type: the shorthand builds a Router and a set of stations, and the interest is in the dispatching policy and in how the response time scales with the pool size.

2.9.1 cl_basic

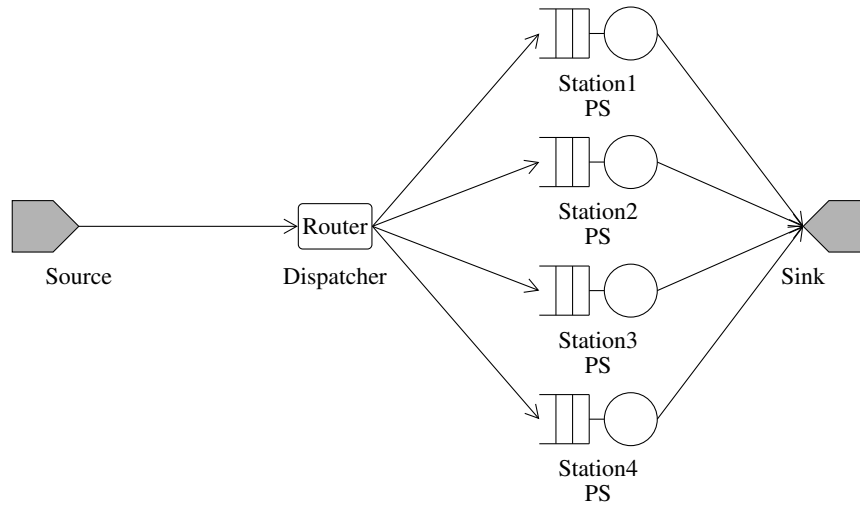
A dispatcher in front of a pool of identical servers, built with the cluster shorthand.

The example is built and solved as follows.

```

Matrix lambda = new Matrix(1, 1);
lambda.set(0, 0, 0.4);
Matrix D = new Matrix(4, 1);
for (int i = 0; i < 4; i++) D.set(i, 0, 1.0); // mean service time = 1
return Network.clusterPs(lambda, D, RoutingStrategy.RAND);

```

Figure 2.97: Topology of `cl_basic`: an open network over 4 queueing stations with a router.

Nodes	7: Source, Queue $\times$ 4, Router, Sink
Classes	1: Class1 (open)
Scheduling	Station1: PS; Station2: PS; Station3: PS; Station4: PS
Arrivals	Source – Class1: Exp(0.4)
Service	Station1 – Class1: Exp(1) Station2 – Class1: Exp(1) Station3 – Class1: Exp(1) Station4 – Class1: Exp(1)
Features	class switching on 7 links
Solvers	MVA

Table 2.97: Features of `cl_basic`.

Running this edition prints

```
jline.lang.Network@69d0a921
```

2.9.2 `cl_closed`

The closed version, with a fixed population of requests.

Nodes	5: Queue $\times$ 3, Delay, Router
Classes	1: Class1 (closed, $N = 8$)
Scheduling	Think: INF; Station1: PS; Station2: PS; Station3: PS
Service	Think – Class1: Exp(1) Station1 – Class1: Exp(1) Station2 – Class1: Exp(1) Station3 – Class1: Exp(1)
Features	class switching on 5 links
Solvers	MVA

Table 2.98: Features of `cl_closed`.

The example is built and solved as follows.

```
Matrix N = new Matrix(1, 1);
N.set(0, 0, 8);
Matrix Z = new Matrix(1, 1);
Z.set(0, 0, 1.0); // think time
Matrix D = new Matrix(3, 1);
for (int i = 0; i < 3; i++) D.set(i, 0, 1.0);
Matrix S = new Matrix(3, 1, 3);
S.fill(1.0);
```

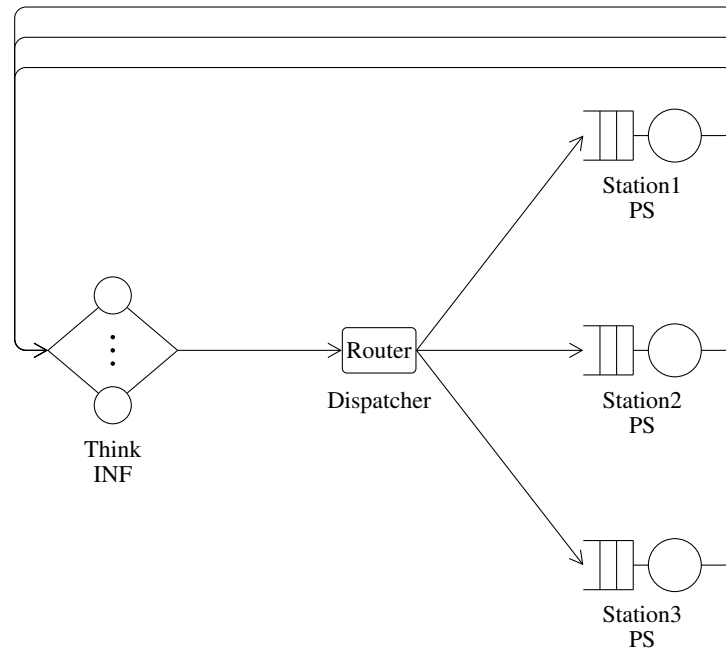


Figure 2.98: Topology of `cl_closed`: a closed network over 4 queueing stations with a router.

```
SchedStrategy[] sched = new SchedStrategy[3];
for (int i = 0; i < 3; i++) sched[i] = SchedStrategy.PS;
return Network.clusterClosed(N, Z, D, sched, S, RoutingStrategy.RAND);
```

Running this edition prints

```
jline.lang.Network@3eb07fd3
```

2.9.3 `cl_compare`

RAND against round-robin dispatching on one cluster.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

Solvers	SSA
---------	-----

Table 2.99: Features of `cl_compare`.

The example is built and solved as follows.

```
System.out.println("\n--- cl_compare: RAND vs RROBIN ---");
Cluster cluster = new Cluster().setNumStations(4).setArrivalRate(1.0).setServiceRate(0.4);
cluster.setScheduling(SchedStrategy.PS);

Map<RoutingStrategy, NetworkAvgTable> results = cluster.compareDispatching(
    m -> new SolverSSA(m, "seed", 23000, "samples", 2000).getAvgTable(),
    RoutingStrategy.RAND, RoutingStrategy.RROBIN);
for (Map.Entry<RoutingStrategy, NetworkAvgTable> e : results.entrySet()) {
    System.out.printf("%n=== Dispatching: %s ===%n", e.getKey());
    e.getValue().print();
}
```


No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.9.4 cl_multiclass

Several request classes through one cluster.

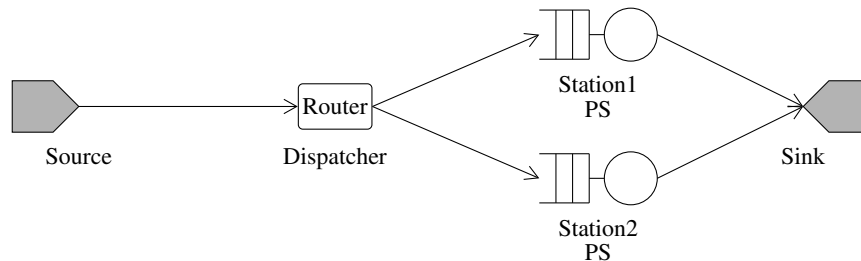


Figure 2.99: Topology of `cl_multiclass`: an open network over 2 queueing stations with a router.

Nodes	5: Source, Queue × 2, Router, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Station1: PS; Station2: PS
Arrivals	Source – Class1: Exp(0.3); Class2: Exp(0.2)
Service	Station1 – Class1: Exp(1); Class2: Exp(2) Station2 – Class1: Exp(1); Class2: Exp(2)
Features	class switching on 5 links
Solvers	MVA

Table 2.100: Features of `cl_multiclass`.

The example is built and solved as follows.

```

Matrix lambda = new Matrix(1, 2);
lambda.set(0, 0, 0.3);
lambda.set(0, 1, 0.2);
Matrix D = new Matrix(2, 2);
D.set(0, 0, 1.0); D.set(0, 1, 0.5);
D.set(1, 0, 1.0); D.set(1, 1, 0.5);
Matrix S = new Matrix(2, 1, 2);
S.fill(1.0);
return Network.clusterPs(lambda, D, S, RoutingStrategy.RAND);

```

Running this edition prints

```
jline.lang.Network@446cdf90
```

2.9.5 cl_scheduling

The same cluster under FCFS and under PS.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

Solvers	SSA
----------------	-----

Table 2.101: Features of `cl_scheduling`.

The example is built and solved as follows.

```

System.out.println("\n--- cl_scheduling: FCFS against PS at SCV 4 ---");
Cluster cluster = new Cluster().setNumStations(3).setArrivalRate(0.9).setServiceRate(0.5)
    .setDispatching(RoutingStrategy.RAND);
cluster.setServiceSCV(4.0);
Map<SchedStrategy, NetworkAvgTable> results = cluster.compareScheduling(
    model -> new SolverSSA(model, "seed", 23000, "samples", 20000).getAvgTable(),
    SchedStrategy.FCFS, SchedStrategy.PS);
for (Map.Entry<SchedStrategy, NetworkAvgTable> e : results.entrySet()) {
    System.out.println("\nScheduling = " + e.getKey());
    e.getValue().print();
}

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.9.6 cl_stations

Clusters whose members are declared station by station.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

Solvers	MVA
---------	-----

Table 2.102: Features of `cl_stations`.

The example is built and solved as follows.

```

System.out.println("\n--- cl_stations: sweep the number of servers ---");
Cluster cluster = new Cluster().setNumStations(1).setArrivalRate(1.6).setServiceRate(1.0)
    .setScheduling(SchedStrategy.PS)
    .setDispatching(RoutingStrategy.RAND);
Map<Integer, NetworkAvgTable> sweep = cluster.sweepNumStations(
    new int[]{2, 3, 4, 6}, SolverMVA.class);
for (Map.Entry<Integer, NetworkAvgTable> e : sweep.entrySet()) {
    System.out.println("\nstations = " + e.getKey());
    e.getValue().print();
}

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.9.7 cl_sweep

A sweep over the cluster size, and the response time curve it traces.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

Solvers	MVA
---------	-----

Table 2.103: Features of `cl_sweep`.

The example is built and solved as follows.

```

System.out.println("\n--- cl_sweep: arrival-rate sweep ---");
Cluster cluster = new Cluster().setNumStations(2).setArrivalRate(0.1).setServiceRate(1.0);
cluster.setScheduling(SchedStrategy.PS).setDispatching(RoutingStrategy.RAND);
Map<Double, NetworkAvgTable> sweep = cluster.sweepArrivalRate(

```

```

    new double[]{0.2, 0.5, 0.9, 1.5}, m -> new MVA(m).getAvgTable());
for (Map.Entry<Double, NetworkAvgTable> e : sweep.entrySet()) {
    System.out.printf("%n== lambda = %g ===%n", e.getKey());
    e.getValue().print();
}

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.9.8 dispatch_closed

The closed version.

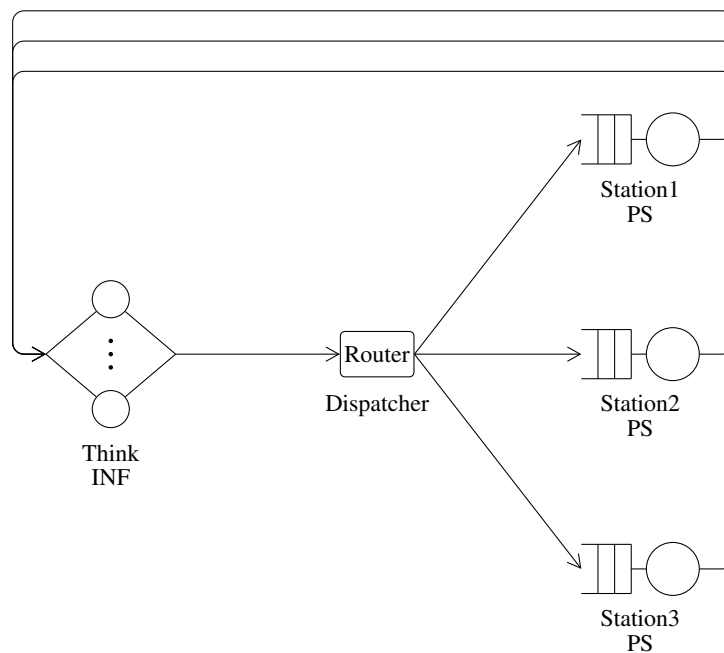


Figure 2.100: Topology of dispatch_closed: a closed network over 4 queueing stations with a router.

Nodes	5: Queue $\times$ 3, Delay, Router
Classes	1: Class1 (closed, $N = 10$)
Scheduling	Think: INF; Station1: PS; Station2: PS; Station3: PS
Service	Think – Class1: Exp(1) Station1 – Class1: Exp(1) Station2 – Class1: Exp(1) Station3 – Class1: Exp(1)
Features	class switching on 5 links
Solvers	MVA, SSA

Table 2.104: Features of dispatch_closed.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.9.9 dispatch_mixed

The mixed version, open and closed classes through the same dispatcher.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

Solvers	MVA, LDES
----------------	-----------

Table 2.105: Features of `dispatch_mixed`.

The example is built and solved as follows.

```
System.out.println("\n--- dispatch_mixed: open + closed classes through the builder ---");
Cluster cluster = new Cluster();
cluster.setNumStations(2);
cluster.setMixed(0.5, 3, 1.0);
cluster.setServiceRates(new double[][]{{2.0, 1.5}, {2.0, 1.5}});
cluster.setScheduling(SchedStrategy.PS);

new MVA(cluster.build()).getAvgTable().print();
new SolverLDES(cluster.build(), "seed", 23000).getAvgTable().print();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.9.10 `dispatch_open`

The dispatcher idiom written out in full, open workload.

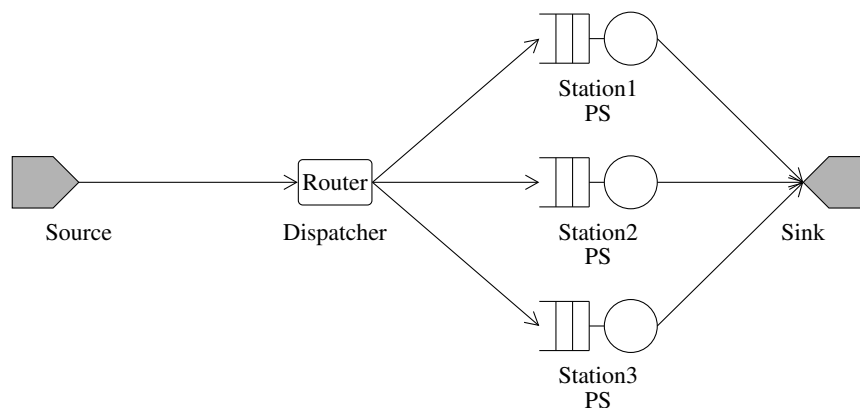


Figure 2.101: Topology of `dispatch_open`: an open network over 3 queueing stations with a router.

Nodes	6: Source, Queue $\times$ 3, Router, Sink
Classes	1: Class1 (open)
Scheduling	Station1: PS; Station2: PS; Station3: PS
Arrivals	Source – Class1: Exp(0.4)
Service	Station1 – Class1: Exp(1) Station2 – Class1: Exp(1) Station3 – Class1: Exp(1)
Features	class switching on 6 links
Solvers	MVA, SSA

Table 2.106: Features of `dispatch_open`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.10 Stochastic Petri nets

A stochastic Petri net is built from `Place` and `Transition` nodes. A transition fires when its input places hold enough tokens, consuming and producing tokens according to the arc weights; timed transitions fire after a random delay, immediate transitions fire at once, and an inhibitor arc disables a transition while its place is marked. A transition may have several firing modes, each with its own rate and arc weights.

Petri nets and queueing models are two notations for the same processes, and two of the scripts make that explicit by building the same system both ways and comparing the generators. Because the firing time need not be exponential, the family also carries the non-Markovian cases.

The figures of this section are drawn in the notation of the net rather than in that of a queueing network. A place is a circle, and the tokens or the number inside it are the marking the model starts from; a transition is a bar, hollow when it fires after a random delay and solid when it fires at once; an arc is plain when it moves one token and labelled with its multiplicity otherwise, and an arc ending in a small circle is an inhibitor, which holds its transition disabled for as long as its place is marked. A transition with more than one firing mode is named with the number of modes, since no single distribution describes it. A circle bisected by a bar is a queueing place, whose tokens are served by a scheduling station embedded in the place.

2.10.1 `spn_basic_closed`

The closed counterpart, a token circulating between two places.

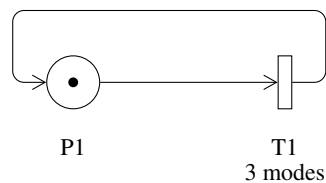


Figure 2.102: Topology of `spn_basic_closed`: a stochastic Petri net over 1 place and 1 transition; a place is a circle holding its initial marking and a transition a bar, hollow when its firing time is random and solid when it fires at once; a transition offering several firing modes is named with their number in place of a distribution.

Nodes	2: Place, Transition
Classes	1: <code>Class1</code> (closed, $N = 1$)
Features	1 transition with 3 firing modes each
Solvers	JMT

Table 2.107: Features of `spn_basic_closed`.

The example is built and solved as follows.

```
public static void spn_basic_closed() throws Exception {
    Network model = StochPetriNetModel.spn_basic_closed();

    // CTMC solver (exact for small Petri nets)
    // try {
    //     CTMC solverCtmc = new CTMC(model);
    //     solverCtmc.getAvgTable().print();
    // } catch (Exception e) {
    //     System.out.println("CTMC solver error: " + e.getMessage());
    // }

    // JMT solver (simulation)
```

```
try {
    JMT solverJmt = new JMT(model, "seed", 23000);
    solverJmt.getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT solver not available: " + e.getMessage());
}
pauseForUser();
```

Running this edition prints

```
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
P1      Class1    1    0  0.35308 0.35308 2.835 2.8322

[Running in non-interactive mode, continuing...]
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by JMT.

2.10.2 spn_basic_open

One place and one timed transition fed by an open class, the Petri net form of an M/M/1 queue.

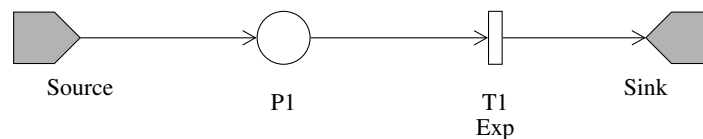


Figure 2.103: Topology of spn_basic_open: a stochastic Petri net over 1 place and 1 transition; a place is a circle holding its initial marking and a transition a bar, hollow when its firing time is random and solid when it fires at once; the net is open, fed by a Source and drained into a Sink.

Nodes	4: Source, Place, Transition, Sink
Classes	1: Class1 (open)
Arrivals	Source – Class1: Exp(1)
Features	1 transition with 1 firing mode each
Solvers	JMT

Table 2.108: Features of spn_basic_open.

The example is built and solved as follows.

```
Network model = StochPetriNetModel.spn_basic_open();
JMT solver = new JMT(model, "seed", 23000);

solver.getAvgTable().print();
pauseForUser();
```

Running this edition prints

```
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Source  Class1    0    0    0    0    0  1.0312
P1      Class1  0.25966  0  0.2503 0.2503 1.0312 1.0427

[Running in non-interactive mode, continuing...]
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by JMT.

2.10.3 spn_closed_fourplaces

Four places with Erlang, Coxian and hyperexponential firing times.

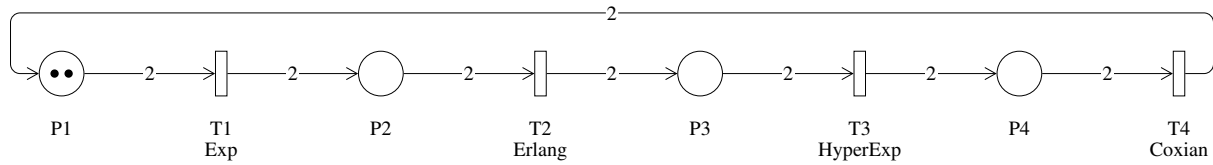


Figure 2.104: Topology of `spn_closed_fourplaces`: a stochastic Petri net over 4 places and 4 transitions; a place is a circle holding its initial marking and a transition a bar, hollow when its firing time is random and solid when it fires at once; an arc that moves more than one token carries its multiplicity.

Nodes	8: Place $\times$ 4, Transition $\times$ 4
Classes	1: Class1 (closed, $N = 2$)
Features	4 transitions with 1 firing mode each
Solvers	JMT

Table 2.109: Features of `spn_closed_fourplaces`.

The example is built and solved as follows.

```
Network model = StochPetriNetModel.spn_closed_fourplaces();

// CTMC solver for exact solution
try {
    CTMC solverCtmc = new CTMC(model);
    solverCtmc.getAvgTable().print();
} catch (Exception e) {
    System.out.println("CTMC solver error: " + e.getMessage());
}

// JMT solver for simulation
try {
    JMT solverJmt = new JMT(model, "seed", 23000);
    solverJmt.getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT solver not available: " + e.getMessage());
}
pauseForUser();
```

Running this edition prints

```
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station  JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
P1        Class1    0.28846 0.28846 0.5      0.5      0.57692 0.57692
P2        Class1    0.76923 0.76923 1.3333  1.3333  0.57692 0.57692
P3        Class1     0.25    0.25   0.43333 0.43333 0.57692 0.57692
P4        Class1    0.69231 0.69231 1.2      1.2      0.57692 0.57692
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station  JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
P1        Class1    0.27748 0      0.48217 0.48217 0.57461 0.57432
P2        Class1    0.77676 0      1.3441  1.3441  0.57432 0.57273
P3        Class1    0.25164 0      0.43035 0.43035 0.57273 0.57294
P4        Class1    0.69386 0      1.2301  1.2301  0.57294 0.5744

[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 2 solvers the script runs (CTMC, JMT), which is the point of the example: the same model read by methods of different cost and different exactness.

2.10.4 spn_closed_twoplaces

A closed net over two places with Erlang firing times.

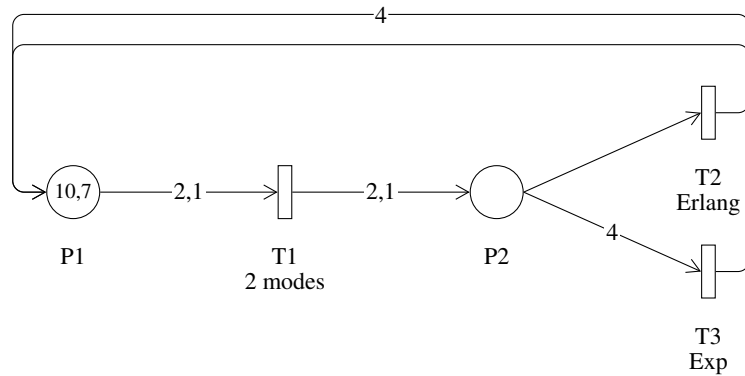


Figure 2.105: Topology of `spn_closed_twoplaces`: a stochastic Petri net over 2 places and 3 transitions; a place is a circle holding its initial marking and a transition a bar, hollow when its firing time is random and solid when it fires at once; an arc that moves more than one token carries its multiplicity; a transition offering several firing modes is named with their number in place of a distribution.

Nodes	5: Place $\times$ 2, Transition $\times$ 3
Classes	2: Class1 (closed, $N = 10$), Class2 (closed, $N = 7$)
Features	3 transitions, 1 to 2 firing modes; class switching on 2 links
Solvers	JMT

Table 2.110: Features of `spn_closed_twoplaces`.

The example is built and solved as follows.

```
Network model = StochPetriNetModel.spn_closed_twoplaces();

// CTMC solver for exact solution
try {
    CTMC solverCtmc = new CTMC(model);
    solverCtmc.getAvgTable().print();
} catch (Exception e) {
    System.out.println("CTMC solver error: " + e.getMessage());
}

// JMT solver for simulation
try {
    JMT solverJmt = new JMT(model, "seed", 23000);
    solverJmt.getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT solver not available: " + e.getMessage());
}

pauseForUser();
```

Running this edition prints

```
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
P1 Class1 0.8893 0.8893 1.1857 1.1857 0.75 0.75
P1 Class2 1.6859 1.6859 1.0928 1.0928 1.5427 1.5427
P2 Class1 9.1107 9.1107 12.148 12.148 0.75 0.75
P2 Class2 5.3141 5.3141 3.4446 3.4446 1.5427 1.5427
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
P1 Class1 0.88109 0 1.1688 1.1688 0.7533 0.75488
P1 Class2 1.708 0 1.1162 1.1162 1.5906 1.6042
```



```

P2      Class1      9.1189      0      12.1      12.1      0.75488      0.75495
P2      Class2      5.292      0      3.3423      3.3423      1.6042      1.5821

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (CTMC, JMT), which is the point of the example: the same model read by methods of different cost and different exactness.

2.10.5 spn_colored_gspn

A colored generalized stochastic Petri net: one firing mode per token color, an immediate transition whose inhibitor arcs make the server exclusive, and firing weights that arbitrate the two colors competing for it.

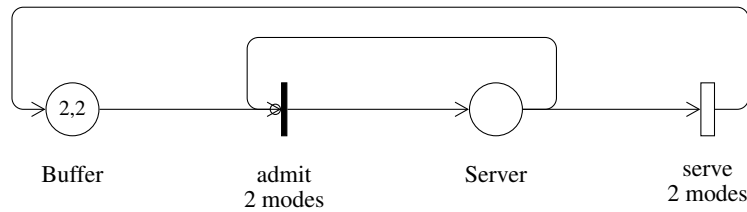


Figure 2.106: Topology of `spn_colored_gspn`: a stochastic Petri net over 2 places and 2 transitions; a place is a circle holding its initial marking and a transition a bar, hollow when its firing time is random and solid when it fires at once; the arc ending in a circle is an inhibitor and holds admit disabled while Server is marked; a transition offering several firing modes is named with their number in place of a distribution.

Nodes	4: Place $\times$ 2, Transition $\times$ 2
Classes	2: Gold (closed, $N = 2$), Silver (closed, $N = 2$)
Features	2 transitions with 2 firing modes each
Solvers	CTMC, SSA

Table 2.111: Features of `spn_colored_gspn`.

The example is built and solved as follows.

```

Network model = StochPetriNetModel.spn_colored_gspn();
new CTMC(model, "cutoff", 4).getAvgTable().print();
new SSA(model, "seed", 23000, "samples", 2e5).getAvgTable().print();
pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.10.6 spn_fluid_dae

Fluid DAE analysis of stochastic Petri nets, including invariant constraints, immediate-transition flows, bounded markings and marking covariances.

Nodes	4: Place $\times$ 2, Transition $\times$ 2
Classes	1: Class1 (closed, $N = 4$)
Features	2 transitions with 1 firing mode each
Solvers	FLD, CTMC

Table 2.112: Features of `spn_fluid_dae`.

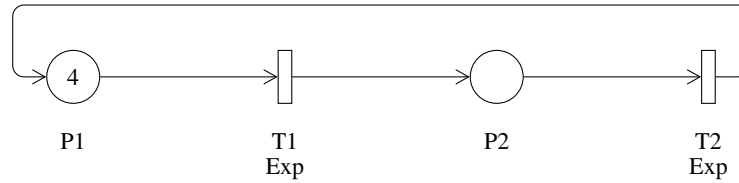


Figure 2.107: Topology of `spn_fluid_dae`: a stochastic Petri net over 2 places and 2 transitions; a place is a circle holding its initial marking and a transition a bar, hollow when its firing time is random and solid when it fires at once.

The example is built and solved as follows.

```
// A closed net whose fluid answer is EXACT.
Network exact = StochPetriNetModel.spn_fluid_exact();
FLD fld = new FLD(exact);
fld.getAvgTable().print();
new CTMC(StochPetriNetModel.spn_fluid_exact(), "cutoff", 6).getAvgTable().print();

// The exact marking is Binomial(4, 3/5), so the variance is
// 4*0.6*0.4 = 0.96.
PetriSolver.PetriReport pr = ((FluidResult) fld.result).petri;
System.out.printf("marking variance: %.6f %.6f (exact 0.96)%n",
    pr.markingVar.get(0, 0), pr.markingVar.get(1, 0));
System.out.printf("invariant \"%s\" = %g, error %.2e%n",
    pr.invariantLabel.get(0), pr.invariantValue[0], pr.invariantError[0]);

// An immediate transition, as an algebraic flow. The vanishing place P2
// holds exactly zero mass and the net answers as the reduced two-place
// net does, which is what the algebraic flow buys: an approximation of
// the immediate transition by a large finite rate would only approach it.
new FLD(StochPetriNetModel.spn_fluid_immediate()).getAvgTable().print();
new FLD(StochPetriNetModel.spn_fluid_reduced()).getAvgTable().print();
pauseForUser();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.10.7 spn_fourmodes

Four firing modes, and the mode selection probabilities.

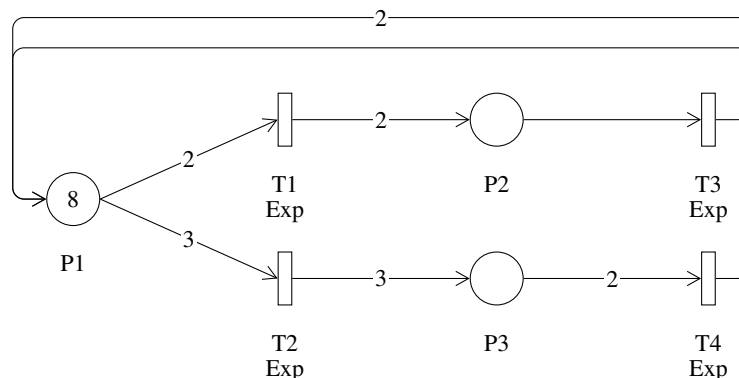


Figure 2.108: Topology of `spn_fourmodes`: a stochastic Petri net over 3 places and 4 transitions; a place is a circle holding its initial marking and a transition a bar, hollow when its firing time is random and solid when it fires at once; an arc that moves more than one token carries its multiplicity.

The example is built and solved as follows.

Nodes	7: Place $\times$ 3, Transition $\times$ 4
Classes	1: Class1 (closed, $N = 8$)
Features	4 transitions with 1 firing mode each
Solvers	JMT

Table 2.113: Features of `spn_fourmodes`.

```

Network model = StochPetriNetModel.spn_fourmodes();
JMT solver = new JMT(model, "seed", 23000);

solver.getAvgTable().print();
pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
P1      Class1   3.4344  0  0.72032 0.72032  4.796 4.7866
P2      Class1   2.5589  0  0.86649 0.86649  2.9984 3.0006
P3      Class1   1.9935  0  1.1405  1.1405  1.8044 1.742

[Running in non-interactive mode, continuing...]

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by JMT.

2.10.8 `spn_inhibiting`

An inhibitor arc, which disables a transition while its place is marked.

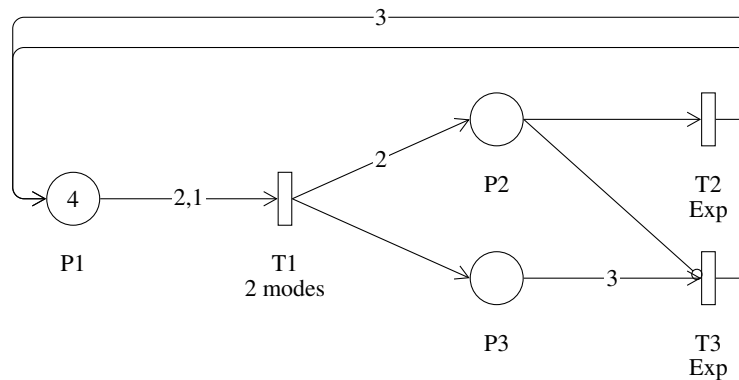


Figure 2.109: Topology of `spn_inhibiting`: a stochastic Petri net over 3 places and 3 transitions; a place is a circle holding its initial marking and a transition a bar, hollow when its firing time is random and solid when it fires at once; an arc that moves more than one token carries its multiplicity; the arc ending in a circle is an inhibitor and holds T3 disabled while P2 is marked; a transition offering several firing modes is named with their number in place of a distribution.

Nodes	6: Place $\times$ 3, Transition $\times$ 3
Classes	1: Class1 (closed, $N = 4$)
Features	3 transitions, 1 to 2 firing modes
Solvers	JMT

Table 2.114: Features of `spn_inhibiting`.

The example is built and solved as follows.

```

Network model = StochPetriNetModel.spn_inhibiting();
JMT solver = new JMT(model, "seed", 23000);

solver.getAvgTable().print();
pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass      QLen  Util    RespT   ResidT    ArvR      Tput
P1      Class1      1.3667    0  0.56802  0.56802  2.4787    2.4778
P2      Class1      0.74496    0  0.43709  0.10927  1.7363    1.7134
P3      Class1      1.8403    0  2.6351   1.3176   0.7158    0.71634

[Running in non-interactive mode, continuing...]

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by JMT.

2.10.9 spn_lpbounds

Linear-programming bounds on a Petri net with SolverBA: the stationary chain is relaxed to a moment polytope and every reported measure minimised and maximised over it, reproducing Table 2 of Liu (1998) on a blocking production line and then bracketing the exact marking of an inhibitor net.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

Solvers	CTMC
----------------	------

Table 2.115: Features of `spn_lpbounds`.

The example is built and solved as follows.

```

// The reference's own Fig. 2b, and its Table 2. The one column not
// reproduced is its u.b.1, the upper side further tightened by the
// subnet-throughput theorems (its Thms 1 and 2); those are not
// implemented, so u.b.2 is the column to compare against.
double[][] mus = {{1, 1.25, 2, 0.5}, {1, 1.25, 2, 2.5}, {1, 1.25, 1.25, 2.5},
                  {1, 1.25, 1.25, 1}, {1.111, 1.111, 1.111, 1.111}};
double[][] pub = {{1.165, 1.951, 2.000, 0.930, 2.000},
                  {1.829, 2.978, 3.529, 1.481, 4.000},
                  {1.581, 2.873, 3.333, 1.333, 4.000},
                  {1.359, 2.757, 3.333, 1.111, 4.000},
                  {1.350, 2.667, 2.963, 1.111, 4.444}};

System.out.println("\nLiu (1998) Table 2: total throughput of the production line");
System.out.printf("%-5s %-32s %-19s %-19s\n",
    "case", "Markovian LP", "published", "operational LP");
System.out.printf("%-5s %9s %9s %9s %9s %9s %9s %9s\n",
    "", "lower", "simul", "upper", "l.b.", "u.b.2", "o.l.b.", "o.u.b.");
for (int c = 0; c < mus.length; c++) {
    NetworkStruct sn = StochPetriNetModel.spn_lpbounds_propline(mus[c]).getStruct();
    // The liveness rows of the reference's Table 1 are OPT-IN, because
    // they hold only on a live net and spn_lpbnd cannot certify
    // liveness. This one is live: a strongly connected marked graph
    // with a token on every cycle. They are the whole of the lower
    // side, so the published l.b. needs them.
    Spn_lpbnd.SpnLpOptions oLo = new Spn_lpbnd.SpnLpOptions();
    oLo.markovian = true;

    ... (38 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

System.out.println("\nsolver.citations():");

```

```

for (LineCitations.Citation c
    : new BA(StochPetriNetModel.spn_inhibiting(), "spnlp2.upper").citations()) {
    System.out.println(c);
}
pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.10.10 spn_open_sevenplaces

A seven-place open net, large enough that the state space matters.

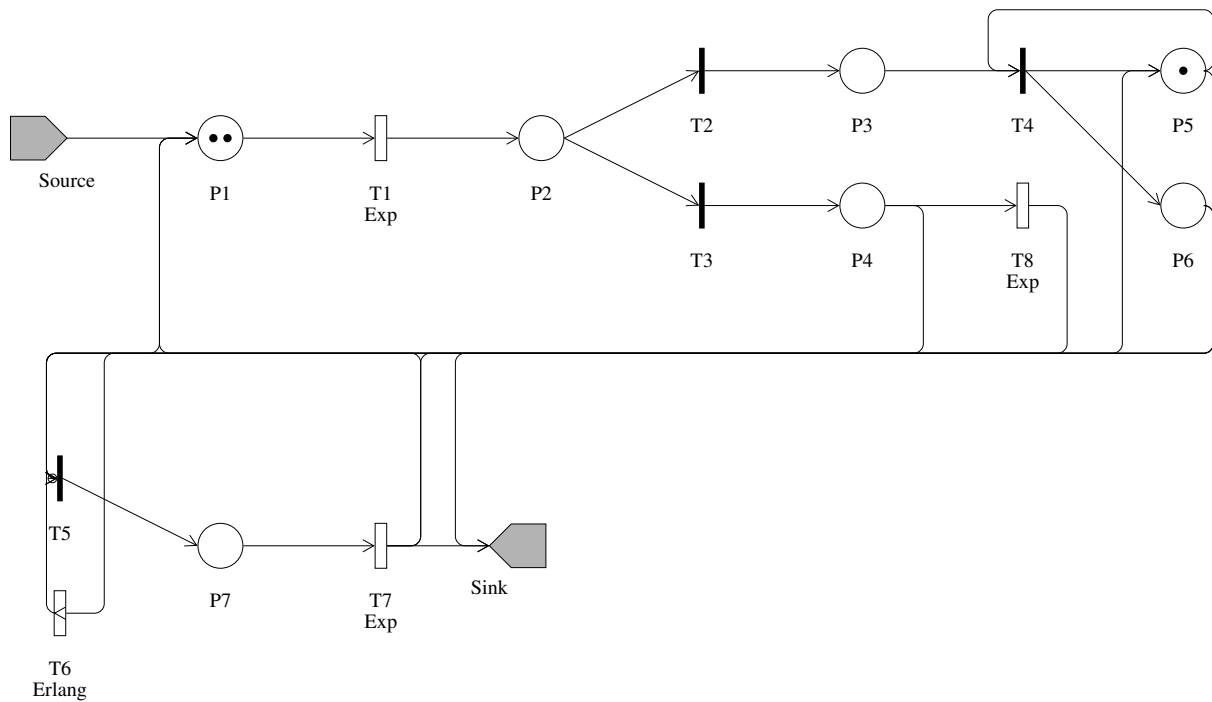


Figure 2.110: Topology of `spn_open_sevenplaces`: a stochastic Petri net over 7 places and 8 transitions; a place is a circle holding its initial marking and a transition a bar, hollow when its firing time is random and solid when it fires at once; the arc ending in a circle is an inhibitor and holds T5 disabled while P6 is marked; the net is open, fed by a Source and drained into a Sink.

Nodes	17: Source, Place $\times$ 7, Transition $\times$ 8, Sink
Classes	1: Class1 (open)
Arrivals	Source – Class1: Exp(1)
Features	8 transitions with 1 firing mode each
Solvers	JMT

Table 2.116: Features of `spn_open_sevenplaces`.

The example is built and solved as follows.

```

Network model = StochPetriNetModel.spn_open_sevenplaces();
JMT solver = new JMT(model, "seed", 23000);

solver.getAvgTable().print();
pauseForUser();

```

Running this edition prints

```
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass   QLen  Util   RespT   ResidT   ArvR   Tput
Source   Class1       0      0        0        0        0    1.0153
P1        Class1   0.73504  0    0.25337  0.48798  2.8389  2.8376
P2        Class1       0      0        0        0        0    2.8376
P3        Class1   0.14404  0    0.10279  0.098981 1.4098  1.4058
P4        Class1   0.50366  0    0.35917  0.17293  1.4223  1.4232
P5        Class1   0.79047  0    0.43293  0.28862  1.8031  1.8066
P6        Class1   1.3832   0    0.98965  0.40319  1.4058  1.4006
P7        Class1   0.2102   0    0.50077  0.77897  0.41759 0.41932

[Running in non-interactive mode, continuing...]
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by JMT.

2.10.11 spn_pareto_service

A Pareto firing time, a heavy-tailed non-Markovian net.

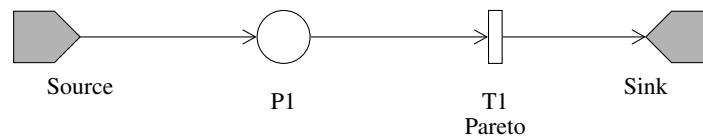


Figure 2.111: Topology of `spn_pareto_service`: a stochastic Petri net over 1 place and 1 transition; a place is a circle holding its initial marking and a transition a bar, hollow when its firing time is random and solid when it fires at once; the net is open, fed by a Source and drained into a Sink.

Nodes	4: Source, Place, Transition, Sink
Classes	1: Class1 (open)
Arrivals	Source – Class1: Exp(0.5)
Features	1 transition with 1 firing mode each
Solvers	JMT

Table 2.117: Features of `spn_pareto_service`.

The example is built and solved as follows.

```
Network model = StochPetriNetModel.spn_pareto_service();

try {
    new jline.solvers.wrappers.jmt.JMT(model, "seed", 23000, "samples", 10000)
        .getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT failed: " + e.getMessage());
}
```

Running this edition prints

```
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass   QLen  Util   RespT   ResidT   ArvR   Tput
Source   Class1       0      0        0        0        0    0.51382
P1        Class1   2.7513   0    5.007   5.007   0.51382 0.51422
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by JMT.

2.10.12 spn_productform_nc

Product-form analysis of cyclic and fork-join stochastic Petri nets with SolverNC, compared against an explicit CTMC and accompanied by the structural certificate and marking invariants.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

Solvers	NC, CTMC
---------	----------

Table 2.118: Features of `spn_productform_nc`.

The example is built and solved as follows.

```
// A closed cycle, where the exact CTMC gives the reference.
System.out.println("\n--- 3-place cyclic net, N = 4 ---");
new NC(StochPetriNetModel.spn_productform_cyclic()).getAvgTable().print();

System.out.println("The same net through the explicit generator, for comparison:");
new CTMC(StochPetriNetModel.spn_productform_cyclic()).getAvgTable().print();

// The certificate the derivation produced. Deficiency zero plus weak
// reversibility is what Feinberg's theorem needs for a positive
// complex-balanced point to exist at ANY choice of rates.
Spn_pf.SpnPfOptions pfOpt = new Spn_pf.SpnPfOptions();
pfOpt.verbose = true;
Spn_pf.SpnPfResult pf =
    Spn_pf.spn_pf(StochPetriNetModel.spn_productform_cyclic(), pfOpt);
System.out.printf("product form: %s, deficiency %d, %d linkage classes, rank %d\n",
    pf.kind, pf.deficiency, pf.linkage, pf.srank);

// A fork-join net, which has no queueing-network form at all.
Network fj = StochPetriNetModel.spn_productform_forkjoin();
System.out.println("\n--- fork-join net, P0 -> P1+P2 -> P3 -> P0, 3 tokens at P0 ---");
new NC(fj).getAvgTable().print();

Spn_pf.SpnPfResult pfj = Spn_pf.spn_pf(StochPetriNetModel.spn_productform_forkjoin());
Spn_metrics.SpnMetricsResult met =
    Spn_metrics.spn_metrics(pfj.spn.mdds, pfj.g, pfj.spn.info);
// Every token that forks must later join and return, so the three modes

    ... (7 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

double inv = 0;
for (int i = 0; i < w.length && i < met.tokens.length; i++) {
    inv += w[i] * met.tokens[i];
}
System.out.printf("place invariant 2*m0 + m1 + m2 + 2*m3 = %.6f\n", inv);
pauseForUser();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.10.13 spn_queueing_place

A queueing station and a Petri net in one model, the two formalisms side by side.

The example is built and solved as follows.

```
int N = 4;
// Queueing places are simulated by LDES.
new LDES(StochPetriNetModel.spn_queueing_place(N), "seed", 23000, "samples", 2e5)
    .getAvgTable().print();
// Exact cross-check: the equivalent finite-population Delay + M/M/1.
new MVA(StochPetriNetModel.spn_queueing_place_ref(N)).getAvgTable().print();
pauseForUser();
```

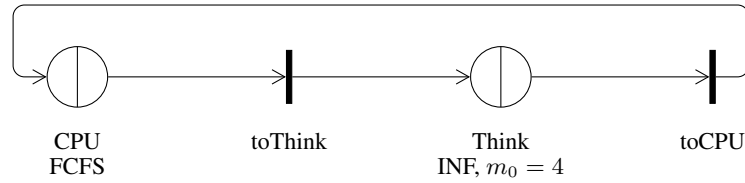


Figure 2.112: Topology of `spn_queueing_place`: a stochastic Petri net over 2 places and 2 transitions; a place is a circle holding its initial marking and a transition a bar, hollow when its firing time is random and solid when it fires at once; the 2 bisected circles are queueing places, whose tokens are served by an embedded station.

Nodes	4: Place $\times$ 2, Transition $\times$ 2
Classes	1: Jobs (closed, $N = 4$)
Service	CPU – Jobs: Exp(1.5) Think – Jobs: Exp(0.5)
Features	2 transitions with 1 firing mode each
Solvers	MVA, LDES

Table 2.119: Features of `spn_queueing_place`.

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.10.14 `spn_twomodes`

A transition with two firing modes.

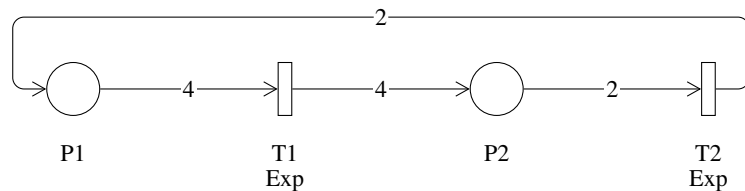


Figure 2.113: Topology of `spn_twomodes`: a stochastic Petri net over 2 places and 2 transitions; a place is a circle holding its initial marking and a transition a bar, hollow when its firing time is random and solid when it fires at once; an arc that moves more than one token carries its multiplicity.

The example is built and solved as follows.

```
Network model = StochPetriNetModel.spn_twomodes();
JMT solver = new JMT(model, "seed", 23000);

solver.getAvgTable().print();
pauseForUser();
```

Running this edition prints

```
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
P1      Class1   4.6632  0  0.89764 0.89764 5.0348 4.9991
P2      Class1   5.3368  0  1.0474  1.0474 4.9991 5.0357

[Running in non-interactive mode, continuing...]
```


Nodes	4: Place $\times$ 2, Transition $\times$ 2
Classes	1: Class1 (closed, $N = 10$)
Features	2 transitions with 1 firing mode each
Solvers	JMT

Table 2.120: Features of `spn_twomodes`.

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by JMT.

2.10.15 `test_spn_nrm_open`

The open net solved by the next-reaction method, against the exact generator.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

Solvers	JMT
----------------	-----

Table 2.121: Features of `test_spn_nrm_open`.

The example is built and solved as follows.

```
final double RTOL = 0.04;
final double SAMPLES = 3e5;
final int SEED = 23000;

// Net 1: M/M/1 SPN, Source Exp(0.5) -> P1 -> T1 Exp(1.0) -> Sink.
double lambda = 0.5;
double mu = 1.0;
double rho = lambda / mu;
double qExact = rho / (1 - rho); // = 1.0
SSA solver = new SSA(StochPetriNetModel.spn_nrm_mml(lambda, mu),
    "method", "nrm", "samples", SAMPLES, "seed", SEED);
solver.getAvg();
Matrix Qn = solver.result.QN;
Matrix Tn = solver.result.TN;
require(solver.result.method != null && solver.result.method.contains("nrm"),
    "net1 did not run NRM");
// stations: Source(0), P1(1)
require(Math.abs(Qn.get(1, 0) - qExact) / qExact < RTOL,
    String.format("net1 P1 tokens %g vs exact %g", Qn.get(1, 0), qExact));
require(Math.abs(Tn.get(0, 0) - lambda) / lambda < RTOL,
    String.format("net1 Source tput %g vs %g", Tn.get(0, 0), lambda));
require(Math.abs(Tn.get(1, 0) - lambda) / lambda < RTOL,
    String.format("net1 P1 tput %g vs %g", Tn.get(1, 0), lambda));

// Cross-check mean tokens against JMT's simulation of the same net.
JMT jmt = new JMT(StochPetriNetModel.spn_nrm_mml(lambda, mu),

    ... (24 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

    String.format("net2 P1 tput %g vs %g", Tn2.get(1, 0), lambda));
require(Math.abs(Tn2.get(2, 0) - lambda) / lambda < RTOL,
    String.format("net2 P2 tput %g vs %g", Tn2.get(2, 0), lambda));

System.out.println("test_spn_nrm_open passed");
pauseForUser();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.11 Workflow models

The workflow models express the standard control-flow patterns – sequence, probabilistic branch, parallel split and join, loop – as activity graphs, and solve them for the completion time of the whole workflow rather than for station metrics.

2.11.1 wf_aph

Acyclic phase-type activity durations.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.11.2 wf_branch

A probabilistic branch and its merge.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.11.3 wf_complex

Nested branches, forks and loops in one workflow.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.11.4 wf_erlang

Erlang activity durations.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.11.5 wf_loop

A loop with a repetition count.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.11.6 wf_parallel

An AND-fork, parallel activities and a join.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

2.11.7 wf_serial

A sequence of activities.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

Chapter 3

Advanced models

The advanced examples cover the features that leave the product-form setting: load dependence, blocking, state-dependent routing, rewards, random environments and the disciplines that no closed form covers. They live under `jline.examples.java.advanced`.

Each of these features narrows the set of solvers that can be used, and the scripts say which. A model that a solver cannot represent is refused rather than approximated silently, so the feature set of a solver is discoverable from the model itself.

3.1 Load-dependent stations

A load-dependent station has a service rate that is a function of the number of jobs it holds, declared through `setLoadDependence` as a rate vector $\mu(n)$. A multiserver station is the special case $\mu(n) = \min(n, c)\mu$, and writing it out as a load-dependent station is how the scripts check the two against each other. Class dependence, declared through `setClassDependence`, lets the rate depend on the class mix rather than on the total only.

The same mechanism supports the flow-equivalent server construction: solve a subnetwork in isolation for every population, tabulate its throughput, and install that table as the rate vector of a single station that replaces it.

3.1.1 fes_aggregation

Aggregating a subnetwork into one station and checking the aggregate against the full model.

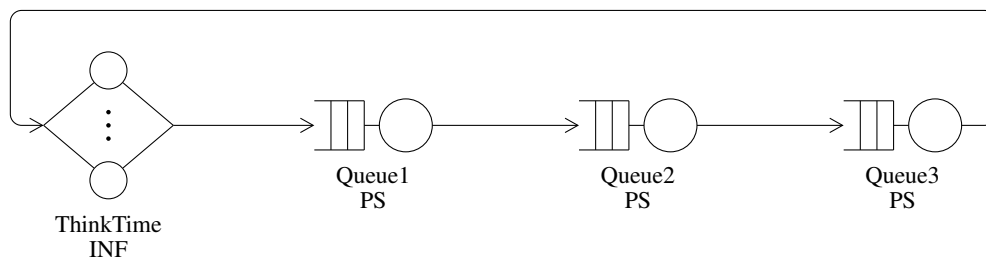


Figure 3.1: Topology of `fes_aggregation`: a closed network over 4 queueing stations.

The example is built and solved as follows.

```
Network model = LoadDependentModel.fes_aggregation();
System.out.println("MVA (original):");
new MVA(model, "method", "exact").getAvgTable().print();

FESResult fes = ModelAdapter.aggregateFES(
```

Nodes	4: Queue $\times$ 3, Delay
Classes	2: Class1 (closed, $N = 3$), Class2 (closed, $N = 2$)
Scheduling	ThinkTime: INF; Queue1: PS; Queue2: PS; Queue3: PS
Service	ThinkTime – Class1: Exp(0.2); Class2: Exp(0.25) Queue1 – Class1: Exp(0.6667); Class2: Exp(0.5) Queue2 – Class1: Exp(1); Class2: Exp(0.8333) Queue3 – Class1: Exp(1.25); Class2: Exp(1)
Solvers	MVA

Table 3.1: Features of fes_aggregation.

```

model, subsetOf(model, new String[] { "Queue1", "Queue2" }));
System.out.println("MVA (FES model):");
new MVA(fes.fesModel, "method", "exact").getAvgTable().print();
pauseForUser();

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

3.1.2 fes_single_class

The flow-equivalent construction step by step: solve the subnetwork, tabulate its throughput, install it as a load-dependent station.

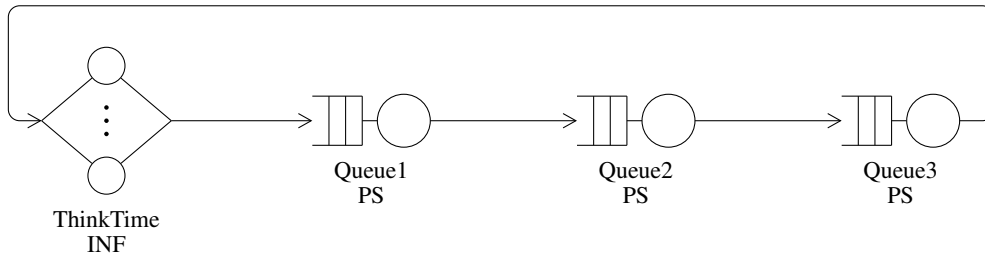


Figure 3.2: Topology of fes_single_class: a closed network over 4 queueing stations.

Nodes	4: Queue $\times$ 3, Delay
Classes	1: Class1 (closed, $N = 5$)
Scheduling	ThinkTime: INF; Queue1: PS; Queue2: PS; Queue3: PS
Service	ThinkTime – Class1: Exp(0.2) Queue1 – Class1: Exp(0.6667) Queue2 – Class1: Exp(1) Queue3 – Class1: Exp(1.25)
Solvers	MVA

Table 3.2: Features of fes_single_class.

The example is built and solved as follows.

```

Network model = LoadDependentModel.fes_single_class();
System.out.println("MVA (original):");
new MVA(model, "method", "exact").getAvgTable().print();

FESResult fes = ModelAdapter.aggregateFES(
    model, subsetOf(model, new String[] { "Queue1", "Queue2" }));
System.out.println("MVA (FES model):");
new MVA(fes.fesModel, "method", "exact").getAvgTable().print();
pauseForUser();

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

3.1.3 ld_class_dependence

Service rates that depend on the class mix at the station, not only on the total population.

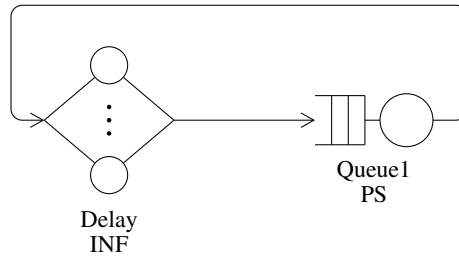


Figure 3.3: Topology of ld_class_dependence: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 16$), Class2 (closed, $N = 8$)
Scheduling	Delay: INF; Queue1: PS
Service	Delay – Class1: Exp(1); Class2: Exp(0.5) Queue1 – Class1: Exp(0.6667); Class2: Exp(0.4)
Solvers	CTMC, MVA

Table 3.3: Features of ld_class_dependence.

The example is built and solved as follows.

```
Network model = LoadDependentModel.ld_class_dependence();

// JMT is not solved here, as in ld_class_dependence.m: the JSIM writer
// has no representation for the class-dependence handle, so SolverJMT
// rejects the model rather than silently solving it unscaled.
new CTMC(model, "exact").getAvgTable().print();
new MVA(model, "method", "qd").getAvgTable().print();

pauseForUser();
```

Running this edition prints

```
CTMC analysis [method: exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay   Class1     0.88196 0.88196    1       1 0.88196 0.88196
Delay   Class2     0.27082 0.27082    2       2 0.13541 0.13541
Queue1  Class1    15.118 0.66147 17.141 17.141 0.88196 0.88196
Queue1  Class2     7.7292 0.33853 57.079 57.079 0.13541 0.13541
MVA analysis [method: qd; type: approximate, deterministic; lang: java; env: python] completed in <t>s.
Iterations: 47.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay   Class1     0.88016 0.88016    1       1 0.88016 0.88016
Delay   Class2     0.27023 0.27023    2       2 0.13512 0.13512
Queue1  Class1    15.12 0.66012 17.178 17.178 0.88016 0.88016
Queue1  Class2     7.7298 0.33779 57.208 57.208 0.13512 0.13512

[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 2 solvers the script runs (CTMC, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

3.1.4 ld_fes_multiclass

The multiclass flow-equivalent server.

The example is built and solved as follows.

```
Network model = LoadDependentModel.ld_fes_multiclass();
System.out.println("MVA (original):");
```

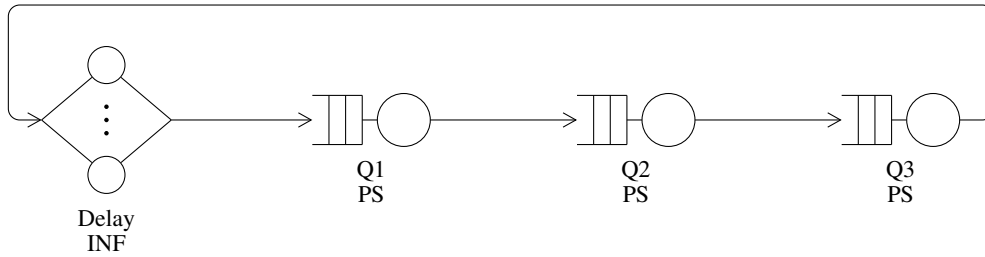


Figure 3.4: Topology of ld_fes_multiclass: a closed network over 4 queueing stations.

Nodes	4: Queue $\times$ 3, Delay
Classes	2: Class1 (closed, $N = 3$), Class2 (closed, $N = 2$)
Scheduling	Delay: INF; Q1: PS; Q2: PS; Q3: PS
Service	Delay – Class1: Exp(1); Class2: Exp(0.6667) Q1 – Class1: Exp(2); Class2: Exp(1.25) Q2 – Class1: Exp(3.333); Class2: Exp(1.667) Q3 – Class1: Exp(2.5); Class2: Exp(1.429)
Solvers	MVA, NC

Table 3.4: Features of ld_fes_multiclass.

```
new MVA(model, "method", "exact").getAvgTable().print();

FESResult fes = ModelAdapter.aggregateFES(
    model, subsetOf(model, new String[] { "Q1", "Q2", "Q3" }));
System.out.println("NC (FES model):");
new NC(fes.fesModel, "method", "exact").getAvgTable().print();
pauseForUser();
```

Running this edition prints

```
MVA (original):
MVA analysis [method: exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay   Class1      0.89626 0.89626      1      1      0.89626 0.89626
Delay   Class2      0.54062 0.54062      1.5    1.5    0.36042 0.36042
Q1      Class1      0.97971 0.44813  1.0931  1.0931  0.89626 0.89626
Q1      Class2      0.63639 0.28833  1.7657  1.7657  0.36042 0.36042
Q2      Class1      0.4441  0.26888  0.4955  0.4955  0.89626 0.89626
Q2      Class2      0.34845 0.21625  0.96679 0.96679  0.36042 0.36042
Q3      Class1      0.67992 0.3585  0.75862 0.75862  0.89626 0.89626
Q3      Class2      0.47454 0.25229  1.3166  1.3166  0.36042 0.36042
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
... (7 captured-output lines omitted; rerun the source example for the full output)

NC analysis [method: conv; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay   Class1      0.89626 0.89626      1      1      0.89626 0.89626
Delay   Class2      0.54062 0.54062      1.5    1.5    0.36042 0.36042
FES     Class1      2.1037  0.51797  2.3472  2.3472  0.89626 0.89626
FES     Class2      1.4594  0.20829  4.0491  4.0491  0.36042 0.36042

[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 2 solvers the script runs (MVA, NC), which is the point of the example: the same model read by methods of different cost and different exactness.

3.1.5 ld_fes_singleclass

A flow-equivalent server replacing a subnetwork, single class.

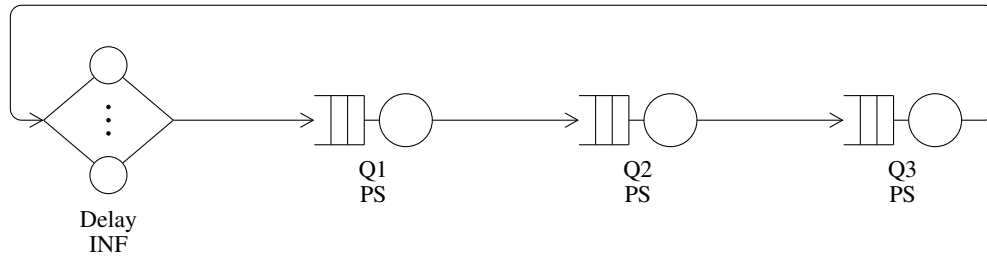


Figure 3.5: Topology of ld_fes_singleclass: a closed network over 4 queueing stations.

Nodes	4: Queue $\times$ 3, Delay
Classes	1: Class1 (closed, $N = 5$)
Scheduling	Delay: INF; Q1: PS; Q2: PS; Q3: PS
Service	Delay – Class1: Exp(0.2) Q1 – Class1: Exp(0.6667) Q2 – Class1: Exp(1) Q3 – Class1: Exp(1.25)
Solvers	MVA, NC

Table 3.5: Features of ld_fes_singleclass.

The example is built and solved as follows.

```
Network model = LoadDependentModel.ld_fes_singleclass();
System.out.println("MVA (original):");
new MVA(model, "method", "exact").getAvgTable().print();

FESResult fes = ModelAdapter.aggregateFES(
    model, subsetOf(model, new String[] { "Q1", "Q2", "Q3" }));
System.out.println("NC (FES model):");
new NC(fes.fesModel, "method", "exact").getAvgTable().print();
pauseForUser();
```

Running this edition prints

```
MVA (original):
MVA analysis [method: exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 2.3333 2.3333 5 5 0.46666 0.46666
Q1 Class1 1.4069 0.69999 3.0148 3.0148 0.46666 0.46666
Q2 Class1 0.72959 0.46666 1.5634 1.5634 0.46666 0.46666
Q3 Class1 0.53021 0.37333 1.1362 1.1362 0.46666 0.46666
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
NC (FES model):
NC analysis [method: conv; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 2.3333 2.3333 5 5 0.46666 0.46666
FES Class1 2.6667 0.7705 5.7145 5.7145 0.46666 0.46666

[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 2 solvers the script runs (MVA, NC), which is the point of the example: the same model read by methods of different cost and different exactness.

3.1.6 ld_global_dependence

A globally state-dependent (Whittle) model: `setGlobalDependence` scales the service rate by a function of the FULL station-class population matrix rather than of the population local to one station. The scaling used here satisfies the Whittle balance property, so the chain is reversible, keeps a product form and is insensitive to the service distribution beyond its mean.

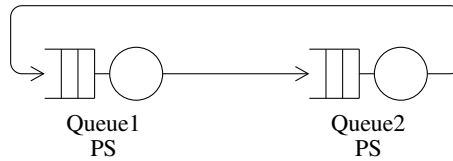


Figure 3.6: Topology of `ld_global_dependence`: a closed network over 2 queueing stations.

Nodes	2: Queue $\times$ 2
Classes	1: Class1 (closed, $N = 3$)
Scheduling	Queue1: PS; Queue2: PS
Service	Queue1 – Class1: Exp(1) Queue2 – Class1: Exp(2)
Solvers	CTMC, SSA

Table 3.6: Features of `ld_global_dependence`.

The example is built and solved as follows.

```

int N = 3;

Network gdmodel = new Network("model");

Queue node1 = new Queue(gdmodel, "Queue1", SchedStrategy.PS);
Queue node2 = new Queue(gdmodel, "Queue2", SchedStrategy.PS);

ClosedClass jobclass1 = new ClosedClass(gdmodel, "Class1", N, node1, 0);
node1.setService(jobclass1, Exp.fitMean(1.0));
node2.setService(jobclass1, Exp.fitMean(0.5));

RoutingMatrix routingMatrix = gdmodel.initRoutingMatrix();
routingMatrix.set(jobclass1, jobclass1, node1, node2, 1.0);
routingMatrix.set(jobclass1, jobclass1, node2, node1, 1.0);
gdmodel.link(routingMatrix);

final int M = gdmodel.getNumberOfStations();
final int K = gdmodel.getNumberOfClasses();
// phi as SolverCTMC sees it: an (nstations x nclasses) matrix of scalings
SerializableFunction<Matrix, Matrix> phi =
    n -> {
        Matrix v = Matrix.ones(M, K);
        double tot = 0;
        for (int i = 0; i < M; i++) {
            for (int r = 0; r < K; r++) tot += n.get(i, r);
        }

        ... (9 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

        // peak scaling is 1: no station ever receives more than the whole link
        Matrix gdPeak = new Matrix(1, 1);
        gdPeak.set(0, 0, 1.0);
        gdmodel.setGlobalDependence(phi, gdPeak);

    return gdmodel;

```

Running this edition prints

```
jline.lang.Network@108c4c35
```

3.1.7 ld_joint_dependence

Joint dependence on the populations of several classes.

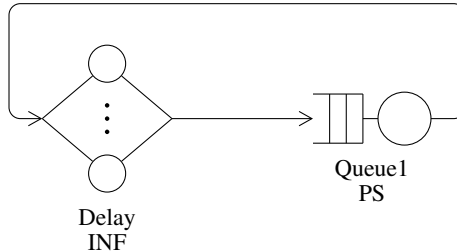


Figure 3.7: Topology of ld_joint_dependence: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 16$), Class2 (closed, $N = 8$)
Scheduling	Delay: INF; Queue1: PS
Service	Delay – Class1: Exp(1); Class2: Exp(0.5) Queue1 – Class1: Exp(0.6667); Class2: Exp(0.4)
Solvers	CTMC, MVA

Table 3.7: Features of ld_joint_dependence.

The example is built and solved as follows.

```
Network model = LoadDependentModel.ld_joint_dependence();

// JMT is not solved here, as in ld_joint_dependence.m: the JSIM writer
// has no representation for the joint-dependence handle.
new CTMC(model, "exact").getAvgTable().print();
new MVA(model, "method", "qd").getAvgTable().print();

pauseForUser();
```

Running this edition prints

```
CTMC analysis [method: exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.89231 0.89231 1 1 0.89231 0.89231
Delay Class2 0.52923 0.52923 2 2 0.26461 0.26461
Queue1 Class1 15.108 0.66923 16.931 16.931 0.89231 0.89231
Queue1 Class2 7.4708 0.33077 28.233 28.233 0.26461 0.26461
MVA analysis [method: qd; type: approximate, deterministic; lang: java; env: python] completed in <t>s.
Iterations: 47.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.88996 0.88996 1 1 0.88996 0.88996
Delay Class2 0.52788 0.52788 2 2 0.26394 0.26394
Queue1 Class1 15.11 0.66747 16.978 16.978 0.88996 0.88996
Queue1 Class2 7.4721 0.32992 28.31 28.31 0.26394 0.26394

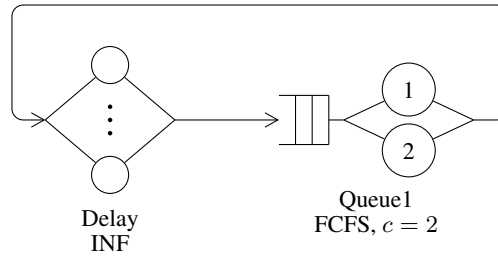
[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 2 solvers the script runs (CTMC, MVA), which is the point of the example: the same model read by methods of different cost and different exactness.

3.1.8 ld_multiserver_fcfs

The FCFS counterpart, with class-dependent scaling on top.

The example is built and solved as follows.

Figure 3.8: Topology of `ld_multiserver_fcfs`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 16$)
Scheduling	Delay: INF; Queue1: FCFS ($c = 2$)
Service	Delay – Class1: Exp(1) Queue1 – Class1: Exp(0.6667)
Solvers	NC, CTMC, MVA, JMT

Table 3.8: Features of `ld_multiserver_fcfs`.

```

Network model = LoadDependentModel.ld_multiserver_fcfs();

// The reference's order on the load-dependent model: CTMC, then exact MVA,
// then JMT at seed 23000. Its golden holds the FIRST table each solver
// produced, so the order and the pinned method are part of the answer.
try {
    new CTMC(model).getAvgTable().print();
} catch (Exception e) {
    System.out.println("CTMC failed: " + e.getMessage());
}
try {
    new MVA(model, "method", "exact").getAvgTable().print();
} catch (Exception e) {
    System.out.println("MVA failed: " + e.getMessage());
}
try {
    new JMT(model, "seed", 23000, "samples", 100000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT failed: " + e.getMessage());
}

pauseForUser();

```

Running this edition prints

```

CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util  RespT  ResidT   ArvR   Tput
Delay   Class1     1.3333 1.3333    1      1   1.3333 1.3333
Queue1  Class1    14.667    1    11     11   1.3333 1.3333
MVA analysis [method: exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util  RespT  ResidT   ArvR   Tput
Delay   Class1     1.3333 1.3333    1      1   1.3333 1.3333
Queue1  Class1    14.667    1    11     11   1.3333 1.3333
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util  RespT  ResidT   ArvR   Tput
Delay   Class1     1.3535 1.3535  0.99717 0.99717 1.3436 1.3395
Queue1  Class1    14.647    1  10.962  10.962  1.3395 1.3397

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 3 solvers the script runs (CTMC, MVA, JMT), which is the point of the example: the same model read by methods of different cost and different exactness.

3.1.9 ld_multiserver_ps

A multiserver PS station written as a load-dependent station, and the rate vector that reproduces it.

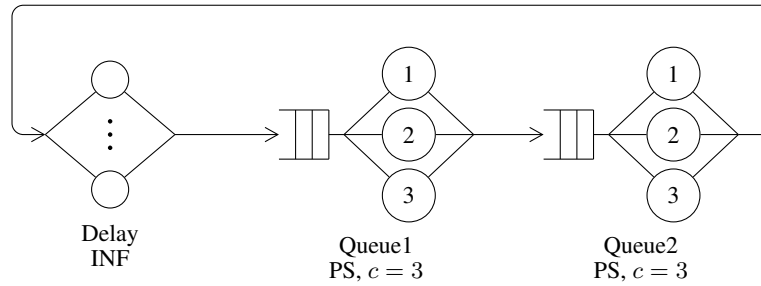


Figure 3.9: Topology of ld_multiserver_ps: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	2: Class1 (closed, $N = 4$), Class2 (closed, $N = 2$)
Scheduling	Delay: INF; Queue1: PS ($c = 3$); Queue2: PS ($c = 3$)
Service	Delay – Class1: Exp(1); Class2: Exp(0.5) Queue1 – Class1: Exp(0.6667); Class2: Exp(0.4) Queue2 – Class1: Exp(0.2857); Class2: Exp(0.2222)
Solvers	MVA, CTMC, NC, JMT

Table 3.9: Features of ld_multiserver_ps.

The example is built and solved as follows.

```
Network model = LoadDependentModel.ld_multiserver_ps();

NetworkSolver[] solvers = new NetworkSolver[] {
    new NC(model, "method", "rd"),
    new MVA(model, "method", "exact"),
    new JMT(model, "seed", 12345)
};

for (NetworkSolver solver : solvers) {
    try {
        if (solver instanceof JMT) {
            // The solver's OWN options, not a fresh defaultOptions(): the latter
            // draws a RANDOM seed in its constructor (SolverOptions ->
            // RandomManager.generateRandomSeed), so replacing the object wholesale
            // discards the seed passed to the constructor above and makes this
            // example irreproducible -- which is what left the ld_multiserver_ps
            // parity row disagreeing with its golden on a Monte-Carlo margin.
            SolverOptions options = solver.getOptions();
            options.samples = 100000;
            ((JMT)solver).setOptions(options);
        }

        solver.getAvgTable().print();
    } catch (Exception e) {
    }
}

pauseForUser();
```

Running this edition prints

```
NC analysis [method: rd; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.43985 0.43985 1 1 0.43985 0.43985
Delay Class2 0.30095 0.30095 2 2 0.15047 0.15047
Queue1 Class1 0.78332 0.21993 1.7809 1.7809 0.43985 0.43985
```

```

Queue1 Class2 0.4437 0.1254 2.9486 2.9486 0.15047 0.15047
Queue2 Class1 2.7768 0.51316 6.3131 6.3131 0.43985 0.43985
Queue2 Class2 1.2554 0.22571 8.3426 8.3426 0.15047 0.15047
MVA analysis [method: exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.54674 0.54674 1 1 0.54674 0.54674
Delay Class2 0.36944 0.36944 2 2 0.18472 0.18472
Queue1 Class1 0.85639 0.27337 1.5664 1.5664 0.54674 0.54674
Queue1 Class2 0.48077 0.15394 2.6027 2.6027 0.18472 0.18472
Queue2 Class1 2.5969 0.63786 4.7498 4.7498 0.54674 0.54674
Queue2 Class2 1.1498 0.27708 6.2244 6.2244 0.18472 0.18472

... (3 captured-output lines omitted; rerun the source example for the full output)

Delay Class1 0.54287 0.54287 1.001 1.001 0.54723 0.54724
Delay Class2 0.37154 0.37154 2.0053 2.0053 0.1845 0.18529
Queue1 Class1 0.85929 0.2743 1.5728 1.5728 0.54724 0.54724
Queue1 Class2 0.48185 0.15345 2.5991 2.5991 0.18529 0.18452
Queue2 Class1 2.597 0.64002 4.7503 4.7503 0.54724 0.54723
Queue2 Class2 1.1497 0.27421 6.184 6.184 0.18452 0.18514

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 3 solvers the script runs (NC, MVA, JMT), which is the point of the example: the same model read by methods of different cost and different exactness.

3.1.10 ld_multiserver_ps_twoclasses

The two-class version.

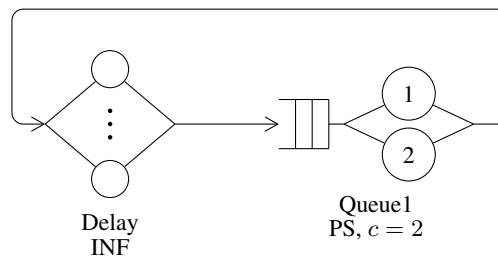


Figure 3.10: Topology of ld_multiserver_ps_twoclasses: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 4$), Class2 (closed, $N = 2$)
Scheduling	Delay: INF; Queue1: PS ($c = 2$)
Service	Delay – Class1: Exp(1); Class2: Exp(0.5) Queue1 – Class1: Exp(0.6667); Class2: Exp(0.4)
Solvers	MVA, CTMC, NC, JMT

Table 3.10: Features of ld_multiserver_ps_twoclasses.

The example is built and solved as follows.

```

Network model = LoadDependentModel.ld_multiserver_ps_twoclasses();

NetworkSolver[] solvers = new NetworkSolver[] {
    new NC(model, "method", "rd"),
    new MVA(model, "method", "exact"),
    new JMT(model, "seed", 12345)
};

for (NetworkSolver solver : solvers) {
    try {

        if (solver instanceof JMT) {
            // The solver's OWN options, not a fresh defaultOptions(): the latter

```

```

        // draws a RANDOM seed in its constructor (SolverOptions ->
        // RandomManager.generateRandomSeed), so replacing the object wholesale
        // discards the seed passed to the constructor above and makes this
        // example irreproducible -- which is what left the ld_multiserver_ps
        // parity row disagreeing with its golden on a Monte-Carlo margin.
        SolverOptions options = solver.getOptions();
        options.samples = 100000;
        ((JMT)solver).setOptions(options);
    }

    solver.getAvgTable().print();
} catch (Exception e) {
}
}

pauseForUser();

```

Running this edition prints

```
jline.lang.Network@736e9adb
```

3.1.11 ld_whittle_bandwidth

An open bandwidth-sharing network in which one route holds several links at once, which no per-station rate scaling can express. Capacity is shared by balanced fairness, whose rate function satisfies the Whittle property by construction, so the stationary law is again of product form and insensitive.

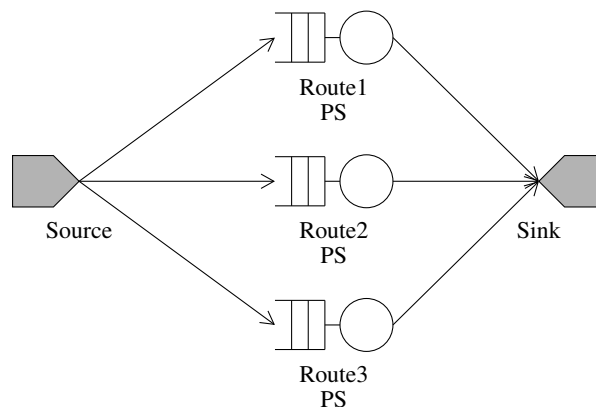


Figure 3.11: Topology of ld_whittle_bandwidth: an open network over 3 queueing stations.

Nodes	5: Source, Queue $\times$ 3, Sink
Classes	3: Route1Flows (open), Route2Flows (open), Route3Flows (open)
Scheduling	Route1: PS; Route2: PS; Route3: PS
Arrivals	Source – Route1Flows: Exp(0.2); Route2Flows: Exp(0.3); Route3Flows: Exp(0.3)
Service	Route1 – Route1Flows: Exp(1); Route2Flows: Disabled; Route3Flows: Disabled Route2 – Route1Flows: Disabled; Route2Flows: Exp(1); Route3Flows: Disabled Route3 – Route1Flows: Disabled; Route2Flows: Disabled; Route3Flows: Exp(1)
Solvers	CTMC

Table 3.11: Features of ld_whittle_bandwidth.

The example is built and solved as follows.

```

final int[][] A = {{1, 1, 0}, {1, 0, 1}};
final double[] C = {1.0, 1.0};
final double[] nu = {0.20, 0.30, 0.30};
final double[] mu = {1.00, 1.00, 1.00};

```

```

final int CUTOFF = 3;
final int S = 3;
final int base = CUTOFF + 1;
final double[] Phi = bfBalanceFunction(A, C, CUTOFF);

Network model = new Network("model");
Source source = new Source(model, "Source");
Queue[] routes = new Queue[S];
for (int s = 0; s < S; s++) {
    routes[s] = new Queue(model, "Route" + (s + 1), SchedStrategy.PS);
}
Sink sink = new Sink(model, "Sink");
OpenClass[] classes = new OpenClass[S];
for (int s = 0; s < S; s++) {
    classes[s] = new OpenClass(model, "Route" + (s + 1) + "Flows");
    source.setArrival(classes[s], new Exp(nu[s]));
}
for (int s = 0; s < S; s++) {
    for (int t = 0; t < S; t++) {
        if (s == t) {
            routes[t].setService(classes[s], new Exp(mu[s]));
        } else {
            ... (33 source lines omitted; full implementation: jar/src/main/java/jline/examples/)
        }
    }
}
return v;
}, Matrix.ones(1, 1));

return model;

```

Running this edition prints

```
jline.lang.Network@3c5a99da
```

3.2 Finite capacity regions

A finite capacity region caps the number of jobs inside a set of stations. A job that would breach the cap is either dropped or made to wait outside the region, per class, and per-class caps may be declared alongside the global one. The region is a property of the model, not of a station, which is what lets it span several stations.

The flagship comparison is `fcr_mm1kdrop`: a region of capacity K around a single queue, with dropping, is an M/M/1/K queue, and the script builds both models and solves them side by side.

```

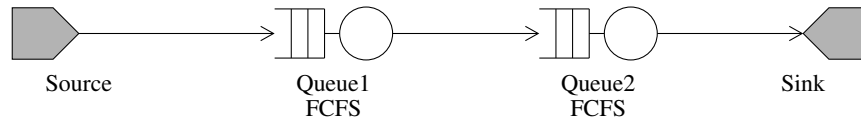
// Model 1: one queue inside a region of capacity K, with dropping
Network modell = new Network("FCR MM1K Dropping");
Source source = new Source(modell, "Source");
Queue queue = new Queue(modell, "Queue", SchedStrategy.FCFS);
Sink sink = new Sink(modell, "Sink");
OpenClass jobclass = new OpenClass(modell, "Class1", 0);
source.setArrival(jobclass, Exp.fitRate(0.8));
queue.setService(jobclass, Exp.fitRate(1.0));
modell.link(Network.serialRouting(source, queue, sink));

modell.addRegion(Arrays.asList(queue));
Region fcr = modell.getRegions().get(0);
fcr.setGlobalMaxJobs(K);
fcr.setDropRule(jobclass, true); // true = drop, false = wait outside

```

3.2.1 fcr_constraints

Per-class capacities inside one region, and the interaction with the global limit.

Figure 3.12: Topology of `fcr_constraints`: an open network over 2 queueing stations.

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	2: HighPriority (open), LowPriority (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	Source – HighPriority: Exp(0.3); LowPriority: Exp(0.5)
Service	Queue1 – HighPriority: Exp(1); LowPriority: Exp(0.8) Queue2 – HighPriority: Exp(1.2); LowPriority: Exp(1)
Solvers	JMT

Table 3.12: Features of `fcr_constraints`.

The example is built and solved as follows.

```
System.out.println("=== FCR Constraint Types Demo ===\n");
System.out.println("Region covers: Queue1, Queue2");
System.out.println("Global max jobs: 2");
System.out.println("HighPriority max: 2 (dropping)");
System.out.println("LowPriority max: 2 (dropping)\n");

Network model = FCRegionModel.fcr_constraints();
JMT solver = new JMT(model, "seed", 23000, "samples", 100000, "verbose", VerboseLevel.SILENT);
NetworkAvgTable avgTable = solver.getAvgTable();

System.out.println(avgTable);

System.out.println("\nNote: jobs of either class are dropped once the region reaches");
System.out.println("its global limit or that class's own limit.");
```

Running this edition prints

```
=== FCR Constraint Types Demo ===

Region covers: Queue1, Queue2
Global max jobs: 2
HighPriority max: 2 (dropping)
LowPriority max: 2 (dropping)

WARNING: [refreshStruct] Priority classes are specified but no priority-aware scheduling policy (PSPRIO,
DPSPRIO, GPSPRIO, HOL, FCFSPRIO, FCFSPRPRIOR, FCFSPRIOR, LCFSPRIO, LCFSPRPRIOR, LCFSPRIOR, SRPTPRIO)
is used in the model. Priorities will be ignored.
WARNING: [refreshStruct] Message casted more than once, repetitions will not be printed on screen for 60
seconds.
jline.solvers.NetworkAvgTable@3ec300f1

Note: jobs of either class are dropped once the region reaches
its global limit or that class's own limit.
```

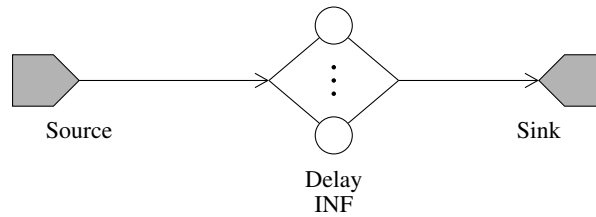
3.2.2 `fcr_lincon`

A linear constraint over the class populations, the general form of a region limit.

The example is built and solved as follows.

```
System.out.println("=== FCR Linear Admission Constraints (LDES) ===\n");

double[][] cases_A = {
    // Case 1: Per-class only (A = I)
    {1.0, 0.0, 0.0, 1.0},
    // Case 2: Global + per-class
```


Figure 3.13: Topology of `fcr_lincon`: an open network over 1 queueing station.

Nodes	3: Source, Delay, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Delay: INF
Arrivals	Source – Class1: Exp(0.3); Class2: Exp(0.2)
Service	Delay – Class1: Exp(1); Class2: Exp(0.8)
Solvers	LDES

Table 3.13: Features of `fcr_lincon`.

```

    {1.0, 0.0, 0.0, 1.0, 1.0, 1.0},
    // Case 3: General weighted
    {2.0, 1.0, 1.0, 3.0}
};
int[] cases_C = {2, 3, 2};
double[][] cases_b = {
    {3.0, 3.0},
    {3.0, 3.0, 5.0},
    {5.0, 7.0}
};
String[] caseNames = {
    "Case 1: Per-class only (A=I)",
    "Case 2: Global + per-class",
    "Case 3: General weighted"
};

for (int c = 0; c < 3; c++) {
    System.out.println("\n--- " + caseNames[c] + " ---");
    int C = cases_C[c];
    int K = 2;

    ... (8 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

    Network model = FCRegionModel.fcr_lincon(A, b);

    SolverLDES solver = new SolverLDES(model, "seed", 23000, "samples", 500000, "verbose",
    VerboseLevel.SILENT);
    NetworkAvgTable avgTable = solver.getAvgTable();
    System.out.println(avgTable);
}

```

Running this edition prints

```

=== FCR Linear Admission Constraints (LDES) ===

--- Case 1: Per-class only (A=I) ---
WARNING: [refreshStruct] Priority classes are specified but no priority-aware scheduling policy (PSPRIO,
DPSPRIO, GPSPRIO, HOL, FCFSPRIO, FCFSPRPRI, FCFSPIPRIO, LCFSPRIO, LCFSPRPRI, LCFSPIPRIO, SRTPRIO)
is used in the model. Priorities will be ignored.
WARNING: [refreshStruct] Message casted more than once, repetitions will not be printed on screen for 60
seconds.
jline.solvers.NetworkAvgTable@2a70a3d8

--- Case 2: Global + per-class ---
jline.solvers.NetworkAvgTable@2de8284b

--- Case 3: General weighted ---
jline.solvers.NetworkAvgTable@18bf3d14

```

3.2.3 fcr_lossn

A loss network: every request that cannot be admitted is lost, and the blocking probabilities are the quantity of interest.

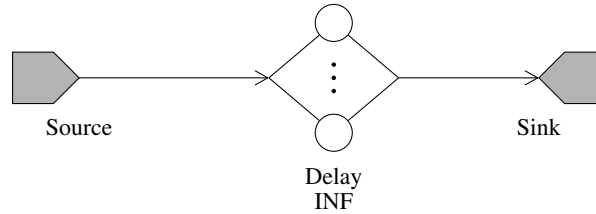


Figure 3.14: Topology of fcr_lossn: an open network over 1 queueing station.

Nodes	3: Source, Delay, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Delay: INF
Arrivals	Source – Class1: Exp(0.3); Class2: Exp(0.2)
Service	Delay – Class1: Exp(1); Class2: Exp(0.8)
Solvers	JMT

Table 3.14: Features of fcr_lossn.

The example is built and solved as follows.

```
System.out.println("=== FCR Loss Network (NC Solver) ===\n");
System.out.println("Model: Single Delay node in FCR with DROP policy");
System.out.println("Global max: 5, Class1 max: 3, Class2 max: 3\n");

Network model = FCRegionModel.fcr_lossn();

// Run NC solver (uses lossn method)
System.out.println("Running NC solver (lossn method)...");
SolverNC solverNC = new SolverNC(model, "verbose", VerboseLevel.SILENT);
NetworkAvgTable avgTableNC = solverNC.getAvgTable();
System.out.println("\nNC Results:");
System.out.println(avgTableNC);

// Run JMT for comparison
System.out.println("\nRunning JMT for comparison...");
JMT solverJMT = new JMT(model, "seed", 23000, "samples", 500000, "verbose", VerboseLevel.SILENT);
NetworkAvgTable avgTableJMT = solverJMT.getAvgTable();
System.out.println("\nJMT Results:");
System.out.println(avgTableJMT);

// Compare throughputs at the Delay node; the NC and JMT tables list
// different station sets, so rows are located by station/class name
System.out.println("\n=== Comparison (Delay node) ===");
System.out.printf("%-15s %12s %12s %12s\n", "Class", "NC Tput", "JMT Tput", "Rel Err");
for (int r = 0; r < 2; r++) {
    String className = "Class" + (r + 1);

    ... (3 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

    System.out.printf("%-15s %12.4f %12.4f %10.2f%%\n",
        className, tputNC, tputJMT, relErr);
}

System.out.println("\nNote: NC uses analytical Erlang fixed-point approximation.");
System.out.println("JMT uses discrete-event simulation.");
```

Running this edition prints

```

=== FCR Loss Network (NC Solver) ===

Model: Single Delay node in FCR with DROP policy
Global max: 5, Class1 max: 3, Class2 max: 3

Running NC solver (lossn method)...
WARNING: [refreshStruct] Priority classes are specified but no priority-aware scheduling policy (PSPRIO,
        DPSPRIO, GPSPRIO, HOL, FCFSPRIO, FCFSPRPRIOR, FCFSPIPRIO, LCFSPRIO, LCFSPRPRIOR, LCFSPIPRIO, SRPTPRIO)
        is used in the model. Priorities will be ignored.
WARNING: [refreshStruct] Message casted more than once, repetitions will not be printed on screen for 60
        seconds.

NC Results:
jline.solvers.NetworkAvgTable@3a883ce7

Running JMT for comparison...

JMT Results:
jline.solvers.NetworkAvgTable@6f3b5d16

=== Comparison (Delay node) ===
Class          NC Tput      JMT Tput      Rel Err
Class1         0.2990         0.2990         0.01%
Class2         0.1996         0.1998         0.12%

Note: NC uses analytical Erlang fixed-point approximation.
JMT uses discrete-event simulation.

```

3.2.4 fcr_mm1kdrop

A region of capacity K around one queue with dropping, which reproduces an M/M/1/K queue.

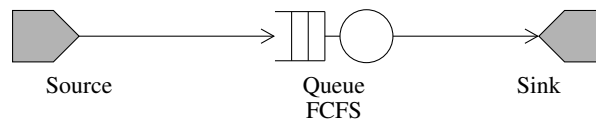


Figure 3.15: Topology of fcr_mm1kdrop: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(0.8)
Service	Queue – Class1: Exp(1)
Solvers	JMT

Table 3.15: Features of fcr_mm1kdrop.

The example is built and solved as follows.

```

int K = 3;
System.out.printf("=== Comparison: FCR Dropping vs M/M/1/K (K=%d) ===%n%n", K);

// FCR dropping model
Network model1 = FCRegionModel.fcr_mm1kdrop();
JMT solver1 = new JMT(model1, "seed", 23000, "samples", 100000, "verbose", VerboseLevel.
    SILENT);
NetworkAvgTable avgTable1 = solver1.getAvgTable();

// Standard M/M/1/K
Network model2 = FCRegionModel.mm1k();
JMT solver2 = new JMT(model2, "seed", 23000, "samples", 100000, "verbose", VerboseLevel.
    SILENT);
NetworkAvgTable avgTable2 = solver2.getAvgTable();

// Compare results (queue is at index 1)

```

```

int queueIdx = 1;
System.out.println("Queue Metrics:");
System.out.printf("FCR Dropping      M/M/1/K%n");
System.out.printf("Queue Length:      %.4f      %.4f%n", avgTable1.getQLen().get (
    queueIdx), avgTable2.getQLen().get (queueIdx));
System.out.printf("Utilization:      %.4f      %.4f%n", avgTable1.getUtil().get (
    queueIdx), avgTable2.getUtil().get (queueIdx));
System.out.printf("Response Time:      %.4f      %.4f%n", avgTable1.getRespT().get (
    queueIdx), avgTable2.getRespT().get (queueIdx));
System.out.printf("Throughput:      %.4f      %.4f%n", avgTable1.getTput().get (
    queueIdx), avgTable2.getTput().get (queueIdx));

// Theoretical M/M/1/K values
double arrivalRate = 0.8;
double serviceRate = 1.0;
double rho = arrivalRate / serviceRate;

    ... (10 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

System.out.printf("Utilization:      %.4f%n", theoretical_U);
System.out.printf("Response Time:      %.4f%n", theoretical_R);
System.out.printf("Throughput:      %.4f%n", theoretical_X);
System.out.printf("Blocking Prob:      %.4f%n", pK);

System.out.println("\n==> FCR with dropping matches M/M/1/K (jobs lost when full)");

```

Running this edition prints

```

=== Comparison: FCR Dropping vs M/M/1/K (K=3) ===

Queue Metrics:
FCR Dropping      M/M/1/K
Queue Length:      1.2175      1.2212
Utilization:      0.6612      0.6612
Response Time:      1.8344      1.8344
Throughput:      0.6621      0.6621

Theoretical M/M/1/K:
Queue Length:      1.2249
Utilization:      0.6612
Response Time:      1.8525
Throughput:      0.6612
Blocking Prob:      0.1734

==> FCR with dropping matches M/M/1/K (jobs lost when full)

```

3.2.5 fcr_mmlwaitq

The same region with waiting instead of dropping, so blocked jobs queue outside the region.

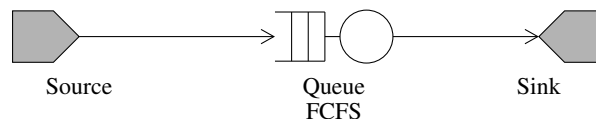


Figure 3.16: Topology of `fcr_mmlwaitq`: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(0.5)
Service	Queue – Class1: Exp(1)
Solvers	JMT

Table 3.16: Features of `fcr_mmlwaitq`.

The example is built and solved as follows.

```
System.out.println("=== Comparison: FCR Blocking vs M/M/1 ===\n");

// FCR blocking model
Network model1 = FCRegionModel.fcr_mm1waitq();
JMT solver1 = new JMT(model1, "seed", 23000, "samples", 100000, "verbose", VerboseLevel.
    SILENT);
// THE NODE TABLE, as the reference prints: its golden holds the FCR1 and
// Sink rows, which no station table carries.
solver1.getAvgNodeTable().print();
NetworkAvgTable avgTable1 = solver1.getAvgTable();

// Standard M/M/1
Network model2 = FCRegionModel.mm1();
JMT solver2 = new JMT(model2, "seed", 23000, "samples", 100000, "verbose", VerboseLevel.
    SILENT);
solver2.getAvgNodeTable().print();
NetworkAvgTable avgTable2 = solver2.getAvgTable();

// Compare results (queue is at index 1)
int queueIdx = 1;
System.out.println("Queue Metrics:");
System.out.printf("FCR Blocking      M/M/1\n");
System.out.printf("Queue Length:      %.4f      %.4f\n", avgTable1.getQLen().get (
    queueIdx), avgTable2.getQLen().get (queueIdx));
System.out.printf("Utilization:      %.4f      %.4f\n", avgTable1.getUtil().get (
    queueIdx), avgTable2.getUtil().get (queueIdx));
System.out.printf("Response Time:    %.4f      %.4f\n", avgTable1.getRespT().get (
    queueIdx), avgTable2.getRespT().get (queueIdx));
System.out.printf("Throughput:      %.4f      %.4f\n", avgTable1.getTput().get (
    queueIdx), avgTable2.getTput().get (queueIdx));

// Theoretical M/M/1 values
... (9 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

System.out.printf("Queue Length:      %.4f\n", theoretical_Q);
System.out.printf("Utilization:      %.4f\n", theoretical_U);
System.out.printf("Response Time:    %.4f\n", theoretical_R);
System.out.printf("Throughput:      %.4f\n", theoretical_X);

System.out.println("\n==> FCR with blocking matches M/M/1 (jobs wait, no loss)");
```

Running this edition prints

```
=== Comparison: FCR Blocking vs M/M/1 ===

Queue Metrics:
FCR Blocking      M/M/1
Queue Length:      0.9784      0.9786
Utilization:      0.4939      0.4939
Response Time:    1.9641      1.9643
Throughput:      0.4992      0.4992

Theoretical M/M/1:
Queue Length:      1.0000
Utilization:      0.5000
Response Time:    2.0000
Throughput:      0.5000

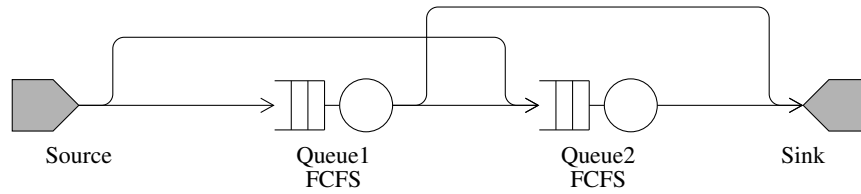
==> FCR with blocking matches M/M/1 (jobs wait, no loss)
```

3.2.6 fcr_oqndrop

A region spanning several stations of an open network, with dropping.

The example is built and solved as follows.

```
System.out.println("=== FCR Multiclass Dropping Example ===\n");
```

Figure 3.17: Topology of `fcr_oqndrop`: an open network over 2 queueing stations.

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	Source – Class1: Exp(0.4); Class2: Exp(0.3)
Service	Queue1 – Class1: Exp(1); Class2: Exp(0.9) Queue2 – Class1: Exp(1.1); Class2: Exp(1)
Solvers	JMT

Table 3.17: Features of `fcr_oqndrop`.

```

Network model = FCRegionModel.fcr_oqndrop();
JMT solver = new JMT(model, "seed", 23000, "samples", 50000, "verbose", VerboseLevel.SILENT);
NetworkAvgTable avgTable = solver.getAvgTable();

System.out.println(avgTable);

```

Running this edition prints

```

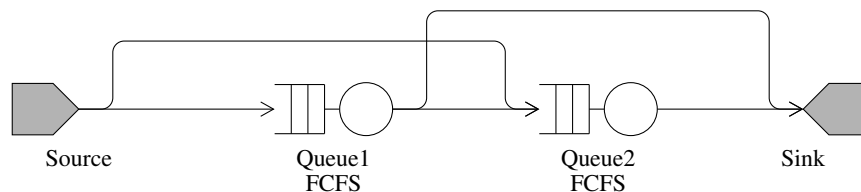
=== FCR Multiclass Dropping Example ===

WARNING: [refreshStruct] Priority classes are specified but no priority-aware scheduling policy (PSPRIO,
DPSPRIO, GPSPRIO, HOL, FCFSPRIO, FCFSPRPRIOR, FCFSPRIOR, LCFSPRIO, LCFSPRPRIOR, LCFSPRIOR, SRPTPRIO)
is used in the model. Priorities will be ignored.
WARNING: [refreshStruct] Message casted more than once, repetitions will not be printed on screen for 60
seconds.
jline.solvers.NetworkAvgTable@3ec300f1

```

3.2.7 `fcr_oqnwaitq`

The same region with waiting.

Figure 3.18: Topology of `fcr_oqnwaitq`: an open network over 2 queueing stations.

The example is built and solved as follows.

```

System.out.println("=== FCR Multiclass Blocking Example ===\n");

Network model = FCRegionModel.fcr_oqnwaitq();
JMT solver = new JMT(model, "seed", 23000, "samples", 10000, "verbose", VerboseLevel.SILENT);
NetworkAvgTable avgTable = solver.getAvgTable();

System.out.println(avgTable);

```

Running this edition prints

```

=== FCR Multiclass Blocking Example ===

```

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	Source – Class1: Exp(0.4); Class2: Exp(0.3)
Service	Queue1 – Class1: Exp(1); Class2: Exp(0.9) Queue2 – Class1: Exp(1.1); Class2: Exp(1)
Solvers	JMT

Table 3.18: Features of `fcr_oqnwaitq`.

```
WARNING: [refreshStruct] Priority classes are specified but no priority-aware scheduling policy (PSPRIO,
DPSPRIO, GPSPRIO, HOL, FCFSPRIO, FCFSPRPRIOR, FCFSPRIOR, LCFSPRIO, LCFSPRPRIOR, LCFSPRIOR, SRTPRIO)
is used in the model. Priorities will be ignored.
WARNING: [refreshStruct] Message casted more than once, repetitions will not be printed on screen for 60
seconds.
jline.solvers.NetworkAvgTable@3ec300f1
```

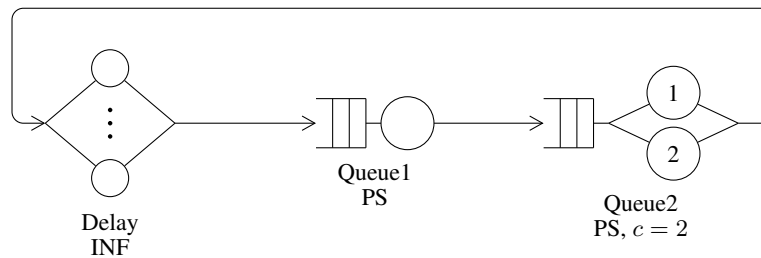
3.3 State probabilities

Beyond the averages, the solvers that build a state space can return probabilities. `getProb` gives the probability of a detailed station state, `getProbAggr` of an aggregate one (the queue length only, with the phases and the in-service order summed out), and the `Sys` variants give the joint state of the whole network.

The detailed state is discipline-dependent: under FCFS it records the order of the jobs in service, under PS it does not, and with non-exponential service it carries the phase of each job. That is why the same script exists in an FCFS and a PS version.

3.3.1 `statepr_aggr`

The marginal probability of an aggregate station state, through `getProbAggr`.

Figure 3.19: Topology of `statepr_aggr`: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	2: Class1 (closed, $N = 2$), Class2 (closed, $N = 0$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS ($c = 2$)
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(3); Class2: Exp(4) Queue2 – Class1: Exp(1); Class2: Exp(3)
Solvers	CTMC, NC

Table 3.19: Features of `statepr_aggr`.

The example is built and solved as follows.

```
Network model = StateProbabilitiesModel.statepr_aggr();
```

```

NetworkSolver solver = new CTMC(model);

try {
    solver.getAvgTable().print();
} catch (Exception e) {
}

pauseForUser();

```

Running this edition prints

```

CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.84 0.84 1 1 0.84 0.84
Queue1 Class1 0.32 0.28 0.38095 0.38095 0.84 0.84
Queue2 Class1 0.84 0.42 1 1 0.84 0.84

[Running in non-interactive mode, continuing...]

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by CTMC.

3.3.2 statepr_aggr_large

The same on a model whose state space is large enough to need the aggregate form.

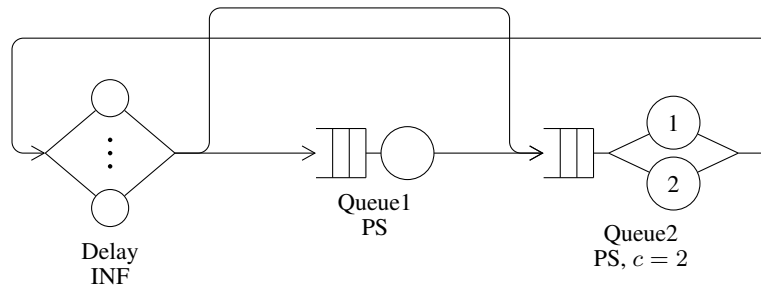


Figure 3.20: Topology of statepr_aggr_large: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	4: Class1 (closed, $N = 1$), Class2 (closed, $N = 0$), Class3 (closed, $N = 4$), Class4 (closed, $N = 0$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS ($c = 2$)
Service	Delay – Class1: Exp(1); Class2: Exp(2); Class3: Exp(1); Class4: Exp(1) Queue1 – Class1: Exp(3); Class2: Exp(4); Class3: Exp(5); Class4: Exp(1) Queue2 – Class1: Exp(1); Class2: Exp(3); Class3: Exp(5); Class4: Exp(2)
Features	class switching on 1 link
Solvers	CTMC, NC

Table 3.20: Features of statepr_aggr_large.

The example is built and solved as follows.

```

Network model = StateProbabilitiesModel.statepr_aggr_large();

NetworkSolver solver = new JMT(model, "seed", 12345);

try {
    // The solver's OWN options, not a fresh defaultOptions(): the latter
    // draws a RANDOM seed in its constructor (SolverOptions ->
    // RandomManager.generateRandomSeed), so replacing the object wholesale
    // discards the seed passed to the constructor above and makes this
    // example irreproducible -- which is what left the ld_multiserver_ps

```



```
// parity row disagreeing with its golden on a Monte-Carlo margin.
SolverOptions options = solver.getOptions();
options.samples = 100000;
((JMT)solver).setOptions(options);

solver.getAvgTable().print();
} catch (Exception e) {
}

pauseForUser();
```

Running this edition prints

```
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.25967 0.25967 1.0024 0.5012 0.26031 0.26021
Delay Class2 0.13087 0.13087 0.50138 0.25069 0.26062 0.26033
Delay Class3 1.2691 1.2691 0.99802 0.49901 1.2699 1.2698
Delay Class4 1.2758 1.2758 1.0048 0.5024 1.2699 1.2699
Queue1 Class1 0.11622 0.08639 0.44491 0.22245 0.26021 0.26021
Queue1 Class2 0.085733 0.064238 0.33284 0.16642 0.26033 0.26061
Queue1 Class3 0.37356 0.25339 0.2901 0.14505 1.2698 1.2698
Queue2 Class1 0.30435 0.1323 1.167 0.58348 0.26021 0.26062
Queue2 Class2 0.10126 0.043418 0.3899 0.19495 0.26061 0.26033
Queue2 Class3 0.30568 0.12745 0.24437 0.12218 1.2698 1.2699
Queue2 Class4 0.77755 0.31689 0.61207 0.30603 1.2699 1.2699

[Running in non-interactive mode, continuing...]
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by JMT.

3.3.3 statepr_allprobs_fcfs

The FCFS version, where the detailed state carries the order of the jobs in service.

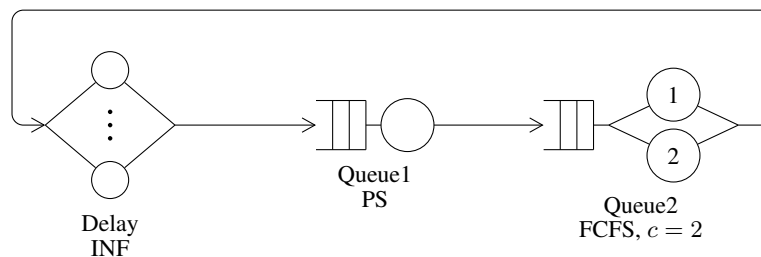


Figure 3.21: Topology of statepr_allprobs_fcfs: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	2: Class1 (closed, $N = 2$), Class2 (closed, $N = 0$)
Scheduling	Delay: INF; Queue1: PS; Queue2: FCFS ($c = 2$)
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(3); Class2: Exp(4) Queue2 – Class1: Exp(3); Class2: Exp(3)
Solvers	CTMC, NC, SSA, JMT

Table 3.21: Features of statepr_allprobs_fcfs.

The example is built and solved as follows.

```
Network model = StateProbabilitiesModel.statepr_allprobs_fcfs();

NetworkSolver solver = new CTMC(model);

try {
```

```

        solver.getAvgTable().print();
    } catch (Exception e) {
    }

    pauseForUser();

```

Running this edition prints

```

CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay   Class1     1.1538  1.1538    1         1  1.1538  1.1538
Queue1   Class1     0.46154 0.38462   0.4       0.4  1.1538  1.1538
Queue2   Class2     0.38462 0.19231  0.33333  0.33333  1.1538  1.1538

[Running in non-interactive mode, continuing...]

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by CTMC.

3.3.4 statepr_allprobs_ps

Every probability API on one PS model: detailed and aggregate, per station and system-wide.

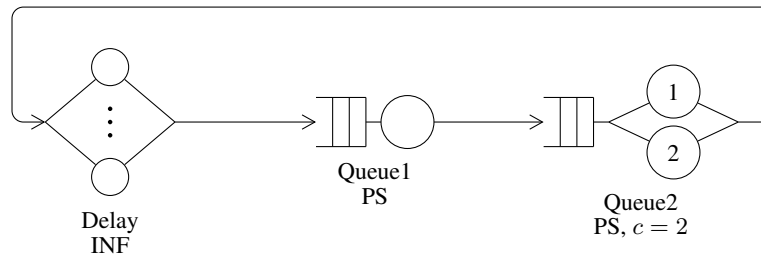


Figure 3.22: Topology of statepr_allprobs_ps: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	2: Class1 (closed, $N = 2$), Class2 (closed, $N = 0$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS ($c = 2$)
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(3); Class2: Exp(4) Queue2 – Class1: Exp(1); Class2: Exp(3)
Solvers	CTMC, NC, SSA, JMT

Table 3.22: Features of statepr_allprobs_ps.

The example is built and solved as follows.

```

Network model = StateProbabilitiesModel.statepr_allprobs_ps();

NetworkSolver solver = new CTMC(model);

try {
    solver.getAvgTable().print();
} catch (Exception e) {
}

pauseForUser();

```

Running this edition prints

```

CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay   Class1     1.1538  1.1538    1         1  1.1538  1.1538
Queue1   Class1     0.46154 0.38462   0.4       0.4  1.1538  1.1538
Queue2   Class2     0.38462 0.19231  0.33333  0.33333  1.1538  1.1538

```

[Running in non-interactive mode, continuing...]

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by CTMC.

3.3.5 statepr_sys_aggr

The joint aggregate state of the whole network, through `getProbSysAggr`.

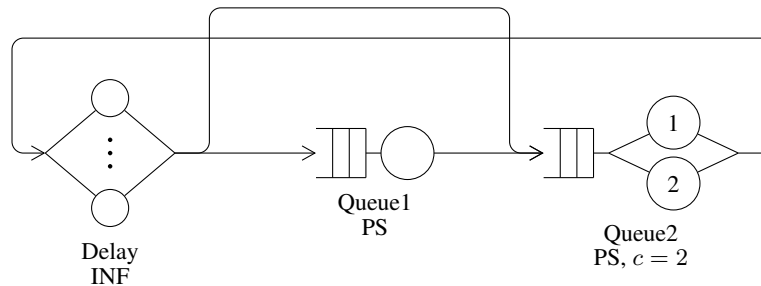


Figure 3.23: Topology of `statepr_sys_aggr`: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	4: Class1 (closed, $N = 1$), Class2 (closed, $N = 0$), Class3 (closed, $N = 3$), Class4 (closed, $N = 0$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS ($c = 2$)
Service	Delay – Class1: Exp(1); Class2: Exp(2); Class3: Exp(1); Class4: Exp(1) Queue1 – Class1: Exp(3); Class2: Exp(4); Class3: Exp(5); Class4: Exp(1) Queue2 – Class1: Exp(1); Class2: Exp(3); Class3: Exp(5); Class4: Exp(2)
Features	class switching on 1 link
Solvers	CTMC, NC, JMT

Table 3.23: Features of `statepr_sys_aggr`.

The example is built and solved as follows.

```

Network model = StateProbabilitiesModel.statepr_sys_aggr();

NetworkSolver solver = new CTMC(model);

try {
    solver.getAvgTable().print();
} catch (Exception e) {
}

pauseForUser();
  
```

Running this edition prints

```

CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.27272 0.27272 1 0.5 0.27272 0.27272
Delay Class2 0.13636 0.13636 0.5 0.25 0.27272 0.27272
Delay Class3 0.98217 0.98217 1 0.5 0.98217 0.98217
Delay Class4 0.98217 0.98217 1 0.5 0.98217 0.98217
Queue1 Class1 0.11211 0.090907 0.41109 0.20554 0.27272 0.27272
Queue1 Class2 0.084084 0.06818 0.30832 0.15416 0.27272 0.27272
Queue1 Class3 0.26672 0.19643 0.27156 0.13578 0.98217 0.98217
Queue2 Class1 0.29604 0.13636 1.0855 0.54276 0.27272 0.27272
Queue2 Class2 0.098681 0.045453 0.36184 0.18092 0.27272 0.27272
Queue2 Class3 0.2197 0.098217 0.22369 0.11184 0.98217 0.98217
Queue2 Class4 0.54925 0.24554 0.55922 0.27961 0.98217 0.98217
  
```

```
[Running in non-interactive mode, continuing...]
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by CTMC.

3.3.6 statepr_sys_aggr_large

The large-model version.

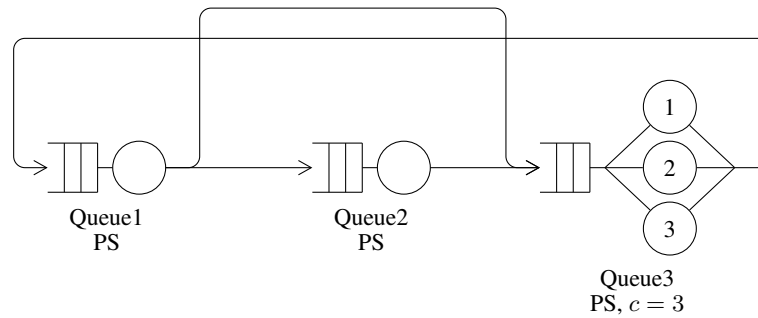


Figure 3.24: Topology of `statepr_sys_aggr_large`: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 3
Classes	4: Class1 (closed, $N = 1$), Class2 (closed, $N = 1$), Class3 (closed, $N = 1$), Class4 (closed, $N = 1$)
Scheduling	Queue1: PS; Queue2: PS; Queue3: PS ($c = 3$)
Service	Queue1 – Class1: Exp(1); Class2: Exp(2); Class3: Exp(1); Class4: Exp(1) Queue2 – Class1: Exp(3); Class2: Exp(4); Class3: Exp(5); Class4: Exp(1) Queue3 – Class1: Exp(1); Class2: Exp(3); Class3: Exp(5); Class4: Exp(2)
Features	class switching on 1 link
Solvers	CTMC

Table 3.24: Features of `statepr_sys_aggr_large`.

The example is built and solved as follows.

```
Network model = StateProbabilitiesModel.statepr_sys_aggr_large();

NetworkSolver solver = new JMT(model, "seed", 12345);

try {
    // The solver's OWN options, not a fresh defaultOptions(): the latter
    // draws a RANDOM seed in its constructor (SolverOptions ->
    // RandomManager.generateRandomSeed), so replacing the object wholesale
    // discards the seed passed to the constructor above and makes this
    // example irreproducible -- which is what left the ld_multiserver_ps
    // parity row disagreeing with its golden on a Monte-Carlo margin.
    SolverOptions options = solver.getOptions();
    options.samples = 100000;
    ((JMT)solver).setOptions(options);

    solver.getAvgTable().print();
} catch (Exception e) {
}

pauseForUser();
```

Running this edition prints

```
jline.lang.Network@6438a396
```

3.3.7 statepr_sys_marg

The joint law of the per-station TOTAL queue lengths, all classes summed out, through `getProbSysMarg`.

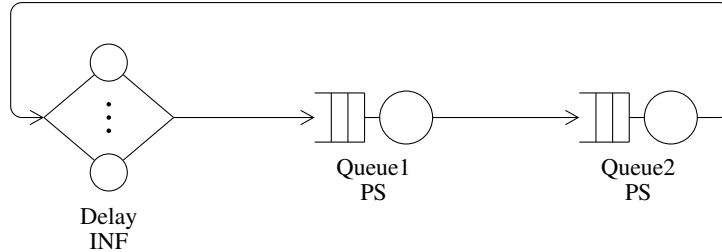


Figure 3.25: Topology of `statepr_sys_marg`: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	2: Class1 (closed, $N = 2$), Class2 (closed, $N = 1$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS
Service	Delay – Class1: Exp(0.6667); Class2: Exp(0.5) Queue1 – Class1: Exp(1.429); Class2: Exp(2.5) Queue2 – Class1: Exp(3.333); Class2: Exp(1.111)
Solvers	CTMC

Table 3.25: Features of `statepr_sys_marg`.

The example is built and solved as follows.

```
int[] N = {2, 1};

Network model = new Network("model");
Delay delay = new Delay(model, "Delay");
Queue queue1 = new Queue(model, "Queue1", SchedStrategy.PS);
Queue queue2 = new Queue(model, "Queue2", SchedStrategy.PS);

ClosedClass class1 = new ClosedClass(model, "Class1", N[0], delay, 0);
ClosedClass class2 = new ClosedClass(model, "Class2", N[1], delay, 0);

delay.setService(class1, Exp.fitMean(1.5));
delay.setService(class2, Exp.fitMean(2.0));
queue1.setService(class1, Exp.fitMean(0.7));
queue1.setService(class2, Exp.fitMean(0.4));
queue2.setService(class1, Exp.fitMean(0.3));
queue2.setService(class2, Exp.fitMean(0.9));

RoutingMatrix P = model.initRoutingMatrix();
P.set(class1, class1, delay, queue1, 1.0);
P.set(class1, class1, queue1, queue2, 1.0);
P.set(class1, class1, queue2, delay, 1.0);
P.set(class2, class2, delay, queue1, 1.0);
P.set(class2, class2, queue1, queue2, 1.0);
P.set(class2, class2, queue2, delay, 1.0);
model.link(P);

... (47 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

err += Math.abs(pb[j] / sb - p[j]) / p[j];
}
System.out.printf("  Bethe engine, mean relative error : %.2f%%\n",
    100.0 * err / states.size());

System.out.println(solver.citations());
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.4 Initial state and warm start

Transient analysis needs an initial state. `initFromMarginal` builds one from marginal queue lengths, filling in the phases and the in-service order in a way consistent with the discipline; a state may also be set explicitly. For closed classes the initial state must conserve the population, which is what makes the ordered form necessary rather than convenient.

A simulation can also be started from the steady state computed by another solver, which removes the warm-up transient from the sample path.

3.4.1 `init_state_fcfs_exp`

An initial state set from marginal queue lengths, and the transient it produces on an FCFS network.

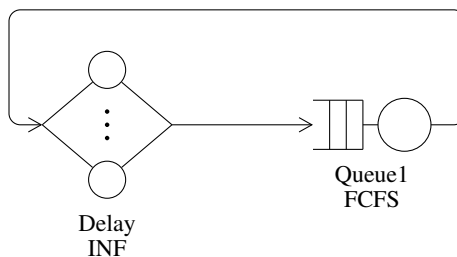


Figure 3.26: Topology of `init_state_fcfs_exp`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 5$)
Scheduling	Delay: INF; Queue1: FCFS
Service	Delay – Class1: Exp(1) Queue1 – Class1: Exp(0.7)

Table 3.26: Features of `init_state_fcfs_exp`.

The example is built and solved as follows.

```
Network model = InitStateModel.init_state_fcfs_exp();

try {
    new CTMC(model, "verbose", 1, "seed", 23000, "samples", 100000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("CTMC failed: " + e.getMessage());
}

try {
    new FLD(model, "verbose", 1, "seed", 23000, "samples", 100000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("FLD failed: " + e.getMessage());
}

pauseForUser();
```

Running this edition prints

```
=== Example 1: Two Initialization Methods ===

Method 1: Default initialization
Initial state: All jobs start at reference station
Fluid solver completed for default initialization
```

```

Method 2: Initialize from marginal distribution [2;3]
Initial state: 2 jobs at station 1, 3 jobs at station 2
Fluid solver completed for marginal initialization
Example 1 completed: Different initial states affect transient behavior
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.69951 0.69951 1 1 0.69951 0.69951
Queue1 Class1 4.3005 0.9993 6.1478 6.1478 0.69951 0.69951
Fluid analysis [method: minnormal; type: approximate, deterministic; lang: java; env: python] completed in
<t>s.
Fluid analysis [method: default/minnormal; type: approximate, deterministic; lang: java; env: python]
completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.69999 0.69999 1 1 0.69999 0.69999
Queue1 Class1 4.3 0.99999 6.1429 6.1429 0.69999 0.69999

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (CTMC, FLD), which is the point of the example: the same model read by methods of different cost and different exactness.

3.4.2 init_state_fcfs_nonexp

The same with non-exponential service, where the phase of the job in service is part of the state.

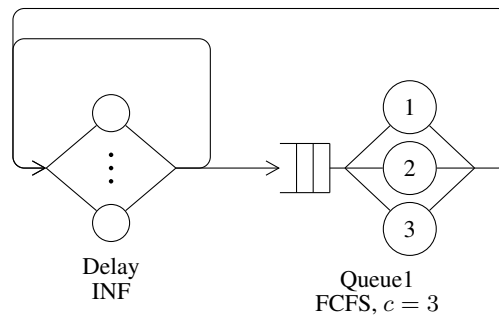


Figure 3.27: Topology of init_state_fcfs_nonexp: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 3$), Class2 (closed, $N = 2$)
Scheduling	Delay: INF; Queue1: FCFS ($c = 3$)
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(1.2); Class2: Erlang(2,2)
Solvers	CTMC, FLD

Table 3.27: Features of init_state_fcfs_nonexp.

The example is built and solved as follows.

```

Network model = InitStateModel.init_state_fcfs_nonexp();

try {
    new CTMC(model, "verbose", 1, "seed", 23000, "samples", 100000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("CTMC failed: " + e.getMessage());
}

try {
    new FLD(model, "verbose", 1, "seed", 23000, "samples", 100000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("FLD failed: " + e.getMessage());
}

pauseForUser();

```

Running this edition prints

```

=== Example 2: Three Initialization Approaches ===

Method 1: Default initialization
Initial state: All jobs start at reference station
Fluid solver completed for default initialization

Method 2: Initialize from marginal distribution [0,0;4,1]
Initial state: 4 jobs of class 1 and 1 job of class 2 at station 2
Fluid solver completed for marginal initialization

Method 3: Uniform prior over states with same job counts
Initial state: Uniform distribution over all valid states with same job counts
Fluid solver completed for uniform prior initialization
Example 2 completed: Different initialization approaches affect transient behavior
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass   QLen      Util      RespT      ResidT      ArvR      Tput
... (5 captured-output lines omitted; rerun the source example for the full output)

Fluid analysis [method: default/minnormal; type: approximate, deterministic; lang: java; env: python]
completed in <t>s.
Station JobClass   QLen      Util      RespT      ResidT      ArvR      Tput
Delay   Class1      2.0273     2.0273      1      1.2821     2.0273     2.0273
Delay   Class2      1.3786     1.3786      1     0.87179    1.3786     1.3786
Queue1  Class1      0.17404    0.056313    0.85847    0.11006    0.20273    0.20273
Queue1  Class2      1.4201     0.45952    1.0302     0.89809    1.3786     1.3786

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (CTMC, FLD), which is the point of the example: the same model read by methods of different cost and different exactness.

3.4.3 init_state_ps

The processor sharing version.

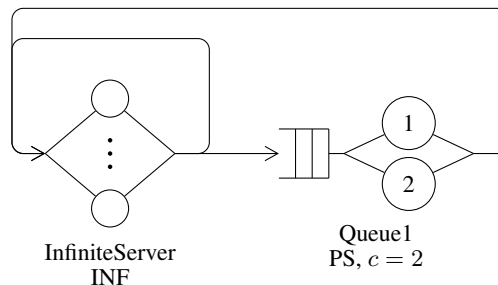


Figure 3.28: Topology of `init_state_ps`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 3$), Class2 (closed, $N = 1$)
Scheduling	InfiniteServer: INF; Queue1: PS ($c = 2$)
Service	InfiniteServer – Class1: Exp(3); Class2: Exp(0.5) Queue1 – Class1: Exp(0.1); Class2: Exp(1)
Solvers	CTMC, JMT, SSA, FLD, MVA, NC

Table 3.28: Features of `init_state_ps`.

The example is built and solved as follows.

```

Network model = InitStateModel.init_state_ps();

try {
    new CTMC(model, "verbose", 1, "seed", 23000, "samples", 100000).getAvgTable().print();
} catch (Exception e) {
}

```



```

        System.out.println("CTMC failed: " + e.getMessage());
    }
    try {
        new JMT(model, "verbose", 1, "seed", 23000, "samples", 100000).getAvgTable().print();
    } catch (Exception e) {
        System.out.println("JMT failed: " + e.getMessage());
    }
    try {
        new SSA(model, "verbose", 1, "seed", 23000, "samples", 100000).getAvgTable().print();
    } catch (Exception e) {
        System.out.println("SSA failed: " + e.getMessage());
    }
    try {
        new FLD(model, "verbose", 1, "seed", 23000, "samples", 100000).getAvgTable().print();
    } catch (Exception e) {
        System.out.println("FLD failed: " + e.getMessage());
    }
    try {
        new MVA(model, "verbose", 1, "seed", 23000, "samples", 100000).getAvgTable().print();
    } catch (Exception e) {
        System.out.println("MVA failed: " + e.getMessage());
    }
    try {
        new NC(model, "verbose", 1, "seed", 23000, "samples", 100000).getAvgTable().print();
    } catch (Exception e) {
        System.out.println("NC failed: " + e.getMessage());
    }
}

pauseForUser();

```

Running this edition prints

```

CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station  JobClass  QLen    Util    RespT    ResidT    ArvR    Tput
InfiniteServer  Class1  0.33653  0.33653  0.33333  0.19841  1.0096  1.0096
InfiniteServer  Class2  1.373   1.373   2        0.80952  0.68652  0.68652
Queue1         Class1  1.3634  0.50479 13.504   0.80381  0.10096  0.10096
Queue1         Class2  0.92708 0.34326 1.3504   0.54659  0.68652  0.68652
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station  JobClass  QLen    Util    RespT    ResidT    ArvR    Tput
InfiniteServer  Class1  0.332   0.332   0.33224  0.19776  0.99645  0.99646
InfiniteServer  Class2  1.3674  1.3674  1.9991   0.80918  0.68249  0.68251
Queue1         Class1  1.3654  0.51204 13.573   0.80794  0.10041  0.10043
Queue1         Class2  0.91601 0.34077 1.3463   0.54493  0.68251  0.67959
SSA analysis [method: default/nrm; type: approximate, randomized; lang: java; env: python] completed in <t>
>s. Iterations: 100000.
Station  JobClass  QLen    Util    RespT    ResidT    ArvR    Tput
InfiniteServer  Class1  0.339   0.339   0.33333  0.19841  1.0077   1.017

... (16 captured-output lines omitted; rerun the source example for the full output)

NC analysis [method: default/exact/gld; type: approximate, deterministic; lang: java; env: python]
completed in <t>s.
Station  JobClass  QLen    Util    RespT    ResidT    ArvR    Tput
InfiniteServer  Class1  0.33653  0.33653  0.33333  0.19841  1.0096  1.0096
InfiniteServer  Class2  1.373   1.373   2        0.80952  0.68652  0.68652
Queue1         Class1  1.3634  0.50479 13.504   0.80381  0.10096  0.10096
Queue1         Class2  0.92708 0.34326 1.3504   0.54659  0.68652  0.68652

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 6 solvers the script runs (CTMC, JMT, SSA, FLD, MVA, NC), which is the point of the example: the same model read by methods of different cost and different exactness.

3.4.4 ldes_warmstart

Seeding a simulation from the steady state of another solver, so the warm-up phase can be skipped. The example is built and solved as follows.

```
int nA = 40;
```

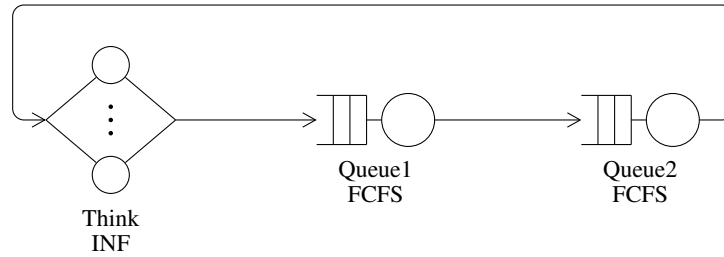


Figure 3.29: Topology of ldes_warmstart: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	1: Jobs (closed, $N = 100$)
Scheduling	Think: INF; Queue1: FCFS; Queue2: FCFS
Service	Think – Jobs: Exp(1) Queue1 – Jobs: Exp(1) Queue2 – Jobs: Exp(0.98)
Solvers	MVA, CTMC, LDES

Table 3.29: Features of ldes_warmstart.

```

System.out.println("PART A: sample efficiency, near-balanced tandem, N=" + nA);
System.out.println();

Matrix exactA = new SolverCTMC(buildModel(nA), "exact", "verbose", false).getAvgQLen();
System.out.println("Exact CTMC mean queue lengths: " + rowToString(exactA.transpose()));

Network m1 = buildModel(nA);
Object[] wMva = warmPlacement(m1, new SolverMVA(m1, "exact", "verbose", false));
Network m2 = buildModel(nA);
Object[] wCtmc = warmPlacement(m2, new SolverCTMC(m2, "exact", "verbose", false));
System.out.printf("Warm placement from SolverMVA  (%.3fs, rounded mean qlen): %s\n",
    (Double) wMva[1], rowToString((Matrix) wMva[0]));
System.out.printf("Warm placement from SolverCTMC (%.3fs, distribution mode): %s\n",
    (Double) wCtmc[1], rowToString((Matrix) wCtmc[0]));
System.out.println();

int[] gridA = {5000, 10000, 20000, 50000, 100000, 200000};
System.out.println("COLD start (default init, MSER-5 warmup removal):");
double[] coldA = timeToAccuracy(nA, exactA, gridA, null, 0.0);
System.out.println("WARM-MVA start (init from MVA mean queue lengths, no warmup):");
double[] warmMvaA = timeToAccuracy(nA, exactA, gridA, (Matrix) wMva[0], (Double) wMva[1]);
System.out.println("WARM-CTMC start (init from mode of exact stationary distribution, no
    warmup):");
double[] warmCtmcA = timeToAccuracy(nA, exactA, gridA, (Matrix) wCtmc[0], (Double) wCtmc[1]);
System.out.println();
System.out.println("Samples needed for " + (int) (100 * TOL) + "% relative error (Part A):");

    ... (28 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

report("COLD    ", coldB);
report("WARM-MVA", warmB);
if (coldB[2] > 0 && warmB[2] > 0 && coldB[0] > 0 && warmB[0] > 0) {
    System.out.printf("  Speedup WARM-MVA vs COLD: %.2fx (%.0fx fewer samples)\n",
        coldB[2] / warmB[2], coldB[0] / warmB[0]);
}

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.5 State-dependent routing

Under RAND or probabilistic routing the destination is drawn from a fixed distribution. State-dependent policies read the network state instead: JSQ sends the job to the shortest queue, k -choices samples k

queues and picks the shortest of those, and round-robin and its weighted form cycle deterministically. Because the routing depends on the state, the network is no longer product-form, and JSQ in particular is only available to the simulators.

3.5.1 sdroute_closed

The closed counterpart.

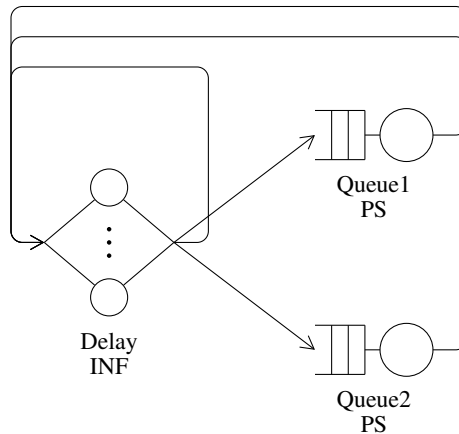


Figure 3.30: Topology of sdroute_closed: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	1: Class1 (closed, $N = 1$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS
Service	Delay – Class1: HyperExp(0.99,0.01;1.534,0.02819) Queue1 – Class1: Exp(1) Queue2 – Class1: Exp(2)
Features	class switching on 1 link
Solvers	CTMC, JMT, SSA

Table 3.30: Features of sdroute_closed.

The example is built and solved as follows.

```
Network model = StateDepRoutingModel.sdroute_closed();

try {
    new CTMC(model, "keep", true).getAvgTable().print();
} catch (Exception e) {
    System.out.println("CTMC failed: " + e.getMessage());
}

try {
    new JMT(model, "samples", 100000, "seed", 23000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT failed: " + e.getMessage());
}

try {
    new SSA(model, "verbose", true, "samples", 10000, "seed", 23000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("SSA failed: " + e.getMessage());
}

pauseForUser();
```

Running this edition prints

```
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
```

```

Delay   Class1   0.66667  0.66667    1      1  0.66667  0.66667
Queue1  Class1   0.22222  0.22222    1  0.33333  0.22222  0.22222
Queue2  Class1   0.11111  0.11111    0.5  0.16667  0.22222  0.22222
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass   QLen    Util    RespT    ResidT    ArvR    Tput
Delay   Class1     0.6634  0.6634  0.99642  0.99642  0.67275  0.66595
Queue1  Class1     0.2227  0.2227  0.99408  0.33136  0.22136  0.22132
Queue2  Class1     0.11295  0.11295  0.50054  0.16685  0.22136  0.22136
SSA analysis [method: default/nrm; type: approximate, randomized; lang: java; env: python] completed in <t>
>s. Iterations: 10000.
Station JobClass   QLen    Util    RespT    ResidT    ArvR    Tput
Delay   Class1     0.61875  0.61875  0.83313  0.83313  0.7518  0.74269
Queue1  Class1     0.25825  0.25825    1  0.33333  0.24756  0.25825
Queue2  Class1     0.12299  0.12299    0.5  0.16667  0.24756  0.24599

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 3 solvers the script runs (CTMC, JMT, SSA), which is the point of the example: the same model read by methods of different cost and different exactness.

3.5.2 sdroute_jsq

Join-the-shortest-queue dispatching over a pool of FCFS queues.

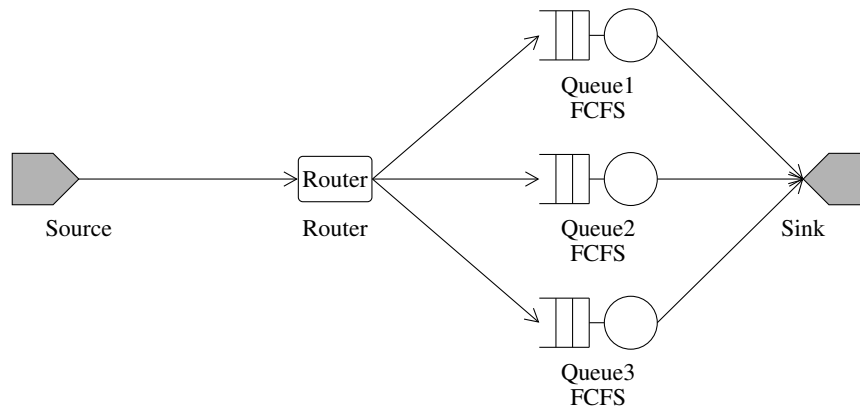


Figure 3.31: Topology of `sdroute_jsq`: an open network over 3 queueing stations with a router.

Nodes	6: Source, Queue × 3, Router, Sink		
Classes	1: Class1 (open)		
Scheduling	Queue1: FCFS; Queue2: FCFS; Queue3: FCFS		
Arrivals	Source – Class1: Exp(2.1)		
Service	Queue1 – Class1: Exp(1)	Queue2 – Class1: Exp(1)	Queue3 – Class1: Exp(1)
Features	class switching on 6 links		
Solvers	JMT, LDES		

Table 3.31: Features of `sdroute_jsq`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.5.3 sdroute_krzesinski

Adaptive routing with balanced loading (Krzesinski 1987): a central server and four single-center branches whose routing probabilities depend on the branch populations, and which keep a product form.

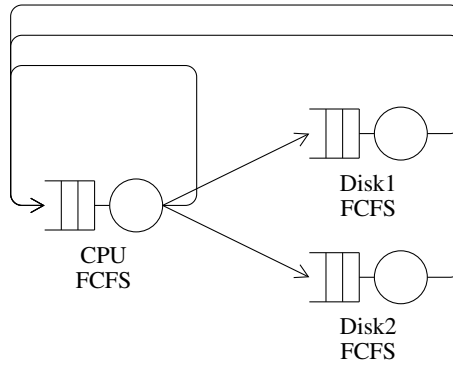


Figure 3.32: Topology of sdroute_krzesinski: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 3
Classes	1: Class1 (closed, $N = 3$)
Scheduling	CPU: FCFS; Disk1: FCFS; Disk2: FCFS
Service	CPU – Class1: Exp(1) Disk1 – Class1: Exp(0.8) Disk2 – Class1: Exp(0.5)
Solvers	NC, CTMC

Table 3.32: Features of sdroute_krzesinski.

The example is built and solved as follows.

```
double[] mu = {1.0, 0.8, 0.5};
int N = 3;

Network model = new Network("sdrCentralServer");
Queue cpu = new Queue(model, "CPU", SchedStrategy.FCFS);
Queue disk1 = new Queue(model, "Disk1", SchedStrategy.FCFS);
Queue disk2 = new Queue(model, "Disk2", SchedStrategy.FCFS);
ClosedClass jobclass = new ClosedClass(model, "Class1", N, cpu, 0);
cpu.setService(jobclass, new Exp(mu[0]));
disk1.setService(jobclass, new Exp(mu[1]));
disk2.setService(jobclass, new Exp(mu[2]));

model.addLink(cpu, cpu);          // denied entry: the busy form of waiting
model.addLink(cpu, disk1);
model.addLink(cpu, disk2);
model.addLink(disk1, cpu);
model.addLink(disk2, cpu);
disk1.setProbRouting(jobclass, cpu, 1.0);
disk2.setProbRouting(jobclass, cpu, 1.0);

// Branch index 1 is the complement M-V and stays empty
List<List<Node>> branches = new ArrayList<List<Node>>();
branches.add(new ArrayList<Node>());
branches.add(new ArrayList<Node>(Arrays.asList((Node) disk1)));
branches.add(new ArrayList<Node>(Arrays.asList((Node) disk2)));
double[][] d = {{0, 1, 2}, {0, 0, 3}};

... (15 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

new SolverNC(model).getAvgTable().print();
System.out.println("Section 4 MVA and convolution, same model:");
jline.solvers.SolverOptions mvaOpt = new SolverNC(model).getOptions();
mvaOpt.method = "sdr.mva";
new SolverNC(model, mvaOpt).getAvgTable().print();
new SolverCTMC(model).getAvgTable().print();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.5.4 sdroute_multibranch

The same product form on branches holding several centers, where the coefficients are the branch traffic equations.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

Solvers	CTMC
---------	------

Table 3.33: Features of `sdroute_multibranch`.

The example is built and solved as follows.

```
System.out.println();
System.out.println("==== SDR with multi-center branches ====");
runCase("A: branch 2 = 2a -> 2b (series)", 0.0);
runCase("B: branch 2 = 2a -> 2b, 2b -> 2a w.p. 0.5 (feedback onto the branch departure)",
0.5);
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.5.5 sdroute_open

Round-robin dispatching from a Router node in an open network.

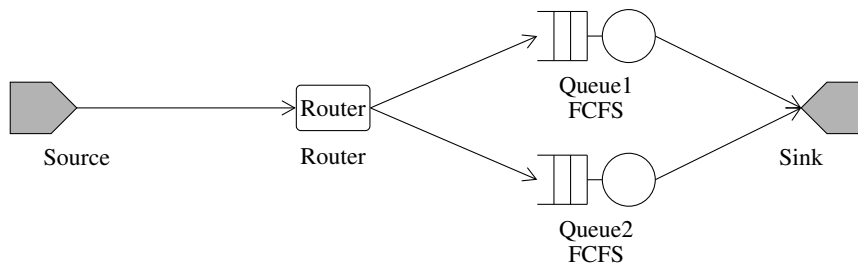


Figure 3.33: Topology of `sdroute_open`: an open network over 2 queueing stations with a router.

Nodes	5: Source, Queue × 2, Router, Sink
Classes	1: Class1 (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	Source – Class1: Exp(1)
Service	Queue1 – Class1: Exp(2) Queue2 – Class1: Exp(2)
Features	class switching on 5 links
Solvers	JMT, CTMC

Table 3.34: Features of `sdroute_open`.

The example is built and solved as follows.

```
Network model = StateDepRoutingModel.sdroute_open();

try {
    new JMT(model, "seed", 23000).getAvgNodeTable().print();
} catch (Exception e) {
    System.out.println("JMT failed: " + e.getMessage());
}

try {
```

```

    new CTMC(model, "cutoff", 5).getAvgNodeTable().print();
} catch (Exception e) {
    System.out.println("CTMC failed: " + e.getMessage());
}

pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Node  JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
Source Class1    0     0     0     0     0    1.012
Router Class1    0     0     0     0    1.012    1.012
Queue1 Class1  0.3024 0.26218 0.57611 0.28806 0.50351 0.50533
Queue2 Class1  0.29317 0.25361 0.56951 0.28476 0.50531 0.50719
Sink   Class1    0     0     0     0    1.012     0
WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 5 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Node  JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
Source Class1    0     0     0     0     0  0.99844
Router Class1    0     0     0     0     0  0.99844  0.99844
Queue1 Class1  0.28701 0.24961 0.57492 0.28746 0.49922 0.49922
Queue2 Class1  0.28701 0.24961 0.57492 0.28746 0.49922 0.49922
Sink   Class1    0     0     0     0     0  0.99844     0

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, CTMC), which is the point of the example: the same model read by methods of different cost and different exactness.

3.5.6 sdroute_twoclasses_closed

Two classes with MAP, MMPP(2) and phase-type service under round-robin dispatching.

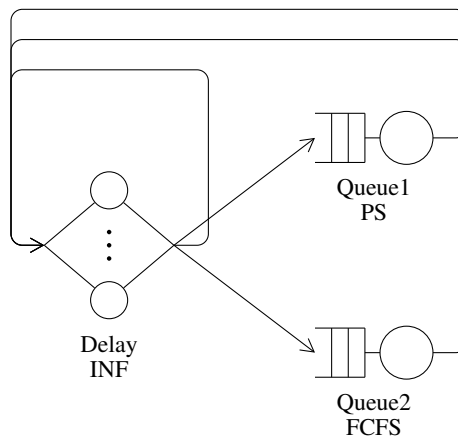


Figure 3.34: Topology of `sdroute_twoclasses_closed`: a closed network over 3 queueing stations.

The example is built and solved as follows.

```

Network model = StateDepRoutingModel.sdroute_twoclasses_closed();

// The reference's own seed and run length. They go through the
// constructor: setOptions(defaultOptions()) would REPLACE the seed with
// 0, which is drawn at random, and the row would differ run to run.
NetworkSolver solver = new JMT(model, "seed", 23000, "samples", 100000);

try {
    solver.getAvgTable().print();
} catch (Exception e) {

```

Nodes	3: Queue $\times$ 2, Delay
Classes	2: Class1 (closed, $N = 1$), Class2 (closed, $N = 2$)
Scheduling	Delay: INF; Queue1: PS; Queue2: FCFS
Service	Delay – Class1: HyperExp(0.99,0.01;1.534,0.02819); Class2: HyperExp(0.99,0.01;1.534,0.02819) Queue1 – Class1: APH; Class2: PH Queue2 – Class1: MAP; Class2: MAP
Features	class switching on 1 link
Solvers	CTMC, JMT, SSA

Table 3.35: Features of `sdroute_twoclasses_closed`.

```
}
pauseForUser();
```

Running this edition prints

```
jline.lang.Network@73035e27
```

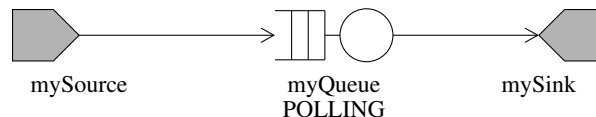
3.6 Cyclic polling

A polling server visits its queues in a cycle. Under gated service it serves the jobs that were waiting when it arrived, under exhaustive service it stays until the queue empties, and under k -limited service it serves at most k jobs per visit. The discipline is declared as `SchedStrategy.POLLING` and the variant is a parameter of the station.

The polling server parks at a queue rather than roving: the visit order and the ratio between the queue lengths are what a polling model must be validated on, not the aggregate throughput, which is insensitive to the discipline.

3.6.1 polling_exhaustive_det

Exhaustive service with deterministic (immediate) switchover.

Figure 3.35: Topology of `polling_exhaustive_det`: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: myClass1 (open), myClass2 (open)
Scheduling	myQueue: POLLING
Arrivals	mySource – myClass1: Det(1); myClass2: Det(1)
Service	myQueue – myClass1: Det(0.001); myClass2: Det(0.001)
Solvers	JMT

Table 3.36: Features of `polling_exhaustive_det`.

The example is built and solved as follows.

```
Network model = CyclicPollingModel.polling_exhaustive_det();

// The reference runs MVA first and JMT second, at this seed and run length.
try {
```



```

        new MVA(model).getAvgTable().print();
    } catch (Exception e) {
        System.out.println("MVA failed: " + e.getMessage());
    }
    try {
        new JMT(model, "seed", 23000, "samples", 100000).getAvgTable().print();
    } catch (Exception e) {
        System.out.println("JMT failed: " + e.getMessage());
    }
    pauseForUser();

```

Running this edition prints

```

MVA analysis [method: default/stationtime; type: approximate, deterministic; lang: java; env: python]
completed in <t>s.
Station  JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
mySource  myClass1           0         0         0         0         0         1
mySource  myClass2           0         0         0         0         0         1
myQueue   myClass1  0.0010011  0.001  0.0010011  0.0010011  1         1
myQueue   myClass2  0.0010011  0.001  0.0010011  0.0010011  1         1
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station  JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
mySource  myClass1           0         0         0         0         0         1
mySource  myClass2           0         0         0         0         0         1
myQueue   myClass1  0.00099993  0.00099993  0.001  0.001  1         1
myQueue   myClass2  0.0019999  0.00099993  0.002  0.002  1         1
[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (MVA, JMT), which is the point of the example: the same model read by methods of different cost and different exactness.

3.6.2 polling_exhaustive_exp

Exhaustive service with exponential switchover times.

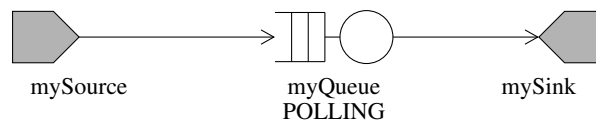


Figure 3.36: Topology of polling_exhaustive_exp: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: myClass1 (open), myClass2 (open)
Scheduling	myQueue: POLLING
Arrivals	mySource – myClass1: Exp(0.1); myClass2: Exp(0.1)
Service	myQueue – myClass1: Exp(1); myClass2: Exp(1.5)
Solvers	JMT

Table 3.37: Features of polling_exhaustive_exp.

The example is built and solved as follows.

```

Network model = CyclicPollingModel.polling_exhaustive_exp();

// The reference runs MVA first and JMT second, at this seed and run length.
try {
    new MVA(model).getAvgTable().print();
} catch (Exception e) {
    System.out.println("MVA failed: " + e.getMessage());
}
try {

```

```

    new JMT(model, "seed", 23000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT failed: " + e.getMessage());
}

pauseForUser();

```

Running this edition prints

```

MVA analysis [method: default/stationtime; type: approximate, deterministic; lang: java; env: python]
  completed in <t>s.
Station  JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
mySource  myClass1         0         0         0         0         0      0.1
mySource  myClass2         0         0         0         0         0      0.1
myQueue   myClass1      0.11685         0.1      1.1685      1.1685      0.1      0.1
myQueue   myClass2      0.084724  0.066667  0.84724  0.84724      0.1      0.1
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station  JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
mySource  myClass1         0         0         0         0         0  0.099696
mySource  myClass2         0         0         0         0         0  0.10064
myQueue   myClass1      0.11377  0.099441  1.1502  1.1502  0.099696  0.099232
myQueue   myClass2      0.082842  0.064638  0.8347  0.8347  0.10064  0.09965

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (MVA, JMT), which is the point of the example: the same model read by methods of different cost and different exactness.

3.6.3 polling_gated

A gated polling server: at each visit it serves the jobs that were waiting when it arrived.

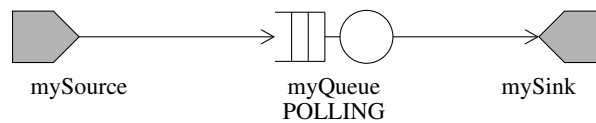


Figure 3.37: Topology of polling_gated: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: myClass1 (open), myClass2 (open)
Scheduling	myQueue: POLLING
Arrivals	mySource – myClass1: Exp(1); myClass2: Exp(0.8)
Service	myQueue – myClass1: Exp(4); myClass2: Exp(1.5)
Solvers	JMT

Table 3.38: Features of polling_gated.

The example is built and solved as follows.

```

Network model = CyclicPollingModel.polling_gated();

// The reference runs MVA first and JMT second, at this seed and run length.
try {
    new MVA(model).getAvgTable().print();
} catch (Exception e) {
    System.out.println("MVA failed: " + e.getMessage());
}
try {
    new JMT(model, "seed", 23000, "samples", 1000000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT failed: " + e.getMessage());
}

```

```
pauseForUser();
```

Running this edition prints

```
MVA analysis [method: default/stationtime; type: approximate, deterministic; lang: java; env: python]
  completed in <t>s.
Station  JobClass  QLen    Util   RespT   ResidT   ArvR   Tput
mySource  myClass1      0        0        0        0        0        1
mySource  myClass2      0        0        0        0        0      0.8
myQueue   myClass1    11.22     0.25    11.22    11.22        1        1
myQueue   myClass2    11.403   0.53333  14.254   14.254    0.8      0.8
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station  JobClass  QLen    Util   RespT   ResidT   ArvR   Tput
mySource  myClass1      0        0        0        0        0    1.0012
mySource  myClass2      0        0        0        0        0    0.80026
myQueue   myClass1    11.31   0.24989   11.34   11.34    1.0012    1.0008
myQueue   myClass2    11.312  0.53337   14.149   14.149   0.80026   0.79995

[Running in non-interactive mode, continuing...]
```

The rows repeat for each of the 2 solvers the script runs (MVA, JMT), which is the point of the example: the same model read by methods of different cost and different exactness.

3.6.4 polling_klimited

k -limited service, at most k jobs per visit.

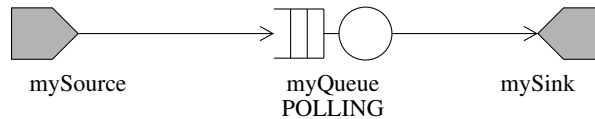


Figure 3.38: Topology of polling_klimited: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: myClass1 (open), myClass2 (open)
Scheduling	myQueue: POLLING
Arrivals	mySource – myClass1: Exp(0.2); myClass2: Exp(0.3)
Service	myQueue – myClass1: Exp(1); myClass2: Exp(1.5)
Solvers	JMT

Table 3.39: Features of polling_klimited.

The example is built and solved as follows.

```
Network model = CyclicPollingModel.polling_klimited();

// The reference runs MVA first and JMT second, at this seed and run length.
try {
    new MVA(model).getAvgTable().print();
} catch (Exception e) {
    System.out.println("MVA failed: " + e.getMessage());
}
try {
    new JMT(model, "seed", 23000, "samples", 100000).getAvgTable().print();
} catch (Exception e) {
    System.out.println("JMT failed: " + e.getMessage());
}

pauseForUser();
```

Running this edition prints

```

MVA analysis [method: default/stationtime; type: approximate, deterministic; lang: java; env: python]
  completed in <t>s.
Station  JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
mySource myClass1    0    0    0    0    0    0.2
mySource myClass2    0    0    0    0    0    0.3
myQueue  myClass1  0.66667  0.2  3.3333  3.3333  0.2    0.2
myQueue  myClass2  1.1333  0.2  3.7778  3.7778  0.3    0.3
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station  JobClass  QLen  Util  RespT  ResidT  ArvR  Tput
mySource myClass1    0    0    0    0    0  0.19924
mySource myClass2    0    0    0    0    0  0.30098
myQueue  myClass1  0.78887  0.19856  3.9795  3.9795  0.19924  0.20042
myQueue  myClass2  1.3909  0.2001  4.7392  4.7392  0.30098  0.30096

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (MVA, JMT), which is the point of the example: the same model read by methods of different cost and different exactness.

3.7 Switchover times

Moving the server between queues costs time. A switchover distribution is declared per arc of the polling cycle, and its mean adds directly to the cycle time, so the waiting time grows even though no extra work is done.

3.7.1 switchover_basic

Pairwise switchover metadata on an ordinary FCFS queue. The JMT run warns that non-polling queues do not consume these times and solves the underlying saturated two-class queue without them; use the cyclic-polling examples for honoured switchover delays.

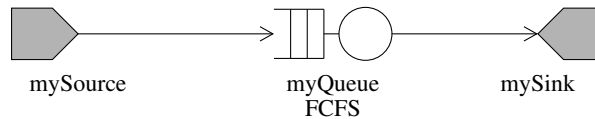


Figure 3.39: Topology of switchover_basic: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: myClass1 (open), myClass2 (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass1: Exp(0.2); myClass2: Exp(0.8)
Service	myQueue – myClass1: Exp(0.1); myClass2: Exp(1.5)
Solvers	JMT

Table 3.40: Features of switchover_basic.

The example is built and solved as follows.

```

Network model = SwitchoverTimesModel.switchover_basic();

// The reference's own seed and its own run length, which is the engine
// default: setOptions(defaultOptions()) would REPLACE the seed set here
// with 0, and a zero seed is drawn at random, so the row would not repeat.
NetworkSolver solver = new JMT(model, "seed", 23000, "keep", true);

try {
    solver.getAvgTable().print();
} catch (Exception e) {
}

```

```
pauseForUser();
```

Running this edition prints

```
jline.lang.Network@443b7951
```

3.8 Response time distribution

`getCdfRespT` returns the cumulative distribution of the response time, per station and per class, rather than its mean. The distribution is what a percentile service level objective is written against, and it is far more sensitive to the service distribution than the mean is: two models with the same mean response time can have very different tails.

3.8.1 `cdf_respt_closed`

The response time CDF at a station of a closed PS network.

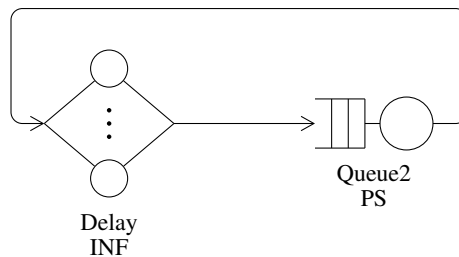


Figure 3.40: Topology of `cdf_respt_closed`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 1$)
Scheduling	Delay: INF; Queue2: PS
Service	Delay – Class1: Exp(10) Queue2 – Class1: Erlang(3,3)
Solvers	JMT, FLD

Table 3.41: Features of `cdf_respt_closed`.

The example is built and solved as follows.

```
Network model = CDFRespTModel.cdf_respt_closed();

NetworkSolver[] solvers = new NetworkSolver[] {
    new JMT(model, "seed", 12345),
    new FLD(model)
};

for (NetworkSolver solver : solvers) {
    try {
        if (solver instanceof JMT) {
            JMT jmtSolver = (JMT) solver;
            // CDF analysis for JMT solver
            // Note: getCdfRespT() returns DistributionResult, not Matrix
            // Example shows general structure for CDF analysis
        } else if (solver instanceof FLD) {
            FLD fluidSolver = (FLD) solver;
            // CDF analysis for Fluid solver
            // Note: getCdfRespT() returns DistributionResult, not Matrix
        }
    }
}
```

```

        // Example shows general structure for CDF analysis
    }

    solver.getAvgTable().print();
} catch (Exception e) {
}
}

pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.08793 0.08793 0.099239 0.099239 0.91269 0.9124
Queue2 Class1 0.91207 0.91207 0.99558 0.99558 0.9124 0.91224
Fluid analysis [method: minnormal; type: approximate, deterministic; lang: java; env: python] completed in
<t>s.
Fluid analysis [method: default/minnormal; type: approximate, deterministic; lang: java; env: python]
completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Delay Class1 0.090909 0.090909 0.1 0.1 0.90909 0.90909
Queue2 Class1 0.90909 0.90909 1 1 0.90909 0.90909

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, FLD), which is the point of the example: the same model read by methods of different cost and different exactness.

3.8.2 cdf_respt_closed_threeclasses

The three-class version, with Erlang service.

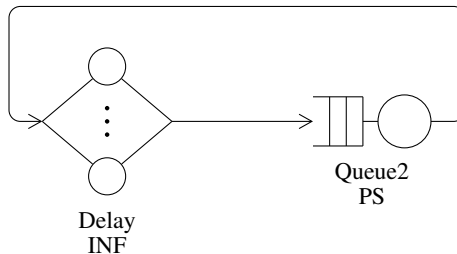


Figure 3.41: Topology of `cdf_respt_closed_threeclasses`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	3: Class1 (closed, $N = 1$), Class2 (closed, $N = 0$), Class3 (closed, $N = 0$)
Scheduling	Delay: INF; Queue2: PS
Service	Delay – Class1: Exp(1); Class2: Exp(1); Class3: Exp(1) Queue2 – Class1: Exp(1); Class2: Erlang(0.5,2); Class3: Exp(100)
Solvers	FLD

Table 3.42: Features of `cdf_respt_closed_threeclasses`.

The example is built and solved as follows.

```

Network model = CDFResPTModel.cdf_respt_closed_threeclasses();

NetworkSolver[] solvers = new NetworkSolver[] {
    new JMT(model, "seed", 12345),
    new FLD(model)
};

```

```

for (NetworkSolver solver : solvers) {
    try {

        if (solver instanceof JMT) {
            JMT jmtSolver = (JMT) solver;
            // The solver's OWN options, not a fresh defaultOptions(): the latter
            // draws a RANDOM seed in its constructor (SolverOptions ->
            // RandomManager.generateRandomSeed), so replacing the object wholesale
            // discards the seed passed to the constructor above and makes this
            // example irreproducible -- which is what left the ld_multiserver_ps
            // parity row disagreeing with its golden on a Monte-Carlo margin.
            SolverOptions options = jmtSolver.getOptions();
            options.samples = 100000;
            jmtSolver.setOptions(options);
        }

        solver.getAvgTable().print();
    } catch (Exception e) {
    }
}

pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.8.3 cdf_respt_distrib

The distribution object returned by `getCdfRespT` and the percentiles read from it.

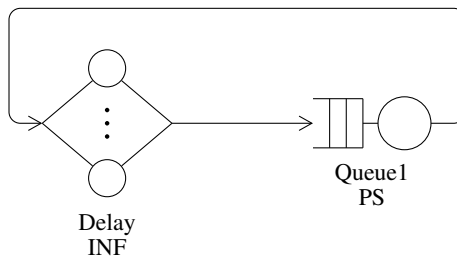


Figure 3.42: Topology of `cdf_respt_distrib`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 1$), Class2 (closed, $N = 3$)
Scheduling	Delay: INF; Queue1: PS
Service	Delay – Class1: Exp(1); Class2: Erlang(0.5,2) Queue1 – Class1: Exp(0.5); Class2: Hyper-Exp(0.99,0.01;0.324,0.005143)
Solvers	FLD, JMT

Table 3.43: Features of `cdf_respt_distrib`.

The example is built and solved as follows.

```

Network model = CDFRespTModel.cdf_respt_distrib();

NetworkSolver[] solvers = new NetworkSolver[] {
    new JMT(model, "seed", 12345),
    new FLD(model)
};

for (NetworkSolver solver : solvers) {
    try {

```

```

DistributionResult cdfResults = null;

if (solver instanceof JMT) {
    JMT jmtSolver = (JMT) solver;
    SolverOptions options = JMT.defaultOptions();
    options.samples = 100000;
    options.seed = 23000;
    jmtSolver.setOptions(options);
    // Use getTranCdfRespT for transient analysis like MATLAB
    cdfResults = jmtSolver.getTranCdfRespT();
    System.out.println("JMT Solver Transient CDF Analysis (Distributions):");
} else if (solver instanceof FLD) {
    FLD fluidSolver = (FLD) solver;
    cdfResults = fluidSolver.getCdfRespT();
    System.out.println("Fluid Solver CDF Analysis (Distributions):");
}

... (4 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

    solver.getAvgTable().print();
} catch (Exception e) {
}
}

pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
JMT Solver Transient CDF Analysis (Distributions):

=== JMT CDF Statistical Analysis ===
Station 1, Class 1: Mean=1.0005, Variance=1.0064, SCV=1.0054
Station 1, Class 2: Mean=3.9898, Variance=8.0429, SCV=0.5052
Station 2, Class 1: Mean=6.4328, Variance=43.7579, SCV=1.0575
Station 2, Class 2: Mean=16.5309, Variance=7664.7234, SCV=28.0482

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station  JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Delay   Class1      0.13469  0.13469  1.0011  1.0011  0.13486  0.13486
Delay   Class2      0.5882   0.5882   3.9895  3.9895  0.14254  0.14592
Queue1  Class1      0.86531  0.26808  6.4659  6.4659  0.13486  0.13484

... (10 captured-output lines omitted; rerun the source example for the full output)

Station  JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Delay   Class1      0.13398  0.13398      1          1  0.13398  0.13398
Delay   Class2      0.58492  0.58492      4          4  0.14623  0.14623
Queue1  Class1      0.86602  0.26797  6.4637  6.4637  0.13398  0.13398
Queue1  Class2      2.4151   0.73115 16.516 16.516  0.14623  0.14623

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, FLD), which is the point of the example: the same model read by methods of different cost and different exactness.

3.8.4 cdf_respt_open_twoclasses

The open two-class version, on FCFS stations.

The example is built and solved as follows.

```

Network model = CDFRespTModel.cdf_respt_open_twoclasses();

NetworkSolver[] solvers = new NetworkSolver[] {
    new JMT(model, "seed", 12345),
    new FLD(model)
};

```

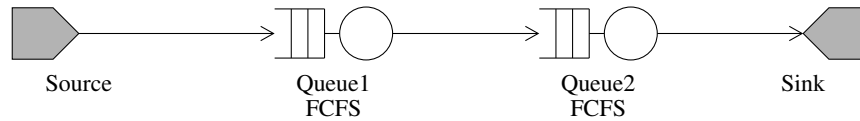



Figure 3.43: Topology of `cdf_respt_open_twoclasses`: an open network over 2 queueing stations.

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	Source – Class1: Exp(0.25); Class2: Exp(0.25)
Service	Queue1 – Class1: Exp(1); Class2: Exp(1) Queue2 – Class1: Exp(1); Class2: Exp(1)
Solvers	FLD, JMT

Table 3.44: Features of `cdf_respt_open_twoclasses`.

```

for (NetworkSolver solver : solvers) {
    try {
        DistributionResult cdfResults = null;

        if (solver instanceof JMT) {
            JMT jmtSolver = (JMT) solver;
            SolverOptions options = JMT.defaultOptions();
            options.samples = 10000;
            options.seed = 23000;
            jmtSolver.setOptions(options);
            // Use getTranCdfRespT for transient analysis like MATLAB
            cdfResults = jmtSolver.getTranCdfRespT();
            System.out.println("JMT Solver Transient CDF Analysis (Open Two Classes):");
        } else if (solver instanceof FLD) {
            FLD fluidSolver = (FLD) solver;
            SolverOptions options = FLD.defaultOptions();
            options.iter_max = 300;
            fluidSolver.setOptions(options);
            cdfResults = fluidSolver.getCdfRespT();

            ... (7 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

            solver.getAvgTable().print();
        } catch (Exception e) {
        }
    }
}

pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
JMT Solver Transient CDF Analysis (Open Two Classes):

=== JMT CDF Statistical Analysis ===
Station 2, Class 1: Mean=1.9357, Variance=3.7872, SCV=1.0107
Station 2, Class 2: Mean=1.9502, Variance=3.8886, SCV=1.0225
Station 3, Class 1: Mean=2.0904, Variance=4.4437, SCV=1.0169
Station 3, Class 2: Mean=2.0688, Variance=4.2944, SCV=1.0034

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
Station JobClass      QLen      Util      RespT      ResidT      ArvR      Tput
Source   Class1         0         0         0         0         0  0.24805
Source   Class2         0         0         0         0         0  0.25423
Queue1   Class1    0.48083    0.25248    1.9043    0.95215    0.24805  0.24804

... (14 captured-output lines omitted; rerun the source example for the full output)

```

```

Source   Class1      0      0      0      0      0      0.25
Source   Class2      0      0      0      0      0      0.25
Queue1   Class1    0.37514  0.25  1.5005  0.75027  0.25  0.25
Queue1   Class2    0.37514  0.25  1.5005  0.75027  0.25  0.25
Queue2   Class1    0.37514  0.25  1.5005  0.75027  0.25  0.25
Queue2   Class2    0.37514  0.25  1.5005  0.75027  0.25  0.25

```

[Running in non-interactive mode, continuing...]

The rows repeat for each of the 2 solvers the script runs (JMT, FLD), which is the point of the example: the same model read by methods of different cost and different exactness.

3.8.5 cdf_respt_populations

How the response time distribution moves as the closed population grows.

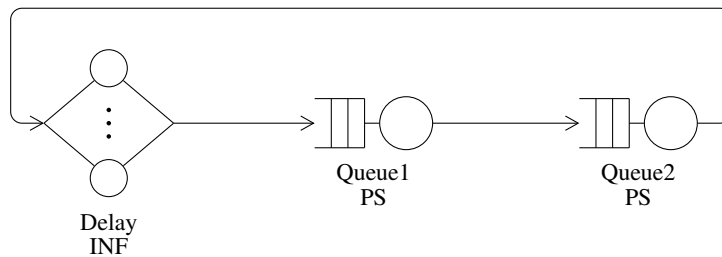


Figure 3.44: Topology of cdf_respt_populations: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	1: Class1 (closed, $N = 8$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS
Service	Delay – Class1: Exp(1) Queue1 – Class1: Exp(0.5) Queue2 – Class1: Exp(0.5)
Solvers	FLD

Table 3.45: Features of cdf_respt_populations.

The example is built and solved as follows.

```

Network model = CDFRespTModel.cdf_respt_populationsN4();

NetworkSolver[] solvers = new NetworkSolver[] {
    new JMT(model, "seed", 12345),
    new FLD(model)
};

for (NetworkSolver solver : solvers) {
    try {
        DistributionResult cdfResults = null;

        if (solver instanceof JMT) {
            JMT jmtSolver = (JMT) solver;
            SolverOptions options = JMT.defaultOptions();
            options.seed = 12345;
            jmtSolver.setOptions(options);
            cdfResults = jmtSolver.getCdfRespT();
            System.out.println("JMT Solver CDF Analysis (Populations):");
        } else if (solver instanceof FLD) {
            FLD fluidSolver = (FLD) solver;
            SolverOptions options = FLD.defaultOptions();
            options.iter_max = 100;
            fluidSolver.setOptions(options);
            cdfResults = fluidSolver.getCdfRespT();
            System.out.println("Fluid Solver CDF Analysis (Populations):");
        }
    }
}

```

```

... (5 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

    solver.getAvgTable().print();
} catch (Exception e) {
}
}

pauseForUser();

```

Running this edition prints

```

JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
JMT Model: <tmp>
JMT analysis [method: default; type: approximate, randomized; lang: java; env: python] completed in <t>s.
JMT Solver CDF Analysis (Populations):

=== JMT CDF Statistical Analysis ===
Station 1, Class 1: Mean=1.0043, Variance=0.9943, SCV=0.9859
Station 2, Class 1: Mean=4.6754, Variance=26.0549, SCV=1.1919
Station 3, Class 1: Mean=4.6939, Variance=26.6422, SCV=1.2092

Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay   Class1     0.38464 0.38464 1.0045  1.0045  0.38528 0.38519
Queue1  Class1     1.8326  0.77879 4.6705  4.6705  0.38519 0.38806
Queue2  Class1     1.7829  0.76221 4.6443  4.6443  0.38806 0.38539
Fluid analysis [method: dae; type: approximate, deterministic; lang: java; env: python] completed in <t>s.

... (6 captured-output lines omitted; rerun the source example for the full output)

Station 3, Class 1: Mean=4.2375, Variance=19.2421, SCV=1.0716

Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Delay   Class1     0.3986  0.3986  1         1       0.3986 0.3986
Queue1  Class1     1.8007  0.79721 4.5175  4.5175  0.3986 0.3986
Queue2  Class1     1.8007  0.79721 4.5175  4.5175  0.3986 0.3986

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (JMT, FLD), which is the point of the example: the same model read by methods of different cost and different exactness.

3.9 Busy periods

A busy period of order n runs from the instant an arrival raises the number of jobs held by the target to n until it falls back below n . The target may be a station, a station-class pair, or a declared set of stations, and the order n ranges over $1..K$. The analytical reference is the busy period formula of Daduna, exposed by the normalizing constant solver; the simulator measures the same quantity from the sample path.

3.9.1 busyp_subnetwork

Mean busy periods of order n for a station, for a station-class pair and for a declared subnetwork of stations.

Nodes	3: Queue $\times$ 3
Classes	1: Class1 (closed, $N = 5$)
Scheduling	Queue1: FCFS; Queue2: FCFS; Queue3: FCFS
Service	Queue1 – Class1: Exp(1.5) Queue2 – Class1: Exp(0.9) Queue3 – Class1: Exp(2)
Solvers	LDES

Table 3.46: Features of busyp_subnetwork.

The example is built and solved as follows.

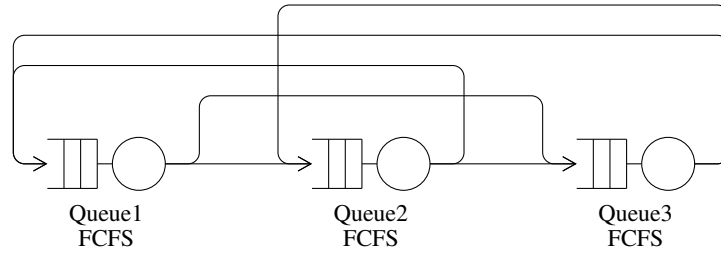


Figure 3.45: Topology of busyp_subnetwork: a closed network over 3 queueing stations.

```
int N = 5;
double[] rate = {1.5, 0.9, 2.0};
double[][] Pc = {{0, 0.6, 0.4}, {0.7, 0, 0.3}, {0.5, 0.5, 0}};

Network model = new Network("busyPeriodModel");
Queue[] q = new Queue[3];
for (int i = 0; i < 3; i++) {
    q[i] = new Queue(model, "Queue" + (i + 1), SchedStrategy.FCFS);
}
ClosedClass jobclass = new ClosedClass(model, "Class1", N, q[0], 0);
for (int i = 0; i < 3; i++) {
    q[i].setService(jobclass, new Exp(rate[i]));
}
RoutingMatrix routingMatrix = model.initRoutingMatrix();
for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        if (Pc[i][j] > 0) {
            routingMatrix.set(jobclass, jobclass, q[i], q[j], Pc[i][j]);
        }
    }
}
model.link(routingMatrix);

int[][] subnets = {{0}, {1}, {2}, {0, 1}};
int[] orders = {1, 3, 5};

... (19 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

for (int k = 0; k < orders.length; k++) {
    b[k] = ldesSolver.getAvgBusyPeriod(subnet, -1, orders[k]);
}
System.out.printf("%-18s %10.4f %10.4f %10.4f\n",
    java.util.Arrays.toString(subnet), b[0], b[1], b[2]);
}
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.10 Stochastic network calculus

Every other family in these examples reports a mean. The stochastic network calculus reports a *tail*: given a violation probability ε it returns a delay or a queue length that is exceeded with probability at most ε , under any work-conserving scheduling policy at the station. Each flow is carried by a moment-generating-function envelope and each server by the counting process of its service; the min-plus operations compose them across a feed-forward network, at the cost of the burstiness a server adds to the flow it forwards. The bound on the mean that follows by integration is loose, and the decay rate of the tail is asymptotically exact, so the quantile is what these examples read.

3.10.1 snc_delay_quantile

A delay quantile with a certified violation probability, and how it tightens deep in the tail.

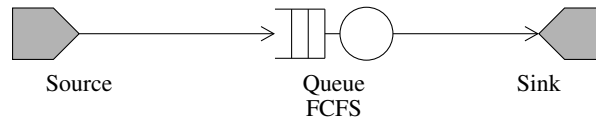


Figure 3.46: Topology of snc_delay_quantile: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(0.6)
Service	Queue – Class1: Exp(1)

Table 3.47: Features of snc_delay_quantile.

The example is built and solved as follows.

```

final double lambda = 0.6;
final double mu = 1.0;
Network model = new Network("SncQuantile");
Source source = new Source(model, "Source");
Queue queue = new Queue(model, "Queue", SchedStrategy.FCFS);
Sink sink = new Sink(model, "Sink");
OpenClass jobclass = new OpenClass(model, "Class1");
source.setArrival(jobclass, new Exp(lambda));
queue.setService(jobclass, new Exp(mu));
RoutingMatrix P = model.initRoutingMatrix();
P.set(jobclass, jobclass, Network.serialRouting(source, queue, sink));
model.link(P);

SolverBA solver = new SolverBA(model, "snc.upper");

// getPercTable is to the snc family what getAvgTable is to a mean
// solver: one row per station and class, reporting the response-time
// and queue-length quantiles at the requested violation probability.
NetworkPercTable percTable = solver.getPercTable(1e-3);
percTable.print();

// The exact M/M/1 sojourn tail is exp(-(mu-lambda)*d) and the exact
// queue-length tail is rho^(n+1). The bound reproduces both DECAY RATES
// exactly and pays a constant prefactor, so the ratio of the bounded
// quantile to the exact one falls towards 1 as eps is tightened: the
// family is at its best exactly where simulation is at its worst.

... (35 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

System.out.printf("    the optimal theta approaches log(mu/lambda) = %.4f, which is%n",
    Math.log(mu / lambda));
System.out.printf("    what makes the backlog decay rate exact%n");
SncResult back = Snc_bound_delay.snc_bound_delay(arv, srv, d.value);
System.out.printf("api: P{D > %.4f} <= %.3e (theta = %.4f), exact tail %.3e%n",
    d.value, back.value, back.theta, Math.exp(-(mu - lambda) * d.value));
  
```

Running this edition prints

Station	JobClass	RespTPerc	QLenPerc			
Queue	Class1	29.608	23.645			
eps	d bound	d exact	ratio	n bound	n exact	ratio
1e-02	23.3454	11.5129	2.028	18.7266	8.0152	2.336
1e-03	29.6082	17.2694	1.714	23.6449	12.5227	1.888
1e-06	47.9527	34.5388	1.388	38.0313	26.0455	1.460

```

1e-09      65.9597    51.8082    1.273      52.1418    39.5682    1.318
1e-12      83.7930    69.0776    1.213      66.1120    53.0909    1.245
BA analysis [method: snc.upper; type: upper bound, deterministic; lang: java; env: python] completed in <t
>s.
Station  JobClass    QLen  Util   RespT   ResidT   ArvR   Tput
Source   Class1         0     0       0       0       0    0.6
Queue    Class1     7.9231  0.6   13.205  13.205   0.6    0.6

exact M/M/1: R = 2.5000, Q = 1.5000
snc.upper : R = 13.2051 (5.3x), Q = 7.9231 (5.3x)

api: d(1e-3) = 29.6082 at theta = 0.4632
      the optimal theta approaches log(mu/lambda) = 0.5108, which is
      what makes the backlog decay rate exact
api: P{D > 29.6082} <= 1.000e-03 (theta = 0.4632), exact tail 7.187e-06

```

3.10.2 snc_tandem_multiclass

Envelope propagation across a feed-forward tandem, pay-bursts-only-once, blind multiplexing, and the model classes the family refuses.

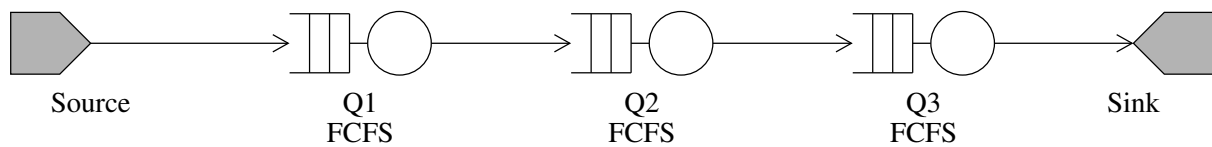


Figure 3.47: Topology of snc_tandem_multiclass: an open network over 3 queueing stations.

Nodes	5: Source, Queue $\times$ 3, Sink
Classes	1: Class1 (open)
Scheduling	Q1: FCFS; Q2: FCFS; Q3: FCFS
Arrivals	Source – Class1: Exp(0.6)
Service	Q1 – Class1: Exp(1.5) Q2 – Class1: Exp(1.2) Q3 – Class1: Exp(1)

Table 3.48: Features of snc_tandem_multiclass.

The example is built and solved as follows.

```

final double lambda = 0.6;
final double[] rates = {1.5, 1.2, 1.0};
Network model = new Network("SncTandem");
Source source = new Source(model, "Source");
Queue q1 = new Queue(model, "Q1", SchedStrategy.FCFS);
Queue q2 = new Queue(model, "Q2", SchedStrategy.FCFS);
Queue q3 = new Queue(model, "Q3", SchedStrategy.FCFS);
Sink sink = new Sink(model, "Sink");
OpenClass jobclass = new OpenClass(model, "Class1");
source.setArrival(jobclass, new Exp(lambda));
q1.setService(jobclass, new Exp(rates[0]));
q2.setService(jobclass, new Exp(rates[1]));
q3.setService(jobclass, new Exp(rates[2]));
RoutingMatrix P = model.initRoutingMatrix();
P.set(jobclass, jobclass, Network.serialRouting(source, q1, q2, q3, sink));
model.link(P);

NetworkAvgTable tandemTable = new SolverBA(model, "snc.upper").getAvgTable();
tandemTable.print();

// Each hop replaces the arrival envelope by the DEPARTURE envelope of
// the station upstream, which carries the burst the server has added.
// The exact answer does not degrade this way -- by Burke's theorem the
// departure process of an M/M/1 is again Poisson -- so the ratio to the
// exact response time grows hop by hop. The bound stays valid; it is the
// price of assuming nothing about the departure process.

```

```

... (55 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

SolverBA sharedSolver = new SolverBA(shared, "snc.upper");
sharedSolver.getAvgTable().print();
sharedSolver.getPercTable(1e-3).print();
System.out.printf("exact per-class response time (aggregate M/M/1, lambda=0.6, mu=1): %.4f%n"
    , 1.0 / (1.0 - 0.6));

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.11 Random environments

A random environment is a Markov process over stages, with one model per stage. The environment moves between stages according to a transition process, and the network parameters change with it; the metrics reported are the environment average, weighted by the stage probabilities, but the transient handoff between stages matters when the stage sojourn is comparable with the relaxation time of the network.

Server breakdown and repair is the smallest instance: two stages, one with the server up and one with it down. The case studies at the end of the family are larger, and are the models on which the environment solver is validated.

3.11.1 example_mapqn2renv

Converting a queueing network with MMPP(2) service into an equivalent random environment model.

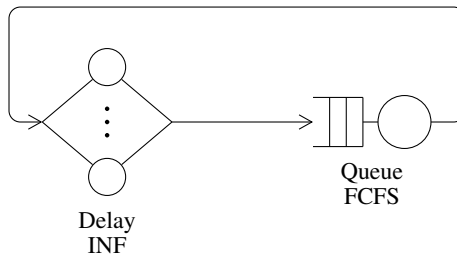


Figure 3.48: Topology of example_mapqn2renv: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 5$)
Scheduling	Delay: INF; Queue: FCFS
Service	Delay – Class1: Exp(1) Queue – Class1: MMPP2
Solvers	JMT

Table 3.49: Features of example_mapqn2renv.

The example is built and solved as follows.

```

System.out.println("=== MMPP2 Closed QN to Random Environment Transformation ===\n");
Network model = RandomEnvironmentModel.mmppClosed("MMPP_ClosedQN");
System.out.println("Original Network:");
System.out.println("  Topology: Delay -> MMPP2 Queue -> Delay (cyclic)");
System.out.println("  Population: N = 5 jobs");
System.out.println("  Think time: Exp(1.0)");
System.out.println("  Service: MMPP2 with phases 0,1\n");

```

```

Environment envModel = MAPQN2RENV.mapqn2renv(model);
System.out.println("Environment stages:");
envModel.printStageTable();

int E = envModel.getNumberOfStages();

// The fluid coupling, over a finite stage horizon: the mean-field exit
// metrics are a Riemann-Stieltjes sum of the stage trajectory against
// the holding-time CDF, so the horizon is part of the answer.
SolverOptions envOptions = new SolverOptions(SolverType.ENV);
envOptions.iter_max = 100;
envOptions.iter_tol = 0.01;
envOptions.method = "default";
envOptions.verbose = VerboseLevel.SILENT;

SolverOptions fldOptions = new SolverOptions(SolverType.FLUID);
fldOptions.timespan = new double[]{0, 100};
fldOptions.verbose = VerboseLevel.SILENT;

    ... (51 source lines omitted; full implementation: jar/src/main/java/jline/examples/)
}
} catch (Exception ex) {
    System.out.println("Stage solver error: " + ex.getMessage());
}
System.out.println("\n=== Transformation Complete ===");
pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.11.2 `renv_basic`

The smallest random environment: two stages, one model per stage, and the stage transition process. No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

Solvers	FLD, MVA
---------	----------

Table 3.50: Features of `renv_basic`.

The example is built and solved as follows.

```

// Block 1: Create base network model
Network baseModel = new Network("BaseModel");
Delay delay = new Delay(baseModel, "ThinkTime");
Queue queue = new Queue(baseModel, "Fast/Slow Server", SchedStrategy.FCFS);

// Closed class with 5 jobs
int N = 5;
ClosedClass jobclass = new ClosedClass(baseModel, "Jobs", N, delay, 0);
delay.setService(jobclass, new Exp(1.0)); // Think time = 1.0
queue.setService(jobclass, new Exp(2.0)); // Placeholder service rate

// Connect nodes in a cycle
baseModel.link(Network.serialRouting(delay, queue));

// Block 2: Create the random environment
int E = 2;
Environment env = new Environment("ServerModes", E);

// Stage 0: Fast mode (service rate = 4.0)
Network fastModel = baseModel.copy();
Queue fastQueue = (Queue) fastModel.getNodeByName("Fast/Slow Server");

```



```

fastQueue.setService(fastModel.getClasses().get(0), new Exp(4.0));
env.addStage(0, "Fast", "operational", fastModel);

// Stage 1: Slow mode (service rate = 1.0)
Network slowModel = baseModel.copy();

... (37 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

System.out.printf("%nStage: %s (prob = %.4f)%n",
    env.getStageName(e), env.probEnv.get(0, e));
new MVA(env.getModel(e), printing(SolverType.MVA)).getAvgTable().print();
}

return env;

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.11.3 renv_container_terminal

A container terminal case study: MAP arrivals, multiserver and load-dependent stations, several environment stages.

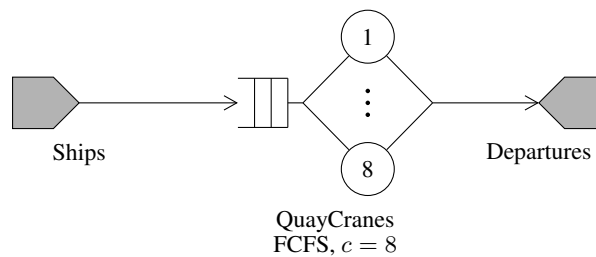


Figure 3.49: Topology of renv_container_terminal: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Containers (open)
Scheduling	QuayCranes: FCFS ($c = 8$)
Arrivals	Ships – Containers: MAP
Service	QuayCranes – Containers: Exp(16)

Table 3.51: Features of renv_container_terminal.

The example is built and solved as follows.

```

double[] dailyRateHr = {6, 30, 40, 62, 76, 79, 119, 164, 152, 130, 79, 70,
    57, 57, 113, 130, 162, 202, 148, 118, 92, 62, 36, 8};
double[] dailyDurationHr = {0.51, 0.84, 0.76, 0.98, 0.67, 2.33, 1.80, 0.62,
    0.36, 1.30, 1.02, 0.98, 0.86, 1.56, 1.30, 1.57,
    0.34, 0.22, 0.47, 1.33, 1.56, 0.93, 0.71, 1.11};
int E = dailyRateHr.length; // hourly environment stages
int N = 6; // straddle carriers circulating (the closed tokens)
int nCranes = 2; // quay cranes, a multi-server FCFS station
double craneRate = 16; // moves per hour served by one crane

Network[] stage = new Network[E];
for (int h = 0; h < E; h++) {
    // The per-token yard completion rate of hour h: the closed-network
    // image of that hour's external demand intensity.
    stage[h] = RandomEnvironmentModel.terminalModel(dailyRateHr[h] / N, craneRate, nCranes, N);
}
Environment env = dailyCycle("RotterdamDailyCycle", stage, dailyDurationHr);

```

```

System.out.printf("Rotterdam container terminal: %d hourly stages, %d carriers, %d cranes.%n"
    ,
    E, N, nCranes);

// The stage solver is an exact CTMC over a 12-hour transient horizon,
// which covers the hour-duration CDF past 99.9% for every stage, so the
// quadrature is not truncating the blend.
double T = 12;
SolverOptions statevecOpt = envOptions("statevec");
SolverOptions meanfieldOpt = envOptions("meanfield");

    ... (97 source lines omitted; full implementation: jar/src/main/java/jline/examples/)
}
double meanLambda = num / den;
System.out.printf("MMPP mean arrival = %.2f/hr, %d cranes @ %.0f/hr, rho = %.3f%n",
    meanLambda, cOpen, craneRate, meanLambda / (cOpen * craneRate));
new MAM(openModel).getAvgTable().print();
pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.11.4 renv_fourstages_repairmen

The four-stage version.

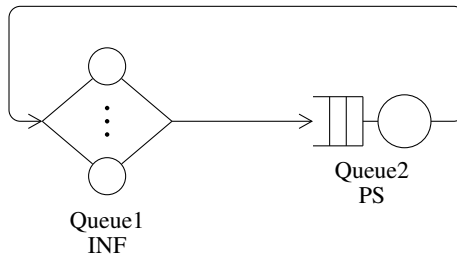


Figure 3.50: Topology of renv_fourstages_repairmen: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 30$)
Scheduling	Queue1: INF; Queue2: PS
Service	Queue1 – Class1: Exp(4) Queue2 – Class1: Exp(1)
Solvers	ENV

Table 3.52: Features of renv_fourstages_repairmen.

The example is built and solved as follows.

```

Environment envModel = RandomEnvironmentModel.renv_fourstages_repairmen();
int E = envModel.getEnsemble().size();

// THIS FIXED POINT IS TOLERANCE-DEPENDENT: run to convergence it lands
// on Queue1 Tput 0.97136 where the reference's iter_tol = 0.05 stops it
// at 0.9716, and the golden holds the latter. So the controls are stated
// rather than left at the engine default.
SolverOptions options = new SolverOptions(SolverType.ENV);
options.timespan = new double[]{0, Double.POSITIVE_INFINITY};
options.iter_max = 100;
options.iter_tol = 0.05;
options.method = "default";
options.verbose = VerboseLevel.STD;

```

```

SolverOptions fluidOptions = new SolverOptions(SolverType.FLUID);
fluidOptions.timespan = new double[]{0, Double.POSITIVE_INFINITY};
fluidOptions.verbose = VerboseLevel.SILENT;

// Create solvers for each stage
NetworkSolver[] solvers = new NetworkSolver[E];
for (int e = 0; e < E; e++) {
    solvers[e] = new FLD(envModel.getModel(e));
    solvers[e].options = fluidOptions.copy();
}

// Create environment solver

... (3 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

solver.getAvgTable().print();
} catch (Exception e) {
    e.printStackTrace();
}

pauseForUser();

```

Running this edition prints

```
jline.lang.Environment@4f2410ac
```

3.11.5 renv_genqn

A randomly generated network placed in an environment.

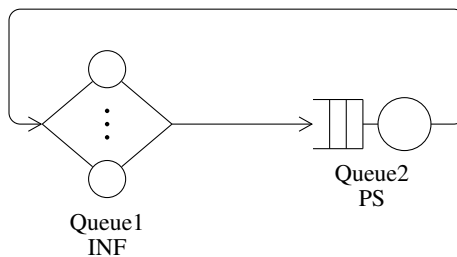


Figure 3.51: Topology of renv_genqn: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 4$)
Scheduling	Queue1: INF; Queue2: PS
Service	Queue1 – Class1: Exp(1) Queue2 – Class1: Exp(0.5)
Solvers	MVA

Table 3.53: Features of renv_genqn.

The example is built and solved as follows.

```

Network model = RandomEnvironmentModel.renv_genqn(1.0, 0.5, 4);
new MVA(model).getAvgTable().print();
pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.11.6 `renv_lqn_twostages`

A layered model in a two-stage environment, including its transient.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows.

```
LayeredNetwork upModel = buildLQN("LQN_UP", 0.8); // fast DB activity
LayeredNetwork downModel = buildLQN("LQN_DOWN", 3.0); // slow DB activity (degraded)

Environment env = new Environment("DBReliability", 2);
env.addStage(0, "UP", "operational", upModel);
env.addStage(1, "DOWN", "degraded", downModel);
env.addTransition(0, 1, new Exp(0.2)); // mean UP time = 5
env.addTransition(1, 0, new Exp(1.0)); // mean DOWN time = 1
env.init();

double T = 50;
SolverENV envSolver = lqnEnvSolver(env, T, 20, 0.02);
envSolver.getEnsembleAvg();
Matrix QN = envSolver.result.QN;
Matrix TN = envSolver.result.TN;
System.out.printf("ENV over LQN ran: aggregate size = %d x %d\n",
    QN.getNumRows(), QN.getNumCols());
envSolver.getAvgTable().print();

Matrix[] up = stageAggregate(upModel, T);
Matrix[] down = stageAggregate(downModel, T);

// (1) The solver runs and returns finite, population-conserving metrics.
for (int i = 0; i < QN.getNumRows(); i++) {
    for (int k = 0; k < QN.getNumCols(); k++) {
        if (Double.isNaN(QN.get(i, k)) || Double.isInfinite(QN.get(i, k))) {

            ... (42 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

        }
    }
}

if (x3 < lo - tol || x3 > hi + tol) {
    throw new RuntimeException("Three-stage ENV-averaged throughput is not bracketed");
}

System.out.println("PASS: ENV-over-LQN meanfield ran; throughput bracketed, monotone in P(UP)
    , 3-stage bracketed.");
pauseForUser();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.11.7 `renv_map_fallback`

An MMPP(2) stage transition process, and the fallback taken when it cannot be represented exactly.

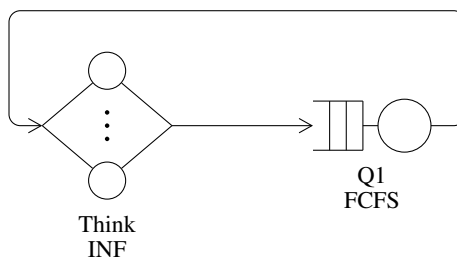


Figure 3.52: Topology of `renv_map_fallback`: a closed network over 2 queueing stations.

The example is built and solved as follows.

Nodes	2: Queue, Delay
Classes	1: C1 (closed, $N = 5$)
Scheduling	Think: INF; Q1: FCFS
Service	Think – C1: Exp(1) Q1 – C1: MMPP2
Solvers	MVA, ENV, CTMC

Table 3.54: Features of `renv_map_fallback`.

```

Network model = RandomEnvironmentModel.mmppClosed("mmppClosed");

// The environment image the solver builds for itself, shown here.
SolverOptions imgOptions = new SolverOptions(SolverType.MVA);
MAPQN2RENV.RenvImage image = MAPQN2RENV.map2renvImage(model, imgOptions);
System.out.printf("Random-environment image: %d stages, phase orders %s, MMPP image: %b%n",
    image.nstages, java.util.Arrays.toString(image.orders), image.isMMPP);
image.env.printStageTable();

// MVA reaches the answer THROUGH the fallback; CTMC takes the MMPP
// natively and is exact, so it is the reference the fallback is read
// against.
System.out.println("\n--- MVA (through the random-environment fallback) ---");
new MVA(model).getAvgTable().print();
System.out.println("\n--- CTMC (native MMPP, exact) ---");
new CTMC(model).getAvgTable().print();

// The two environment LIMITS, forced. 'dec' is the quasi-stationary
// limit of a SLOW environment (each phase reaches its own steady state
// before the next switch); 'avg' is the rate-averaged limit of a FAST
// one (the phases blur into their mean rate).
for (String limit : new String[]{"dec", "avg"}) {
    SolverOptions options = new SolverOptions(SolverType.MVA);
    options.config.map_env_method = limit;
    System.out.println("\n--- MVA, map_env_method = '" + limit + "' ---");
    new MVA(model, options).getAvgTable().print();

    ... (7 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

    new MVA(model, off).getAvgTable();
    System.out.println("map_env='off': the model was accepted, which it should not be");
} catch (Exception ex) {
    System.out.println("\nmap_env='off': " + ex.getMessage());
}
pauseForUser();

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.11.8 `renv_node_breakdown`

Server breakdown and repair as an environment over a queueing model.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

Solvers	FLD, ENV
----------------	----------

Table 3.55: Features of `renv_node_breakdown`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run

the identified implementation to obtain its results.

3.11.9 `renv_rotterdam_blending`

The Rotterdam blending case study.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows.

```
double[] rateHr = {6, 30, 40, 62, 76, 79, 119, 164, 152, 130, 79, 70,
                  57, 57, 113, 130, 162, 202, 148, 118, 92, 62, 36, 8};
double[] durHr = {0.51, 0.84, 0.76, 0.98, 0.67, 2.33, 1.80, 0.62,
                  0.36, 1.30, 1.02, 0.98, 0.86, 1.56, 1.30, 1.57,
                  0.34, 0.22, 0.47, 1.33, 1.56, 0.93, 0.71, 1.11};
int K = rateHr.length;
double[] lambda = new double[K];
double[] durMin = new double[K];
double[] fracW = new double[K];
double totDur = 0;
for (int h = 0; h < K; h++) {
    lambda[h] = rateHr[h] * 0.5 / 60.0; // arrivals per MINUTE
    durMin[h] = durHr[h] * 60.0;
    totDur += durMin[h];
}
for (int h = 0; h < K; h++) {
    fracW[h] = durMin[h] / totDur;
}

// The study's simulated ground truth, N = 24..34 (Table 13).
double[] simW = {154.0661, 118.3936, 99.962, 88.2379, 80.9272, 75.0887,
                 70.745, 67.4326, 64.9719, 62.7913, 61.2534};
double[] simQex = {89.8508, 63.3970, 49.3650, 40.8783, 35.3546, 30.6543,
                  27.4111, 24.7218, 22.3622, 20.3893, 18.9252};
int tailFactor = 15; // M_trunc = N*(1 + tailFactor)
SoqnParams p = new SoqnParams();

... (11 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

int j = N - 24;
System.out.printf("%4d | %10.4f %10.4f %6.2f | %10.4f %10.4f %6.2f%n", N, r[0], simW[j],
                  100 * Math.abs(r[0] - simW[j]) / simW[j], r[1], simQex[j],
                  100 * Math.abs(r[1] - simQex[j]) / simQex[j]);
}
pauseForUser();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.11.10 `renv_threestages_repairmen`

The three-stage version.

Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 2$)
Scheduling	Queue1: INF; Queue2: PS
Service	Queue1 – Class1: Exp(3) Queue2 – Class1: Exp(1)
Solvers	ENV

Table 3.56: Features of `renv_threestages_repairmen`.

The example is built and solved as follows.

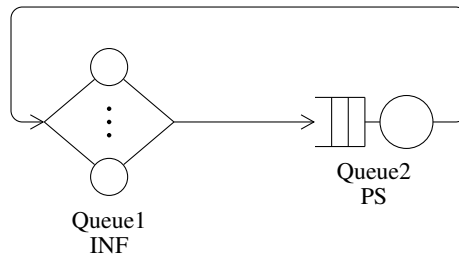


Figure 3.53: Topology of `renv_threestages_repairmen`: a closed network over 2 queueing stations.

```
Environment envModel = RandomEnvironmentModel.renv_threestages_repairmen();
int E = envModel.getEnsemble().size();

// CTMC STAGES, as the reference pins: each stage is solved exactly and
// the horizon is the whole line, so the stage solve is a steady state.
SolverOptions envOptions = new SolverOptions(SolverType.ENV);
envOptions.timespan = new double[]{0, Double.POSITIVE_INFINITY};
envOptions.iter_max = 100;
envOptions.iter_tol = 0.05;
envOptions.method = "default";

// The stage horizon stays at the CTMC default, which is open at both
// ends: SolverENV reads timespan[1] to choose between a steady-state and
// a transient stage solve, and an open horizon is the steady state the
// reference asks each stage for.
SolverOptions ctmcOptions = new SolverOptions(SolverType.CTMC);
ctmcOptions.stiff = false;
ctmcOptions.verbose = VerboseLevel.SILENT;

// Create solvers for each stage
NetworkSolver[] solvers = new NetworkSolver[E];
for (int e = 0; e < E; e++) {
    solvers[e] = new CTMC(envModel.getModel(e));
    solvers[e].options = ctmcOptions.copy();
}

... (4 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

solver.getAvgTable().print();
} catch (Exception e) {
    e.printStackTrace();
}

pauseForUser();
```

Running this edition prints

```
ENV analysis [method: default; type: approximate, deterministic; lang: java; env: python] completed in <t>
s. Iterations: 3.
Station JobClass QLen Util RespT ResidT ArvR Tput
Queue1 Class1 0.83071 0.83071 0.62313 0 0 1.3331
Queue2 Class1 1.1693 0.75799 0.857 0 0 1.3644

[Running in non-interactive mode, continuing...]
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by ENV.

3.11.11 `renv_twostages_repairmen`

A repairman model in a two-stage random environment.

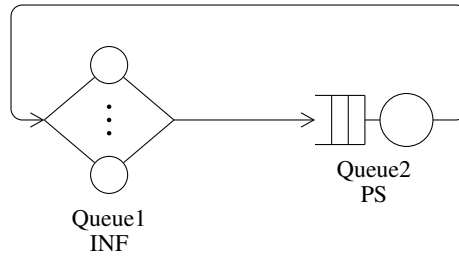


Figure 3.54: Topology of `renv_twestages_repairmen`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 1$)
Scheduling	Queue1: INF; Queue2: PS
Service	Queue1 – Class1: Exp(2) Queue2 – Class1: Exp(1)
Solvers	ENV

Table 3.57: Features of `renv_twestages_repairmen`.

The example is built and solved as follows.

```
Environment envModel = RandomEnvironmentModel.renv_twestages_repairmen();
int E = envModel.getEnsemble().size();

// THE ITERATION CONTROLS AND THE STAGE HORIZON ARE PART OF THE GOLDEN.
// An environment couples TRANSIENT stage solves, so a row given a
// different horizon or integrator answers a different question: the
// reference states timespan [0, 1e3] on the fluid solver and leaves the
// integrator at its default, and overriding `stiff` or the ODE max step
// walks a different trajectory.
SolverOptions options = new SolverOptions(SolverType.ENV);
options.timespan = new double[]{0, Double.POSITIVE_INFINITY};
options.iter_max = 100;
options.iter_tol = 0.01;
options.method = "default";
options.verbose = VerboseLevel.STD;

SolverOptions fluidOptions = new SolverOptions(SolverType.FLUID);
fluidOptions.timespan = new double[]{0, 1000};
fluidOptions.verbose = VerboseLevel.SILENT;

// Create solvers for each stage
NetworkSolver[] solvers = new NetworkSolver[E];
for (int e = 0; e < E; e++) {
    solvers[e] = new FLD(envModel.getModel(e));
    solvers[e].options = fluidOptions.copy();
}

... (5 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

solver.getAvgTable().print();
} catch (Exception e) {
    e.printStackTrace();
}

pauseForUser();
```

Running this edition prints

```
ENV analysis [method: default; type: approximate, deterministic; lang: java; env: python] completed in <t>
s. Iterations: 5.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Queue1   Class1   0.55553 0.55553 0.80361      0      0   0.6913
Queue2   Class1   0.44447 0.44447 0.64288      0      0   0.69137
```


[Running in non-interactive mode, continuing...]

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by ENV.

3.12 Layered cache-queueing models

A `CacheTask` inside a layered model exposes `ItemEntry` points, and a call to one is a cache request whose hit or miss outcome selects the downstream branch. The replacement policy is simulated directly, so hit and miss are exact sample-path events rather than a hit-ratio parameter.

3.12.1 `lcq_async_cache`

An ASYNCHRONOUS (non-blocking) call into the cache task, as a prefetch or a warm-up makes: the client continues without waiting for the reply while the request still resolves through `POST_CACHE` hit/miss branching.

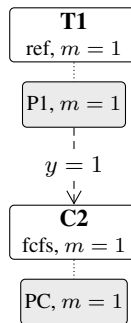


Figure 3.55: Call graph of `lcq_async_cache`: 2 tasks over 2 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	2: P1 (ps, $m = 1$), PC (ps, $m = 1$)
Tasks	2: T1 (ref, $m = 1$), C2 (fcfs, $m = 1$)
Entries	2: E1, I2
Activities	4, host demands A1: 0.0, AC2: 0.0, AC2h: 1.0, AC2m: 2.0
Calls	1 (0 synchronous, 1 asynchronous)

Table 3.58: Features of `lcq_async_cache`.

The example is built and solved as follows.

```

LayeredNetwork model = LayeredCQModel.lcq_async_cache();

SolverOptions lnoptions = LN.defaultOptions();
lnoptions.verbose = jline.VerboseLevel.STD;
LN solver = new LN(model, (subModel) -> new jline.solvers.mva.MVA(subModel, "verbose",
    jline.VerboseLevel.SILENT), lnoptions);

solver.getAvgTable().print();

pauseForUser();
  
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.12.2 lcq_singlehost

A cache task and its item entries inside a layered model on one processor.

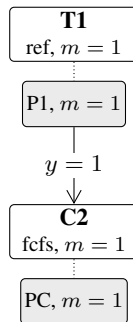


Figure 3.56: Call graph of `lcq_singlehost`: 2 tasks over 2 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	2: P1 (ps, $m = 1$), PC (ps, $m = 1$)
Tasks	2: T1 (ref, $m = 1$), C2 (fcfs, $m = 1$)
Entries	2: E1, I2
Activities	4, host demands A1: 0.0, AC2: 0.0, AC2h: 1.0, AC2m: 2.0
Calls	1 (1 synchronous, 0 asynchronous)

Table 3.59: Features of `lcq_singlehost`.

The example is built and solved as follows.

```

LayeredNetwork model = LayeredCQModel.lcq_singlehost();

// MVA LAYERS, as the reference pins: MVA layers and NC layers are
// different fixed points, and on this model they part company by 4.5%.
LN solver = new LN(model, (subModel) -> new jline.solvers.mva.MVA(subModel, "verbose",
    jline.VerboseLevel.SILENT));

try {
    solver.getAvgTable().print();
} catch (Exception e) {
}

pauseForUser();
  
```

Running this edition prints

```

Iter 1. Analyze time: 0.171s. Update time: 0.013s. Runtime: 0.172s.
Iter 2. Analyze time: 0.030s. Update time: 0.009s. Runtime: 0.215s.
Iter 3. Analyze time: 0.029s. Update time: 0.009s. Runtime: 0.254s.
Iter 4. Analyze time: 0.026s. Update time: 0.010s. Runtime: 0.290s.
Iter 5. Analyze time: 0.028s. Update time: 0.010s. Runtime: 0.329s.
Iter 6. Analyze time: 0.027s. Update time: 0.018s. Runtime: 0.368s.
Iter 7. Analyze time: 0.028s. Update time: 0.009s. Runtime: 0.414s.
Iter 8. Analyze time: 0.026s. Update time: 0.008s. Runtime: 0.451s.
Iter 9. Analyze time: 0.028s. Update time: 0.010s. Runtime: 0.489s.
Iter 10. Analyze time: 0.023s. Update time: 0.008s. Runtime: 0.522s.
Iter 11. Analyze time: 0.022s. Update time: 0.006s. Runtime: 0.554s.
Iter 12. Analyze time: 0.017s. Update time: 0.007s. Runtime: 0.578s.
Iter 13. Analyze time: 0.020s. Update time: 0.006s. Runtime: 0.607s.
Iter 14. Analyze time: 0.016s. Update time: 0.006s. Runtime: 0.630s.
Iter 15. Analyze time: 0.017s. Update time: 0.006s. Runtime: 0.653s.
Iter 16. Analyze time: 0.019s. Update time: 0.006s. Runtime: 0.679s.

... (42 captured-output lines omitted; rerun the source example for the full output)

E1   Entry           1           0       1.5      NaN      NaN    0.66667
I2   Entry           1           1       1.5      NaN      NaN    0.66667
  
```

A1	Activity	1	0	1.5	0	NaN	0.66667
AC2	Activity	0	0	0	0	NaN	0
AC2h	Activity	0.33333	0.33333	1	0.5	NaN	0.33333
AC2m	Activity	0.66667	0.66667	2	1	NaN	0.33333

[Running in non-interactive mode, continuing...]

3.12.3 lcq_threehosts

The three-processor version, where the cache and its clients are on separate hosts.

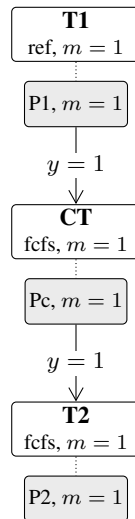


Figure 3.57: Call graph of `lcq_threehosts`: 3 tasks over 3 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	3: P1 (ps, $m = 1$), Pc (ps, $m = 1$), P2 (ps, $m = 1$)
Tasks	3: T1 (ref, $m = 1$), CT (fcfs, $m = 1$), T2 (fcfs, $m = 1$)
Entries	3: E1, IE, E2
Activities	5, host demands A1: 0.0, Ac: 0.0, Ac_hit: 1.0, Ac_miss: 2.0, A2: 0.2
Calls	2 (2 synchronous, 0 asynchronous)

Table 3.60: Features of `lcq_threehosts`.

The example is built and solved as follows.

```

LayeredNetwork model = LayeredCQModel.lcq_threehosts();

// NC layers FIRST, then MVA layers: the reference runs both, and the
// golden holds the first table (goldens/corpus.json multiModelExamples).
LN ncLayers = new LN(model, (subModel) -> new jline.solvers.nc.NC(subModel, "verbose",
    jline.VerboseLevel.SILENT));
LN mvaLayers = new LN(model, (subModel) -> new jline.solvers.mva.MVA(subModel, "verbose",
    jline.VerboseLevel.SILENT));

try {
    ncLayers.getAvgTable().print();
    mvaLayers.getAvgTable().print();
} catch (Exception e) {
}

pauseForUser();
  
```

Running this edition prints

```

Iter 1. Analyze time: 0.386s. Update time: 0.016s. Runtime: 0.388s.
Iter 2. Analyze time: 0.202s. Update time: 0.018s. Runtime: 0.606s.
Iter 3. Analyze time: 0.173s. Update time: 0.013s. Runtime: 0.798s.
Iter 4. Analyze time: 0.154s. Update time: 0.013s. Runtime: 0.967s.
Iter 5. Analyze time: 0.199s. Update time: 0.022s. Runtime: 1.179s.
Iter 6. Analyze time: 0.222s. Update time: 0.016s. Runtime: 1.426s.
Iter 7. Analyze time: 0.173s. Update time: 0.012s. Runtime: 1.616s.
Iter 8. Analyze time: 0.171s. Update time: 0.017s. Runtime: 1.801s.
Iter 9. Analyze time: 0.180s. Update time: 0.015s. Runtime: 1.999s.
Iter 10. Analyze time: 0.221s. Update time: 0.011s. Runtime: 2.236s.
Iter 11. Analyze time: 0.206s. Update time: 0.010s. Runtime: 2.454s.
Iter 12. Analyze time: 0.186s. Update time: 0.009s. Runtime: 2.651s.
Iter 13. Analyze time: 0.140s. Update time: 0.009s. Runtime: 2.801s.
Iter 14. Analyze time: 0.111s. Update time: 0.009s. Runtime: 2.922s.
Iter 15. Analyze time: 0.103s. Update time: 0.007s. Runtime: 3.035s.
Iter 16. Analyze time: 0.093s. Update time: 0.008s. Runtime: 3.136s.

... (115 captured-output lines omitted; rerun the source example for the full output)

E2      Entry      0.0625  0.0625  0.2      NaN      NaN      0.3125
A2      Activity    0.0625  0.0625  0.2      0.2      NaN      0.3125
A1      Activity      1         0      1.6      0      NaN      0.625
Ac      Activity      0         0         0         0      NaN      0
Ac_hit  Activity    0.3125  0.3125  1         0.5     NaN      0.3125
Ac_miss Activity    0.6875  0.625  2.2         1      NaN      0.3125

[Running in non-interactive mode, continuing...]

```

3.13 Pass and swap

Under the pass-and-swap discipline a job that finds an incompatible job in service swaps with it rather than waiting, subject to a compatibility relation between classes. The initial placement of a closed population is therefore part of the model: the state must be set in the order the discipline assumes, and a placement that ignores it changes the answer.

3.13.1 pas_closed_tandem_fig6

The closed tandem of the reference paper, figure 6.

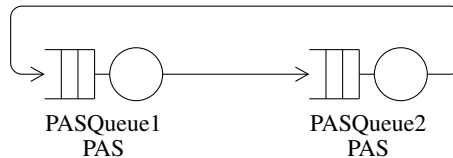


Figure 3.58: Topology of pas_closed_tandem_fig6: a closed network over 2 queueing stations.

Nodes	2: Queue $\times$ 2
Classes	6: Class1 (closed, $N = 1$), Class2 (closed, $N = 1$), Class3 (closed, $N = 1$), Class4 (closed, $N = 1$), Class5 (closed, $N = 1$), Class6 (closed, $N = 1$)
Scheduling	PASQueue1: PAS; PASQueue2: PAS
Service	PASQueue1 – Class1: Exp(1); Class2: Exp(1); Class3: Exp(1); Class4: Exp(1); Class5: Exp(1); Class6: Exp(1) PASQueue2 – Class1: Exp(1.3); Class2: Exp(1.3); Class3: Exp(1.3); Class4: Exp(1.3); Class5: Exp(1.3); Class6: Exp(1.3)
Features	finite buffer 6 at PASQueue1; finite buffer 6 at PASQueue2; class switching on 2 links
Solvers	CTMC

Table 3.61: Features of pas_closed_tandem_fig6.

The example is built and solved as follows.

```
System.out.println("=== Closed tandem of two PAS queues (Fig 5/6), CTMC ===");
ctmc().print();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.13.2 pas_compatibility_5class

Five classes and the compatibility matrix that says which may swap with which.

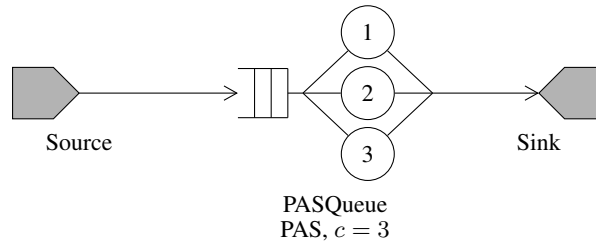


Figure 3.59: Topology of pas_compatibility_5class: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	5: Class1 (open), Class2 (open), Class3 (open), Class4 (open), Class5 (open)
Scheduling	PASQueue: PAS ($c = 3$)
Arrivals	Source – Class1: Exp(0.5); Class2: Exp(0.4); Class3: Exp(0.3); Class4: Exp(0.2); Class5: Exp(0.1)
Service	PASQueue – Class1: Exp(1); Class2: Exp(1); Class3: Exp(1); Class4: Exp(2); Class5: Exp(1)
Features	finite buffer 3 at PASQueue
Solvers	CTMC

Table 3.62: Features of pas_compatibility_5class.

The example is built and solved as follows.

```
Network model = compatibilityModel();
SolverOptions opt = SolverCTMC.defaultOptions();
opt.method = "exact"; // pin the state-space path
opt.cutoff = Matrix.singleton(3);
return new SolverCTMC(model, opt).getAvgTable();
```

Running this edition prints

```
WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 3 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
CTMC analysis [method: exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.
jline.solvers.NetworkAvgTable@74f0ea28
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by CTMC.

3.13.3 pas_cyclic_ctmc_vs_bruteforce

A cyclic pass-and-swap network solved by CTMC and by brute-force enumeration, as a cross-check. No topology is drawn for this example: the captured run yielded no Network or LayeredNetwork to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows.

Solvers

CTMC, LDES

Table 3.63: Features of pas_cyclic_ctmc_vs_bruteforce.

```

Object[] bf = bruteforce();
double G = (Double) bf[0];
double[] Xbf = (double[]) bf[1];
double[][] Qbf = (double[][]) bf[2];
System.out.printf("Brute-force normalizing constant G = %.12g%n%n", G);

int total = 0;
for (int k : K) total += k;

// (3) CTMC -- exact
SolverOptions opt = SolverCTMC.defaultOptions();
opt.method = "exact"; // pin the state-space path
opt.cutoff = Matrix.singleton(total);
NetworkAvgTable tc = new SolverCTMC(buildModel(), opt).getAvgTable();
tc.print();
double[] Xc = extractX(tc);
double[][] Qc = extractQ(tc);
double errX = 0, errQ = 0;
for (int r = 0; r < R; r++) errX = Math.max(errX, Math.abs(Xbf[r] - Xc[r]));
for (int i = 0; i < 2; i++)
    for (int r = 0; r < R; r++) errQ = Math.max(errQ, Math.abs(Qbf[i][r] - Qc[i][r]));
System.out.printf("%nCTMC: max|dX| = %.3e    max|dQ| = %.3e%n", errX, errQ);
if (errX > 1e-9 || errQ > 1e-9)
    throw new RuntimeException("CTMC vs brute-force mismatch");
System.out.println("PASS: CTMC matches brute-force product form within 1.0e-9.");

... (10 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

for (int r = 0; r < R; r++) relQ = Math.max(relQ, Math.abs(Qbf[i][r] - Ql[i][r]) / Qbf[i][r]);
System.out.printf("LDES: max rel|dX| = %.2f%%    max rel|dQ| = %.2f%%%n",
    100 * relX, 100 * relQ);
if (relX > 0.02 || relQ > 0.02)
    throw new RuntimeException("LDES vs brute-force exceeds 2%");
System.out.println("PASS: LDES matches brute-force within 2% (simulation noise).");

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.13.4 pas_cyclic_normconst_bruteforce

The same cross-check on the normalizing constant.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows.

```

double[] e1 = new double[R];
double[] e2 = new double[R];
Arrays.fill(e1, 1.0);
Arrays.fill(e2, 1.0);
double G = normConst(K, e1, e2, MU1, MU2);

StringBuilder kstr = new StringBuilder();
for (int r = 0; r < R; r++) {
    if (r > 0) {
        kstr.append(' ');
    }
    kstr.append(K[r]);
}

```

```

}
System.out.printf("Closed cyclic PAS network: R=%d classes, population K=[%s]\n",
    R, kstr);
System.out.printf("Brute-force normalizing constant G = %.15g\n", G);

// X_r = e_r * G(K - l_r) / G(K); with e_r = 1 this is a ratio of
// normalizing constants at populations K-l_r and K.
StringBuilder xstr = new StringBuilder();
for (int r = 0; r < R; r++) {
    double X = 0;
    if (K[r] > 0) {
        int[] Km = K.clone();
        Km[r]--;
        X = normConst(Km, e1, e2, MU1, MU2) / G;

        ... (4 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

        xstr.append(String.format("%.6g", X));
    }
}
System.out.printf("Per-class throughput X = [%s]\n", xstr);

selftestSingleClass();
return G;

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.13.5 pas_mmk

The pass-and-swap discipline on an M/M/k station.

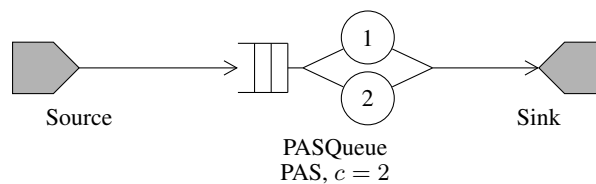


Figure 3.60: Topology of pas_mmk: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	PASQueue: PAS ($c = 2$)
Arrivals	Source – Class1: Exp(0.7); Class2: Exp(0.5)
Service	PASQueue – Class1: Exp(1); Class2: Exp(1)
Features	finite buffer 4 at PASQueue
Solvers	CTMC

Table 3.64: Features of pas_mmk.

The example is built and solved as follows.

```

Network model = mmkModel();
SolverOptions opt = SolverCTMC.defaultOptions();
opt.method = "exact"; // pin the state-space path
opt.cutoff = Matrix.singleton(4);
return new SolverCTMC(model, opt).getAvgTable();

```

Running this edition prints

```

WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 4 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
CTMC analysis [method: exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.

```

```
jline.solvers.NetworkAvgTable@61d47554
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by CTMC.

3.13.6 pas_nc_sampling

The normalizing constant of a pass-and-swap network, by sampling.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows.

```
System.out.println("=== CTMC (exact) ===");
NetworkAvgTable Tc = ctmc();
System.out.println(
    "=== SolverNC method='sampling' (importance sampling, 5e5 samples) ===");
SolverOptions ncOpt = SolverNC.defaultOptions();
ncOpt.method = "sampling";
ncOpt.samples = 500000;
ncOpt.seed = 777;
ncOpt.verbose = VerboseLevel.SILENT;
NetworkAvgTable Tn = new SolverNC(model(), ncOpt).getAvgTable();

List<Double> qc = Tc.getQLen();
List<Double> qn = Tn.getQLen();
List<String> stations = Tc.getStationNames();
List<String> classes = Tc.getClassNames();
System.out.printf("%n station      class      CTMC      NC-samp%n");
double err = 0;
for (int i = 0; i < qc.size(); i++) {
    System.out.printf("  %-9s  %-6s  %9.5f  %9.5f%n",
        stations.get(i), classes.get(i), qc.get(i), qn.get(i));
    err = Math.max(err, Math.abs(qn.get(i) - qc.get(i)));
}
System.out.printf(
    "%nNC-samp vs CTMC:  max|dQ| = %.3e (importance-sampling noise)%n", err);
if (err > 5e-2) {
    throw new RuntimeException(
        "NC sampling does not match the exact CTMC within IS tolerance");
}
System.out.println("PASS: SolverNC 'sampling' matches the exact CTMC within "
    + "importance-sampling noise.");
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.13.7 pas_saturation

The saturated regime, where the swap rule sets the throughput.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

Solvers	CTMC, LDES
---------	------------

Table 3.65: Features of pas_saturation.

The example is built and solved as follows.

```
// compatibility comp(s,i) = 1 iff server s serves class i; capacities cap_s
final int[][] comp = {
```



```

        {1, 0},    // server 1 -> class 1
        {1, 1},    // server 2 -> classes 1, 2 (shared)
        {0, 1}};   // server 3 -> class 2
final double[] capS = {2, 1, 2};
final double[] lambda = {1.5, 1.0};
final int I = 2;
final int S = 3;

// Rank function: total capacity of servers compatible with a present class.
ServiceRate muRank = c -> {
    boolean[] present = new boolean[I];
    for (int j = 0; j < c.getNumCols(); j++) {
        int r = (int) c.get(0, j);
        if (r >= 0 && r < I) {
            present[r] = true;
        }
    }
    double total = 0;
    for (int s = 0; s < S; s++) {
        for (int i = 0; i < I; i++) {
            if (present[i] && comp[s][i] == 1) {
                total += capS[s];
                break;
            }
        }
    }

    ... (39 source lines omitted; full implementation: jar/src/main/java/jline/examples/)
} catch (RuntimeException e) {
    System.out.println(" " + e.getMessage());
}

System.out.println("\nPASS: PAS saturation handling matches CTMC on a large buffer "
    + "and errors cleanly on infinite buffers.");

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.13.8 pas_selfloop

A self-looping class under pass-and-swap.

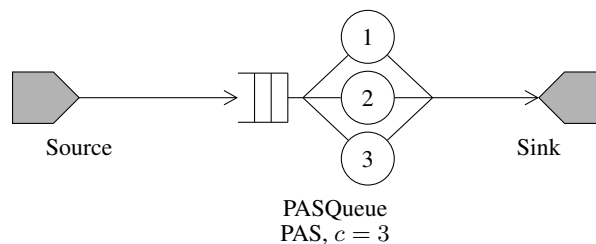


Figure 3.61: Topology of pas_selfloop: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	3: Class1 (open), Class2 (open), Class3 (open)
Scheduling	PASQueue: PAS ($c = 3$)
Arrivals	Source – Class1: Exp(0.6); Class2: Exp(0.4); Class3: Exp(0.3)
Service	PASQueue – Class1: Exp(1); Class2: Exp(1); Class3: Exp(1)
Features	finite buffer 3 at PASQueue
Solvers	CTMC

Table 3.66: Features of pas_selfloop.

The example is built and solved as follows.

```

Network model = selfloopModel();
SolverOptions opt = SolverCTMC.defaultOptions();
opt.method = "exact"; // pin the state-space path
opt.cutoff = Matrix.singleton(3);
return new SolverCTMC(model, opt).getAvgTable();

```

Running this edition prints

```

WARNING: [runAnalyzerBody] CTMC solver using state space cutoff = 3 for open/mixed model. State space
truncation may cause inaccurate results. Consider varying cutoff to assess sensitivity.
CTMC analysis [method: exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.
jline.solvers.NetworkAvgTable@69b794e2

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by CTMC.

3.14 Reward models

A reward model attaches a rate to each state, or a value to each transition, and reports the expected accumulated reward. Cost per waiting job, revenue per completion and penalty per loss are the usual three, and together they turn a performance model into an economic one. Rewards may be declared over aggregate states when the detailed state space is too large to enumerate.

3.14.1 rewardModel_aggregation

Rewards over aggregated states, when the detailed state is too large to enumerate.

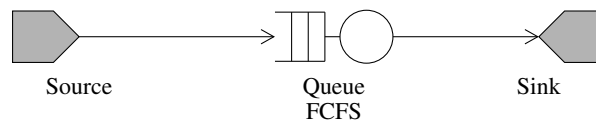


Figure 3.62: Topology of rewardModel_aggregation: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: HighPriority (open), LowPriority (open)
Scheduling	Queue: FCFS
Arrivals	Source – HighPriority: Exp(1); LowPriority: Exp(0.8)
Service	Queue – HighPriority: Exp(3); LowPriority: Exp(3)
Features	finite buffer 4 at Queue

Table 3.67: Features of rewardModel_aggregation.

The example is built and solved as follows.

```

Node[] n = new Node[3];
Network model = rewardModel("RewardAggregationExample", SchedStrategy.FCFS, 1, 4, n);
final Queue queue = (Queue) n[1];
OpenClass c1 = new OpenClass(model, "HighPriority", 0);
OpenClass c2 = new OpenClass(model, "LowPriority", 0);
((Source) n[0]).setArrival(c1, new Exp(1.0));
((Source) n[0]).setArrival(c2, new Exp(0.8));
queue.setService(c1, new Exp(3.0));
queue.setService(c2, new Exp(3.0));
model.link(Network.serialRouting(model.getClasses(), n[0], n[1], n[2]));

final int i1 = model.getJobClassIndex(c1);
final int i2 = model.getJobClassIndex(c2);
model.setReward("TotalJobs", Reward.queueLength(queue));

```

```

model.setReward("MaxClass", new RewardFunction() {
    private static final long serialVersionUID = 1L;

    public double compute(Matrix s, NetworkStruct sn) {
        return Math.max(at(s, sn, queue, i1), at(s, sn, queue, i2));
    }
});
model.setReward("ClassCount", new RewardFunction() {
    private static final long serialVersionUID = 1L;

    public double compute(Matrix s, NetworkStruct sn) {
        return (at(s, sn, queue, i1) > 0 ? 1.0 : 0.0)

        ... (8 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

    public double compute(Matrix s, NetworkStruct sn) {
        return 2.0 * at(s, sn, queue, i1) + at(s, sn, queue, i2);
    }
});

reportSteadyState(new SolverCTMC(model));

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.14.2 rewardModel_mm1k

State rewards on an M/M/1/K queue: cost per waiting job, revenue per completion, penalty per loss.

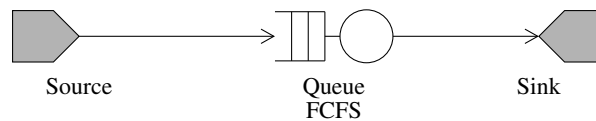


Figure 3.63: Topology of rewardModel_mm1k: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(2)
Service	Queue – Class1: Exp(3)
Features	finite buffer 3 at Queue
Solvers	CTMC

Table 3.68: Features of rewardModel_mm1k.

The example is built and solved as follows.

```

Node[] n = new Node[3];
Network model = rewardModel("RewardExample", SchedStrategy.FCFS, 1, 3, n);
final Queue queue = (Queue) n[1];
final OpenClass oclass = new OpenClass(model, "Class1", 0);
((Source) n[0]).setArrival(oclass, new Exp(2.0));
queue.setService(oclass, new Exp(3.0));
model.link(Network.serialRouting(n[0], n[1], n[2]));

final int c = model.getJobClassIndex(oclass);
model.setReward("QueueLength", Reward.queueLength(queue, oclass));
model.setReward("Utilization", Reward.utilization(queue, oclass));
model.setReward("BlockingProb", Reward.blocking(queue));
model.setReward("QueueCost", new RewardFunction() {
    private static final long serialVersionUID = 1L;

```

```

    public double compute(Matrix state, NetworkStruct sn) {
        double jobs = at(state, sn, queue, c);
        return jobs * jobs;
    }
});

SolverCTMC solver = new SolverCTMC(model);
reportSteadyState(solver);

Print.section("CTMC transient, timespan [0, 5]");
SolverOptions topt = SolverCTMC.defaultOptions();

    ... (25 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

}
Map<String, Double> R = solver.getAvgReward();
compare("QueueLength", R.get("QueueLength"), L);
compare("Utilization", R.get("Utilization"), 1.0 - pi[0]);
compare("BlockingProb", R.get("BlockingProb"), pi[kbuf]);
compare("QueueCost", R.get("QueueCost"), cost);

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.14.3 rewardModel_multiclass

Per-class rewards on a multiclass PS station.

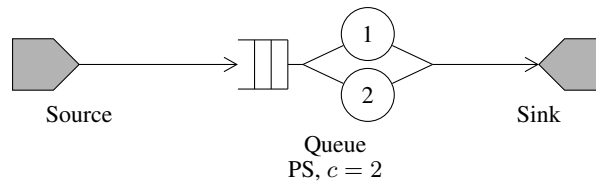


Figure 3.64: Topology of rewardModel_multiclass: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Interactive (open), Batch (open)
Scheduling	Queue: PS ($c = 2$)
Arrivals	Source – Interactive: Exp(2); Batch: Exp(1.5)
Service	Queue – Interactive: Exp(0.5); Batch: Exp(1)
Features	finite buffer 6 at Queue

Table 3.69: Features of rewardModel_multiclass.

The example is built and solved as follows.

```

Node[] n = new Node[3];
Network model = rewardModel("MultiClassRewardExample", SchedStrategy.PS, 2, 6, n);
final Queue queue = (Queue) n[1];
OpenClass ci = new OpenClass(model, "Interactive", 0);
OpenClass cb = new OpenClass(model, "Batch", 0);
((Source) n[0]).setArrival(ci, new Exp(2.0));
((Source) n[0]).setArrival(cb, new Exp(1.5));
queue.setService(ci, new Exp(0.5));
queue.setService(cb, new Exp(1.0));
model.link(Network.serialRouting(model.getClasses(), n[0], n[1], n[2]));

final int i1 = model.getJobClassIndex(ci);
final int i2 = model.getJobClassIndex(cb);
model.setReward("Interactive_QLen", Reward.queueLength(queue, ci));
model.setReward("Interactive_Util", Reward.utilization(queue, ci));

```

```

model.setReward("Batch_QLen", Reward.queueLength(queue, cb));
model.setReward("Batch_Util", Reward.utilization(queue, cb));
model.setReward("Total_QLen", Reward.queueLength(queue));
model.setReward("Total_Util", Reward.utilization(queue));
model.setReward("Interactive_Ratio", new RewardFunction() {
    private static final long serialVersionUID = 1L;

    public double compute(Matrix s, NetworkStruct sn) {
        return at(s, sn, queue, i1) / Math.max(at(s, sn, queue, i2), 0.001);
    }
});

... (30 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

if (R.get("Total_QLen") > 0) {
    Print.kv("Interactive share (%)", 100.0 * R.get("Interactive_QLen") / R.get("Total_QLen"));
}
Print.kv("Batch share (%)", 100.0 * R.get("Batch_QLen") / R.get("Total_QLen"));
Print.kv("Interactive resp. time", R.get("Interactive_QLen") / lambdaInt);
Print.kv("Batch resp. time", R.get("Batch_QLen") / lambdaBatch);

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.14.4 rewardModel_templates

The reward templates that cover the common cases without writing a reward function.

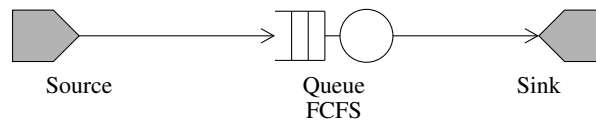


Figure 3.65: Topology of rewardModel_templates: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(1.5)
Service	Queue – Class1: Exp(2)
Features	finite buffer 10 at Queue

Table 3.70: Features of rewardModel_templates.

The example is built and solved as follows.

```

Node[] n = new Node[3];
Network model = rewardModel("RewardTemplatesExample", SchedStrategy.FCFS, 1, 10, n);
Queue queue = (Queue) n[1];
OpenClass oclass = new OpenClass(model, "Class1", 0);
((Source) n[0]).setArrival(oclass, new Exp(1.5));
queue.setService(oclass, new Exp(2.0));
model.link(Network.serialRouting(n[0], n[1], n[2]));

model.setReward("QueueLength", Reward.queueLength(queue));
model.setReward("QueueLength_Class1", Reward.queueLength(queue, oclass));
model.setReward("Utilization", Reward.utilization(queue));
model.setReward("Utilization_Class1", Reward.utilization(queue, oclass));
model.setReward("BlockingProb", Reward.blocking(queue));

SolverCTMC solver = new SolverCTMC(model);
reportSteadyState(solver);

```

```
Print.section("M/M/1/K analytical");
double rho = 1.5 / 2.0;
int kbuf = 10;
double[] pi = mmlkPi(rho, kbuf);
double L = 0;
for (int j = 0; j <= kbuf; j++) {
    L += j * pi[j];
}
Map<String, Double> R = solver.getAvgReward();
compare("QueueLength", R.get("QueueLength"), L);
compare("Utilization", R.get("Utilization"), 1.0 - pi[0]);
compare("BlockingProb", R.get("BlockingProb"), pi[kbuf]);
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.15 Agent models

The agent interface builds a network from the point of view of the entities that move through it rather than from the stations. The resulting models are ordinary networks – Jackson networks, closed multiclass networks, G-networks with their negative customers – and are solved by the same solvers.

3.15.1 ag_closed_network

The closed counterpart.

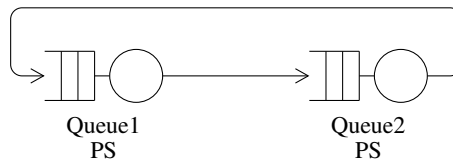


Figure 3.66: Topology of ag_closed_network: a closed network over 2 queueing stations.

Nodes	2: Queue × 2
Classes	1: Class1 (closed, $N = 10$)
Scheduling	Queue1: PS; Queue2: PS
Service	Queue1 – Class1: Exp(2) Queue2 – Class1: Exp(1)
Solvers	MVA, CTMC

Table 3.71: Features of ag_closed_network.

The example is built and solved as follows.

```
System.out.println("=== Closed Network (2 PS Queues, 10 jobs) ===\n");

Network model = AgentModel.closedNetwork();

// AG with INAP method
System.out.println("AG (method=inap):");
NetworkSolver solverInap = new AG(model, "inap");
solverInap.getAvgTable().print();

// AG with exact method
System.out.println("\nAG (method=exact):");
NetworkSolver solverExact = new AG(model, "exact");
solverExact.getAvgTable().print();

// MVA for comparison
System.out.println("\nMVA:");
```

```

NetworkSolver solverMVA = new MVA(model);
solverMVA.getAvgTable().print();

// CTMC for exact results
System.out.println("\nCTMC (exact):");
NetworkSolver solverCTMC = new CTMC(model);
solverCTMC.getAvgTable().print();

pauseForUser();

```

Running this edition prints

```

=== Closed Network (2 PS Queues, 10 jobs) ===

AG (method=inap):
AG analysis [method: inap; type: approximate, deterministic; lang: java; env: python] completed in <t>s.
  Iterations: 91.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Queue1  Class1     2.4134  0.72839  1.6567  1.6567  0.99485  1.4568
Queue2  Class1     8.1985  0.99485  8.2409  8.2409  1.4568  0.99485

AG (method=exact):
WARNING: [solver_ag] AutoCAT (exact method) is not available. Falling back to INAP.
AG analysis [method: inap; type: approximate, deterministic; lang: java; env: python] completed in <t>s.
  Iterations: 91.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Queue1  Class1     2.4134  0.72839  1.6567  1.6567  0.99485  1.4568
Queue2  Class1     8.1985  0.99485  8.2409  8.2409  1.4568  0.99485

MVA:

... (4 captured-output lines omitted; rerun the source example for the full output)

CTMC (exact):
CTMC analysis [method: default; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Queue1  Class1     0.99463  0.49976  0.99511  0.99511  0.99951  0.99951
Queue2  Class1     9.0054  0.99951  9.0098  9.0098  0.99951  0.99951

[Running in non-interactive mode, continuing...]

```

The rows repeat for each of the 2 solvers the script runs (MVA, CTMC), which is the point of the example: the same model read by methods of different cost and different exactness.

3.15.2 ag_gnetwork

A G-network, with the negative customers that remove jobs on arrival.

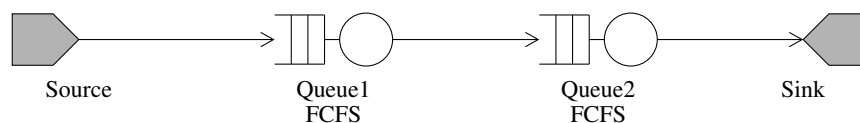


Figure 3.67: Topology of ag_gnetwork: an open network over 2 queueing stations.

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	2: Positive (open), Negative (signal)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	Source – Positive: Exp(1); Negative: Exp(0.3)
Service	Queue1 – Positive: Exp(2); Negative: Exp(2) Queue2 – Positive: Exp(3); Negative: Exp(3)

Table 3.72: Features of ag_gnetwork.

The example is built and solved as follows.

```

System.out.println("=== G-Network (Gelenbe Network) with Negative Customers ===");

```

```

System.out.println("Positive arrival rate: 1.0");
System.out.println("Negative signal rate: 0.3");
System.out.println("Service rates: mu1=2.0, mu2=3.0\n");

Network model = AgentModel.gNetwork();

// AG with INAP method
System.out.println("AG (method=inap):");
NetworkSolver solverInap = new AG(model, "inap");
solverInap.getAvgTable().print();

// Theoretical insight
System.out.println("\nNote: In G-networks, negative customers reduce the effective");
System.out.println("      load at target queues by removing jobs upon arrival.");
System.out.println("      The utilization at Queue2 is lower than it would be");
System.out.println("      without negative signals due to job removals.");

pauseForUser();

```

Running this edition prints

```

=== G-Network (Gelenbe Network) with Negative Customers ===
Positive arrival rate: 1.0
Negative signal rate: 0.3
Service rates: mu1=2.0, mu2=3.0

AG (method=inap):
AG analysis [method: inap; type: approximate, deterministic; lang: java; env: python] completed in <t>s.
  Iterations: 2.

```

Station	JobClass	QLen	Util	RespT	ResidT	ArvR	Tput
Source	Positive	0	0	0	0	0	1
Source	Negative	0	0	0	0	0	0.3
Queue1	Positive	0.76923	0.43478	0.88462	0.88462	1	0.86957
Queue1	Negative	0	0	0	0	0.3	0
Queue2	Positive	0.40816	0.28986	0.46939	0.46939	0.86957	0.86957

```

Note: In G-networks, negative customers reduce the effective
      load at target queues by removing jobs upon arrival.
      The utilization at Queue2 is lower than it would be
      without negative signals due to job removals.

[Running in non-interactive mode, continuing...]

```

3.15.3 ag_jackson_network

A Jackson network built agent by agent.

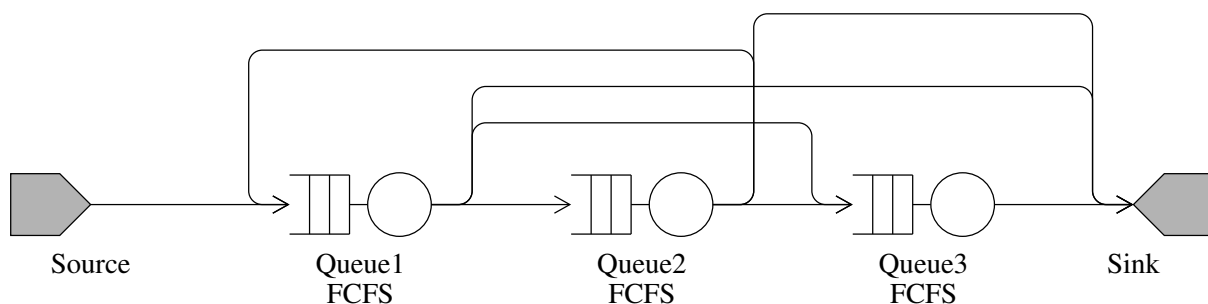


Figure 3.68: Topology of ag_jackson_network: an open network over 3 queueing stations.

The example is built and solved as follows.

```

System.out.println("=== Jackson Network (3 Queues) ===\n");

Network model = AgentModel.jacksonNetwork();

// AG with INAP method
System.out.println("AG (method=inap):");

```


Nodes	5: Source, Queue $\times$ 3, Sink
Classes	1: Class1 (open)
Scheduling	Queue1: FCFS; Queue2: FCFS; Queue3: FCFS
Arrivals	Source – Class1: Exp(1)
Service	Queue1 – Class1: Exp(2) Queue2 – Class1: Exp(3) Queue3 – Class1: Exp(2.5)
Solvers	MVA

Table 3.73: Features of ag_jackson_network.

```

NetworkSolver solverInap = new AG(model, "inap");
solverInap.getAvgTable().print();

// AG with exact method
System.out.println("\nAG (method=exact):");
NetworkSolver solverExact = new AG(model, "exact");
solverExact.getAvgTable().print();

// MVA for comparison
System.out.println("\nMVA:");
NetworkSolver solverMVA = new MVA(model);
solverMVA.getAvgTable().print();

pauseForUser();

```

Running this edition prints

```

=== Jackson Network (3 Queues) ===

AG (method=inap):
AG analysis [method: inap; type: approximate, deterministic; lang: java; env: python] completed in <t>s.
  Iterations: 8.
Station  JobClass    QLen    Util    RespT    ResidT    ArvR    Tput
Source   Class1        0        0        0        0        0        1
Queue1   Class1      1.1905    0.54349  1.0953    1.1905    1.087    1.087
Queue2   Class1      0.16949  0.14492  0.38983  0.16949  0.43479  0.43477
Queue3   Class1      0.22342  0.18262  0.48937  0.22341  0.45653  0.45655

AG (method=exact):
WARNING: [solver_ag] AutoCAT (exact method) is not available. Falling back to INAP.
AG analysis [method: inap; type: approximate, deterministic; lang: java; env: python] completed in <t>s.
  Iterations: 8.
Station  JobClass    QLen    Util    RespT    ResidT    ArvR    Tput
Source   Class1        0        0        0        0        0        1
Queue1   Class1      1.1905    0.54349  1.0953    1.1905    1.087    1.087

... (4 captured-output lines omitted; rerun the source example for the full output)

MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 54.
Station  JobClass    QLen    Util    RespT    ResidT    ArvR    Tput
Source   Class1        0        0        0        0        0        1
Queue1   Class1      1.1905    0.54348  1.0952    1.1905    1.087    1.087
Queue2   Class1      0.16949  0.14493  0.38983  0.16949  0.43478  0.43478
Queue3   Class1      0.2234    0.18261  0.48936  0.2234    0.45652  0.45652

[Running in non-interactive mode, continuing...]

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

3.15.4 ag_multiclass_closed

The multiclass closed version.

The example is built and solved as follows.

```

System.out.println("=== Multiclass Closed Network ===");
System.out.println("Class 1: 5 jobs, Class 2: 3 jobs\n");

Network model = AgentModel.multiclassClosed();

```

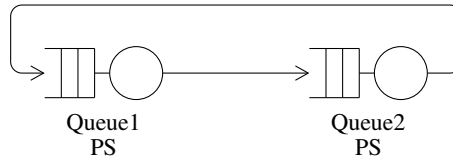


Figure 3.69: Topology of ag_multiclass_closed: a closed network over 2 queueing stations.

Nodes	2: Queue × 2
Classes	2: Class1 (closed, $N = 5$), Class2 (closed, $N = 3$)
Scheduling	Queue1: PS; Queue2: PS
Service	Queue1 – Class1: Exp(2); Class2: Exp(1.5) Queue2 – Class1: Exp(1); Class2: Exp(0.8)
Solvers	MVA

Table 3.74: Features of ag_multiclass_closed.

```

// AG with INAP method
System.out.println("AG (method=inap):");
NetworkSolver solverInap = new AG(model, "inap");
solverInap.getAvgTable().print();

// AG with exact method
System.out.println("\nAG (method=exact):");
NetworkSolver solverExact = new AG(model, "exact");
solverExact.getAvgTable().print();

// MVA for comparison
System.out.println("\nMVA:");
NetworkSolver solverMVA = new MVA(model);
solverMVA.getAvgTable().print();

pauseForUser();

```

Running this edition prints

```

=== Multiclass Closed Network ===
Class 1: 5 jobs, Class 2: 3 jobs

AG (method=inap):
AG analysis [method: inap; type: approximate, deterministic; lang: java; env: python] completed in <t>s.
Iterations: 49.
Station JobClass   QLen    Util    RespT    ResidT    ArvR    Tput
Queue1  Class1     1.6243  0.67984  1.1946   1.1946   0.95678  1.3597
Queue1  Class2     1.1225  0.62483  1.1977   1.1977   0.70154  0.93725
Queue2  Class1     3.6418  0.95678  3.8062   3.8062   1.3597   0.95678
Queue2  Class2     1.9846  0.87692  2.8289   2.8289   0.93725  0.70154

AG (method=exact):
WARNING: [solver_ag] AutoCAT (exact method) is not available. Falling back to INAP.
AG analysis [method: inap; type: approximate, deterministic; lang: java; env: python] completed in <t>s.
Iterations: 49.
Station JobClass   QLen    Util    RespT    ResidT    ArvR    Tput
Queue1  Class1     1.6243  0.67984  1.1946   1.1946   0.95678  1.3597

... (5 captured-output lines omitted; rerun the source example for the full output)

MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
Station JobClass   QLen    Util    RespT    ResidT    ArvR    Tput
Queue1  Class1     0.63167  0.31287  1.0095   1.0095   0.62574  0.62574
Queue1  Class2     0.39711  0.19837  1.3346   1.3346   0.29755  0.29755
Queue2  Class1     4.3683   0.62574  6.981    6.981    0.62574  0.62574
Queue2  Class2     2.6029   0.37194  8.7476   8.7476   0.29755  0.29755

[Running in non-interactive mode, continuing...]

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and

throughput of each station-class pair, as returned by MVA.

3.15.5 ag_tandem_open

An open tandem written through the agent interface.

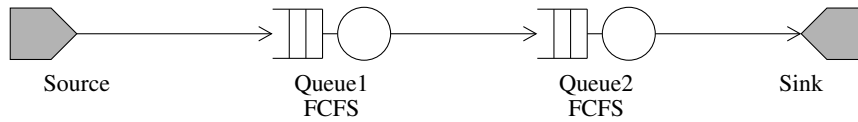


Figure 3.70: Topology of ag_tandem_open: an open network over 2 queueing stations.

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	1: Class1 (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	Source – Class1: Exp(0.5)
Service	Queue1 – Class1: Exp(1) Queue2 – Class1: Exp(1.5)
Solvers	MVA

Table 3.75: Features of ag_tandem_open.

The example is built and solved as follows.

```
System.out.println("=== Open Tandem Queue (M/M/1 -> M/M/1) ===\n");

Network model = AgentModel.tandemOpen();

// Analytical solution for comparison
double lambda = 0.5;
double mu1 = 1.0;
double mu2 = 1.5;
double U1_exact = lambda / mu1;
double U2_exact = lambda / mu2;
double Q1_exact = U1_exact / (1 - U1_exact);
double Q2_exact = U2_exact / (1 - U2_exact);

System.out.println("Analytical (M/M/1):");
System.out.printf(" Queue1: U=%.4f, Q=%.4f\n", U1_exact, Q1_exact);
System.out.printf(" Queue2: U=%.4f, Q=%.4f\n", U2_exact, Q2_exact);

// AG with INAP method
System.out.println("AG (method=inap):");
NetworkSolver solverInap = new AG(model, "inap");
solverInap.getAvgTable().print();

// AG with exact method
System.out.println("\nAG (method=exact):");
NetworkSolver solverExact = new AG(model, "exact");
solverExact.getAvgTable().print();

// MVA for comparison
System.out.println("\nMVA:");
NetworkSolver solverMVA = new MVA(model);
solverMVA.getAvgTable().print();

pauseForUser();
```

Running this edition prints

```
=== Open Tandem Queue (M/M/1 -> M/M/1) ===

Analytical (M/M/1):
 Queue1: U=0.5000, Q=1.0000
 Queue2: U=0.3333, Q=0.5000
```

```

AG (method=inap):
AG analysis [method: inap; type: approximate, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Source Class1 0 0 0 0 0 0.5
Queue1 Class1 1 0.5 2 2 0.5 0.5
Queue2 Class1 0.5 0.33333 1 1 0.5 0.5

AG (method=exact):
WARNING: [solver_ag] AutoCAT (exact method) is not available. Falling back to INAP.
AG analysis [method: inap; type: approximate, deterministic; lang: java; env: python] completed in <t>s.

... (5 captured-output lines omitted; rerun the source example for the full output)

MVA:
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 49.
Station JobClass QLen Util RespT ResidT ArvR Tput
Source Class1 0 0 0 0 0 0.5
Queue1 Class1 1 0.5 2 2 0.5 0.5
Queue2 Class1 0.5 0.33333 1 1 0.5 0.5

[Running in non-interactive mode, continuing...]

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

3.15.6 ag_tandem_phasetype

An open tandem whose two queues share a mean service time and differ only in variability (Erlang-2 against a hyperexponential), solved by the RCAT methods, which carry the phase-type law exactly where an M/M/1 reading sees no difference.

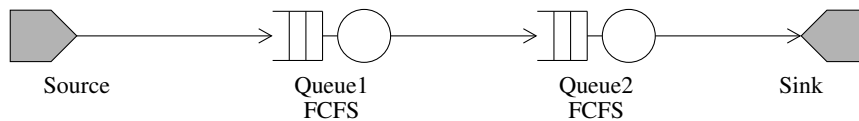


Figure 3.71: Topology of ag_tandem_phasetype: an open network over 2 queueing stations.

Nodes	4: Source, Queue × 2, Sink
Classes	1: Class1 (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	Source – Class1: Exp(0.5)
Service	Queue1 – Class1: Erlang(2,2) Queue2 – Class1: HyperExp(0.99,0.01;1.14,0.07584)

Table 3.76: Features of ag_tandem_phasetype.

The example is built and solved as follows.

```

System.out.println("=== Open Tandem with Phase-Type Service ===\n");

Network model = AgentModel.tandemPhaseType();

double lambda = 0.5, meanS = 1.0, scv1 = 0.5;
double rho = lambda * meanS;
double pk1 = rho + rho * rho * (1 + scv1) / (2 * (1 - rho));
System.out.println("Queue1 is an isolated M/Er2/1, so its exact mean queue length is the");
System.out.printf("Pollaczek-Khinchine value %.6f. The M/M/1 reading would be %.6f.%n",
    pk1, rho / (1 - rho));

System.out.println("AG (method=inap):");
new AG(model, "inap").getAvgTable().print();

// 'inapinf' additionally drops the maxStates truncation, solving each

```

```
// open component on its infinite state space through Neuts' rate
// matrix R. That matters most at Queue2, whose service law has the
// heavier tail.
System.out.println("\nAG (method=inapinf):");
new AG(model, "inapinf").getAvgTable().print();

pauseForUser();
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.16 Large-scale models

When the state space cannot be enumerated, the exact solvers are out and the question becomes how much accuracy the approximations retain. The script here is a large closed PS network solved by several approximate methods.

3.16.1 largescale_tbi

A large closed PS network, and the accuracy the approximate solvers retain on it.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows.

```
final int M = 10;    // queueing stations; the model has M + M*(M-1) in total
final int N = 250;  // closed population (vehicles)
final int kph = 16; // Erlang phases per transit delay
final double Tend = 20;

Network model = new Network("tbi_largescale");
// the reference seeds MATLAB's twister; the demands here are drawn from
// the same ranges rather than from the same stream
Random rng = new Random(1);
Queue[] Q = new Queue[M];
for (int i = 0; i < M; i++) {
    Q[i] = new Queue(model, "Q" + (i + 1), SchedStrategy.PS);
}
Delay[][] D = new Delay[M][M];
for (int i = 0; i < M; i++) {
    for (int j = 0; j < M; j++) {
        if (i != j) {
            D[i][j] = new Delay(model, "D" + (i + 1) + "_" + (j + 1));
        }
    }
}
ClosedClass job = new ClosedClass(model, "C1", N, Q[0]);
double[][] lambda = new double[M][M];
for (int i = 0; i < M; i++) {
    Q[i].setService(job, Erlang.fitMeanAndOrder(1.0 / (1 + 3 * rng.nextDouble()), 4));
    for (int j = 0; j < M; j++) {

        ... (63 source lines omitted; full implementation: jar/src/main/java/jline/examples/)
    }
}

// For comparison, the undecomposed solution of the same model,
// options.method = "closing";
// takes several minutes on the same machine (about 80 seconds already at
// M=8, and beyond 10 minutes at M=12).
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.17 Custom solvers

A solver is a class with an analyzer and a feature set. The feature set is a claim about what the solver can represent, and the dispatcher uses it to decide whether a model may be given to it; the analyzer computes the metrics. These two scripts show the smallest pair that the dispatcher will accept.

3.17.1 solver_custom

A user-defined solver registered with the dispatcher.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows.

```
int M = sn.nstations;           // number of stations
int K = sn.nclasses;           // number of classes
Matrix N = sn.njobs;           // job populations
Matrix rates = sn.rates;       // arrival and service rates
java.util.Map<Integer, Matrix> V = sn.visits; // visits

SolverResult result = new SolverResult();
result.QN = new Matrix(M, K);
result.UN = new Matrix(M, K);
result.RN = new Matrix(M, K);
result.TN = new Matrix(M, K);
result.CN = new Matrix(1, K);
result.XN = new Matrix(1, K);

System.out.println("The solution algorithm needs to be implemented in "
    + "SolverCustomAlgorithm.java: returning with no result.");
return result;
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

3.17.2 solver_custom_analyzer

Its analyzer, the function that actually computes the metrics.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows.

```
long tstart = System.nanoTime();

System.out.println("Any activity prior or after launching the solution algorithm "
    + "needs to be implemented in SolverCustomAnalyzer.java.");
SolverResult result = SolverCustomAlgorithm.solver_custom(sn, options);

zeroNaNs(result.QN);
zeroNaNs(result.UN);
zeroNaNs(result.RN);
zeroNaNs(result.TN);
zeroNaNs(result.CN);
zeroNaNs(result.XN);

result.runtime = (System.nanoTime() - tstart) / 1e9;
return result;
```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

Chapter 4

Model gallery

The gallery is a catalogue of small named models, one function each, returning a model rather than solving it. It exists twice over: as documentation, because a name in Kendall notation is the fastest way to find the model one means, and as the cross-codebase parity suite, because the same 71 functions exist in all four codebases and their metrics are compared numerically.

The models live in `jline.examples.java.Gallery`, one static method per model. Calling a gallery function builds and returns the model without solving it, so the caller chooses the solver. Running a Python gallery file as a script prints a short confirmation of the model it built.

```
public static Network gallery_mml() {
    Network model = new Network("M/M/1");
    Source source = new Source(model, "mySource");
    Queue queue = new Queue(model, "myQueue", SchedStrategy.FCFS);
    Sink sink = new Sink(model, "mySink");
    OpenClass oclass = new OpenClass(model, "myClass");
    source.setArrival(oclass, new Exp(1));
    queue.setService(oclass, new Exp(2));
    model.link(Network.serialRouting(source, queue, sink));
    return model;
}
```

4.1 Reading the names

The single-station entries are named in Kendall notation, with the distributions abbreviated: `m` exponential, `d` deterministic, `erl` Erlang, `hyp` hyperexponential, `aph` acyclic phase-type, `cox` Coxian, `me` matrix-exponential, `map` Markovian arrival process, `mmap` marked MAP, `gam` Gamma, `par` Pareto, `u` uniform. So `gallery_erlml` is an Erl/M/1 queue and `gallery_mhypk` an M/HyperExp/k. A trailing `_ps` replaces FCFS by processor sharing, `k` in place of `1` makes the station multiserver, and the suffixes `_tandem`, `_linear`, `_reentrant`, `_feedback`, `_multiclass` and `_prio` describe the topology or the workload around the station.

The remaining entries are named after what they model rather than after a queueing notation: `gallery_cqn` and `gallery_repairmen` for the closed archetypes, `gallery_fj_open` and `gallery_fj_closed` for fork-join, `gallery_fcr` for a finite capacity region, `gallery_cache_lru` and `gallery_cache_routing` for caches, `gallery_renv_breakdown` for a random environment, `gallery_lqn_basic`, `gallery_lqn_workflows` and `gallery_lqn_random` for layered models, `gallery_multitier` and `gallery_multitier_storage` for multitier applications, `gallery_lukumar_reentrant` for the Lu-Kumar re-entrant line, and `gallery_qn_random` for a generated network.

4.2 The catalogue

The complete list, in the order in which the models are registered:

gallery_aphm1	gallery_merl1_linear
gallery_cache_lru	gallery_merl1_reentrant
gallery_cache_routing	gallery_merl1_tandem
gallery_coxm1	gallery_merlk
gallery_cqn	gallery_mhyp1
gallery_cqn_multiclass	gallery_mhyp1_linear
gallery_detm1	gallery_mhyp1_reentrant
gallery_dm1	gallery_mhyp1_tandem
gallery_erldk	gallery_mhypk
gallery_erlerl1	gallery_mm1
gallery_erlerl1_reentrant	gallery_mm1_feedback
gallery_erlm1	gallery_mm1_linear
gallery_erlm1_ps	gallery_mm1_multiclass
gallery_erlm1_reentrant	gallery_mm1_prio
gallery_erlm1ps	gallery_mm1_ps
gallery_fcr	gallery_mm1_ps_feedback
gallery_fj_closed	gallery_mm1_ps_multiclass
gallery_fj_open	gallery_mm1_ps_reentrant
gallery_fj_quorum	gallery_mm1_reentrant
gallery_gamml	gallery_mm1_tandem
gallery_hyperl1_feedback	gallery_mm1_tandem_multiclass
gallery_hyperl1_reentrant	gallery_mmlk
gallery_hyperlk	gallery_mmap1
gallery_hyphyp1_linear	gallery_mmap1_multiclass
gallery_hyphyp1_reentrant	gallery_mmapk
gallery_hyphyp1_tandem	gallery_mmk
gallery_hypm1	gallery_mpar1
gallery_hypm1_reentrant	gallery_multitier
gallery_lqn_basic	gallery_multitier_storage
gallery_lqn_random	gallery_parm1
gallery_lqn_workflows	gallery_qn_random
gallery_lukumar_reentrant	gallery_renv_breakdown
gallery_mapm1	gallery_repairmen
gallery_mapmk	gallery_replayerm1
gallery_mdk	gallery_um1
gallery_merl1	

4.2.1 gallery_aphm1

The APH/M/1 queue: acyclic phase-type interarrival times, exponential service, and a single server.

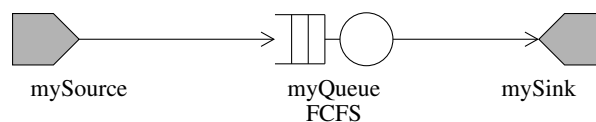


Figure 4.1: Topology of gallery_aphm1: an open network over 1 queueing station.

The example is built and solved as follows.

```

Network model = new Network("APH/M/1");
Source source = new Source(model, "mySource");
Queue queue = new Queue(model, "myQueue", SchedStrategy.FCFS);
Sink sink = new Sink(model, "mySink");
OpenClass oclass = new OpenClass(model, "myClass");
source.setArrival(oclass, APH.fitCentral(1, 0.99, 1.999));
queue.setService(oclass, new Exp(2));
  
```


Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass: APH
Service	myQueue – myClass: Exp(2)

Table 4.2: Features of gallery_aphm1.

```
model.link(Network.serialRouting(source, queue, sink));
return model;
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.2 gallery_cache_lru

A cache node under least-recently-used replacement, read by a closed class.

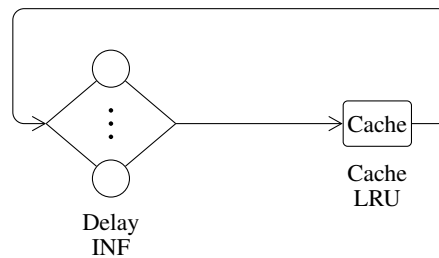


Figure 4.2: Topology of gallery_cache_lru: a closed network over 1 queueing station with a cache node.

Nodes	2: Delay, Cache
Classes	3: JobClass (closed, $N = 1$), HitClass (closed, $N = 0$), MissClass (closed, $N = 0$)
Scheduling	Delay: INF
Service	Delay – JobClass: Exp(1); HitClass: Disabled; MissClass: Disabled
Features	drop rule {'HitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at Delay; cache: LRU replacement, 5 items, capacity 2; class switching on 2 links

Table 4.3: Features of gallery_cache_lru.

The example is built and solved as follows.

```
public static Network gallery_cache_lru() {
    Network model = new Network("Cache-LRU");
    int n = 5; // number of items
    int m = 2; // cache capacity
    Delay delay = new Delay(model, "Delay");
    Cache cacheNode = new Cache(model, "Cache", n, m, ReplacementStrategy.LRU);
    ClosedClass jobClass = new ClosedClass(model, "JobClass", 1, delay, 0);
    ClosedClass hitClass = new ClosedClass(model, "HitClass", 0, delay, 0);
    ClosedClass missClass = new ClosedClass(model, "MissClass", 0, delay, 0);
    delay.setService(jobClass, new Exp(1));
    Matrix pmatrix = new Matrix(1, n);
    for (int i = 0; i < n; i++) {
        pmatrix.set(0, i, 1.0 / n);
    }
    DiscreteSampler pAccess = new DiscreteSampler(pmatrix);
    cacheNode.setRead(jobClass, pAccess);
    cacheNode.setHitClass(jobClass, hitClass);
    cacheNode.setMissClass(jobClass, missClass);
}
```

```

RoutingMatrix P = model.initRoutingMatrix();
P.set(jobClass, jobClass, delay, cacheNode, 1.0);
P.set(hitClass, jobClass, cacheNode, delay, 1.0);
P.set(missClass, jobClass, cacheNode, delay, 1.0);
model.link(P);
return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.3 gallery_cache_routing

A cache whose hit and miss outcomes route the job to different downstream stations.

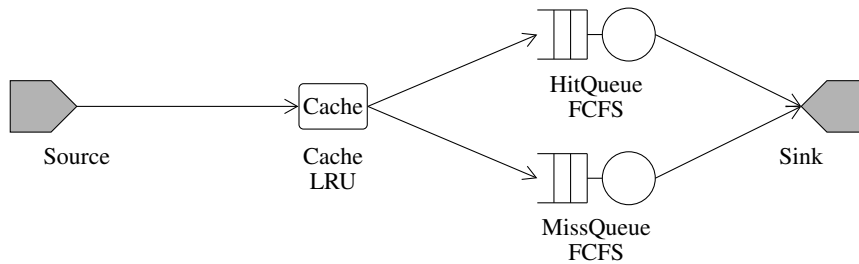


Figure 4.3: Topology of gallery_cache_routing: an open network over 2 queueing stations with a cache node.

Nodes	5: Source, Queue × 2, Cache, Sink
Classes	3: InitClass (open), HitClass (open), MissClass (open)
Scheduling	HitQueue: FCFS; MissQueue: FCFS
Arrivals	Source – InitClass: Exp(1); HitClass: Disabled; MissClass: Disabled
Service	HitQueue – InitClass: Disabled; HitClass: Exp(2); MissClass: Disabled MissQueue – InitClass: Disabled; HitClass: Disabled; MissClass: Exp(1)
Features	cache: LRU replacement, 4 items, capacity 2; drop rule {'InitClass': 'waitingQueue', 'MissClass': 'waitingQueue'} at HitQueue; drop rule {'InitClass': 'waitingQueue', 'HitClass': 'waitingQueue'} at MissQueue; class switching on 4 links

Table 4.4: Features of gallery_cache_routing.

The example is built and solved as follows.

```

public static Network gallery_cache_routing() {
    Network model = new Network("Cache-Routing");
    int n = 4; // number of items
    int m = 2; // cache capacity
    Source source = new Source(model, "Source");
    Cache cacheNode = new Cache(model, "Cache", n, m, ReplacementStrategy.LRU);
    Queue hitQueue = new Queue(model, "HitQueue", SchedStrategy.FCFS);
    Queue missQueue = new Queue(model, "MissQueue", SchedStrategy.FCFS);
    Sink sink = new Sink(model, "Sink");
    OpenClass jobClass = new OpenClass(model, "InitClass", 0);
    OpenClass hitClass = new OpenClass(model, "HitClass", 0);
    OpenClass missClass = new OpenClass(model, "MissClass", 0);
    source.setArrival(jobClass, new Exp(1));
    hitQueue.setService(hitClass, new Exp(2.0));
    missQueue.setService(missClass, new Exp(1.0));
    Matrix pmatrix = new Matrix(1, n);
    for (int i = 0; i < n; i++) {
        pmatrix.set(0, i, 1.0 / n);
    }
    DiscreteSampler pAccess = new DiscreteSampler(pmatrix);
    cacheNode.setRead(jobClass, pAccess);
}

```

```

cacheNode.setHitClass(jobClass, hitClass);
cacheNode.setMissClass(jobClass, missClass);
RoutingMatrix P = model.initRoutingMatrix();
P.set(jobClass, jobClass, source, cacheNode, 1.0);
P.set(hitClass, hitClass, cacheNode, hitQueue, 1.0);
P.set(hitClass, hitClass, hitQueue, sink, 1.0);
P.set(missClass, missClass, cacheNode, missQueue, 1.0);
P.set(missClass, missClass, missQueue, sink, 1.0);
model.link(P);
return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.4 gallery_coxm1

The Coxian/M/1 queue: Coxian interarrival times, exponential service, and a single server.

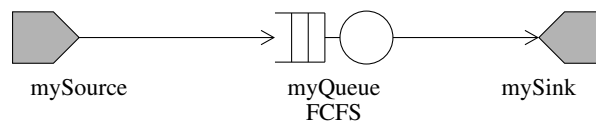


Figure 4.4: Topology of gallery_coxm1: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass: Coxian
Service	myQueue – myClass: Exp(2)

Table 4.5: Features of gallery_coxm1.

The example is built and solved as follows.

```

Network model = new Network("Cox/M/1");
Source source = new Source(model, "mySource");
Queue queue = new Queue(model, "myQueue", SchedStrategy.FCFS);
Sink sink = new Sink(model, "mySink");
OpenClass oclass = new OpenClass(model, "myClass");
source.setArrival(oclass, Coxian.fitCentral(1, 0.99, 1.999));
queue.setService(oclass, new Exp(2));
model.link(Network.serialRouting(source, queue, sink));
return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.5 gallery_cqn

The closed queueing network archetype: a fixed job population circulating between a Delay station and a queue.

Nodes	2: Queue × 2
Classes	1: Class1 (closed, $N = 3$)
Scheduling	Queue1: PS; Queue2: PS
Service	Queue1 – Class1: Exp(0.8271) Queue2 – Class1: Exp(0.4381)

Table 4.6: Features of gallery_cqn.

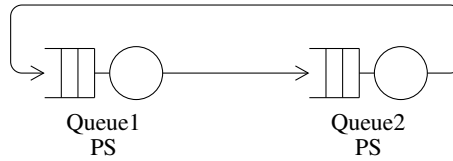


Figure 4.5: Topology of `gallery_cqn`: a closed network over 2 queueing stations.

The example is built and solved as follows.

```
return gallery_cqn(2, false, 2300);
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.6 `gallery_cqn_multiclass`

The closed archetype with several classes, each with its own population.

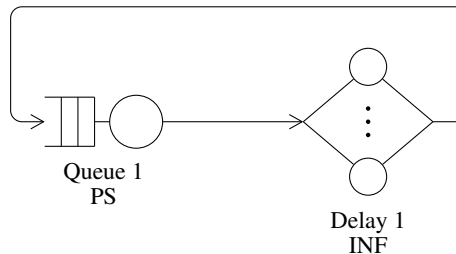


Figure 4.6: Topology of `gallery_cqn_multiclass`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 5$), Class2 (closed, $N = 5$)
Scheduling	Queue 1: PS; Delay 1: INF
Service	Queue 1 – Class1: Exp(0.03226); Class2: Exp(0.07143) Delay 1 – Class1: Exp(0.04762); Class2: Exp(0.08333)

Table 4.7: Features of `gallery_cqn_multiclass`.

The example is built and solved as follows.

```
public static Network gallery_cqn_multiclass() {
    return gallery_cqn_multiclass(2, 2, false, 2300);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.7 `gallery_detm1`

The D/M/1 queue: deterministic interarrival times, exponential service, and a single server.

The example is built and solved as follows.

```
public static Network gallery_detm1() {
    Network model = new Network("D/M/1");
    Source source = new Source(model, "mySource");
    Queue queue = new Queue(model, "myQueue", SchedStrategy.FCFS);
    Sink sink = new Sink(model, "mySink");
    OpenClass oclass = new OpenClass(model, "myClass");
}
```

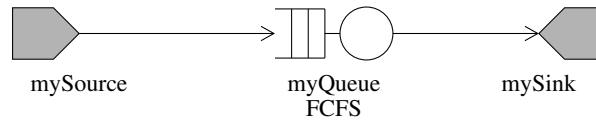


Figure 4.7: Topology of gallery_detm1: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass: Det(1)
Service	myQueue – myClass: Exp(2)

Table 4.8: Features of gallery_detm1.

```

source.setArrival(oclass, new Det(1));
queue.setService(oclass, new Exp(2));
model.link(Network.serialRouting(source, queue, sink));
return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.8 gallery_dm1

The D/M/1 queue: deterministic interarrival times, exponential service, and a single server.

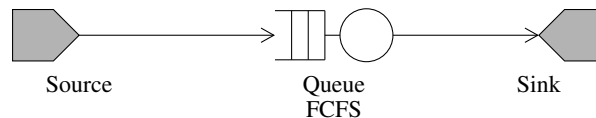


Figure 4.8: Topology of gallery_dm1: an open network over 1 queueing station.

The example is built and solved as follows.

```

public static Network gallery_dm1() {
    Network model = new Network("D/M/1");
    Source source = new Source(model, "Source");
    Queue queue = new Queue(model, "Queue", SchedStrategy.FCFS);
    Sink sink = new Sink(model, "Sink");
    OpenClass oclass = new OpenClass(model, "Class1");
    source.setArrival(oclass, new Det(1));
    queue.setService(oclass, new Exp(2));
    model.link(Network.serialRouting(source, queue, sink));
    return model;
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.9 gallery_erldk

The Erl/D/k queue: Erlang interarrival times, deterministic service, and k parallel servers.

The example is built and solved as follows.

```

public static Network gallery_erldk() {
    return gallery_erldk(2);
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Det(1)
Service	Queue – Class1: Exp(2)

Table 4.9: Features of gallery_dm1.

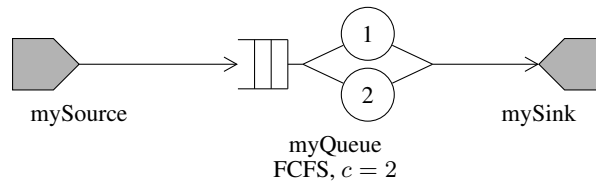


Figure 4.9: Topology of gallery_erldk: an open network over 1 queueing station.

4.2.10 gallery_erlerl1

The Erl/Erl/1 queue: Erlang interarrival times, Erlang service, and a single server.

The example is built and solved as follows.

```
public static Network gallery_erlerl1() {
    return gallery_erlerl1(5);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.11 gallery_erlerl1_reentrant

An Erl/Erl/1 queue with re-entrant visits.

The example is built and solved as follows.

```
public static Network gallery_erlerl1_reentrant() {
    return gallery_erlerl1_reentrant(5);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.12 gallery_erlm1

The Erl/M/1 queue: Erlang interarrival times, exponential service, and a single server.

The example is built and solved as follows.

```
public static Network gallery_erlm1() {
    Network model = new Network("Erl/M/1");
    Source source = new Source(model, "mySource");
    Queue queue = new Queue(model, "myQueue", SchedStrategy.FCFS);
    Sink sink = new Sink(model, "mySink");
}
```

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS ($c = 2$)
Arrivals	mySource – myClass: Erlang(5,5)
Service	myQueue – myClass: Det(1)

Table 4.10: Features of gallery_erldk.

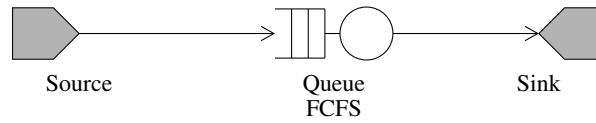


Figure 4.10: Topology of gallery_erlerl1: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Erlang(5,5); Class2: Disabled
Service	Queue – Class1: Erlang(10,5); Class2: Exp(3)
Features	class switching on 2 links

Table 4.11: Features of gallery_erlerl1.

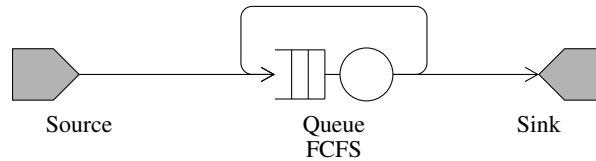


Figure 4.11: Topology of gallery_erlerl1_reentrant: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Erlang(5,5); Class2: Disabled
Service	Queue – Class1: Erlang(50,5); Class2: Exp(10)
Features	class switching on 1 link

Table 4.12: Features of gallery_erlerl1_reentrant.

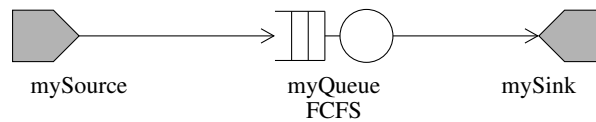


Figure 4.12: Topology of gallery_erlml1: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass: Erlang(5,5)
Service	myQueue – myClass: Exp(2)

Table 4.13: Features of gallery_erlml1.

```

OpenClass oclass = new OpenClass(model, "myClass");
source.setArrival(oclass, Erlang.fitMeanAndOrder(1.0, 5));
queue.setService(oclass, new Exp(2));
model.link(Network.serialRouting(source, queue, sink));
return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.13 gallery_erlm1_ps

The Erl/M/1 queue: Erlang interarrival times, exponential service, and a single server. In this variant, the station serves under processor sharing rather than FCFS.

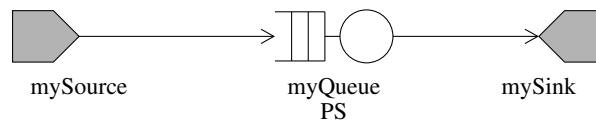


Figure 4.13: Topology of gallery_erlm1_ps: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: PS
Arrivals	mySource – myClass: Erlang(5,5)
Service	myQueue – myClass: Exp(2)

Table 4.14: Features of gallery_erlm1_ps.

The example is built and solved as follows.

```

public static Network gallery_erlm1_ps() {
    Network model = new Network("Erl/M/1");
    Source source = new Source(model, "mySource");
    Queue queue = new Queue(model, "myQueue", SchedStrategy.PS);
    Sink sink = new Sink(model, "mySink");
    OpenClass oclass = new OpenClass(model, "myClass");
    source.setArrival(oclass, Erlang.fitMeanAndOrder(1.0, 5));
    queue.setService(oclass, new Exp(2));
    model.link(Network.serialRouting(source, queue, sink));
    return model;
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.14 gallery_erlm1_reentrant

The Erl/M/1 queue: Erlang interarrival times, exponential service, and a single server. In this variant, jobs re-enter the station, so a visit is counted more than once.

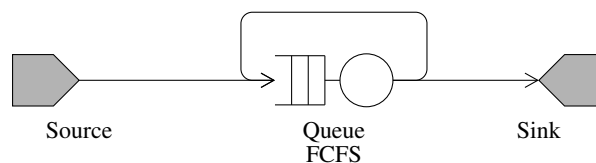


Figure 4.14: Topology of gallery_erlm1_reentrant: an open network over 1 queueing station.

The example is built and solved as follows.

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Erlang(5,5); Class2: Disabled
Service	Queue – Class1: Exp(2); Class2: Exp(3)
Features	class switching on 1 link

Table 4.15: Features of gallery_erlm1_reentrant.

```

public static Network gallery_erlm1_reentrant() {
    Network model = new Network("Erl/M/1-Reentrant");

    // Block 1: nodes
    Source node1 = new Source(model, "Source");
    Queue node2 = new Queue(model, "Queue", SchedStrategy.FCFS);
    Sink node3 = new Sink(model, "Sink");

    // Block 2: classes
    OpenClass jobclass1 = new OpenClass(model, "Class1", 0);
    OpenClass jobclass2 = new OpenClass(model, "Class2", 0);

    node1.setArrival(jobclass1, Erlang.fitMeanAndOrder(1, 5)); // (Source,Class1)
    node1.setArrival(jobclass2, Disabled.getInstance()); // (Source,Class2)
    node2.setService(jobclass1, new Exp(2)); // (Queue,Class1)
    node2.setService(jobclass2, new Exp(3)); // (Queue,Class2)

    // Block 3: topology
    RoutingMatrix routingMatrix = model.initRoutingMatrix();

    routingMatrix.set(jobclass1, jobclass1, node1, node2, 1.00); // (Source,Class1) -> (Queue,Class1)
    routingMatrix.set(jobclass1, jobclass2, node2, node2, 1.00); // (CS_Queue_to_Queue,Class1) -> (Queue,Class2)
    routingMatrix.set(jobclass2, jobclass2, node2, node3, 1.00); // (Queue,Class2) -> (Sink,Class2)

    model.link(routingMatrix);

    return model;
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.15 gallery_erlm1ps

The Erl/M/1 queue: Erlang interarrival times, exponential service, and a single server. In this variant, the station serves under processor sharing rather than FCFS.

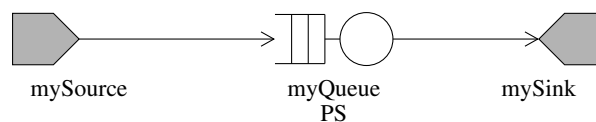


Figure 4.15: Topology of gallery_erlm1ps: an open network over 1 queueing station.

The example is built and solved as follows.

```

public static Network gallery_erlm1ps() {
    Network model = new Network("Er/M/1-PS");
    Source source = new Source(model, "mySource");
    Queue queue = new Queue(model, "myQueue", SchedStrategy.PS);
    Sink sink = new Sink(model, "mySink");
    OpenClass oclass = new OpenClass(model, "myClass");
}

```

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: PS
Arrivals	mySource – myClass: Erlang(5,5)
Service	myQueue – myClass: Exp(2)

Table 4.16: Features of gallery_erlmlps.

```
source.setArrival(oclass, Erlang.fitMeanAndOrder(1.0, 5));
queue.setService(oclass, new Exp(2));
model.link(Network.serialRouting(source, queue, sink));
return model;
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.16 gallery_fcr

A finite capacity region, which caps the number of jobs admitted to a set of stations irrespective of their individual buffers.

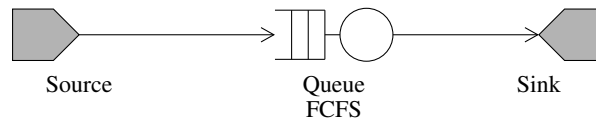


Figure 4.16: Topology of gallery_fcr: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(0.8)
Service	Queue – Class1: Exp(1)

Table 4.17: Features of gallery_fcr.

The example is built and solved as follows.

```
public static Network gallery_fcr() {
    return gallery_fcr(3);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.17 gallery_fj_closed

The closed fork-join counterpart, with a fixed population circulating through the block.

The example is built and solved as follows.

```
public static Network gallery_fj_closed() {
    Network model = new Network("Fork-Join-Closed");
    Delay delay = new Delay(model, "Delay");
    Queue queue1 = new Queue(model, "Queue1", SchedStrategy.PS);
    Queue queue2 = new Queue(model, "Queue2", SchedStrategy.PS);
    Fork fork = new Fork(model, "Fork");
    Join join = new Join(model, "Join", fork);
    ClosedClass oclass = new ClosedClass(model, "class1", 5, delay);
    delay.setService(oclass, new Exp(1.0));
}
```

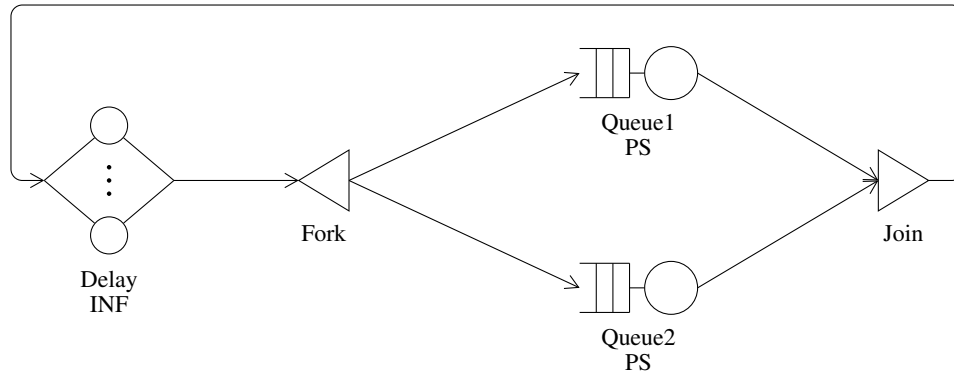


Figure 4.17: Topology of `gallery_fj_closed`: a closed network over 3 queueing stations with a fork-join block.

Nodes	5: Queue $\times$ 2, Delay, Fork, Join
Classes	1: class1 (closed, $N = 5$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS
Service	Delay – class1: Exp(1) Queue1 – class1: Exp(1) Queue2 – class1: Exp(1)

Table 4.18: Features of `gallery_fj_closed`.

```

queue1.setService(oclass, new Exp(1.0));
queue2.setService(oclass, new Exp(1.0));
RoutingMatrix P = model.initRoutingMatrix();
P.set(oclass, oclass, delay, fork, 1.0);
P.set(oclass, oclass, fork, queue1, 1.0);
P.set(oclass, oclass, fork, queue2, 1.0);
P.set(oclass, oclass, queue1, join, 1.0);
P.set(oclass, oclass, queue2, join, 1.0);
P.set(oclass, oclass, join, delay, 1.0);
model.link(P);
return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.18 `gallery_fj_open`

An open fork-join block: a job forks into parallel tasks and its siblings are joined before departure.

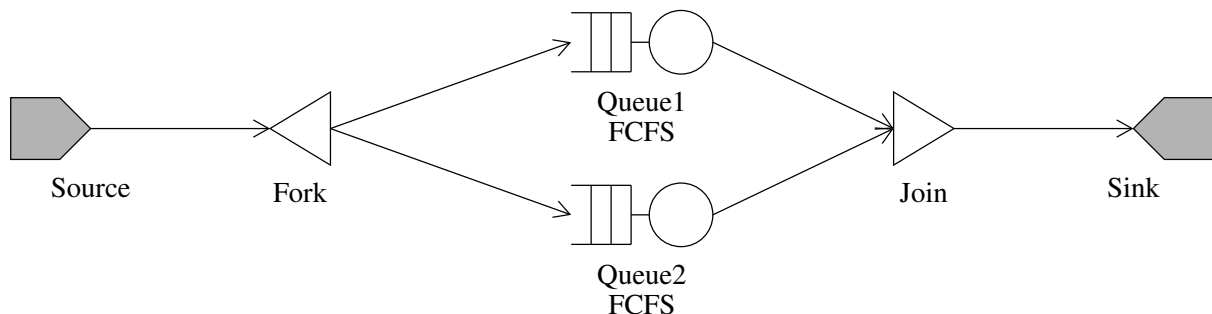


Figure 4.18: Topology of `gallery_fj_open`: an open network over 2 queueing stations with a fork-join block.

The example is built and solved as follows.

Nodes	6: Source, Queue × 2, Fork, Join, Sink
Classes	1: class1 (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	Source – class1: Exp(0.05)
Service	Queue1 – class1: Exp(1) Queue2 – class1: Exp(2)

Table 4.19: Features of gallery_fj_open.

```

public static Network gallery_fj_open() {
    Network model = new Network("Fork-Join-Open");
    Source source = new Source(model, "Source");
    Queue queue1 = new Queue(model, "Queue1", SchedStrategy.FCFS);
    Queue queue2 = new Queue(model, "Queue2", SchedStrategy.FCFS);
    Fork fork = new Fork(model, "Fork");
    Join join = new Join(model, "Join", fork);
    Sink sink = new Sink(model, "Sink");
    OpenClass oclass = new OpenClass(model, "class1");
    source.setArrival(oclass, new Exp(0.05));
    queue1.setService(oclass, new Exp(1.0));
    queue2.setService(oclass, new Exp(2.0));
    RoutingMatrix P = model.initRoutingMatrix();
    P.set(oclass, oclass, source, fork, 1.0);
    P.set(oclass, oclass, fork, queue1, 1.0);
    P.set(oclass, oclass, fork, queue2, 1.0);
    P.set(oclass, oclass, queue1, join, 1.0);
    P.set(oclass, oclass, queue2, join, 1.0);
    P.set(oclass, oclass, join, sink, 1.0);
    model.link(P);
    return model;
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.19 gallery_fj_quorum

A fork-join block whose join fires on a quorum of the siblings, the later ones being discarded when they arrive.

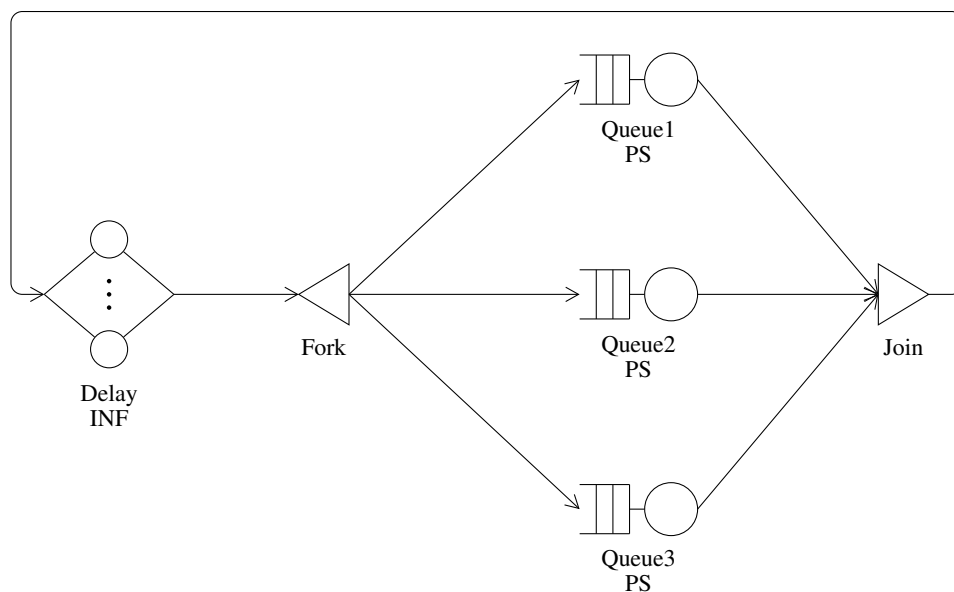


Figure 4.19: Topology of gallery_fj_quorum: a closed network over 4 queueing stations with a fork-join block.

Nodes	6: Queue $\times$ 3, Delay, Fork, Join
Classes	1: class1 (closed, $N = 5$)
Scheduling	Delay: INF; Queue1: PS; Queue2: PS; Queue3: PS
Service	Delay – class1: Exp(1) Queue1 – class1: Exp(2) Queue2 – class1: Exp(2) Queue3 – class1: Exp(2)

Table 4.20: Features of gallery_fj_quorum.

The example is built and solved as follows.

```

Network model = new Network("Fork-Join-Quorum");
Delay delay = new Delay(model, "Delay");
Queue queue1 = new Queue(model, "Queue1", SchedStrategy.PS);
Queue queue2 = new Queue(model, "Queue2", SchedStrategy.PS);
Queue queue3 = new Queue(model, "Queue3", SchedStrategy.PS);
Fork fork = new Fork(model, "Fork");
Join join = new Join(model, "Join", fork);
ClosedClass oclass = new ClosedClass(model, "class1", 5, delay);
delay.setService(oclass, new Exp(1.0));
queue1.setService(oclass, new Exp(2.0));
queue2.setService(oclass, new Exp(2.0));
queue3.setService(oclass, new Exp(2.0));
join.setStrategy(oclass, JoinStrategy.PARTIAL);
join.setRequired(oclass, 2);
RoutingMatrix P = model.initRoutingMatrix();
P.set(oclass, oclass, delay, fork, 1.0);
P.set(oclass, oclass, fork, queue1, 1.0);
P.set(oclass, oclass, fork, queue2, 1.0);
P.set(oclass, oclass, fork, queue3, 1.0);
P.set(oclass, oclass, queue1, join, 1.0);
P.set(oclass, oclass, queue2, join, 1.0);
P.set(oclass, oclass, queue3, join, 1.0);
P.set(oclass, oclass, join, delay, 1.0);
model.link(P);
return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.20 gallery_gamm1

The Gamma/M/1 queue: Gamma interarrival times, exponential service, and a single server.

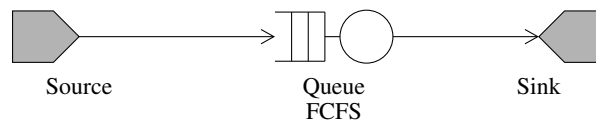


Figure 4.20: Topology of gallery_gamm1: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Gamma(5,0.2)
Service	Queue – Class1: Exp(2)

Table 4.21: Features of gallery_gamm1.

The example is built and solved as follows.

```

public static Network gallery_gamm1() {
    Network model = new Network("Gam/M/1");

```

```

Source source = new Source(model, "mySource");
Queue queue = new Queue(model, "myQueue", SchedStrategy.FCFS);
Sink sink = new Sink(model, "mySink");
OpenClass oclass = new OpenClass(model, "myClass");
source.setArrival(oclass, Gamma.fitMeanAndSCV(1.0, 1.0 / 5));
queue.setService(oclass, new Exp(2));
model.link(Network.serialRouting(source, queue, sink));
return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.21 gallery_hyperl1_feedback

A HyperExp/Erl/1 queue with probabilistic feedback.

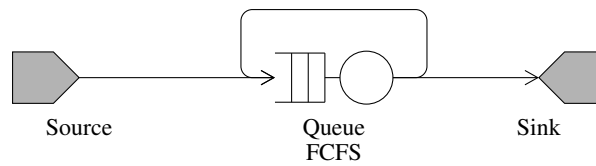


Figure 4.21: Topology of gallery_hyperl1_feedback: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: HyperExp(0.99,0.01;2.294,0.01759)
Service	Queue – Class1: Erlang(100,5)

Table 4.22: Features of gallery_hyperl1_feedback.

The example is built and solved as follows.

```

public static Network gallery_hyperl1_feedback() {
    Network model = new Network("Hyper/Erl/1-Feedback");
    Source source = new Source(model, "Source");
    Queue queue = new Queue(model, "Queue", SchedStrategy.FCFS);
    Sink sink = new Sink(model, "Sink");
    OpenClass oclass1 = new OpenClass(model, "Class1");
    source.setArrival(oclass1, HyperExp.fitMeanAndSCV(1, 64));
    queue.setService(oclass1, Erlang.fitMeanAndOrder(0.05, 5));
    RoutingMatrix P = model.initRoutingMatrix();
    P.set(oclass1, oclass1, source, queue, 1.0);
    P.set(oclass1, oclass1, queue, queue, 0.9);
    P.set(oclass1, oclass1, queue, sink, 0.1);
    model.link(P);
    return model;
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.22 gallery_hyperl1_reentrant

The HyperExp/Erl/1 queue: hyperexponential interarrival times, Erlang service, and a single server. In this variant, jobs re-enter the station, so a visit is counted more than once.

The example is built and solved as follows.

```

public static Network gallery_hyperl1_reentrant() {
    Network model = new Network("Hyper/Erl/1-Reentrant");

```

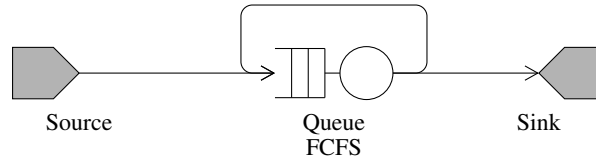


Figure 4.22: Topology of `gallery_hyperl1_reentrant`: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: HyperExp(0.99,0.01;2.294,0.01759); Class2: Disabled
Service	Queue – Class1: Erlang(10,5); Class2: Exp(3)
Features	class switching on 1 link

Table 4.23: Features of `gallery_hyperl1_reentrant`.

```
// Block 1: nodes
Source node1 = new Source(model, "Source");
Queue node2 = new Queue(model, "Queue", SchedStrategy.FCFS);
Sink node3 = new Sink(model, "Sink");

// Block 2: classes
OpenClass jobclass1 = new OpenClass(model, "Class1", 0);
OpenClass jobclass2 = new OpenClass(model, "Class2", 0);

node1.setArrival(jobclass1, HyperExp.fitMeanAndSCV(1, 64)); // (Source,Class1)
node1.setArrival(jobclass2, Disabled.getInstance()); // (Source,Class2)
node2.setService(jobclass1, Erlang.fitMeanAndOrder(0.5, 5)); // (Queue,Class1)
node2.setService(jobclass2, new Exp(3)); // (Queue,Class2)

// Block 3: topology
RoutingMatrix routingMatrix = model.initRoutingMatrix();

routingMatrix.set(jobclass1, jobclass1, node1, node2, 1.00); // (Source,Class1) -> (Queue,Class1)
routingMatrix.set(jobclass1, jobclass2, node2, node2, 1.00); // (CS_Queue_to_Queue,Class1) -> (Queue,Class2)
routingMatrix.set(jobclass2, jobclass2, node2, node3, 1.00); // (Queue,Class2) -> (Sink,Class2)

model.link(routingMatrix);

return model;
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.23 `gallery_hyperl1k`

The HyperExp/Erl/ k queue: hyperexponential interarrival times, Erlang service, and k parallel servers.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS ($c = 2$)
Arrivals	mySource – myClass: HyperExp(0.1127,0.8873;0.4057,3.194)
Service	myQueue – myClass: Erlang(4,4)

Table 4.24: Features of `gallery_hyperl1k`.

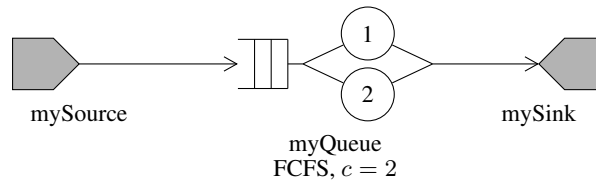


Figure 4.23: Topology of `gallery_hyperlk`: an open network over 1 queueing station.

The example is built and solved as follows.

```
public static Network gallery_hyperlk() {
    return gallery_hyperlk(2);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.24 `gallery_hyphyp1_linear`

The HyperExp/HyperExp/1 queue: hyperexponential interarrival times, hyperexponential service, and a single server. In this variant, the model is a linear chain of such stations.

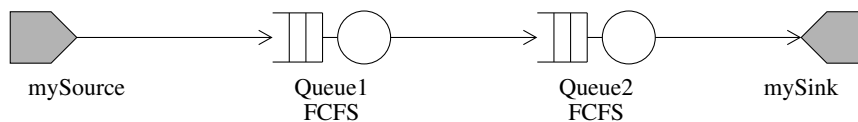


Figure 4.24: Topology of `gallery_hyphyp1_linear`: an open network over 2 queueing stations.

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	1: myClass (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	mySource – myClass: HyperExp(0.99,0.01;1.077,0.1244)
Service	Queue1 – myClass: HyperExp(0.99,0.01;1.196,0.1383) Queue2 – myClass: HyperExp(0.99,0.01;1.235,0.1015)

Table 4.25: Features of `gallery_hyphyp1_linear`.

The example is built and solved as follows.

```
public static Network gallery_hyphyp1_linear() {
    return gallery_hyphyp1_linear(2, 0.9);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.25 `gallery_hyphyp1_reentrant`

The HyperExp/HyperExp/1 queue: hyperexponential interarrival times, hyperexponential service, and a single server. In this variant, jobs re-enter the station, so a visit is counted more than once.

The example is built and solved as follows.

```
public static Network gallery_hyphyp1_reentrant() {
    Network model = new Network("Hyper/Hyper/1-Reentrant");

    // Block 1: nodes
    Source node1 = new Source(model, "Source");
}
```

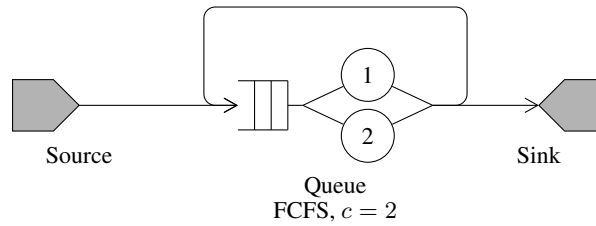



Figure 4.25: Topology of gallery_hyphyp1_reentrant: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue: FCFS ($c = 2$)
Arrivals	Source – Class1: HyperExp(0.99,0.01;2.294,0.01759); Class2: Disabled
Service	Queue – Class1: HyperExp(0.99,0.01;2.281,0.1517); Class2: Exp(3)
Features	class switching on 1 link

Table 4.26: Features of gallery_hyphyp1_reentrant.

```

Queue node2 = new Queue(model, "Queue", SchedStrategy.FCFS);
node2.setNumberOfServers(2);
Sink node3 = new Sink(model, "Sink");

// Block 2: classes
OpenClass jobclass1 = new OpenClass(model, "Class1", 0);
OpenClass jobclass2 = new OpenClass(model, "Class2", 0);

node1.setArrival(jobclass1, HyperExp.fitMeanAndSCV(1, 64)); // (Source,Class1)
node1.setArrival(jobclass2, Disabled.getInstance()); // (Source,Class2)
node2.setService(jobclass1, HyperExp.fitMeanAndSCV(0.5, 4)); // (Queue,Class1)
node2.setService(jobclass2, new Exp(3)); // (Queue,Class2)

// Block 3: topology
RoutingMatrix routingMatrix = model.initRoutingMatrix();

routingMatrix.set(jobclass1, jobclass1, node1, node2, 1.00); // (Source,Class1) -> (Queue,Class1)
routingMatrix.set(jobclass1, jobclass2, node2, node2, 1.00); // (CS_Queue_to_Queue,Class1) -> (Queue,Class2)
routingMatrix.set(jobclass2, jobclass2, node2, node3, 1.00); // (Queue,Class2) -> (Sink,Class2)

model.link(routingMatrix);

return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.26 gallery_hyphyp1_tandem

Two HyperExp/HyperExp/1 queues in series.

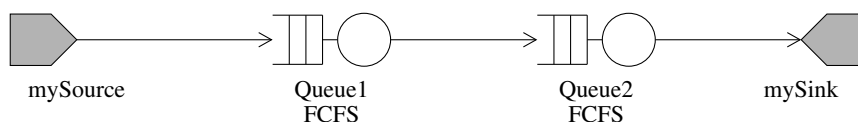


Figure 4.26: Topology of gallery_hyphyp1_tandem: an open network over 2 queueing stations.

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	1: myClass (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	mySource – myClass: HyperExp(0.99,0.01;1.077,0.1244)
Service	Queue1 – myClass: HyperExp(0.99,0.01;1.196,0.1383) Queue2 – myClass: HyperExp(0.99,0.01;1.235,0.1015)

Table 4.27: Features of gallery_hyphyp1_tandem.

The example is built and solved as follows.

```
public static Network gallery_hyphyp1_tandem() {
    return gallery_hyphyp1_linear(2);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.27 gallery_hypm1

The HyperExp/M/1 queue: hyperexponential interarrival times, exponential service, and a single server.

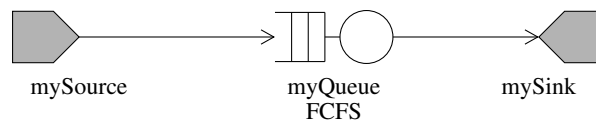


Figure 4.27: Topology of gallery_hypm1: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass: HyperExp(0.99,0.01;2.294,0.01759)
Service	myQueue – myClass: Exp(2)

Table 4.28: Features of gallery_hypm1.

The example is built and solved as follows.

```
public static Network gallery_hypm1() {
    Network model = new Network("H2/M/1");
    Source source = new Source(model, "mySource");
    Queue queue = new Queue(model, "myQueue", SchedStrategy.FCFS);
    Sink sink = new Sink(model, "mySink");
    OpenClass oclass = new OpenClass(model, "myClass");
    source.setArrival(oclass, HyperExp.fitMeanAndSCV(1, 64));
    queue.setService(oclass, new Exp(2));
    model.link(Network.serialRouting(source, queue, sink));
    return model;
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.28 gallery_hypm1_reentrant

The HyperExp/M/1 queue: hyperexponential interarrival times, exponential service, and a single server. In this variant, jobs re-enter the station, so a visit is counted more than once.

The example is built and solved as follows.

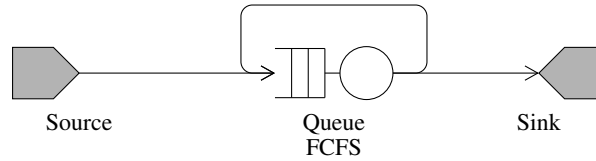


Figure 4.28: Topology of gallery_hypm1_reentrant: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: HyperExp(0.99,0.01;1.14,0.07584); Class2: Disabled
Service	Queue – Class1: Exp(2); Class2: Exp(3)
Features	class switching on 1 link

Table 4.29: Features of gallery_hypm1_reentrant.

```

public static Network gallery_hypm1_reentrant() {
    Network model = new Network("Hyper/Erl/1-Reentrant");

    // Block 1: nodes
    Source node1 = new Source(model, "Source");
    Queue node2 = new Queue(model, "Queue", SchedStrategy.FCFS);
    Sink node3 = new Sink(model, "Sink");

    // Block 2: classes
    OpenClass jobclass1 = new OpenClass(model, "Class1", 0);
    OpenClass jobclass2 = new OpenClass(model, "Class2", 0);

    node1.setArrival(jobclass1, HyperExp.fitMeanAndSCV(1, 4)); // (Source,Class1)
    node1.setArrival(jobclass2, Disabled.getInstance()); // (Source,Class2)
    node2.setService(jobclass1, new Exp(2)); // (Queue,Class1)
    node2.setService(jobclass2, new Exp(3)); // (Queue,Class2)

    // Block 3: topology
    RoutingMatrix routingMatrix = model.initRoutingMatrix();

    routingMatrix.set(jobclass1, jobclass1, node1, node2, 1.00); // (Source,Class1) -> (Queue,Class1)
    routingMatrix.set(jobclass1, jobclass2, node2, node2, 1.00); // (CS_Queue_to_Queue,Class1) -> (Queue,Class2)
    routingMatrix.set(jobclass2, jobclass2, node2, node3, 1.00); // (Queue,Class2) -> (Sink,Class2)

    model.link(routingMatrix);

    return model;
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.29 gallery_lqn_basic

The smallest layered queueing network: one reference task calling one server task.
The example is built and solved as follows.

```

public static LayeredNetwork gallery_lqn_basic() {
    LayeredNetwork model = new LayeredNetwork("LQN-Basic");
    Processor P1 = new Processor(model, "P1", 2, SchedStrategy.PS);
    Processor P2 = new Processor(model, "P2", 3, SchedStrategy.PS);
    Task T1 = new Task(model, "T1", 50, SchedStrategy.REF).on(P1).setThinkTime(new Exp(1.0 / 2));
}

```

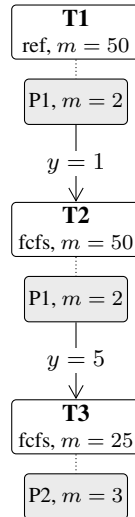


Figure 4.29: Call graph of `gallery_lqn_basic`: 3 tasks over 2 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	2: P1 (ps, $m = 2$), P2 (ps, $m = 3$)
Tasks	3: T1 (ref, $m = 50$), T2 (fcfs, $m = 50$), T3 (fcfs, $m = 25$)
Entries	3: E1, E2, E3
Activities	3, host demands AS1: 0.1, AS2: 0.05, AS3: 0.02
Calls	2 (2 synchronous, 0 asynchronous)
Think times	T1: 2.0

Table 4.30: Features of `gallery_lqn_basic`.

```

Task T2 = new Task(model, "T2", 50, SchedStrategy.FCFS).on(P1).setThinkTime(new Exp(1.0 / 3));
Task T3 = new Task(model, "T3", 25, SchedStrategy.FCFS).on(P2).setThinkTime(new Exp(1.0 / 4));
Entry E1 = new Entry(model, "E1").on(T1);
Entry E2 = new Entry(model, "E2").on(T2);
Entry E3 = new Entry(model, "E3").on(T3);
Activity A1 = new Activity(model, "AS1", new Exp(10)).on(T1).boundTo(E1).synchCall(E2, 1);
Activity A2 = new Activity(model, "AS2", new Exp(20)).on(T2).boundTo(E2).synchCall(E3, 5).repliesTo(E2);
Activity A3 = new Activity(model, "AS3", new Exp(50)).on(T3).boundTo(E3).repliesTo(E3);
return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.30 `gallery_lqn_random`

A randomly generated layered model, used to exercise the layered solvers beyond hand-written topologies.

The example is built and solved as follows.

```

public static LayeredNetwork gallery_lqn_random() {
    return gallery_lqn_random(23000L);
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

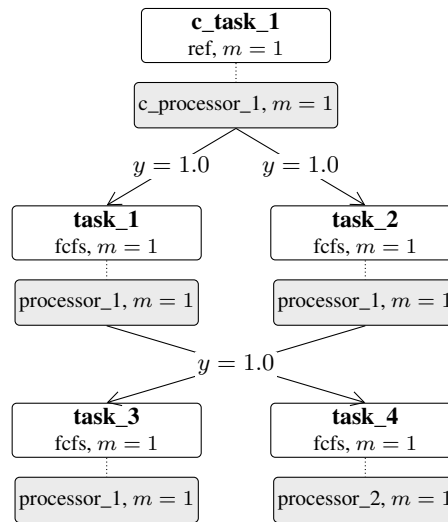


Figure 4.30: Call graph of `gallery_lqn_random`: 5 tasks over 3 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	3: c_processor_1 (inf, $m = 1$), processor_1 (ps, $m = 1$), processor_2 (ps, $m = 1$)
Tasks	5: c_task_1 (ref, $m = 1$), task_1 (fcfs, $m = 1$), task_2 (fcfs, $m = 1$), task_3 (fcfs, $m = 1$), task_4 (fcfs, $m = 1$)
Entries	5: c_entry_1, entry_1, entry_2, entry_3, entry_4
Activities	5, host demands c_activity_1: 1.0, activity_1: 1.0, activity_2: 1.0, activity_3: 1.0, activity_4: 1.0
Calls	4 (4 synchronous, 0 asynchronous)

Table 4.31: Features of `gallery_lqn_random`.

4.2.31 gallery_lqn_workflows

Workflow patterns (sequence, AND-fork and join, loop) as an activity graph in a layered model.

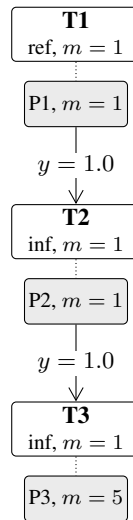


Figure 4.31: Call graph of gallery_lqn_workflows: 3 tasks over 3 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	3: P1 (inf, $m = 1$), P2 (inf, $m = 1$), P3 (ps, $m = 5$)
Tasks	3: T1 (ref, $m = 1$), T2 (inf, $m = 1$), T3 (inf, $m = 1$)
Entries	3: Entry, E2, E3
Activities	14, host demands A1: 1.0, A2: 2.0, A3: 3.0, B1: 0.1, B2: 0.2, B3: 0.3, B4: 0.4, B5: 0.5, ...
Calls	2 (2 synchronous, 0 asynchronous)
Think times	T1: 0.0

Table 4.32: Features of gallery_lqn_workflows.

The example is built and solved as follows.

```

public static LayeredNetwork gallery_lqn_workflows() {
    LayeredNetwork model = new LayeredNetwork("LQN-Workflows");
    Processor P1 = new Processor(model, "P1", Integer.MAX_VALUE, SchedStrategy.INF);
    Task T1 = new Task(model, "T1", 1, SchedStrategy.REF).on(P1);
    T1.setThinkTime(new Immediate());
    Entry E1 = new Entry(model, "Entry").on(T1);

    Processor P2 = new Processor(model, "P2", Integer.MAX_VALUE, SchedStrategy.INF);
    Task T2 = new Task(model, "T2", Integer.MAX_VALUE, SchedStrategy.INF).on(P2).setThinkTime(
        new Immediate());
    Entry E2 = new Entry(model, "E2").on(T2);

    Processor P3 = new Processor(model, "P3", 5, SchedStrategy.PS);
    Task T3 = new Task(model, "T3", Integer.MAX_VALUE, SchedStrategy.INF).on(P3);
    T3.setThinkTime(Exp.fitMean(10));
    Entry E3 = new Entry(model, "E3").on(T3);

    Activity A1 = new Activity(model, "A1", Exp.fitMean(1)).on(T1).boundTo(E1);
    Activity A2 = new Activity(model, "A2", Exp.fitMean(2)).on(T1);
    Activity A3 = new Activity(model, "A3", Exp.fitMean(3)).on(T1).synchCall(E2, 1.0);

    Activity B1 = new Activity(model, "B1", Exp.fitMean(0.1)).on(T2).boundTo(E2);
    Activity B2 = new Activity(model, "B2", Exp.fitMean(0.2)).on(T2);
    Activity B3 = new Activity(model, "B3", Exp.fitMean(0.3)).on(T2);
    Activity B4 = new Activity(model, "B4", Exp.fitMean(0.4)).on(T2);
}

```

```

Activity B5 = new Activity(model, "B5", Exp.fitMean(0.5)).on(T2);
Activity B6 = new Activity(model, "B6", Exp.fitMean(0.6)).on(T2).synchCall(E3, 1.0).
repliesTo(E2);

... (31 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

List<Activity> orJoinSources = new ArrayList<Activity>();
orJoinSources.add(C2);
orJoinSources.add(C3);
orJoinSources.add(C4);
T3.addPrecedence(ActivityPrecedence.OrJoin(orJoinSources, C5));
return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.32 gallery_lukumar_reentrant

The Lu-Kumar re-entrant line, the standard counterexample in which a work-conserving policy destabilizes a network whose stations are individually underloaded.

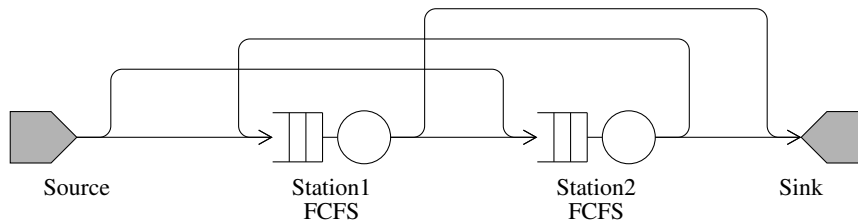


Figure 4.32: Topology of gallery_lukumar_reentrant: an open network over 2 queueing stations.

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	4: Class1 (open), Class2 (open), Class3 (open), Class4 (open)
Scheduling	Station1: FCFS; Station2: FCFS
Arrivals	Source – Class1: Exp(0.08); Class2: Disabled; Class3: Exp(0.08); Class4: Disabled
Service	Station1 – Class1: Exp(0.1); Class2: Disabled; Class3: Disabled; Class4: Exp(1) Station2 – Class1: Disabled; Class2: Exp(1); Class3: Exp(0.1); Class4: Disabled
Features	class switching on 1 link

Table 4.33: Features of gallery_lukumar_reentrant.

The example is built and solved as follows.

```

public static Network gallery_lukumar_reentrant() {
    return gallery_lukumar_reentrant("FCFS");
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.33 gallery_mapm1

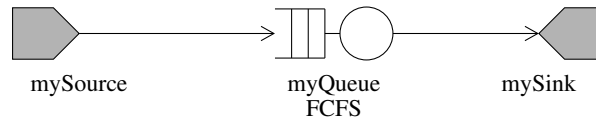
The MAP/M/1 queue: a Markovian arrival process, exponential service, and a single server.

The example is built and solved as follows.

```

public static Network gallery_mapm1() {
    return gallery_mapm1(MAP.rand(2));
}

```

Figure 4.33: Topology of `gallery_mapm1`: an open network over 1 queueing station.

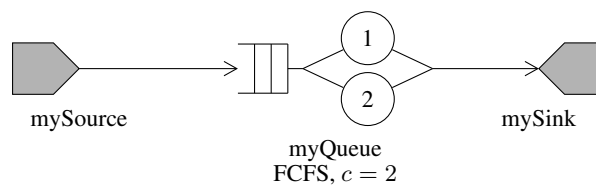
Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass: MAP
Service	myQueue – myClass: Exp(2)

Table 4.34: Features of `gallery_mapm1`.

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.34 `gallery_mapmk`

The MAP/M/ k queue: a Markovian arrival process, exponential service, and k parallel servers.

Figure 4.34: Topology of `gallery_mapmk`: an open network over 1 queueing station.

The example is built and solved as follows.

```
public static Network gallery_mapmk() {
    return gallery_mapmk(MAP.rand(2), 2);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.35 `gallery_mdk`

The M/D/ k queue: Poisson arrivals, deterministic service, and k parallel servers.

The example is built and solved as follows.

```
public static Network gallery_mdk() {
    return gallery_mdk(2);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.36 `gallery_merl1`

The M/Erl/1 queue: Poisson arrivals, Erlang service, and a single server.

The example is built and solved as follows.

```
public static Network gallery_merl1() {
    Network model = new Network("M/E/1");
}
```


Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS ($c = 2$)
Arrivals	mySource – myClass: MAP
Service	myQueue – myClass: Exp(2)

Table 4.35: Features of gallery_mapmk.

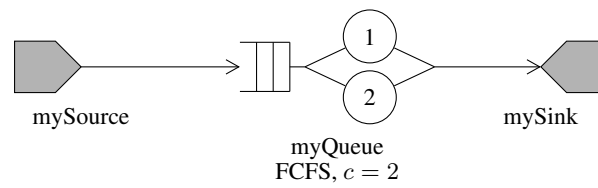


Figure 4.35: Topology of gallery_mdk: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS ($c = 2$)
Arrivals	mySource – myClass: Exp(1)
Service	myQueue – myClass: Det(1)

Table 4.36: Features of gallery_mdk.

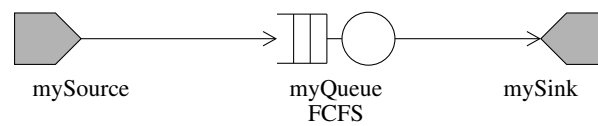


Figure 4.36: Topology of gallery_merl1: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass: Exp(1)
Service	myQueue – myClass: Erlang(4,2)

Table 4.37: Features of gallery_merl1.

```
Source source = new Source(model, "mySource");
Queue queue = new Queue(model, "myQueue", SchedStrategy.FCFS);
Sink sink = new Sink(model, "mySink");
OpenClass oclass = new OpenClass(model, "myClass");
source.setArrival(oclass, new Exp(1));
queue.setService(oclass, Erlang.fitMeanAndOrder(0.5, 2));
model.link(Network.serialRouting(source, queue, sink));
return model;
```

Running this edition prints

```
[Ljava.lang.Object;@46ee7fe8
```

4.2.37 gallery_merl1_linear

The M/Erl/1 queue: Poisson arrivals, Erlang service, and a single server. In this variant, the model is a linear chain of such stations.

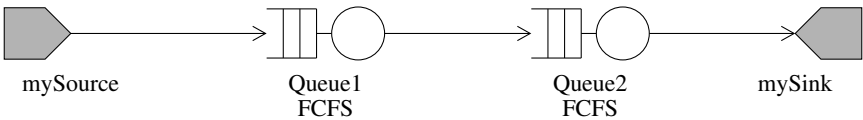


Figure 4.37: Topology of gallery_merl1_linear: an open network over 2 queueing stations.

Nodes	4: Source, Queue × 2, Sink
Classes	1: myClass (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	mySource – myClass: Exp(1)
Service	Queue1 – myClass: Erlang(1.111,1) Queue2 – myClass: Erlang(2.222,2)

Table 4.38: Features of gallery_merl1_linear.

The example is built and solved as follows.

```
public static Network gallery_merl1_linear() {
    return gallery_merl1_linear(2, 0.9);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.38 gallery_merl1_reentrant

The M/Erl/1 queue: Poisson arrivals, Erlang service, and a single server. In this variant, jobs re-enter the station, so a visit is counted more than once.

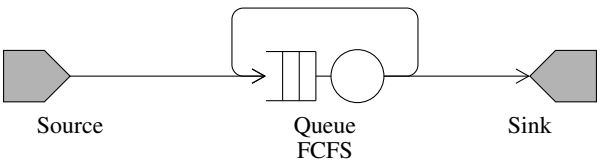


Figure 4.38: Topology of gallery_merl1_reentrant: an open network over 1 queueing station.

The example is built and solved as follows.

```
public static Network gallery_merl1_reentrant() {
    Network model = new Network("M/Erl/1-Reentrant");
}
```

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(1); Class2: Disabled
Service	Queue – Class1: Erlang(10,5); Class2: Exp(3)
Features	class switching on 1 link

Table 4.39: Features of gallery_mer11_reentrant.

```
// Block 1: nodes
Source node1 = new Source(model, "Source");
Queue node2 = new Queue(model, "Queue", SchedStrategy.FCFS);
Sink node3 = new Sink(model, "Sink");

// Block 2: classes
OpenClass jobclass1 = new OpenClass(model, "Class1", 0);
OpenClass jobclass2 = new OpenClass(model, "Class2", 0);

node1.setArrival(jobclass1, Exp.fitMean(1.00)); // (Source,Class1)
node1.setArrival(jobclass2, Disabled.getInstance()); // (Source,Class2)
node2.setService(jobclass1, Erlang.fitMeanAndOrder(0.5, 5)); // (Queue,Class1)
node2.setService(jobclass2, Exp.fitMean(0.333333)); // (Queue,Class2)

// Block 3: topology
RoutingMatrix routingMatrix = model.initRoutingMatrix();

routingMatrix.set(jobclass1, jobclass1, node1, node2, 1.00); // (Source,Class1) -> (Queue,Class1)
routingMatrix.set(jobclass1, jobclass2, node2, node2, 1.00); // (CS_Queue_to_Queue,Class1) -> (Queue,Class2)
routingMatrix.set(jobclass2, jobclass2, node2, node3, 1.00); // (Queue,Class2) -> (Sink,Class2)

model.link(routingMatrix);

return model;
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.39 gallery_mer11_tandem

Two M/Erl/1 queues in series.

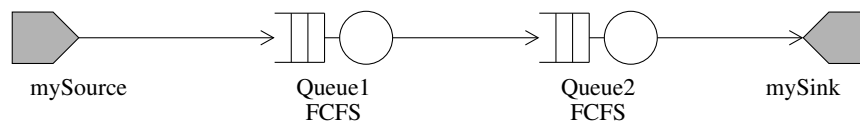


Figure 4.39: Topology of gallery_mer11_tandem: an open network over 2 queueing stations.

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	1: myClass (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	mySource – myClass: Exp(1)
Service	Queue1 – myClass: Erlang(1.111,1) Queue2 – myClass: Erlang(2.222,2)

Table 4.40: Features of gallery_mer11_tandem.

The example is built and solved as follows.

```
public static Network gallery_mer11_tandem() {
    return gallery_mer11_linear(2);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.40 gallery_mer1k

The M/Erl/ k queue: Poisson arrivals, Erlang service, and k parallel servers.

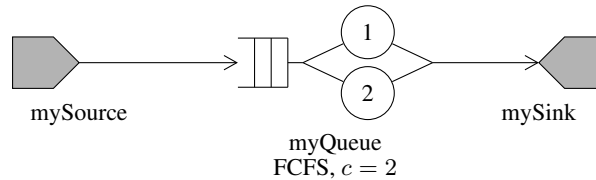


Figure 4.40: Topology of gallery_mer1k: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS ($c = 2$)
Arrivals	mySource – myClass: Exp(1)
Service	myQueue – myClass: Erlang(4,2)

Table 4.41: Features of gallery_mer1k.

The example is built and solved as follows.

```
public static Network gallery_mer1k() {
    return gallery_mer1k(2);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.41 gallery_mhyp1

The M/HyperExp/1 queue: Poisson arrivals, hyperexponential service, and a single server.

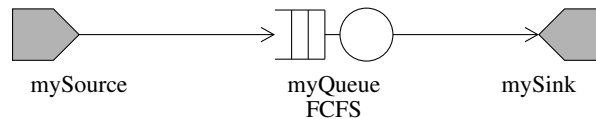


Figure 4.41: Topology of gallery_mhyp1: an open network over 1 queueing station.

The example is built and solved as follows.

```
public static Network gallery_mhyp1() {
    Network model = new Network("M/H/1");
    Source source = new Source(model, "mySource");
    Queue queue = new Queue(model, "myQueue", SchedStrategy.FCFS);
    Sink sink = new Sink(model, "mySink");
    OpenClass oclass = new OpenClass(model, "myClass");
    source.setArrival(oclass, new Exp(1));
    queue.setService(oclass, HyperExp.fitMeanAndSCV(0.5, 4));
    model.link(Network.serialRouting(source, queue, sink));
    return model;
}
```

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass: Exp(1)
Service	myQueue – myClass: HyperExp(0.99,0.01;2.281,0.1517)

Table 4.42: Features of gallery_mhyp1.

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.42 gallery_mhyp1_linear

The M/HyperExp/1 queue: Poisson arrivals, hyperexponential service, and a single server. In this variant, the model is a linear chain of such stations.

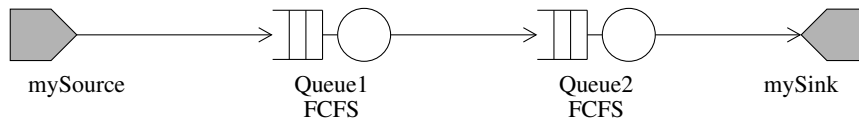


Figure 4.42: Topology of gallery_mhyp1_linear: an open network over 2 queueing stations.

Nodes	4: Source, Queue × 2, Sink
Classes	1: myClass (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	mySource – myClass: Exp(1)
Service	Queue1 – myClass: HyperExp(0.99,0.01;1.196,0.1383) Queue2 – myClass: HyperExp(0.99,0.01;1.196,0.1383)

Table 4.43: Features of gallery_mhyp1_linear.

The example is built and solved as follows.

```
public static Network gallery_mhyp1_linear() {
    return gallery_mhyp1_linear(2, 0.9);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.43 gallery_mhyp1_reentrant

The M/HyperExp/1 queue: Poisson arrivals, hyperexponential service, and a single server. In this variant, jobs re-enter the station, so a visit is counted more than once.

The example is built and solved as follows.

```
public static Network gallery_mhyp1_reentrant() {
    Network model = new Network("M/Hyper/1-Reentrant");

    // Block 1: nodes
    Source node1 = new Source(model, "Source");
    Queue node2 = new Queue(model, "Queue", SchedStrategy.FCFS);
    Sink node3 = new Sink(model, "Sink");

    // Block 2: classes
    OpenClass jobclass1 = new OpenClass(model, "Class1", 0);
    OpenClass jobclass2 = new OpenClass(model, "Class2", 0);
}
```

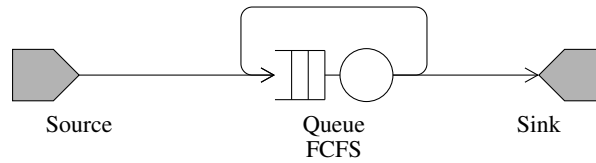


Figure 4.43: Topology of gallery_mhyp1_reentrant: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(1); Class2: Disabled
Service	Queue – Class1: Coxian; Class2: Exp(3)
Features	class switching on 1 link

Table 4.44: Features of gallery_mhyp1_reentrant.

```

node1.setArrival(jobclass1, Exp.fitMean(1.00)); // (Source,Class1)
node1.setArrival(jobclass2, Disabled.getInstance()); // (Source,Class2)
node2.setService(jobclass1, Coxian.fitMeanAndSCV(0.5, 4)); // (Queue,Class1)
node2.setService(jobclass2, Exp.fitMean(0.333333)); // (Queue,Class2)

// Block 3: topology
RoutingMatrix routingMatrix = model.initRoutingMatrix();

routingMatrix.set(jobclass1, jobclass1, node1, node2, 1.00); // (Source,Class1) -> (Queue,Class1)
routingMatrix.set(jobclass1, jobclass2, node2, node2, 1.00); // (CS_Queue_to_Queue,Class1) -> (Queue,Class2)
routingMatrix.set(jobclass2, jobclass2, node2, node3, 1.00); // (Queue,Class2) -> (Sink,Class2)

model.link(routingMatrix);

return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.44 gallery_mhyp1_tandem

Two M/HyperExp/1 queues in series.

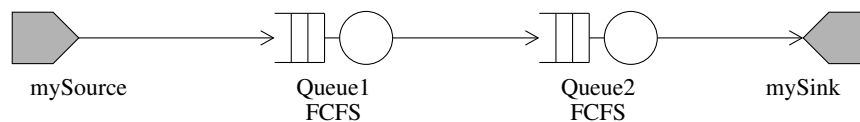


Figure 4.44: Topology of gallery_mhyp1_tandem: an open network over 2 queueing stations.

The example is built and solved as follows.

```

public static Network gallery_mhyp1_tandem() {
    return gallery_mhyp1_linear(2);
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	1: myClass (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	mySource – myClass: Exp(1)
Service	Queue1 – myClass: HyperExp(0.99,0.01;1.196,0.1383) Queue2 – myClass: HyperExp(0.99,0.01;1.196,0.1383)

Table 4.45: Features of gallery_mhyp1_tandem.

4.2.45 gallery_mhypk

The M/HyperExp/ k queue: Poisson arrivals, hyperexponential service, and k parallel servers.

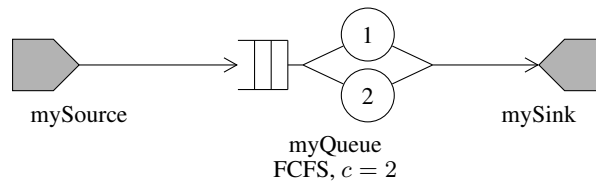


Figure 4.45: Topology of gallery_mhypk: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS ($c = 2$)
Arrivals	mySource – myClass: Exp(1)
Service	myQueue – myClass: Coxian

Table 4.46: Features of gallery_mhypk.

The example is built and solved as follows.

```
public static Network gallery_mhypk() {
    return gallery_mhypk(2);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.46 gallery_mm1

The M/M/1 queue: Poisson arrivals, exponential service, and a single server.

The example is built and solved as follows.

```
Network model = new Network("M/M/1");
Source source = new Source(model, "mySource");
Queue queue = new Queue(model, "myQueue", SchedStrategy.FCFS);
Sink sink = new Sink(model, "mySink");
OpenClass oclass = new OpenClass(model, "myClass");
source.setArrival(oclass, new Exp(1));
queue.setService(oclass, new Exp(2));
model.link(Network.serialRouting(source, queue, sink));
return model;
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

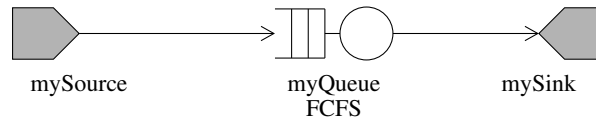


Figure 4.46: Topology of gallery_mm1: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass: Exp(1)
Service	myQueue – myClass: Exp(2)

Table 4.47: Features of gallery_mm1.

4.2.47 gallery_mm1_feedback

The M/M/1 queue: Poisson arrivals, exponential service, and a single server. In this variant, a completing job returns to the station with fixed probability.

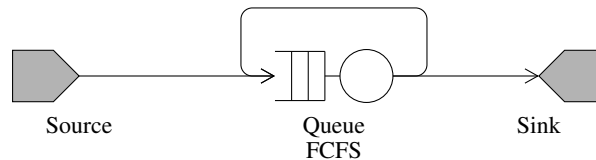


Figure 4.47: Topology of gallery_mm1_feedback: an open network over 1 queueing station.

The example is built and solved as follows.

```
public static Network gallery_mm1_feedback() {
    return gallery_mm1_feedback(1.0 / 3);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.48 gallery_mm1_linear

The M/M/1 queue: Poisson arrivals, exponential service, and a single server. In this variant, the model is a linear chain of such stations.

The example is built and solved as follows.

```
public static Network gallery_mm1_linear() {
    return gallery_mm1_linear(2, 0.9);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.49 gallery_mm1_multiclass

The M/M/1 queue: Poisson arrivals, exponential service, and a single server. In this variant, several open classes share the station.

The example is built and solved as follows.

```
public static Network gallery_mm1_multiclass() {
    Network model = new Network("M[2]/M[2]/1");
}
```


Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(1)
Service	Queue – Class1: Exp(2)

Table 4.48: Features of gallery_mm1_feedback.

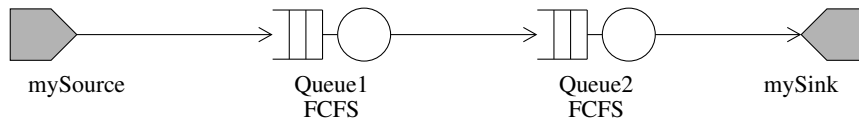


Figure 4.48: Topology of gallery_mm1_linear: an open network over 2 queueing stations.

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	1: myClass (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	mySource – myClass: Exp(1)
Service	Queue1 – myClass: Exp(1.111) Queue2 – myClass: Exp(1.111)

Table 4.49: Features of gallery_mm1_linear.

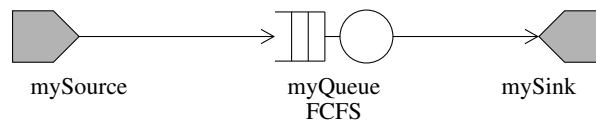


Figure 4.49: Topology of gallery_mm1_multiclass: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: myClass1 (open), myClass2 (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass1: Exp(1); myClass2: Exp(0.5)
Service	myQueue – myClass1: Exp(4); myClass2: Exp(4)

Table 4.50: Features of gallery_mm1_multiclass.

```

Source source = new Source(model, "mySource");
Queue queue = new Queue(model, "myQueue", SchedStrategy.FCFS);
Sink sink = new Sink(model, "mySink");

OpenClass oclass1 = new OpenClass(model, "myClass1");
source.setArrival(oclass1, new Exp(1));
queue.setService(oclass1, new Exp(4));

OpenClass oclass2 = new OpenClass(model, "myClass2");
source.setArrival(oclass2, new Exp(0.5));
queue.setService(oclass2, new Exp(4));

RoutingMatrix P = model.initRoutingMatrix();
P.set(oclass1, Network.serialRouting(source, queue, sink));
P.set(oclass2, Network.serialRouting(source, queue, sink));
model.link(P);
return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.50 gallery_mm1_prio

The M/M/1 queue: Poisson arrivals, exponential service, and a single server. In this variant, the classes carry priorities, served head-of-line.

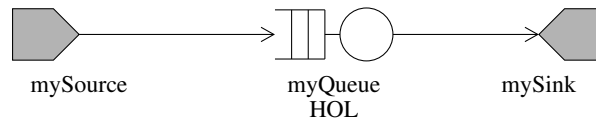


Figure 4.50: Topology of gallery_mm1_prio: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: myClass1 (open), myClass2 (open)
Scheduling	myQueue: HOL
Arrivals	mySource – myClass1: Exp(1); myClass2: Exp(0.5)
Service	myQueue – myClass1: Exp(4); myClass2: Exp(4)

Table 4.51: Features of gallery_mm1_prio.

The example is built and solved as follows.

```

public static Network gallery_mm1_prio() {
    Network model = new Network("M[2]/M[2]/1");
    Source source = new Source(model, "mySource");
    Queue queue = new Queue(model, "myQueue", SchedStrategy.HOL);
    Sink sink = new Sink(model, "mySink");
    OpenClass oclass1 = new OpenClass(model, "myClass1", 1);
    source.setArrival(oclass1, new Exp(1));
    queue.setService(oclass1, new Exp(4));
    OpenClass oclass2 = new OpenClass(model, "myClass2", 0);
    source.setArrival(oclass2, new Exp(0.5));
    queue.setService(oclass2, new Exp(4));
    RoutingMatrix P = model.initRoutingMatrix();
    P.set(oclass1, Network.serialRouting(source, queue, sink));
    P.set(oclass2, Network.serialRouting(source, queue, sink));
    model.link(P);
    return model;
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.51 gallery_mm1_ps

The M/M/1 queue: Poisson arrivals, exponential service, and a single server. In this variant, the station serves under processor sharing rather than FCFS.

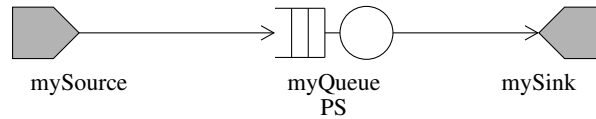


Figure 4.51: Topology of gallery_mm1_ps: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: PS
Arrivals	mySource – myClass: Exp(1)
Service	myQueue – myClass: Exp(2)

Table 4.52: Features of gallery_mm1_ps.

The example is built and solved as follows.

```

public static Network gallery_mm1_ps() {
    Network model = new Network("M/M/1-PS");
    Source source = new Source(model, "mySource");
    Queue queue = new Queue(model, "myQueue", SchedStrategy.PS);
    Sink sink = new Sink(model, "mySink");
    OpenClass oclass = new OpenClass(model, "myClass");
    source.setArrival(oclass, new Exp(1));
    queue.setService(oclass, new Exp(2));
    model.link(Network.serialRouting(source, queue, sink));
    return model;
}
  
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.52 gallery_mm1_ps_feedback

The M/M/1 queue: Poisson arrivals, exponential service, and a single server. In this variant, the station serves under processor sharing rather than FCFS and a completing job returns to the station with fixed probability.

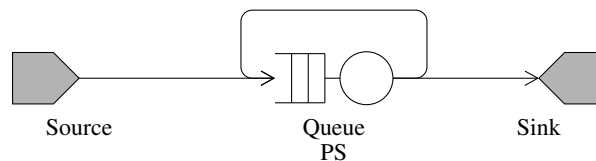


Figure 4.52: Topology of gallery_mm1_ps_feedback: an open network over 1 queueing station.

The example is built and solved as follows.

```

public static Network gallery_mm1_ps_feedback() {
    return gallery_mm1_ps_feedback(1.0 / 3.0);
}
  
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: PS
Arrivals	Source – Class1: Exp(1)
Service	Queue – Class1: Exp(2)

Table 4.53: Features of gallery_mm1_ps_feedback.

4.2.53 gallery_mm1_ps_multiclass

The M/M/1 queue: Poisson arrivals, exponential service, and a single server. In this variant, the station serves under processor sharing rather than FCFS and several open classes share the station.

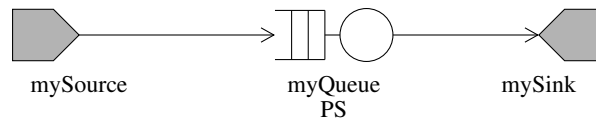


Figure 4.53: Topology of gallery_mm1_ps_multiclass: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: myClass1 (open), myClass2 (open)
Scheduling	myQueue: PS
Arrivals	mySource – myClass1: Exp(1); myClass2: Exp(0.5)
Service	myQueue – myClass1: Exp(4); myClass2: Exp(4)

Table 4.54: Features of gallery_mm1_ps_multiclass.

The example is built and solved as follows.

```

public static Network gallery_mm1_ps_multiclass() {
    Network model = new Network("M[2]/M[2]/1");

    Source source = new Source(model, "mySource");
    Queue queue = new Queue(model, "myQueue", SchedStrategy.PS);
    Sink sink = new Sink(model, "mySink");

    OpenClass oclass1 = new OpenClass(model, "myClass1");
    source.setArrival(oclass1, new Exp(1));
    queue.setService(oclass1, new Exp(4));

    OpenClass oclass2 = new OpenClass(model, "myClass2");
    source.setArrival(oclass2, new Exp(0.5));
    queue.setService(oclass2, new Exp(4));

    RoutingMatrix P = model.initRoutingMatrix();
    P.set(oclass1, Network.serialRouting(source, queue, sink));
    P.set(oclass2, Network.serialRouting(source, queue, sink));
    model.link(P);
    return model;
}
  
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.54 gallery_mm1_ps_reentrant

The M/M/1 queue: Poisson arrivals, exponential service, and a single server. In this variant, the station serves under processor sharing rather than FCFS and jobs re-enter the station, so a visit is counted more than once.

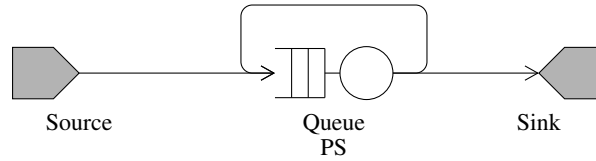


Figure 4.54: Topology of gallery_mm1_ps_reentrant: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue: PS
Arrivals	Source – Class1: Exp(1); Class2: Disabled
Service	Queue – Class1: Exp(2); Class2: Exp(3)
Features	class switching on 1 link

Table 4.55: Features of gallery_mm1_ps_reentrant.

The example is built and solved as follows.

```
public static Network gallery_mm1_ps_reentrant() {
    Network model = new Network("M/M/1-PS-Reentrant");

    // Block 1: nodes
    Source node1 = new Source(model, "Source");
    Queue node2 = new Queue(model, "Queue", SchedStrategy.PS);
    Sink node3 = new Sink(model, "Sink");

    // Block 2: classes
    OpenClass jobclass1 = new OpenClass(model, "Class1", 0);
    OpenClass jobclass2 = new OpenClass(model, "Class2", 0);

    node1.setArrival(jobclass1, Exp.fitMean(1.00)); // (Source,Class1)
    node1.setArrival(jobclass2, Disabled.getInstance()); // (Source,Class2)
    node2.setService(jobclass1, Exp.fitMean(0.50)); // (Queue,Class1)
    node2.setService(jobclass2, Exp.fitMean(0.333333)); // (Queue,Class2)

    // Block 3: topology
    RoutingMatrix routingMatrix = model.initRoutingMatrix();

    routingMatrix.set(jobclass1, jobclass1, node1, node2, 1.00); // (Source,Class1) -> (Queue,Class1)
    routingMatrix.set(jobclass1, jobclass2, node2, node2, 1.00); // (CS_Queue_to_Queue,Class1) -> (Queue,Class2)
    routingMatrix.set(jobclass2, jobclass2, node2, node3, 1.00); // (Queue,Class2) -> (Sink,Class2)

    model.link(routingMatrix);

    return model;
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.55 gallery_mm1_reentrant

The M/M/1 queue: Poisson arrivals, exponential service, and a single server. In this variant, jobs re-enter the station, so a visit is counted more than once.

The example is built and solved as follows.

```
public static Network gallery_mm1_reentrant() {
    Network model = new Network("M/M/1-Reentrant");

    // Block 1: nodes
```

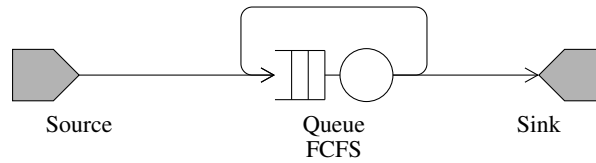


Figure 4.55: Topology of gallery_mm1_reentrant: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(1); Class2: Disabled
Service	Queue – Class1: Exp(2); Class2: Exp(3)
Features	class switching on 1 link

Table 4.56: Features of gallery_mm1_reentrant.

```

Source node1 = new Source(model, "Source");
Queue node2 = new Queue(model, "Queue", SchedStrategy.FCFS);
Sink node3 = new Sink(model, "Sink");

// Block 2: classes
OpenClass jobclass1 = new OpenClass(model, "Class1", 0);
OpenClass jobclass2 = new OpenClass(model, "Class2", 0);

node1.setArrival(jobclass1, Exp.fitMean(1.00)); // (Source,Class1)
node1.setArrival(jobclass2, Disabled.getInstance()); // (Source,Class2)
node2.setService(jobclass1, Exp.fitMean(0.50)); // (Queue,Class1)
node2.setService(jobclass2, Exp.fitMean(0.333333)); // (Queue,Class2)

// Block 3: topology
RoutingMatrix routingMatrix = model.initRoutingMatrix();

routingMatrix.set(jobclass1, jobclass1, node1, node2, 1.00); // (Source,Class1) -> (Queue,Class1)
routingMatrix.set(jobclass1, jobclass2, node2, node2, 1.00); // (CS_Queue_to_Queue,Class1) -> (Queue,Class2)
routingMatrix.set(jobclass2, jobclass2, node2, node3, 1.00); // (Queue,Class2) -> (Sink,Class2)

model.link(routingMatrix);

return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.56 gallery_mm1_tandem

Two M/M/1 queues in series.

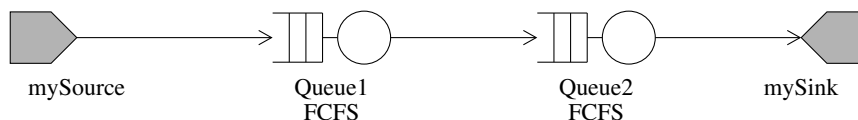


Figure 4.56: Topology of gallery_mm1_tandem: an open network over 2 queueing stations.

The example is built and solved as follows.

```

public static Network gallery_mm1_tandem() {
    return gallery_mm1_linear(2, 0.9);
}

```

Nodes	4: Source, Queue × 2, Sink
Classes	1: myClass (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	mySource – myClass: Exp(1)
Service	Queue1 – myClass: Exp(1.111) Queue2 – myClass: Exp(1.111)

Table 4.57: Features of gallery_mm1_tandem.

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.57 gallery_mm1_tandem_multiclass

Two M/M/1 queues in series under a multiclass workload.

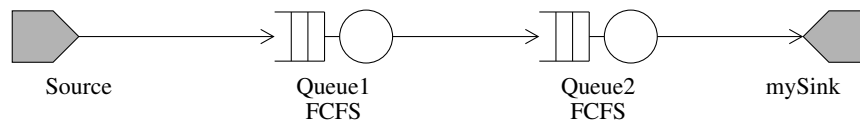


Figure 4.57: Topology of gallery_mm1_tandem_multiclass: an open network over 2 queueing stations.

Nodes	4: Source, Queue × 2, Sink
Classes	2: myClass1 (open), myClass2 (open)
Scheduling	Queue1: FCFS; Queue2: FCFS
Arrivals	Source – myClass1: Exp(1); myClass2: Exp(0.5)
Service	Queue1 – myClass1: Exp(4); myClass2: Exp(2) Queue2 – myClass1: Exp(6); myClass2: Exp(6)

Table 4.58: Features of gallery_mm1_tandem_multiclass.

The example is built and solved as follows.

```

public static Network gallery_mm1_tandem_multiclass() {
    Network model = new Network("M[2]/M[2]/1 -> -/M[2]/1");

    Source source = new Source(model, "Source");
    Queue queue1 = new Queue(model, "Queue1", SchedStrategy.FCFS);
    Queue queue2 = new Queue(model, "Queue2", SchedStrategy.FCFS);
    Sink sink = new Sink(model, "mySink");

    OpenClass oclass1 = new OpenClass(model, "myClass1");
    source.setArrival(oclass1, new Exp(1));
    queue1.setService(oclass1, new Exp(4));
    queue2.setService(oclass1, new Exp(6));

    OpenClass oclass2 = new OpenClass(model, "myClass2");
    source.setArrival(oclass2, new Exp(0.5));
    queue1.setService(oclass2, new Exp(2));
    queue2.setService(oclass2, new Exp(6));

    RoutingMatrix P = model.initRoutingMatrix();
    P.set(oclass1, Network.serialRouting(source, queue1, queue2, sink));
    P.set(oclass2, Network.serialRouting(source, queue1, queue2, sink));
    model.link(P);

    return model;
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.58 gallery_mm1k

The M/M/1/K queue: a single exponential server behind a buffer of K places, so an arrival finding the buffer full is lost.

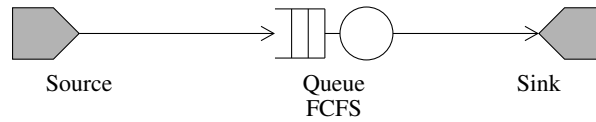


Figure 4.58: Topology of gallery_mm1k: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(0.8)
Service	Queue – Class1: Exp(1)
Features	finite buffer 3 at Queue

Table 4.59: Features of gallery_mm1k.

The example is built and solved as follows.

```
public static Network gallery_mm1k() {
    return gallery_mm1k(3);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.59 gallery_mmap1

The M/MAP/1 queue: Poisson arrivals, MAP-modulated service, and a single server.

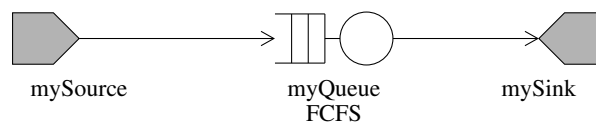


Figure 4.59: Topology of gallery_mmap1: an open network over 1 queueing station.

The example is built and solved as follows.

```
public static Network gallery_mmap1() {
    MAP map = MAP.rand();
    map.setMean(0.5);
    return gallery_mmap1(map);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.60 gallery_mmap1_multiclass

The M/MAP/1 queue: Poisson arrivals, MAP-modulated service, and a single server. In this variant, several open classes share the station.

The example is built and solved as follows.

```
public static Network gallery_mmap1_multiclass() {
    MAP map1 = MAP.rand();
}
```


Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass: Exp(1)
Service	myQueue – myClass: MAP

Table 4.60: Features of gallery_mmap1.

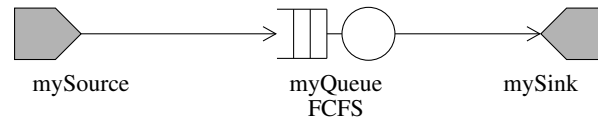


Figure 4.60: Topology of gallery_mmap1_multiclass: an open network over 1 queueing station.

```

map1.setMean(0.5);
MAP map2 = MAP.rand();
map2.setMean(0.5);
return gallery_mmap1_multiclass(map1, map2);

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.61 gallery_mmapk

The M/MAP/ k queue: Poisson arrivals, MAP-modulated service, and k parallel servers.
The example is built and solved as follows.

```

public static Network gallery_mmapk() {
    return gallery_mmapk(MAP.rand(), 2);
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.62 gallery_mmk

The M/M/ k queue: Poisson arrivals, exponential service, and k parallel servers.
The example is built and solved as follows.

```

public static Network gallery_mmk() {
    return gallery_mmk(2);
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.63 gallery_mpar1

The M/Pareto/1 queue: Poisson arrivals, Pareto service, and a single server.

Nodes	3: Source, Queue, Sink
Classes	2: myClass1 (open), myClass2 (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass1: Exp(0.7); myClass2: Exp(0.3)
Service	myQueue – myClass1: MAP; myClass2: MAP

Table 4.61: Features of gallery_mmap1_multiclass.

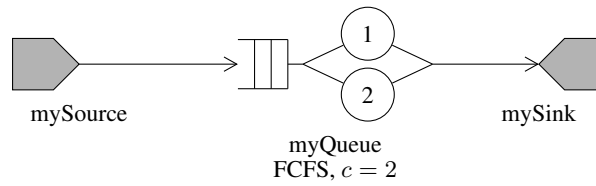


Figure 4.61: Topology of gallery_mmapk: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS ($c = 2$)
Arrivals	mySource – myClass: Exp(1)
Service	myQueue – myClass: MAP

Table 4.62: Features of gallery_mmapk.

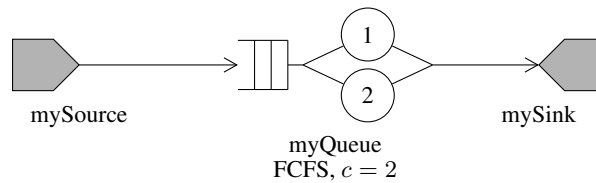


Figure 4.62: Topology of gallery_mmk: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS ($c = 2$)
Arrivals	mySource – myClass: Exp(1)
Service	myQueue – myClass: Exp(2)

Table 4.63: Features of gallery_mmk.

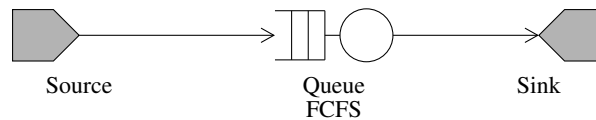


Figure 4.63: Topology of gallery_mpar1: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(1)
Service	Queue – Class1: Pareto(2.008,0.251)

Table 4.64: Features of gallery_mpar1.

The example is built and solved as follows.

```
public static Network gallery_mpar1() {
    Network model = new Network("M/Par/1");
    Source source = new Source(model, "Source");
    Queue queue = new Queue(model, "Queue", SchedStrategy.FCFS);
    Sink sink = new Sink(model, "Sink");
    OpenClass oclass = new OpenClass(model, "Class1");
    source.setArrival(oclass, new Exp(1));
    queue.setService(oclass, Pareto.fitMeanAndSCV(0.5, 64));
    model.link(Network.serialRouting(source, queue, sink));
    return model;
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.64 gallery_multitier

A multitier application: presentation, application and database tiers as a closed network.

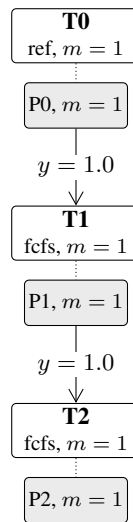


Figure 4.64: Call graph of `gallery_multitier`: 3 tasks over 3 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	3: P0 (ps, $m = 1$), P1 (ps, $m = 1$), P2 (ps, $m = 1$)
Tasks	3: T0 (ref, $m = 1$), T1 (fcfs, $m = 1$), T2 (fcfs, $m = 1$)
Entries	9: E0, E10, E11, E12, E13, E20, E21, E22, E23
Activities	22, host demands A0: 1.0, A1: 1.0, A2: 1.0, A3: 1.0, B0: 1.0, B1: 1.0, B2: 1.0, B3: 1.0, ...
Calls	8 (8 synchronous, 0 asynchronous)

Table 4.65: Features of `gallery_multitier`.

The example is built and solved as follows.

```
public static LayeredNetwork gallery_multitier() {
    LayeredNetwork model = new LayeredNetwork("testLQN3");

    // Layer 1: client
    Processor P0 = new Processor(model, "P0", 1, SchedStrategy.PS);
    Task T0 = new Task(model, "T0", 1, SchedStrategy.REF).on(P0);
    Entry E0 = new Entry(model, "E0").on(T0);

    // Layer 2: application server
}
```

```

Processor P1 = new Processor(model, "P1", 1, SchedStrategy.PS);
Task T1 = new Task(model, "T1", 1, SchedStrategy.FCFS).on(P1);
Entry E10 = new Entry(model, "E10").on(T1);
Entry E11 = new Entry(model, "E11").on(T1);
Entry E12 = new Entry(model, "E12").on(T1);
Entry E13 = new Entry(model, "E13").on(T1);

// Layer 3: database
Processor P2 = new Processor(model, "P2", 1, SchedStrategy.PS);
Task T2 = new Task(model, "T2", 1, SchedStrategy.FCFS).on(P2);
Entry E20 = new Entry(model, "E20").on(T2);
Entry E21 = new Entry(model, "E21").on(T2);
Entry E22 = new Entry(model, "E22").on(T2);
Entry E23 = new Entry(model, "E23").on(T2);

// Client activities
Activity A0 = new Activity(model, "A0", new Exp(1.0)).on(T0).boundTo(E0).synchCall(E12,
1.0);

... (40 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

b7sources.add(B7b);
T1.addPrecedence(ActivityPrecedence.OrJoin(b7sources, B8));
T2.addPrecedence(ActivityPrecedence.Serial(C0, C1));
T2.addPrecedence(ActivityPrecedence.Serial(C3, C4, C5));

return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.65 gallery_multitier_storage

The multitier application with an explicit storage tier behind the database.

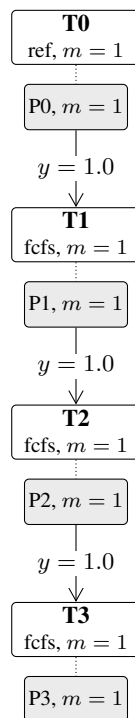


Figure 4.65: Call graph of gallery_multitier_storage: 4 tasks over 4 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	4: P0 (ps, $m = 1$), P1 (ps, $m = 1$), P2 (ps, $m = 1$), P3 (ps, $m = 1$)
Tasks	4: T0 (ref, $m = 1$), T1 (fcfs, $m = 1$), T2 (fcfs, $m = 1$), T3 (fcfs, $m = 1$)
Entries	10: E0, E10, E11, E12, E13, E20, E21, E22, E23, E3
Activities	24, host demands A0: 1.0, A1: 1.0, A2: 1.0, A3: 1.0, B0: 1.0, B1: 1.0, B2: 1.0, B3: 1.0, ...
Calls	10 (10 synchronous, 0 asynchronous)

Table 4.66: Features of gallery_multitier_storage.

The example is built and solved as follows.

```
public static LayeredNetwork gallery_multitier_storage() {
    LayeredNetwork model = new LayeredNetwork("testLQN3_Cache");

    // Layer 1: client
    Processor P0 = new Processor(model, "P0", 1, SchedStrategy.PS);
    Task T0 = new Task(model, "T0", 1, SchedStrategy.REF).on(P0);
    Entry E0 = new Entry(model, "E0").on(T0);

    // Layer 2: application server
    Processor P1 = new Processor(model, "P1", 1, SchedStrategy.PS);
    Task T1 = new Task(model, "T1", 1, SchedStrategy.FCFS).on(P1);
    Entry E10 = new Entry(model, "E10").on(T1);
    Entry E11 = new Entry(model, "E11").on(T1);
    Entry E12 = new Entry(model, "E12").on(T1);
    Entry E13 = new Entry(model, "E13").on(T1);

    // Layer 3: database
    Processor P2 = new Processor(model, "P2", 1, SchedStrategy.PS);
    Task T2 = new Task(model, "T2", 1, SchedStrategy.FCFS).on(P2);
    Entry E20 = new Entry(model, "E20").on(T2);
    Entry E21 = new Entry(model, "E21").on(T2);
    Entry E22 = new Entry(model, "E22").on(T2);
    Entry E23 = new Entry(model, "E23").on(T2);

    // Layer 4: cache (10 items, capacity 2, LRU, uniform access)
    int totalitems = 10;

    ... (57 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

    List<Activity> cacheTargets = new ArrayList<Activity>();
    cacheTargets.add(D1a);
    cacheTargets.add(D1b);
    T3.addPrecedence(ActivityPrecedence.CacheAccess(D0, cacheTargets));

    return model;
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.66 gallery_parm1

The Pareto/M/1 queue: Pareto interarrival times, exponential service, and a single server.

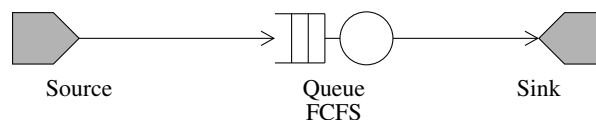


Figure 4.66: Topology of gallery_parm1: an open network over 1 queueing station.

The example is built and solved as follows.

```
public static Network gallery_parm1() {
    Network model = new Network("Par/M/1");
    Source source = new Source(model, "Source");
}
```

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Pareto(2.008,0.5019)
Service	Queue – Class1: Exp(2)

Table 4.67: Features of gallery_parm1.

```

Queue queue = new Queue(model, "Queue", SchedStrategy.FCFS);
Sink sink = new Sink(model, "Sink");
OpenClass oclass = new OpenClass(model, "Class1");
source.setArrival(oclass, Pareto.fitMeanAndSCV(1, 64));
queue.setService(oclass, new Exp(2));
model.link(Network.serialRouting(source, queue, sink));
return model;

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.67 gallery_qn_random

A randomly generated queueing network, used as a generator-driven parity fixture.

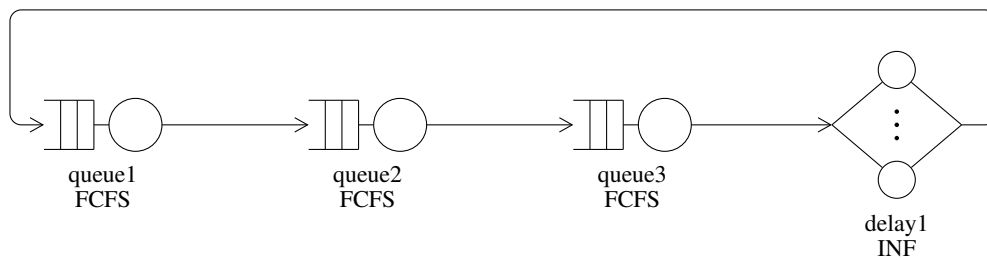


Figure 4.67: Topology of gallery_qn_random: a closed network over 4 queueing stations.

Nodes	4: Queue $\times$ 3, Delay
Classes	2: CClass1 (closed, $N = 14$), CClass2 (closed, $N = 11$)
Scheduling	queue1: FCFS; queue2: FCFS; queue3: FCFS; delay1: INF
Service	queue1 – CClass1: Exp(1); CClass2: Exp(1) queue2 – CClass1: Exp(1); CClass2: Exp(1) queue3 – CClass1: Exp(1); CClass2: Exp(1) delay1 – CClass1: Exp(1); CClass2: Exp(1)

Table 4.68: Features of gallery_qn_random.

The example is built and solved as follows.

```

public static Network gallery_qn_random() {
    return gallery_qn_random(23000L);
}

```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.68 gallery_renv_breakdown

A station in a random environment that alternates between an operational and a failed stage.

No topology is drawn for this example: the captured run yielded no `Network` or `LayeredNetwork` to draw one from, either because the script builds a model of another kind (an environment or a workflow) or because it builds it inside a helper it does not expose.

The example is built and solved as follows.

```
public static Environment gallery_renv_breakdown() {
    Network model = new Network("ServerWithFailures");
    Source source = new Source(model, "Arrivals");
    Queue queue = new Queue(model, "Server", SchedStrategy.FCFS);
    Sink sink = new Sink(model, "Departures");
    OpenClass jobclass = new OpenClass(model, "Jobs");
    source.setArrival(jobclass, new Exp(0.8));
    queue.setService(jobclass, new Exp(2.0));
    queue.setNumberOfServers(1);
    RoutingMatrix P = model.initRoutingMatrix();
    P.set(jobclass, jobclass, source, queue, 1.0);
    P.set(jobclass, jobclass, queue, sink, 1.0);
    model.link(P);
    Environment env = new Environment("ServerEnv");
    env.addNodeFailureRepair(model, queue, new Exp(0.1), new Exp(1.0), new Exp(0.5));
    env.init();
    return env;
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.69 gallery_repairmen

The machine repairman model: machines think, fail, and queue for one repair station.

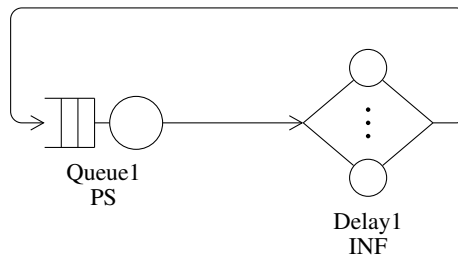


Figure 4.68: Topology of gallery_repairmen: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 8$)
Scheduling	Queue1: PS; Delay1: INF
Service	Queue1 – Class1: Exp(0.7685) Delay1 – Class1: Exp(0.5)

Table 4.69: Features of gallery_repairmen.

The example is built and solved as follows.

```
public static Network gallery_repairmen() {
    return gallery_repairmen(2, 23000);
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.70 gallery_replayerm1

A single queue whose interarrival times are replayed from a measurement trace.

The example is built and solved as follows.

```
public static Network gallery_replayerm1() {
    return gallery_replayerm1(ExampleData.path("/example_trace.txt"));
}
```

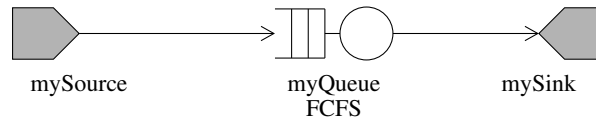


Figure 4.69: Topology of gallery_replayerm1: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: myClass (open)
Scheduling	myQueue: FCFS
Arrivals	mySource – myClass: Replayer(trace)
Service	myQueue – myClass: Exp(29.68)

Table 4.70: Features of gallery_replayerm1.

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

4.2.71 gallery_um1

The U/M/1 queue: uniform interarrival times, exponential service, and a single server.

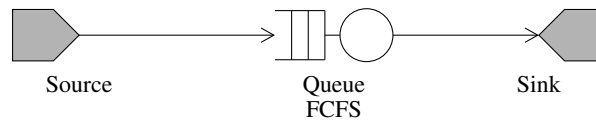


Figure 4.70: Topology of gallery_um1: an open network over 1 queueing station.

The example is built and solved as follows.

```
public static Network gallery_um1() {
    Network model = new Network("U/M/1");
    Source source = new Source(model, "Source");
    Queue queue = new Queue(model, "Queue", SchedStrategy.FCFS);
    Sink sink = new Sink(model, "Sink");
    OpenClass oclass = new OpenClass(model, "Class1");
    source.setArrival(oclass, new Uniform(1, 2));
    queue.setService(oclass, new Exp(2));
    model.link(Network.serialRouting(source, queue, sink));
    return model;
}
```

The captured run did not complete successfully. No result is reproduced, and this edition makes no completion claim.

Each of these is also a parity fixture: the four codebases build the same model and the harness compares the metrics solver by solver, so a divergence in any one of them is caught here before it reaches a larger model.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Uniform(1,2)
Service	Queue – Class1: Exp(2)

Table 4.71: Features of `gallery_um1`.

Chapter 5

Discrete-time models

In a slotted model time advances in slots of fixed length, and every sampled interarrival and service time must fall on the slot lattice. A sample that does not is an error rather than something to round: refusing is what keeps a discrete-time result exact. Within a slot the events are ordered by a fixed phase – completions, internal movement, arrivals, bookkeeping – so that a departure and an arrival in the same slot cannot be confused.

Geometric interarrival and service times give the Geo/Geo/1 queue, whose exact closed form is known, and it is against that form that the solvers are checked. The models live under `jline.examples.java.discrete`, and slotted mode is requested through the solver options rather than by the model.

```
Network model = new Network("GeoGeo1");
Source source = new Source(model, "Source");
Queue queue = new Queue(model, "Queue", SchedStrategy.FCFS);
Sink sink = new Sink(model, "Sink");
OpenClass jobClass = new OpenClass(model, "Class1");
source.setArrival(jobClass, new Geometric(a));
queue.setService(jobClass, new Geometric(s));
model.link(Network.serialRouting(source, queue, sink));

new SolverNC(model, slotted()).getAvgTable().print();
System.out.printf("closed form: E[N] = %g, E[T] = %g slots%n",
    a * (1 - a) / (s - a), (1 - a) / (s - a));
```

5.1 The slotted models

5.1.1 dt_bernoulli_loaddep

A load-dependent Bernoulli server, $p(n) = s \min(n, c)$, the discrete-time multiserver approximation.

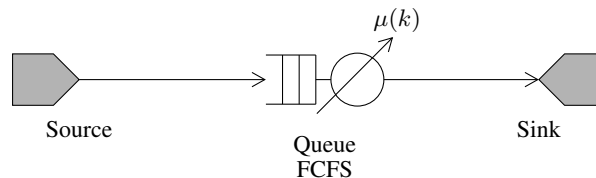


Figure 5.1: Topology of `dt_bernoulli_loaddep`: an open network over 1 queueing station.

The example is built and solved as follows.

```
System.out.println("\n--- ex3_bernoulli_loaddep: example 2.10 and theorem 2.11 ---");
double a = 0.6;
double s = 0.3;
```

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Geometric(0.6)
Service	Queue – Class1: Geometric(0.3)
Features	finite buffer 20 at Queue; load-dependent rates at Queue
Solvers	NC

Table 5.1: Features of dt_bernoulli_loaddep.

```

int c = 3;
int L = 20;
Network model = DiscreteModel.dt_bernoulli_loaddep(a, s, c, L);
new SolverNC(model, slotted()).getAvgTable().print();
double[] p = new double[L];
for (int n = 1; n <= L; n++) {
    p[n - 1] = s * Math.min(n, c);
}
Bernoulli1Result r = Dqsys_bernoulli1.dqsys_bernoulli1(new double[]{a}, p, L);
System.out.println("time-stationary law pi(0..6) : "
    + Arrays.toString(Arrays.copyOf(r.pmf, 7)));
System.out.println("arrival law pi_1(0..6): "
    + Arrays.toString(Arrays.copyOf(r.arrivalPmf, 7)));
System.out.println("no PASTA in discrete time: the two rows above are different laws");

```

Running this edition prints

```

--- ex3_bernoulli_loaddep: example 2.10 and theorem 2.11 ---
NC analysis [method: dt.bernoulli1; type: exact, deterministic; lang: java; env: python] completed in <t>s
.
Station JobClass QLen Util RespT ResidT ArvR Tput
Source Class1 0 0 0 0 0 0.6
Queue Class1 2.0644 0.95402 3.4406 3.4406 0.6 0.6
time-stationary law pi(0..6) : [0.04597701149425346, 0.22988505747126736, 0.40229885057471787,
0.2681992337164785, 0.04469987228607978, 0.0074499787143466325, 0.001241663119057774]
arrival law pi_1(0..6): [0.11494252873563435, 0.40229885057472026, 0.40229885057472026,
0.06704980842912008, 0.011174968071520018, 0.0018624946785866705, 3.104157797644452E-4]
no PASTA in discrete time: the two rows above are different laws

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by NC.

5.1.2 dt_cycle

A closed cycle of slotted stations.

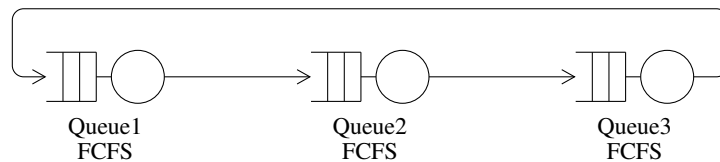


Figure 5.2: Topology of dt_cycle: a closed network over 3 queueing stations.

The example is built and solved as follows.

```

System.out.println("\n--- ex4_cycle: closed cycle, corollary 3.4 ---");
double[] p = DiscreteModel.defaultCycleRates();
int N = 5;
Network model = DiscreteModel.dt_cycle(p, N);
new SolverNC(model, slotted()).getAvgTable().print();
DpfqnNcResult nc = Dpfqn_nc.dpfqn_nc(p, N);
System.out.printf("log G(N,J) = %g%n", nc.lG);
System.out.printf("throughput = %g jobs per slot%n", nc.throughput());

```

Nodes	3: Queue $\times$ 3
Classes	1: Jobs (closed, $N = 5$)
Scheduling	Queue1: FCFS; Queue2: FCFS; Queue3: FCFS
Service	Queue1 – Jobs: Geometric(0.5) Queue2 – Jobs: Geometric(0.25) Queue3 – Jobs: Geometric(0.7)
Solvers	NC

Table 5.2: Features of dt_cycle.

```
double[] util = new double[p.length];
for (int j = 0; j < p.length; j++) {
    util[j] = nc.throughput() / p[j];
}
System.out.println("utilizations = " + Arrays.toString(util));
```

Running this edition prints

```
--- ex4_cycle: closed cycle, corollary 3.4 ---
NC analysis [method: dt.cycle; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Queue1 Jobs 0.72191 0.49656 2.9076 2.9076 0.24828 0.24828
Queue2 Jobs 3.8668 0.99313 15.574 15.574 0.24828 0.24828
Queue3 Jobs 0.41125 0.35469 1.6564 1.6564 0.24828 0.24828
log G(N,J) = 6.90882
throughput = 0.248282 jobs per slot
utilizations = [0.4965639088467044, 0.9931278176934089, 0.3546885063190746]
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by NC.

5.1.3 dt_cycle_loaddep

The cycle with a load-dependent slotted server.

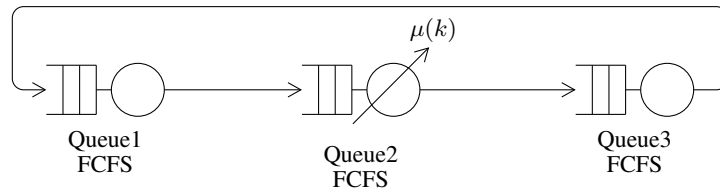


Figure 5.3: Topology of dt_cycle_loaddep: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 3
Classes	1: Jobs (closed, $N = 5$)
Scheduling	Queue1: FCFS; Queue2: FCFS; Queue3: FCFS
Service	Queue1 – Jobs: Geometric(0.5) Queue2 – Jobs: Geometric(0.25) Queue3 – Jobs: Geometric(0.7)
Features	load-dependent rates at Queue2
Solvers	NC

Table 5.3: Features of dt_cycle_loaddep.

The example is built and solved as follows.

```
System.out.println("\n--- ex5_cycle_loaddep: closed cycle, theorem 3.2 ---");
double[] p = DiscreteModel.defaultCycleRates();
int N = 5;
Network model = DiscreteModel.dt_cycle_loaddep(p, N, 1, 2);
new SolverNC(model, slotted()).getAvgTable().print();
```

```
double[][] P = new double[p.length][N];
for (int j = 0; j < p.length; j++) {
    for (int n = 1; n <= N; n++) {
        P[j][n - 1] = (j == 1) ? p[j] * Math.min(n, 2) : p[j];
    }
}
DpfqnNcLdResult nc = Dpfqn_ncld.dpfqn_ncld(P, N);
System.out.println("P(X_2 = 0..N) = " + Arrays.toString(nc.marginal(1)));
```

Running this edition prints

```
--- ex5_cycle_loaddep: closed cycle, theorem 3.2 ---
NC analysis [method: dt.cycleld; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Queue1 Jobs 1.7936 0.84515 4.2445 4.2445 0.42258 0.42258
Queue2 Jobs 2.3912 0.94757 5.6586 5.6586 0.42258 0.42258
Queue3 Jobs 0.81523 0.60368 1.9292 1.9292 0.42258 0.42258
P(X_2 = 0..N) = [0.05243257509316642, 0.20483277208263767, 0.29010780910885847, 0.25011132807362985,
0.15678620565809628, 0.045729309983611396]
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by NC.

5.1.4 dt_cycle_multiclass

The multiclass cycle.

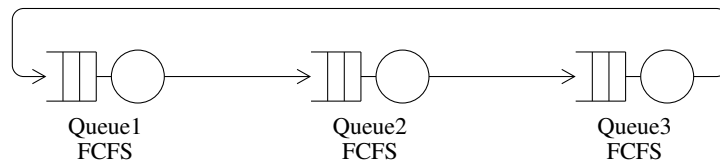


Figure 5.4: Topology of dt_cycle_multiclass: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 3
Classes	2: Chain1 (closed, $N = 3$), Chain2 (closed, $N = 2$)
Scheduling	Queue1: FCFS; Queue2: FCFS; Queue3: FCFS
Service	Queue1 – Chain1: Geometric(0.5); Chain2: Geometric(0.5) Queue2 – Chain1: Geometric(0.25); Chain2: Geometric(0.25) Queue3 – Chain1: Geometric(0.7); Chain2: Geometric(0.7)
Solvers	NC

Table 5.4: Features of dt_cycle_multiclass.

The example is built and solved as follows.

```
System.out.println("\n--- ex6_cycle_multiclass: multichain cycle, section 3.2 ---");
double[] p = DiscreteModel.defaultCycleRates();
int[] pops = new int[]{3, 2};
Network model = DiscreteModel.dt_cycle_multiclass(p, pops);
new SolverNC(model, slotted()).getAvgTable().print();
DpfqnNcResult nc = Dpfqn_nc.dpfqn_nc(p, pops[0] + pops[1]);
double X = nc.throughput();
System.out.printf("aggregate throughput = %g jobs per slot\n", X);
System.out.printf("per-chain throughput = %g and %g\n",
    X * pops[0] / (pops[0] + pops[1]), X * pops[1] / (pops[0] + pops[1]));
```

Running this edition prints

```
--- ex6_cycle_multiclass: multichain cycle, section 3.2 ---
NC analysis [method: dt.cycle; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Station JobClass QLen Util RespT ResidT ArvR Tput
Queue1 Chain1 0.43315 0.29794 2.9076 2.9076 0.14897 0.14897
Queue1 Chain2 0.28876 0.19863 2.9076 2.9076 0.099313 0.099313
```

Queue2	Chain1	2.3201	0.59588	15.574	15.574	0.14897	0.14897
Queue2	Chain2	1.5467	0.39725	15.574	15.574	0.099313	0.099313
Queue3	Chain1	0.24675	0.21281	1.6564	1.6564	0.14897	0.14897
Queue3	Chain2	0.1645	0.14188	1.6564	1.6564	0.099313	0.099313
aggregate throughput		= 0.248282 jobs per slot					
per-chain throughput		= 0.148969 and 0.0993128					

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by NC.

5.1.5 dt_geogeo1

The Geo/Geo/1 queue: Bernoulli arrivals and Bernoulli service on the slot lattice, against the closed form of Daduna.

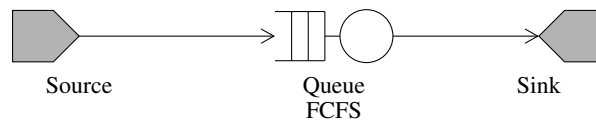


Figure 5.5: Topology of dt_geogeo1: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Geometric(0.2)
Service	Queue – Class1: Geometric(0.5)
Solvers	NC

Table 5.5: Features of dt_geogeo1.

The example is built and solved as follows.

```
System.out.println("\n--- ex1_geogeo1: Geo/Geo/1 on the slot lattice ---");
double a = 0.2;
double s = 0.5;
Network model = DiscreteModel.dt_geogeo1(a, s);
new SolverNC(model, slotted()).getAvgTable().print();
System.out.printf("closed form: E[N] = %g, E[T] = %g slots\n",
    a * (1 - a) / (s - a), (1 - a) / (s - a));
```

Running this edition prints

```
--- ex1_geogeo1: Geo/Geo/1 on the slot lattice ---
NC analysis [method: dt.bernoullil1; type: exact, deterministic; lang: java; env: python] completed in <t>s
.
Station JobClass    QLen  Util   RespT  ResidT  ArvR  Tput
Source  Class1        0      0       0       0       0    0.2
Queue   Class1      0.53333  0.4    2.6667  2.6667  0.2  0.2
closed form: E[N] = 0.533333, E[T] = 2.66667 slots
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by NC.

5.1.6 dt_geogeo1_loss

The Geo/Geo/1/L loss system, and the loss probability.

The example is built and solved as follows.

```
System.out.println("\n--- ex2_geogeo1_loss: finite buffer, corollary 2.8 ---");
double a = 0.2;
double s = 0.5;
```

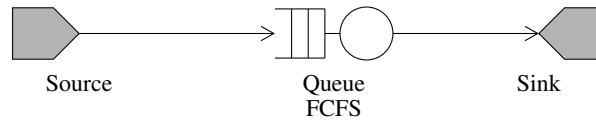


Figure 5.6: Topology of dt_geogeol_loss: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Class1 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Geometric(0.2)
Service	Queue – Class1: Geometric(0.5)
Features	finite buffer 4 at Queue
Solvers	NC

Table 5.6: Features of dt_geogeol_loss.

```

int L = 4;
Network model = DiscreteModel.dt_geogeol_loss(a, s, L);
new SolverNC(model, slotted()).getAvgTable().print();
BernoulliResult r = Dqsys_bernoulli1.dqsys_bernoulli1(new double[] {a}, new double[] {s}, L);
System.out.println("queue length law : " + Arrays.toString(r.pmf));
System.out.printf("loss probability : %g\n", r.lossProb);
System.out.printf("carried throughput : %g of the %g offered per slot\n", r.throughput, a);

```

Running this edition prints

```

--- ex2_geogeol_loss: finite buffer, corollary 2.8 ---
NC analysis [method: dt.bernoulli1; type: exact, deterministic; lang: java; env: python] completed in <t>s
.
Station JobClass   QLen   Util   RespT   ResidT   ArvR   Tput
Source   Class1      0       0       0       0       0       0.2
Queue    Class1      0.52256 0.3985  2.6226  2.6226  0.2    0.19925
queue length law : [0.6015037593984962, 0.3007518796992481, 0.07518796992481204, 0.018796992481203006,
0.0037593984962406026]
loss probability : 0.00375940
carried throughput : 0.199248 of the 0.200000 offered per slot

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by NC.

Superposing two slotted streams produces batches, not a slotted stream of the same kind, so a tandem of slotted queues is the first model in which a discrete-time implementation can go wrong while the single queue still looks right. dt_cycle exists for that reason.

Chapter 6

Solver-specific examples

Most examples in this manual are written around a model and let the solver vary. The scripts of this chapter do the opposite: they are written around one solver, and print what it builds rather than only what it concludes. They live under the solver example classes.

The CTMC solver is the natural subject, because its intermediate objects are the model: the state space, the infinitesimal generator and the stationary distribution are all printable, and the averages are read off them. Printing them is how one checks that a model means what it was intended to mean.

6.0.1 `ctmc_spn_mm1`

The same M/M/1 system written as a queueing model and as a Petri net, and the two generators compared.

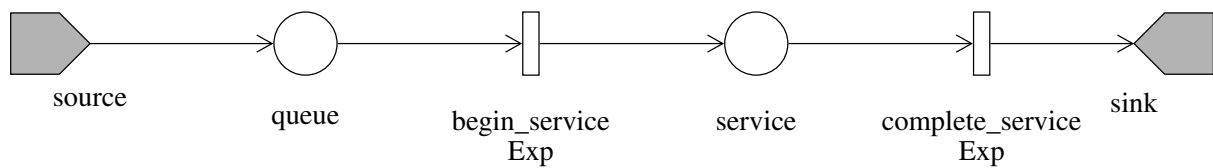


Figure 6.1: Topology of `ctmc_spn_mm1`: a stochastic Petri net over 2 places and 2 transitions; a place is a circle holding its initial marking and a transition a bar, hollow when its firing time is random and solid when it fires at once; the net is open, fed by a Source and drained into a Sink.

Nodes	6: Source, Place $\times$ 2, Transition $\times$ 2, Sink
Classes	1: jobs (open)
Arrivals	source – jobs: Exp(0.8)
Features	2 transitions with 1 firing mode each
Solvers	CTMC

Table 6.1: Features of `ctmc_spn_mm1`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

6.0.2 `ctmc_spn_vs_nc_closed_qn`

A closed network solved exactly by CTMC over its Petri net form and by the normalizing constant method.

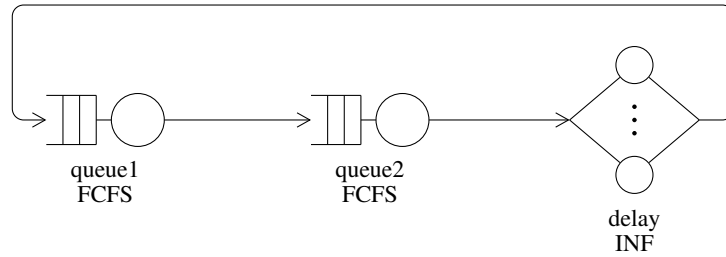


Figure 6.2: Topology of `ctmc_spn_vs_nc_closed_qn`: a closed network over 3 queueing stations.

Nodes	3: Queue $\times$ 2, Delay
Classes	1: jobs (closed, $N = 3$)
Scheduling	queue1: FCFS; queue2: FCFS; delay: INF
Service	queue1 – jobs: Exp(1) queue2 – jobs: Exp(0.8) delay – jobs: Exp(0.5)

Table 6.2: Features of `ctmc_spn_vs_nc_closed_qn`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

6.0.3 `ctmc_tandem_mm1`

The generator of a two-station tandem, its state space and its stationary distribution, printed in full.

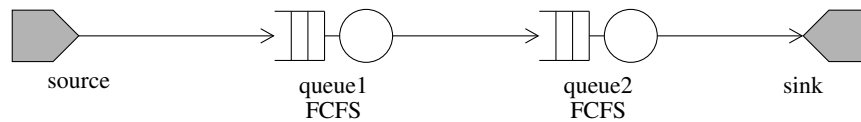


Figure 6.3: Topology of `ctmc_tandem_mm1`: an open network over 2 queueing stations.

The example is built and solved as follows.

```

System.out.println("=====
");
System.out.println("Tandem M/M/1 Queues (Series of Two Stations)");
System.out.println("=====
");

Network model = new Network("tandem_mm1");
Source source = new Source(model, "source");
Queue queue1 = new Queue(model, "queue1", SchedStrategy.FCFS);
Queue queue2 = new Queue(model, "queue2", SchedStrategy.FCFS);
Sink sink = new Sink(model, "sink");
OpenClass jobclass = new OpenClass(model, "jobs");

source.setArrival(jobclass, Exp.fitMean(1 / 0.6));
queue1.setService(jobclass, Exp.fitMean(1.0));
queue2.setService(jobclass, Exp.fitMean(1 / 1.2));

RoutingMatrix R = model.initRoutingMatrix();
R.set(jobclass, jobclass, source, queue1, 1.0);
R.set(jobclass, jobclass, queue1, queue2, 1.0);
R.set(jobclass, jobclass, queue2, sink, 1.0);
model.link(R);

System.out.println("\nSolving tandem M/M/1 system with CTMC...");
System.out.println("Parameters:");

```

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	1: jobs (open)
Scheduling	queue1: FCFS; queue2: FCFS
Arrivals	source – jobs: Exp(0.6)
Service	queue1 – jobs: Exp(1) queue2 – jobs: Exp(1.2)
Solvers	CTMC

Table 6.3: Features of ctmc_tandem_mm1.

```

System.out.println("  Station 1: lambda=0.6, mu=1.0, rho=0.6");
System.out.println("  Station 2: lambda=0.6, mu=1.2, rho=0.5\n");

... (3 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

System.out.println("\n
=====");
System.out.println("CTMC Results Summary:");
System.out.println("=====
");
report(avgTable, "queue1", "Queue 1");
report(avgTable, "queue2", "Queue 2");

```

No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

Two of the three scripts build the same system twice, once as a queueing model and once as a stochastic Petri net, and compare the generators. That is the strongest available check on either notation: the two descriptions are independent, and if they agree state by state then neither has been misunderstood.

Chapter 7

Model interchange

A model can be written to, and read back from, a JSON document. The format is the interchange between the codebases and the wire protocol of the LDES engine, which is driven entirely by it: the client serializes `model.json`, the engine solves it and returns `result.json`.

The examples under the reader and writer tests are round-trip fixtures: each builds a model that uses one group of wire keys, writes it, reads it back and checks that the reconstructed model has the same structure. What a round trip must preserve is the declared state of the model; what it must not carry is the derived state, which is recomputed on load. The JSON examples have no counterpart in the `jline.examples` tree: the Java side exercises the format through its reader and writer tests instead.

```
Network model = JsonModel.heteroServers();
LineModelIO.save(model, "model.json");
Network reloaded = LineModelIO.load("model.json");
```

7.1 The fixtures

7.1.1 json_balking_retrial

Balking thresholds and retrial orbits over the wire.

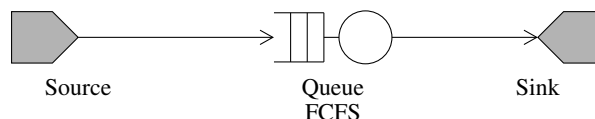


Figure 7.1: Topology of `json_balking_retrial`: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(1); Class2: Exp(0.5)
Service	Queue – Class1: Exp(2); Class2: Exp(3)
Features	finite buffer 15 at Queue; drop rule { 'Class1': 'retrialWithLimit', 'Class2': 'retrialWithLimit' } at Queue

Table 7.1: Features of `json_balking_retrial`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

7.1.2 json_classcap_droprule

Per-class capacities and drop rules.

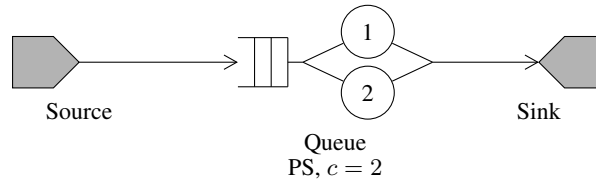


Figure 7.2: Topology of json_classcap_droprule: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue: PS ($c = 2$)
Arrivals	Source – Class1: Exp(1); Class2: Exp(0.5)
Service	Queue – Class1: Exp(3); Class2: Exp(2)
Features	finite buffer 20 at Queue; load-dependent rates at Queue; drop rule {'Class1': 'drop', 'Class2': 'blockingAfterService'} at Queue

Table 7.2: Features of json_classcap_droprule.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

7.1.3 json_fcr_details

The finite capacity region block, with its members, limits and drop rules.

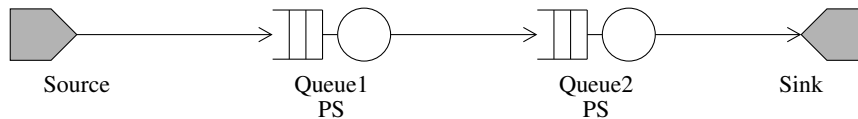


Figure 7.3: Topology of json_fcr_details: an open network over 2 queueing stations.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

7.1.4 json_hetero_servers

Heterogeneous server types at one station, their per-type service rates and the policy that orders them. The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

7.1.5 json_join_deadline

Join deadlines and the quorum they imply.

Nodes	4: Source, Queue $\times$ 2, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Queue1: PS; Queue2: PS
Arrivals	Source – Class1: Exp(1); Class2: Exp(0.5)
Service	Queue1 – Class1: Exp(3); Class2: Exp(2) Queue2 – Class1: Exp(4); Class2: Exp(3)

Table 7.3: Features of json_fcr_details.

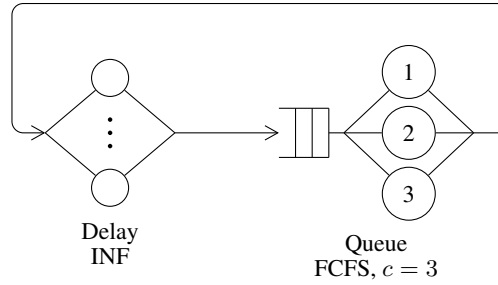


Figure 7.4: Topology of json_hetero_servers: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 3$), Class2 (closed, $N = 2$)
Scheduling	Delay: INF; Queue: FCFS ($c = 3$)
Service	Delay – Class1: Exp(1); Class2: Exp(2) Queue – Class1: Exp(3); Class2: Exp(2)

Table 7.4: Features of json_hetero_servers.

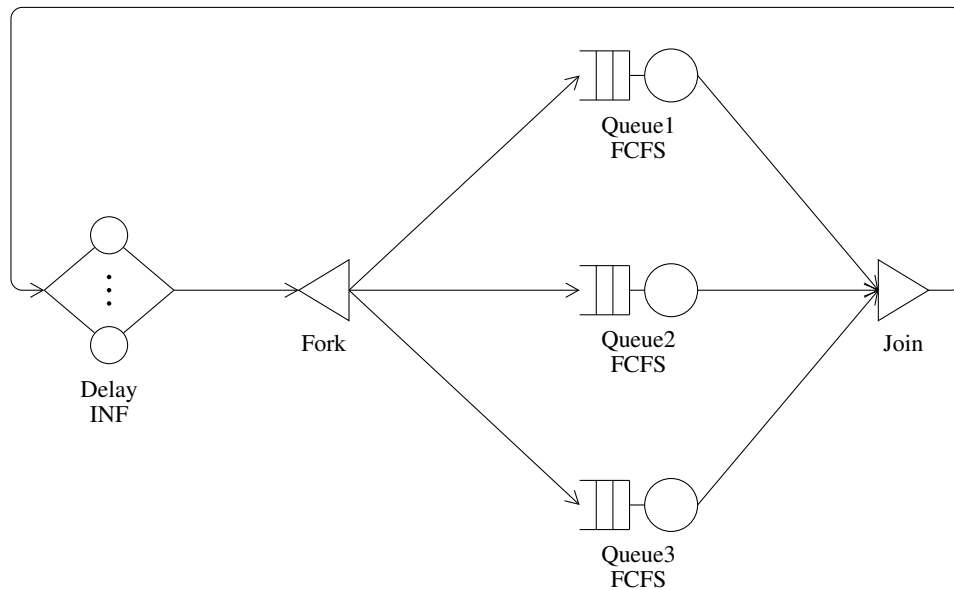


Figure 7.5: Topology of json_join_deadline: a closed network over 4 queueing stations with a fork-join block.

Nodes	6: Queue $\times$ 3, Delay, Fork, Join
Classes	1: Class1 (closed, $N = 5$)
Scheduling	Delay: INF; Queue1: FCFS; Queue2: FCFS; Queue3: FCFS
Service	Delay – Class1: Exp(1) Queue1 – Class1: Exp(3) Queue2 – Class1: Exp(4) Queue3 – Class1: Exp(2)

Table 7.5: Features of json_join_deadline.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

7.1.6 json_signal_classes

G-network signal classes.

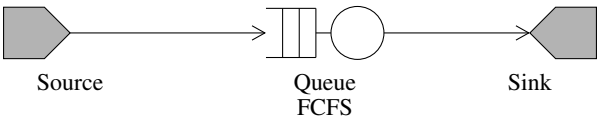


Figure 7.6: Topology of json_signal_classes: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	2: Class1 (open), Signal1 (signal)
Scheduling	Queue: FCFS
Arrivals	Source – Class1: Exp(2); Signal1: Exp(0.5)
Service	Queue – Class1: Exp(5); Signal1: Immediate

Table 7.6: Features of json_signal_classes.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

The fixtures are grouped by wire key rather than by model family on purpose: a key that no fixture exercises is a key that no round trip tests, and grouping them this way makes the gap visible.

Chapter 8

Inference and estimation

Inference runs the modeling process backwards: given measurements of a system – arrival rates, utilizations, response times, or a raw trace – it estimates the model parameters that would produce them. The unknown parameters are marked in the model by a distribution with a NaN parameter, and the estimator fills them in.

The estimators divide into three families: likelihood-based (MLE and its quasi-likelihood variant), recursive (the extended Kalman filter and its smoother, which track a parameter that drifts), and Bayesian (sampling and variational, which return a posterior rather than a point estimate). The changepoint scripts test all three against a workload that changes abruptly, which is where a recursive estimator earns its cost. The reference material is in the Inference chapter of the LINE user manual. Of the inference examples only the variational one has a counterpart in the `jline.examples` tree; the Java side exercises the other estimators through the inference API and its tests.

The scripts live under `jline.examples.java.inference` and the inference API, and each one generates a synthetic dataset from known parameters, so the estimate can be compared against the truth.

```
// The unknown demands are declared as NaN and estimated from the data
queue.setService(jobclass1, new Exp(Double.NaN)); // true demand 0.1
queue.setService(jobclass2, new Exp(Double.NaN)); // true demand 0.3
```

8.1 The estimators

8.1.1 `est_ekf_changepoint`

The filter under a changepoint in the workload.

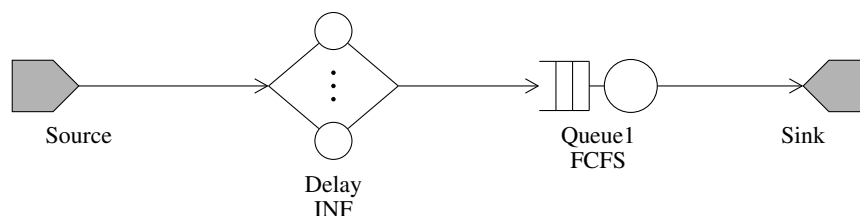


Figure 8.1: Topology of `est_ekf_changepoint`: an open network over 2 queueing stations.

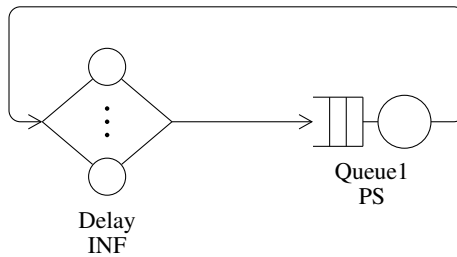
The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

Nodes	4: Source, Queue, Delay, Sink
Classes	1: Class1 (open)
Scheduling	Delay: INF; Queue1: FCFS
Arrivals	Source – Class1: Exp(1)
Service	Delay – Class1: Exp(1) Queue1 – Class1: Exp(3.616)

Table 8.1: Features of `est_ekf_changepoint`.

8.1.2 `est_ekf_closed`

The closed-network filter.

Figure 8.2: Topology of `est_ekf_closed`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 2$)
Scheduling	Delay: INF; Queue1: PS
Service	Delay – Class1: Exp(1) Queue1 – Class1: Exp(2.5)
Solvers	MVA

Table 8.2: Features of `est_ekf_closed`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

8.1.3 `est_ekf_open`

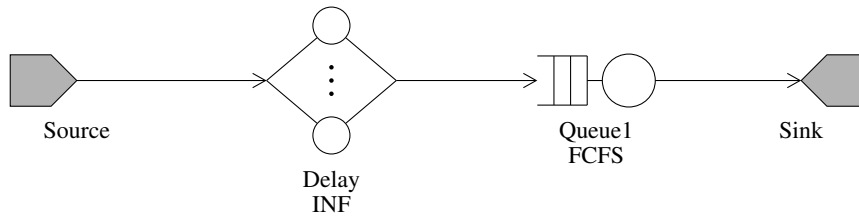
An extended Kalman filter tracking the demands of an open network online.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

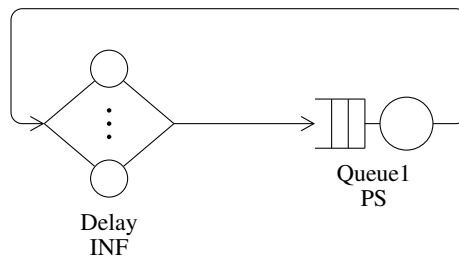
8.1.4 `est_erps_closed`

The closed-network smoother.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

Figure 8.3: Topology of `est_ekf_open`: an open network over 2 queueing stations.

Nodes	4: Source, Queue, Delay, Sink
Classes	1: Class1 (open)
Scheduling	Delay: INF; Queue1: FCFS
Arrivals	Source – Class1: Exp(1)
Service	Delay – Class1: Exp(1) Queue1 – Class1: Exp(4.123)
Solvers	MVA

Table 8.3: Features of `est_ekf_open`.Figure 8.4: Topology of `est_erps_closed`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 1$), Class2 (closed, $N = 3$)
Scheduling	Delay: INF; Queue1: PS
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(10); Class2: Exp(3.333)
Solvers	MVA

Table 8.4: Features of `est_erps_closed`.

8.1.5 est_erps_open

The Rauch-Tung-Striebel smoother on an open network.

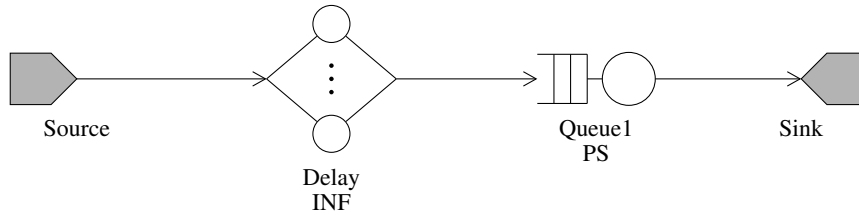


Figure 8.5: Topology of `est_erps_open`: an open network over 2 queueing stations.

Nodes	4: Source, Queue, Delay, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Delay: INF; Queue1: PS
Arrivals	Source – Class1: Exp(0.1); Class2: Exp(0.05)
Service	Delay – Class1: HyperExp(0.5,0.5;3,10); Class2: HyperExp(0.5,0.5;2,8) Queue1 – Class1: Exp(10); Class2: Exp(3.333)
Solvers	MVA

Table 8.5: Features of `est_erps_open`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

8.1.6 est_mcmc_closed

Markov chain Monte Carlo sampling of the posterior over the demands.

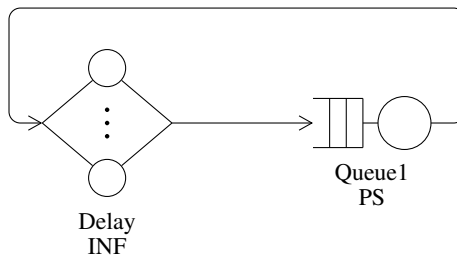


Figure 8.6: Topology of `est_mcmc_closed`: a closed network over 2 queueing stations.

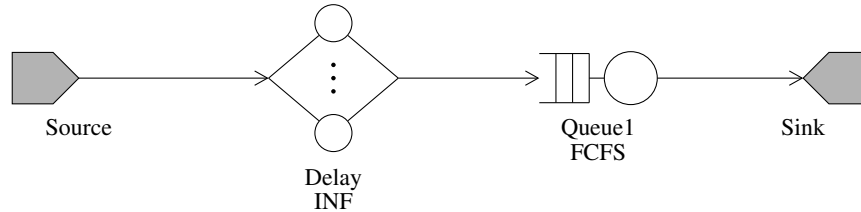
The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

8.1.7 est_mle_open

Maximum likelihood estimation of the service demands of an open network from arrival rate, utilization and response time samples.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 2$), Class2 (closed, $N = 3$)
Scheduling	Delay: INF; Queue1: PS
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(0.5703); Class2: Exp(0.5703)
Solvers	MVA

Table 8.6: Features of `est_mcmc_closed`.Figure 8.7: Topology of `est_mle_open`: an open network over 2 queueing stations.

for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

8.1.8 `est_qmle_closed`

Quasi-likelihood estimation on a closed network.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

8.1.9 `est_trace_closed`

Estimation from a raw arrival and completion trace rather than from aggregate samples.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

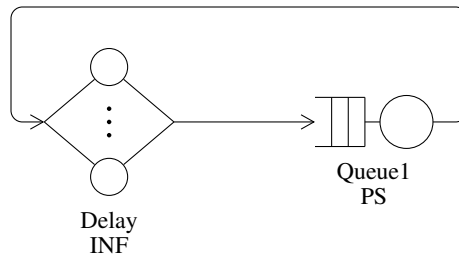
8.1.10 `est_ubo_changepoint`

The variational estimator under a changepoint.

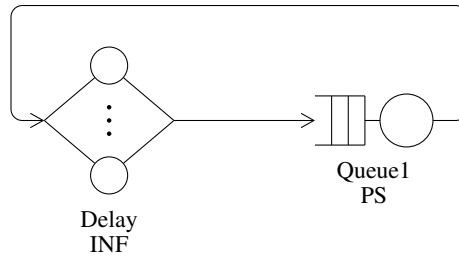
The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

Nodes	4: Source, Queue, Delay, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Delay: INF; Queue1: FCFS
Arrivals	Source – Class1: Exp(1); Class2: Exp(0.5)
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(7.962); Class2: Exp(4.77)
Solvers	MVA

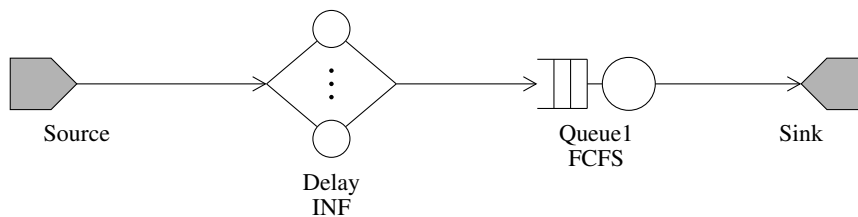
Table 8.7: Features of `est_mle_open`.

Figure 8.8: Topology of `est_qmle_closed`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 2$), Class2 (closed, $N = 3$)
Scheduling	Delay: INF; Queue1: PS
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(9.553); Class2: Exp(4.837)
Solvers	MVA

Table 8.8: Features of `est_qmle_closed`.Figure 8.9: Topology of `est_trace_closed`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 5$)
Scheduling	Delay: INF; Queue1: PS
Service	Delay – Class1: Exp(1) Queue1 – Class1: Exp(5.43)
Solvers	MVA, CTMC

Table 8.9: Features of `est_trace_closed`.Figure 8.10: Topology of `est_ubo_changepoint`: an open network over 2 queueing stations.

Nodes	4: Source, Queue, Delay, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Delay: INF; Queue1: FCFS
Arrivals	Source – Class1: Exp(1); Class2: Exp(0.5)
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(3.338); Class2: Exp(5.012)

Table 8.10: Features of `est_ubo_changepoint`.

8.1.11 `est_ubo_closed`

The closed-network version.

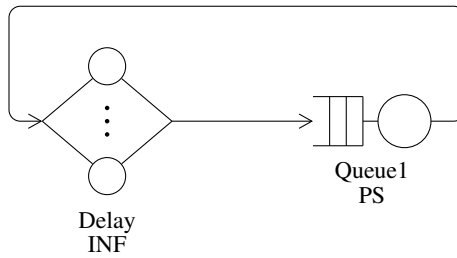


Figure 8.11: Topology of `est_ubo_closed`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 1$), Class2 (closed, $N = 2$)
Scheduling	Delay: INF; Queue1: PS
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(10); Class2: Exp(3.333)
Solvers	MVA

Table 8.11: Features of `est_ubo_closed`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

8.1.12 `est_ubo_open`

Variational Bayesian estimation on an open network.

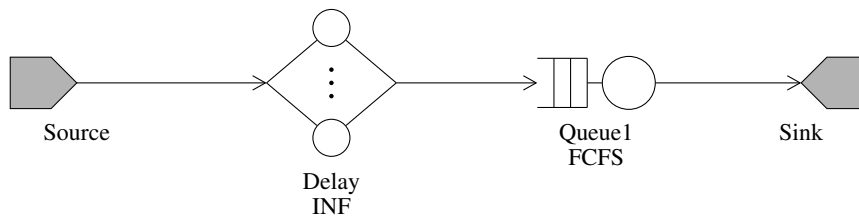


Figure 8.12: Topology of `est_ubo_open`: an open network over 2 queueing stations.

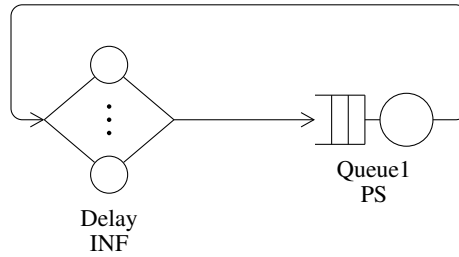
The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

8.1.13 `est_ubo_variants_closed`

The variants of the variational estimator, side by side.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

Nodes	4: Source, Queue, Delay, Sink
Classes	2: Class1 (open), Class2 (open)
Scheduling	Delay: INF; Queue1: FCFS
Arrivals	Source – Class1: Exp(0.1); Class2: Exp(0.05)
Service	Delay – Class1: HyperExp(0.5,0.5;3,10); Class2: HyperExp(0.5,0.5;2,8) Queue1 – Class1: Exp(10); Class2: Exp(3.333)
Solvers	MVA

Table 8.12: Features of `est_ubo_open`.Figure 8.13: Topology of `est_ubo_variants_closed`: a closed network over 2 queueing stations.

8.1.14 `est_ubr_changepoint`

The regression estimator under a changepoint.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

8.1.15 `est_ubr_closed`

The closed-network version.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

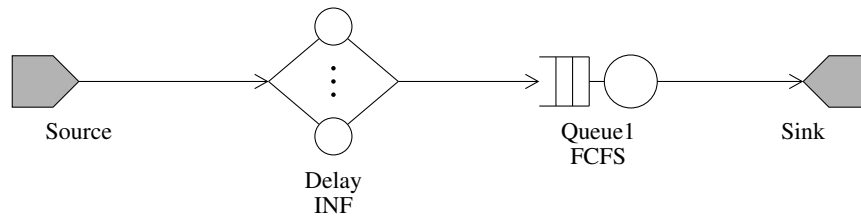
8.1.16 `est_ubr_open`

The regression-based estimator on an open network.

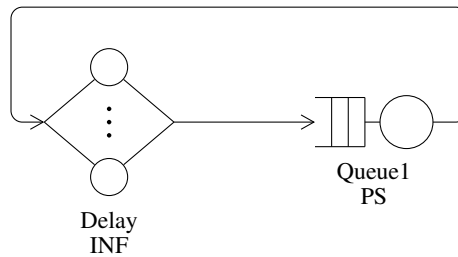
The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 1$), Class2 (closed, $N = 3$)
Scheduling	Delay: INF; Queue1: PS
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(11.21); Class2: Exp(4.547)
Solvers	MVA

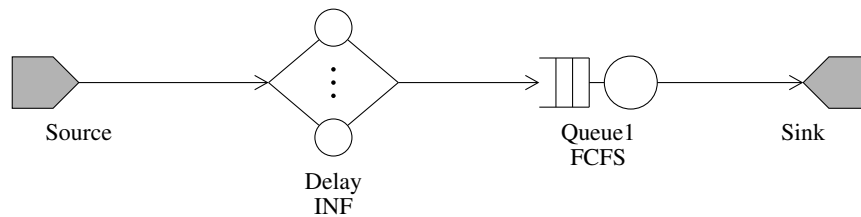
Table 8.13: Features of `est_ubo_variants_closed`.

Figure 8.14: Topology of `est_ubr_changepoint`: an open network over 2 queueing stations.

Nodes	4: Source, Queue, Delay, Sink
Classes	1: Class1 (open)
Scheduling	Delay: INF; Queue1: FCFS
Arrivals	Source – Class1: Exp(1)
Service	Delay – Class1: Exp(1) Queue1 – Class1: Exp(1.665)

Table 8.14: Features of `est_ubr_changepoint`.Figure 8.15: Topology of `est_ubr_closed`: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	2: Class1 (closed, $N = 1$), Class2 (closed, $N = 2$)
Scheduling	Delay: INF; Queue1: PS
Service	Delay – Class1: Exp(1); Class2: Exp(1) Queue1 – Class1: Exp(2.502); Class2: Exp(13.93)
Solvers	MVA

Table 8.15: Features of `est_ubr_closed`.Figure 8.16: Topology of `est_ubr_open`: an open network over 2 queueing stations.

Nodes	4: Source, Queue, Delay, Sink
Classes	1: Class1 (open)
Scheduling	Delay: INF; Queue1: FCFS
Arrivals	Source – Class1: Exp(0.1)
Service	Delay – Class1: HyperExp(0.5,0.5;3,10) Queue1 – Class1: Exp(1.976)
Solvers	MVA

Table 8.16: Features of `est_ubr_open`.

8.1.17 est_vi_closed

Variational inference over the transition counts of a closed loop, from noisy queue-length readings, returning a Gamma posterior per rate rather than a point estimate.

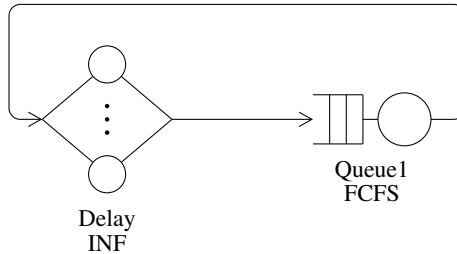


Figure 8.17: Topology of est_vi_closed: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	1: Class1 (closed, $N = 10$)
Scheduling	Delay: INF; Queue1: FCFS
Service	Delay – Class1: Exp(0.5) Queue1 – Class1: Exp(1.994)
Solvers	MVA

Table 8.17: Features of est_vi_closed.

The example is built and solved as follows.

```
int N = 10;

// define model, with the queue rate to be estimated
Network model = new Network("model");
Delay delay = new Delay(model, "Delay");
Queue queue = new Queue(model, "Queue1", SchedStrategy.FCFS);
ClosedClass jobclass = new ClosedClass(model, "Class1", N, delay, 0);
delay.setService(jobclass, new Exp(0.5));
queue.setService(jobclass, new Exp(2.0)); // starting point of the estimate
model.link(model.serialRouting(delay, queue));

// queue-length readings, one per unit time, 10% of them faulty
double[] qlen = {1, 2, 3, 2, 4, 3, 5, 4, 3, 4};
double[] ts = new double[qlen.length];
double[] dlen = new double[qlen.length];
for (int k = 0; k < qlen.length; k++) {
    ts[k] = k + 1;
    dlen[k] = N - qlen[k];
}

// estimate the service rate of the queue
ParamEstimator se = new ParamEstimator(model);
se.options.method = "vi";
se.options.epsilon = 0.1; // probability that a reading is faulty
se.options.priorShape = 2.0; // Gamma prior shape; the rate is set from the model
se.options.variational.ngrid = 51; // time grid of the backward and forward passes

... (14 source lines omitted; full implementation: jar/src/main/java/jline/examples/)

}
System.out.println();

// solve the model the estimate has been written into
System.out.println("SOLVER: MVA");
new SolverMVA(model).getAvgTable().print();
```

Running this edition prints


```

Estimated demand: 0.50152363
posterior service rate ~ Gamma(20.8739, 10.4688), mean 1.9939
evidence lower bound over the iterations: -89.393 -164.046 -163.711 -174.650 -163.234
SOLVER: MVA
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
Station  JobClass    QLen      Util    RespT    ResidT      ArvR      Tput
Delay    Class1      3.9671    3.9671      2         2    1.9835    1.9835
Queue1   Class1      6.0329    0.99479    3.0415    3.0415    1.9835    1.9835

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

Chapter 9

Optimization

Besides evaluating a given model, LINE can solve design problems on queueing networks. While the solvers answer evaluation questions (given a model, what are its performance metrics?), the optimization layer answers design questions: how many servers, what service rates, which routing probabilities, or what job populations optimize a cost or performance objective subject to service-level constraints. The reference material on decision variables, objectives, constraints and solvers is in the Optimization chapter of the LINE user manual; this chapter collects the worked examples.

The relevant scripts are included under the `jline.examples.opt.LineOptExamples` class and each runs in seconds. The general structure of an optimization script consists of four blocks:

1. Definition of the LINE network model
2. Declaration of the decision variables
3. Specification of the objective and constraints
4. Solution

Cost coefficients are supplied as `Map<Station, Double>` instances, and solver options are collected in a `LineOptSolverOptions` object whose setters return the object itself, allowing chained configuration.

9.1 Sizing an M/M/c queue

The M/M/c queue is a classic model of a service center with c parallel servers fed by a common infinite-capacity buffer. In this example, we wish to find the minimum number of servers such that the utilization does not exceed 50%. Jobs arrive at rate $\lambda = 3$ job/s and each server works at rate $\mu = 1$ job/s, hence the utilization with c servers is $\rho = \lambda/(c\mu) = 3/c$. It is known from theory that the optimum is therefore $c^* = 6$; we wish to recover this value. The following script (`LineOptExamples.serverSizing`) solves the problem with the default differential evolution solver, at a cost of 10 units per server:

```
// Block 1: LINE model (Source -> Queue -> Sink)
Network model = new Network("MMc");
Source source = new Source(model, "Arrivals");
Queue queue = new Queue(model, "Server", SchedStrategy.FCFS);
Sink sink = new Sink(model, "Departures");
OpenClass jobs = new OpenClass(model, "Jobs");
source.setArrival(jobs, new Exp(3.0)); // lambda = 3
queue.setService(jobs, new Exp(1.0)); // mu = 1 per server
model.link(Network.serialRouting(source, queue, sink));

// Block 2: decision variables
OptimizationProblem p = new OptimizationProblem(model);
```

```
p.addVariable(new ServerAllocation(queue, 1, 10));

// Block 3: objective and constraints
Map<Station, Double> serverCost = new LinkedHashMap<Station, Double>();
serverCost.put(queue, 10.0);
p.setObjective(new MinimizeCost(serverCost, null, null, null));
p.addConstraint(new UtilizationConstraint(queue, 0.5));

// Block 4: solution
OptimizationResult r = p.solve(new LineOptSolverOptions().setSeed(42));
System.out.println("servers=" + r.getVariableValue("Server_servers")
    + " cost=" + r.objectiveValue + " evals=" + r.modelEvaluations);
```

Obtaining the following output:

```
Optimal servers : 6
Cost           : 60.0
Feasible       : True
LINE evaluations: 10
```

Note that only 10 distinct configurations required a LINE solve: the evaluator caches results at the level of the decoded configuration, so the hundreds of candidate vectors explored by differential evolution collapse onto the 10 possible server counts.

9.2 Exact sizing by bisection

The sizing problem of Tutorial 1 has a special structure: feasibility is monotone in the decision variable, since adding servers can only decrease utilization. In such cases `BisectionSolver` finds the exact optimum in $O(\log n)$ LINE evaluations. Model and problem definition are identical to Tutorial 1 (`LineOptExamples.bisectionSizing`); only the solution block changes:

```
OptimizationResult r = new BisectionSolver(p).solve();
```

Obtaining the following output:

```
Optimal servers : 6
Cost           : 60.0
LINE evaluations: 4 (vs hundreds for DE)
```

9.3 Continuous service rate tuning

We now choose the cheapest service rate, a continuous decision, meeting a response time SLA on an M/M/1 queue. With $\lambda = 3$, the response time is $R = 1/(\mu - 3)$, so $R \leq 0.5$ requires $\mu \geq 5$; the rate is billed at 20 cost units per unit, hence the optimum is $\mu^* = 5$ at cost 100. In the script (`LineOptExamples.serviceRate`), the model is an M/M/1 queue built as in Tutorial 1 with initial rate $\mu = 4$, and the problem is declared as:

```
OptimizationProblem p = new OptimizationProblem(model);
p.addVariable(new ServiceRate(queue, jobs, 3.5, 8.0));
Map<Station, Double> rateCost = new LinkedHashMap<Station, Double>();
rateCost.put(queue, 20.0);
p.setObjective(new MinimizeCost(null, rateCost, null, null));
p.addConstraint(new ResponseTimeConstraint(queue, jobs, 0.5));

OptimizationResult r = p.solve(
    new LineOptSolverOptions().setSeed(42).setMaxIterations(60));
```

Obtaining the following output:

```
Optimal rate    : 5.001 (theory: 5.0)
```

Cost	: 100.0
Feasible	: True

9.4 Load balancing across heterogeneous servers

An arrival stream of rate $\lambda = 2$ must be split between a fast queue ($\mu_1 = 4$) and a slow queue ($\mu_2 = 2.5$) so that the mean end-to-end response time is minimized. Modeling each queue as M/M/1, the mean number of jobs in the system is $N(p) = \frac{\rho_1}{1-\rho_1} + \frac{\rho_2}{1-\rho_2}$ with $\rho_1 = p\lambda/\mu_1$ and $\rho_2 = (1-p)\lambda/\mu_2$, and by Little's law the system response time is $N(p)/\lambda$. Setting $dN/dp = 0$ yields $p^* \approx 0.743$ with response time 0.4264: the optimal policy does not saturate the fast server. The script (`LineOptExamples.loadBalancing`) declares the routing split as the decision variable:

```
OptimizationProblem p = new OptimizationProblem(model);
List<Node> targets = new ArrayList<Node>();
targets.add(fast);
targets.add(slow);
p.addVariable(new RoutingProbabilities(jobs, source, targets));
p.setObjective(new MinimizeSystemResponseTime(jobs));

OptimizationResult r = p.solve(
    new LineOptSolverOptions().setSeed(42).setMaxIterations(60));
double[] probs = (double[]) r.getVariableValue("Jobs_routing_from_S");
```

Obtaining the following output:

Fraction to Fast:	0.736	(theory: 0.743)
Fraction to Slow:	0.264	
System RT	: 0.4248	(theory: 0.4264)

The system response time is computed by Little's law from the chain queue lengths and throughput, which remains correct on parallel topologies.

9.5 Population sizing in a closed network

An interactive system is modeled as a closed network with a Delay station (mean think time 1) and a processor sharing server (rate 2). We seek the largest number of concurrent users the system can sustain while the server response time stays within an SLA of 2 time units. Feasibility is monotone non-increasing in the population, so bisection applies with the `max_feasible` direction (`LineOptExamples.populationSizing`):

```
Network model = new Network("Interactive");
Delay think = new Delay(model, "Think");
Queue server = new Queue(model, "AppServer", SchedStrategy.PS);
ClosedClass users = new ClosedClass(model, "Users", 1, think);
think.setService(users, new Exp(1.0));
server.setService(users, new Exp(2.0));
model.link(Network.serialRouting(think, server));

OptimizationProblem p = new OptimizationProblem(model);
p.addVariable(new JobPopulation(users, 1, 50));
p.setObjective(new MinimizeCost(null, null, null, null));
p.addConstraint(new ResponseTimeConstraint(server, users, 2.0));

OptimizationResult r = new BisectionSolver(p, "max_feasible").solve();
```

Obtaining the following output:

Max users	: 5
Feasible	: True
LINE evaluations:	6

9.6 Robust sizing under workload scenarios

Capacity decisions should often hold under load uncertainty. Here the server pool of Tutorial 1 is sized so that the utilization SLA holds both under the average load ($\lambda = 3$) and under a peak-load scenario ($\lambda = 4.5$). Scenario models share the node and class names of the base model and are registered with `addScenario`; the solver enforces every constraint on all scenarios (`LineOptExamples.robustSizing`):

```
Network base = mmc(3.0); // average load
Network peak = mmc(4.5); // peak load scenario
Queue queue = (Queue) base.getNodes().get(1);

OptimizationProblem p = new OptimizationProblem(base);
p.addVariable(new ServerAllocation(queue, 1, 12));
p.setObjective(new MinimizeCost(serverCost, null, null, null));
p.addConstraint(new UtilizationConstraint(queue, 0.5));
p.addScenario(peak, 1.0);

OptimizationResult r = new BisectionSolver(p).solve();
```

Obtaining the following output:

```
Robust servers : 9 (6 for average load only)
Cost           : 90.0
Feasible       : True
```

The average load alone needs 6 servers; the peak scenario pushes the answer to $\lceil 4.5/0.5 \rceil = 9$.

9.7 Cost-performance tradeoff frontier

How much does tightening the utilization SLA cost? `ParetoSweep` re-solves the sizing problem for a sweep of utilization bounds (epsilon-constraint method) and reports the non-dominated cost frontier, here with each point solved exactly by bisection (`LineOptExamples.paretoFrontier`):

```
double[] eps = {0.3, 0.4, 0.5, 0.6, 0.75, 0.9};
ParetoSweep sweep = new ParetoSweep(p, new ParetoSweep.ConstraintFactory() {
    public Constraint make(double epsilon) {
        return new UtilizationConstraint(queue, epsilon);
    }
}, eps, "bisection");
sweep.solve();
for (ParetoPoint pt : sweep.getFrontier()) {
    System.out.println(" util <= " + pt.epsilon + " -> " + pt.objectiveValue
        + " (" + pt.result.getVariableValue("Server_servers") + ")");
}
```

Obtaining the following output:

```
Utilization bound -> optimal cost (servers)
util <= 0.3 -> 100.0 (10)
util <= 0.4 -> 80.0 (8)
util <= 0.5 -> 60.0 (6)
util <= 0.6 -> 50.0 (5)
util <= 0.75 -> 40.0 (4)
```

The bound 0.9 is absent from the frontier: it requires the same 4 servers as 0.75, so its point is dominated. Frontier points are returned as a list of `ParetoPoint` objects carrying the bound (`epsilon`), the optimal objective (`objectiveValue`) and the underlying `OptimizationResult` (`result`); plotting is left to the caller.

9.8 Decomposing a multi-variable problem

Jointly choosing the number of servers and the service rate couples two variable types. The decomposition workflow splits them into blocks (servers first, then rates), solves the blocks in sequence with the other block's values held fixed, and cycles until the full penalized objective stabilizes, i.e., block coordinate descent (`LineOptExamples.decomposition`):

```
OptimizationProblem p = new OptimizationProblem(model);
p.addVariable(new ServerAllocation(queue, 1, 10));
p.addVariable(new ServiceRate(queue, jobs, 1.0, 4.0));
p.setObjective(new MinimizeCost(serverCost, rateCost, null, null));
p.addConstraint(new ResponseTimeConstraint(queue, jobs, 0.5));

DecompositionWorkflow workflow = p.decompose();
workflow.autoDecompose();
workflow.setSolverOptions(new LineOptSolverOptions().setSeed(42));
WorkflowResult r = workflow.solveSequential(4, 1e-3);
```

Obtaining the following output:

```
Converged      : True after 1 cycle(s)
Servers        : 10
Rate           : 2.045
Objective      : 140.90
```

As with any coordinate descent scheme on a non-convex problem, the workflow converges to a block-wise optimum that may differ from the joint optimum; a joint solve on this instance finds cost 76. Decomposition remains preferable when the joint dimensionality is intractable.

9.9 The models behind the tutorials

The tutorials above are design problems, and each one optimizes a queueing network that the narrative describes but does not draw. This section gives that network for every script of `jline.examples.opt.LineOptExamples`: its topology, its features, and the output the script prints once the optimizer has run.

9.9.1 `opt_bisection_sizing`

The same sizing problem solved exactly by bisection, which is admissible because the constraint is monotone in the number of servers.

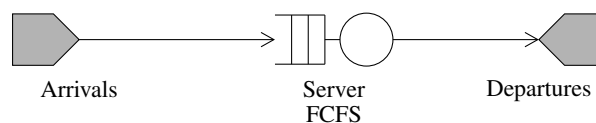


Figure 9.1: Topology of `opt_bisection_sizing`: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Jobs (open)
Scheduling	Server: FCFS
Arrivals	Arrivals – Jobs: Exp(3)
Service	Server – Jobs: Exp(1)
Features	class switching on 3 links
Solvers	MVA

Table 9.1: Features of `opt_bisection_sizing`.

The example is built and solved as follows.

```
Network model = mmc(3.0);
Queue queue = (Queue) model.getNodes().get(1);
OptimizationProblem p = new OptimizationProblem(model);
p.addVariable(new ServerAllocation(queue, 1, 10));
p.setObjective(new MinimizeCost(cost(queue, 10.0), null, null, null));
p.addConstraint(new UtilizationConstraint(queue, 0.5));
OptimizationResult r = new BisectionSolver(p).solve();
System.out.println("[bisection_sizing] servers=" + r.getVariableValue("Server_servers")
    + " cost=" + r.objectiveValue + " probes=" + r.modelEvaluations);
```

Running this edition prints

```
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
[bisection_sizing] servers=6 cost=60.0 probes=4
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

9.9.2 opt_decomposition

Decomposing a multi-variable problem into blocks solved in turn, the coordinate-descent workflow and its blockwise optimum.

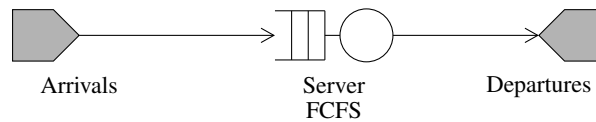


Figure 9.2: Topology of opt_decomposition: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Jobs (open)
Scheduling	Server: FCFS
Arrivals	Arrivals – Jobs: Exp(3)
Service	Server – Jobs: Exp(1)
Features	class switching on 3 links
Solvers	MVA

Table 9.2: Features of opt_decomposition.

The example is built and solved as follows.

```
Network model = mmc(3.0);
Queue queue = (Queue) model.getNodes().get(1);
OpenClass jobs = (OpenClass) model.getClasses().get(0);
OptimizationProblem p = new OptimizationProblem(model);
p.addVariable(new ServerAllocation(queue, 1, 10));
p.addVariable(new ServiceRate(queue, jobs, 1.0, 4.0));
p.setObjective(new MinimizeCost(cost(queue, 10.0), cost(queue, 20.0), null, null));
p.addConstraint(new ResponseTimeConstraint(queue, jobs, 0.5));
DecompositionWorkflow workflow = p.decompose();
```

```

workflow.autoDecompose();
workflow.setSolverOptions(new LineOptSolverOptions().setSeed(42));
WorkflowResult r = workflow.solveSequential(4, 1e-3);
System.out.println("[decomposition] converged=" + r.converged
    + " servers=" + r.getFinalVariableValue("Server_servers")
    + " rate=" + r.getFinalVariableValue("Server_Jobs_rate")
    + " obj=" + r.finalObjective);

```

Running this edition prints

```

MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
WARNING: [getAvg] The model has unstable queues (utilization >= 1); station utilization is reported capped
at 1.0, queue length and response time as Inf.
Successful method execution completed by SolverMVA
WARNING: [getAvg] Message casted more than once, repetitions will not be printed on screen for 60 seconds.

... (1284 captured-output lines omitted; rerun the source example for the full output)

Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
[decomposition] converged=true servers=7 rate=2.000519245058782 obj=110.01038490117564

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

9.9.3 opt_load_balancing

Load balancing across heterogeneous servers: the decision variables are the routing probabilities, which must form a simplex.

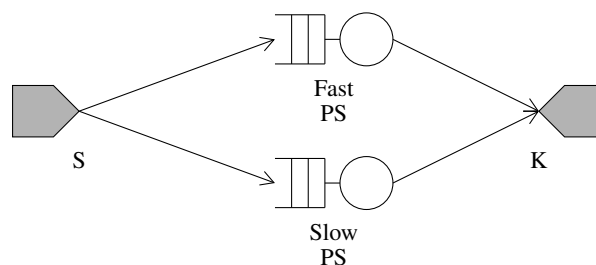


Figure 9.3: Topology of opt_load_balancing: an open network over 2 queueing stations.

The example is built and solved as follows.

```

Network model = new Network("LoadBalance");
Source source = new Source(model, "S");
Queue fast = new Queue(model, "Fast", SchedStrategy.PS);
Queue slow = new Queue(model, "Slow", SchedStrategy.PS);
Sink sink = new Sink(model, "K");
OpenClass jobs = new OpenClass(model, "Jobs");
source.setArrival(jobs, new Exp(2.0));
fast.setService(jobs, new Exp(4.0));

```


Nodes	4: Source, Queue $\times$ 2, Sink
Classes	1: Jobs (open)
Scheduling	Fast: PS; Slow: PS
Arrivals	S – Jobs: Exp(2)
Service	Fast – Jobs: Exp(4) Slow – Jobs: Exp(2.5)
Features	class switching on 4 links
Solvers	MVA

Table 9.3: Features of opt_load_balancing.

```
slow.setService(jobs, new Exp(2.5));
model.addLink(source, fast);
model.addLink(source, slow);
model.addLink(fast, sink);
model.addLink(slow, sink);
OptimizationProblem p = new OptimizationProblem(model);
List<Node> targets = new ArrayList<Node>();
targets.add(fast);
targets.add(slow);
p.addVariable(new RoutingProbabilities(jobs, source, targets));
p.setObjective(new MinimizeSystemResponseTime(jobs));
OptimizationResult r = p.solve(new LineOptSolverOptions().setSeed(42).setMaxIterations(60));
double[] probs = (double[]) r.getVariableValue("Jobs_routing_from_S");
System.out.println("[load_balancing] fast=" + probs[0] + " slow=" + probs[1]);
```

Running this edition prints

```
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 42.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 33.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 46.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 46.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 63.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 85.

... (157 captured-output lines omitted; rerun the source example for the full output)

Successful method execution completed by SolverMVA
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 34.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 41.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
[load_balancing] fast=0.7362332547246861 slow=0.26376674527531385
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

9.9.4 opt_lqn_host_demand

Host demand tuning on a layered model, with the gradient optimizer and layer freezing.

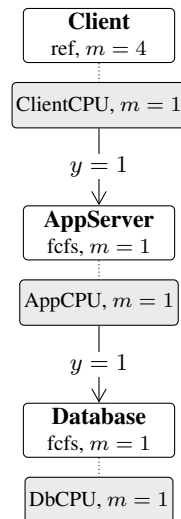


Figure 9.4: Call graph of `opt_lqn_host_demand`: 3 tasks over 3 processors, drawn with the reference task on top and a called task below its caller; the shaded box under each task is the processor it runs on.

Processors	3: ClientCPU (inf, $m = 1$), AppCPU (ps, $m = 1$), DbCPU (ps, $m = 1$)
Tasks	3: Client (ref, $m = 4$), AppServer (fcfs, $m = 1$), Database (fcfs, $m = 1$)
Entries	3: Browse, Render, Query
Activities	3, host demands BrowseAct: 0.5, RenderAct: 0.25, QueryAct: 0.2
Calls	2 (2 synchronous, 0 asynchronous)
Think times	Client: 1.0

Table 9.4: Features of `opt_lqn_host_demand`.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

9.9.5 opt_pareto_frontier

The cost-performance tradeoff frontier, swept rather than solved at one weighting, through ParetoSweep.

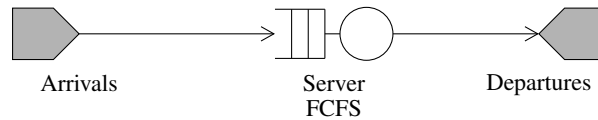


Figure 9.5: Topology of opt_pareto_frontier: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Jobs (open)
Scheduling	Server: FCFS
Arrivals	Arrivals – Jobs: Exp(3)
Service	Server – Jobs: Exp(1)
Features	class switching on 3 links
Solvers	MVA

Table 9.5: Features of opt_pareto_frontier.

The example is built and solved as follows.

```
Network model = mmc(3.0);
final Queue queue = (Queue) model.getNodes().get(1);
OptimizationProblem p = new OptimizationProblem(model);
p.addVariable(new ServerAllocation(queue, 1, 10));
p.setObjective(new MinimizeCost(cost(queue, 10.0), null, null, null));
double[] eps = {0.3, 0.4, 0.5, 0.6, 0.75, 0.9};
ParetoSweep sweep = new ParetoSweep(p, new ParetoSweep.ConstraintFactory() {
    public Constraint make(double epsilon) {
        return new UtilizationConstraint(queue, epsilon);
    }
}, eps, "bisection");
sweep.solve();
System.out.println("[pareto_frontier]");
for (ParetoPoint pt : sweep.getFrontier()) {
    System.out.println("    util <= " + pt.epsilon + " -> " + pt.objectiveValue
        + " (" + pt.result.getVariableValue("Server_servers") + ")");
}
```

Running this edition prints

```
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
```

```
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.

... (44 captured-output lines omitted; rerun the source example for the full output)

Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
[pareto_frontier]
  util <= 0.3 -> 100.0 (10)
  util <= 0.4 -> 80.0 (8)
  util <= 0.5 -> 60.0 (6)
  util <= 0.6 -> 50.0 (5)
  util <= 0.75 -> 40.0 (4)
```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

9.9.6 opt_population_sizing

Population sizing in a closed network: how many jobs to admit so that throughput is maximized without breaching a residence-time bound.

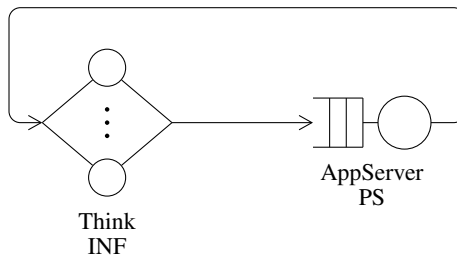


Figure 9.6: Topology of opt_population_sizing: a closed network over 2 queueing stations.

Nodes	2: Queue, Delay
Classes	1: Users (closed, $N = 1$)
Scheduling	Think: INF; AppServer: PS
Service	Think – Users: Exp(1) AppServer – Users: Exp(2)
Features	class switching on 2 links
Solvers	MVA

Table 9.6: Features of opt_population_sizing.

The example is built and solved as follows.

```
Network model = new Network("Interactive");
Delay think = new Delay(model, "Think");
Queue server = new Queue(model, "AppServer", SchedStrategy.PS);
ClosedClass users = new ClosedClass(model, "Users", 1, think);
think.setService(users, new Exp(1.0));
server.setService(users, new Exp(2.0));
model.link(Network.serialRouting(think, server));
OptimizationProblem p = new OptimizationProblem(model);
p.addVariable(new JobPopulation(users, 1, 50));
p.setObjective(new MinimizeCost(null, null, null, null));
p.addConstraint(new ResponseTimeConstraint(server, users, 2.0));
OptimizationResult r = new BisectionSolver(p, "max_feasible").solve();
System.out.println("[population_sizing] users=" + r.getVariableValue("Users_population")
    + " evals=" + r.modelEvaluations);
```

Running this edition prints

```
MVA analysis [method: default/egflin; type: approximate, deterministic; lang: java; env: python] completed
in <t>s. Iterations: 21.
Successful method execution completed by SolverMVA
```

```

Successful method execution completed by SolverMVA
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: exact; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/exact; type: exact, deterministic; lang: java; env: python] completed in <t>
s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
[population_sizing] users=5 evals=6

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

9.9.7 opt_robust_sizing

Robust sizing, in which the constraint must hold across a set of workload scenarios rather than at one nominal load.

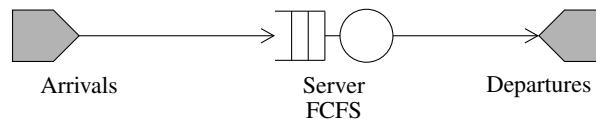


Figure 9.7: Topology of opt_robust_sizing: an open network over 1 queueing station.

Nodes	3: Source, Queue, Sink
Classes	1: Jobs (open)
Scheduling	Server: FCFS
Arrivals	Arrivals – Jobs: Exp(3)
Service	Server – Jobs: Exp(1)
Features	class switching on 3 links
Solvers	MVA

Table 9.7: Features of opt_robust_sizing.

The example is built and solved as follows.

```

Network base = mmc(3.0);
Network peak = mmc(4.5);
Queue queue = (Queue) base.getNodes().get(1);
OptimizationProblem p = new OptimizationProblem(base);
p.addVariable(new ServerAllocation(queue, 1, 12));
p.setObjective(new MinimizeCost(cost(queue, 10.0), null, null, null));
p.addConstraint(new UtilizationConstraint(queue, 0.5));
p.addScenario(peak, 1.0);
OptimizationResult r = new BisectionSolver(p).solve();
System.out.println("[robust_sizing] servers=" + r.getVariableValue("Server_servers")
    + " cost=" + r.objectiveValue);

```

Running this edition prints

```

MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA

```

```

MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
[robust_sizing] servers=9 cost=90.0

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

9.9.8 opt_sensitivity_report

The sensitivity of the optimum to each parameter, reported alongside the solution.

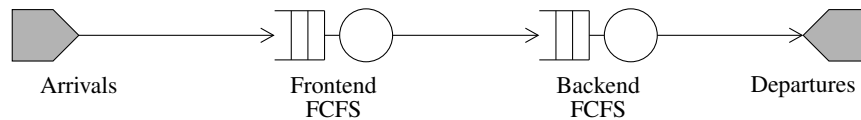


Figure 9.8: Topology of opt_sensitivity_report: an open network over 2 queueing stations.

Nodes	4: Source, Queue × 2, Sink
Classes	1: Jobs (open)
Scheduling	Frontend: FCFS; Backend: FCFS
Arrivals	Arrivals – Jobs: Exp(1)
Service	Frontend – Jobs: Exp(2) Backend – Jobs: Exp(1.5)

Table 9.8: Features of opt_sensitivity_report.

The example is built and solved as follows. This reference example has no source implementation in this edition and is not runnable here. Consult the corresponding language directory in the LINE repository for the current porting status. No execution output has been captured for this language edition. Run the identified implementation to obtain its results.

9.9.9 opt_server_sizing

Sizing an M/M/c queue: choose the number of servers that minimizes cost subject to a response-time bound, by differential evolution over an integer decision variable.

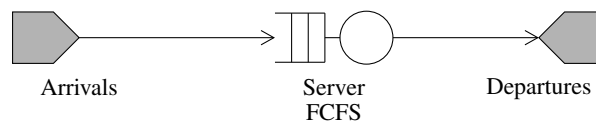


Figure 9.9: Topology of opt_server_sizing: an open network over 1 queueing station.

The example is built and solved as follows.

```

Network model = mmc(3.0);
Queue queue = (Queue) model.getNodes().get(1);

```

Nodes	3: Source, Queue, Sink
Classes	1: Jobs (open)
Scheduling	Server: FCFS
Arrivals	Arrivals – Jobs: Exp(3)
Service	Server – Jobs: Exp(1)
Features	class switching on 3 links
Solvers	MVA

Table 9.9: Features of `opt_server_sizing`.

```

OptimizationProblem p = new OptimizationProblem(model);
p.addVariable(new ServerAllocation(queue, 1, 10));
p.setObjective(new MinimizeCost(cost(queue, 10.0), null, null, null));
p.addConstraint(new UtilizationConstraint(queue, 0.5));
OptimizationResult r = p.solve(new LineOptSolverOptions().setSeed(42));
System.out.println("[server_sizing] servers=" + r.getVariableValue("Server_servers")
    + " cost=" + r.objectiveValue + " evals=" + r.modelEvaluations);

```

Running this edition prints

```

MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
WARNING: [getAvg] The model has unstable queues (utilization >= 1); station utilization is reported capped
    at 1.0, queue length and response time as Inf.
Successful method execution completed by SolverMVA
WARNING: [getAvg] Message casted more than once, repetitions will not be printed on screen for 60 seconds.

... (9 captured-output lines omitted; rerun the source example for the full output)

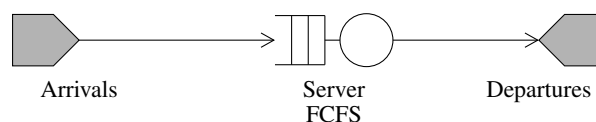
Successful method execution completed by SolverMVA
MVA analysis [method: default/mmk; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
MVA analysis [method: default/mml; type: exact, deterministic; lang: java; env: python] completed in <t>s.
Successful method execution completed by SolverMVA
Successful method execution completed by SolverMVA
[server_sizing] servers=6 cost=60.0 evals=10

```

The columns are the mean queue length, utilization, response time, residence time, arrival rate and throughput of each station-class pair, as returned by MVA.

9.9.10 `opt_service_rate`

Continuous tuning of a service rate, the archetype of a continuous decision variable under a service-level constraint.

Figure 9.10: Topology of `opt_service_rate`: an open network over 1 queueing station.

The example is built and solved as follows.

Index

- ag_closed_network, 213
- ag_gnetwork, 214
- ag_jackson_network, 215
- ag_multiclass_closed, 216
- ag_tandem_open, 218
- ag_tandem_phasetype, 219
- agent models, 213
- bisection, 298
- block coordinate descent, 301
- busy period, 186
- busyp_subnetwork, 186
- Cache node, 97
- cache task, 200
- cache_compare_replc, 97
- cache_itemsize_costcap, 98
- cache_mmap_rr_env, 99
- cache_replc_climb, 100
- cache_replc_fifo, 101
- cache_replc_hlru, 102
- cache_replc_lru, 103
- cache_replc_qlru, 104
- cache_replc_routing, 105
- cache_replc_rr, 107
- cache_rmf_transient, 108
- cachenet_tandem, 109
- caching, 97, 222
- capacity planning, 300
- cdf_respt_closed, 180
- cdf_respt_closed_threeclasses, 181
- cdf_respt_distrib, 182
- cdf_respt_open_twoclasses, 183
- cdf_respt_populations, 185
- cl_basic, 117
- cl_closed, 118
- cl_compare, 119
- cl_multiclass, 120
- cl_scheduling, 120
- cl_stations, 121
- cl_sweep, 121
- class priorities, 43
- class switching, 38
- ClassSwitch node, 38
- closed queueing network, 15
- cluster network, 117
- constraint, 297
- cqn_bas_blocking, 15
- cqn_bcmp_theorem, 15
- cqn_lcfs_multiclass, 17
- cqn_mmp2_service, 17
- cqn_multichain_cs, 18
- cqn_multiserver, 18
- cqn_multiserver_nc, 20
- cqn_online, 20
- cqn_repairmen, 21
- cqn_repairmen_multi, 22
- cqn_scheduling_dps, 24
- cqn_threeclass_hyperl, 25
- cqn_twoclass_erl, 26
- cqn_twoclass_hyperl, 27
- cqn_twoqueues_multi, 28
- cross-codebase parity, 1
- cs_implicit, 39
- cs_multi_diamond, 39
- cs_single_diamond, 40
- cs_transient_class, 41
- CTMC solver, 279
- ctmc_spn_mm1, 279
- ctmc_spn_vs_nc_closed_qn, 279
- ctmc_tandem_mm1, 280
- custom solver, 221
- cyclic polling, 175
- decision variable, 297
- design problem, 297, 301
- differential evolution, 297
- discrete time, 273
- dispatch_closed, 122
- dispatch_mixed, 122
- dispatch_open, 123
- distributions
 - Geometric, 273
- dt_bernoulli_loaddep, 273

[dt_cycle](#), [274](#)
[dt_cycle_loaddep](#), [275](#)
[dt_cycle_multiclass](#), [276](#)
[dt_geogeo1](#), [277](#)
[dt_geogeo1_loss](#), [277](#)
epsilon-constraint method, [300](#)
[est_ekf_changepoint](#), [286](#)
[est_ekf_closed](#), [287](#)
[est_ekf_open](#), [287](#)
[est_erps_closed](#), [287](#)
[est_erps_open](#), [289](#)
[est_mcmc_closed](#), [289](#)
[est_mle_open](#), [289](#)
[est_qmle_closed](#), [290](#)
[est_trace_closed](#), [290](#)
[est_ubo_changepoint](#), [290](#)
[est_ubo_closed](#), [292](#)
[est_ubo_open](#), [292](#)
[est_ubo_variants_closed](#), [292](#)
[est_ubr_changepoint](#), [293](#)
[est_ubr_closed](#), [293](#)
[est_ubr_open](#), [293](#)
[est_vi_closed](#), [295](#)
[example_mapqn2renv](#), [190](#)
Extended Kalman Filter, [286](#)

[fcr_constraints](#), [150](#)
[fcr_lincon](#), [151](#)
[fcr_lossn](#), [153](#)
[fcr_mmlkdrop](#), [154](#)
[fcr_mmlwaitq](#), [155](#)
[fcr_oqndrop](#), [156](#)
[fcr_oqnwaitq](#), [157](#)
feature table, [1](#)
[fes_aggregation](#), [139](#)
[fes_single_class](#), [140](#)
file formats
 JSON, [282](#)
finite capacity region, [150](#), [222](#)
[fj_asymm](#), [49](#)
[fj_basic_closed](#), [50](#)
[fj_basic_nesting](#), [51](#)
[fj_basic_open](#), [52](#)
[fj_complex_serial](#), [54](#)
[fj_cs_multi_visits](#), [55](#)
[fj_cs_postfork](#), [56](#)
[fj_cs_prefork](#), [57](#)
[fj_deep_nesting](#), [58](#)
[fj_delays](#), [60](#)

[fj_mixed_openclosed](#), [61](#)
[fj_nojoin](#), [63](#)
[fj_route_overlap](#), [64](#)
[fj_serialfjs_closed](#), [66](#)
[fj_serialfjs_open](#), [67](#)
[fj_threebranches](#), [68](#)
[fj_tiny_closed](#), [70](#)
[fj_twoclasses_forked](#), [70](#)
[fj_variable_fanout](#), [72](#)
flow-equivalent server, [139](#)
Fork node, [48](#)
fork-join, [48](#), [222](#)

[gallery_aphm1](#), [223](#)
[gallery_cache_lru](#), [224](#)
[gallery_cache_routing](#), [225](#)
[gallery_coxm1](#), [226](#)
[gallery_cqn](#), [226](#)
[gallery_cqn_multiclass](#), [227](#)
[gallery_detm1](#), [227](#)
[gallery_dml](#), [228](#)
[gallery_erldk](#), [228](#)
[gallery_erlerl1](#), [229](#)
[gallery_erlerl1_reentrant](#), [229](#)
[gallery_erlm1](#), [229](#)
[gallery_erlm1_ps](#), [231](#)
[gallery_erlm1_reentrant](#), [231](#)
[gallery_erlm1ps](#), [232](#)
[gallery_fcr](#), [233](#)
[gallery_fj_closed](#), [233](#)
[gallery_fj_open](#), [234](#)
[gallery_fj_quorum](#), [235](#)
[gallery_gamm1](#), [236](#)
[gallery_hyperl1_feedback](#), [237](#)
[gallery_hyperl1_reentrant](#), [237](#)
[gallery_hyperlk](#), [238](#)
[gallery_hyphyp1_linear](#), [239](#)
[gallery_hyphyp1_reentrant](#), [239](#)
[gallery_hyphyp1_tandem](#), [240](#)
[gallery_hypm1](#), [241](#)
[gallery_hypm1_reentrant](#), [241](#)
[gallery_lqn_basic](#), [242](#)
[gallery_lqn_random](#), [243](#)
[gallery_lqn_workflows](#), [245](#)
[gallery_lukumar_reentrant](#), [246](#)
[gallery_mapm1](#), [246](#)
[gallery_mapmk](#), [247](#)
[gallery_mdk](#), [247](#)
[gallery_merl1](#), [247](#)
[gallery_merl1_linear](#), [249](#)

[gallery_merll_reentrant](#), 249
[gallery_merll_tandem](#), 250
[gallery_merlk](#), 251
[gallery_mhyp1](#), 251
[gallery_mhyp1_linear](#), 252
[gallery_mhyp1_reentrant](#), 252
[gallery_mhyp1_tandem](#), 253
[gallery_mhypk](#), 254
[gallery_mm1](#), 254
[gallery_mm1_feedback](#), 255
[gallery_mm1_linear](#), 255
[gallery_mm1_multiclass](#), 255
[gallery_mm1_prio](#), 257
[gallery_mm1_ps](#), 258
[gallery_mm1_ps_feedback](#), 258
[gallery_mm1_ps_multiclass](#), 259
[gallery_mm1_ps_reentrant](#), 259
[gallery_mm1_reentrant](#), 260
[gallery_mm1_tandem](#), 261
[gallery_mm1_tandem_multiclass](#), 262
[gallery_mm1k](#), 263
[gallery_mmap1](#), 263
[gallery_mmap1_multiclass](#), 263
[gallery_mmapk](#), 264
[gallery_mmk](#), 264
[gallery_mpar1](#), 264
[gallery_multitier](#), 266
[gallery_multitier_storage](#), 267
[gallery_parm1](#), 268
[gallery_qn_random](#), 269
[gallery_renv_breakdown](#), 269
[gallery_repairmen](#), 270
[gallery_replayerm1](#), 270
[gallery_um1](#), 271
[Geo/Geo/1 queue](#), 273
[Geometric distribution](#), 273

[HOL scheduling](#), 43

[infinitesimal generator](#), 279
[init_state_fcfs_exp](#), 165
[init_state_fcfs_nonexp](#), 166
[init_state_ps](#), 167

[Join node](#), 48
[JSON format](#), 282
[json_balking_retrial](#), 282
[json_classcap_droprule](#), 283
[json_fcr_details](#), 283
[json_hetero_servers](#), 283
[json_join_deadline](#), 283

[json_signal_classes](#), 285

[Kendall notation](#), 222

[large-scale models](#), 220
[largescale_tbi](#), 220
[layered queueing network](#), 73, 222
[lcq_async_cache](#), 200
[lcq_singlehost](#), 201
[lcq_threehosts](#), 202
[ld_class_dependence](#), 141
[ld_fes_multiclass](#), 141
[ld_fes_singleclass](#), 143
[ld_global_dependence](#), 144
[ld_joint_dependence](#), 145
[ld_multiserver_fcfs](#), 145
[ld_multiserver_ps](#), 147
[ld_multiserver_ps_twoclasses](#), 148
[ld_whittle_bandwidth](#), 149
[ldes_warmstart](#), 168
[load balancing](#), 299, 301
[load dependence](#), 139
[lqn_basic](#), 73
[lqn_bpmn](#), 74
[lqn_bpmn_trace](#), 76
[lqn_flatph](#), 78
[lqn_fork_open_arrival](#), 79
[lqn_init](#), 80
[lqn_jsq](#), 81
[lqn_moment3](#), 82
[lqn_multi_solvers](#), 82
[lqn_ofbiz](#), 84
[lqn_open_arrival](#), 84
[lqn_paramident](#), 86
[lqn_rrobin](#), 86
[lqn_serial](#), 87
[lqn_server_pools](#), 88
[lqn_setup](#), 89
[lqn_sockshop](#), 90
[lqn_srvnph](#), 92
[lqn_transient](#), 93
[lqn_twotasks](#), 94
[lqn_workflows](#), 95
[Lu-Kumar re-entrant line](#), 222

[mam_transient_mapmap1](#), 2
[maximum likelihood](#), 286
[MCMC](#), 286
[mixed network](#), 29
[model features](#)
 [caching](#), 97, 222

- class priorities, [43](#)
- class switching, [38](#)
- cyclic polling, [175](#)
- finite capacity region, [150](#), [222](#)
- fork-join, [48](#), [222](#)
- load balancing, [299](#), [301](#)
- load dependence, [139](#)
- random environment, [190](#), [222](#)
- reward models, [209](#)
- slotted time, [273](#)
- state-dependent routing, [169](#)
- switchover times, [179](#)
- warm start, [165](#)
- model gallery, [222](#)
- model interchange, [282](#)
- mqn_basic, [29](#)
- mqn_multichain_cs, [30](#)
- mqn_multiserver_fcfs, [32](#)
- mqn_multiserver_ps, [33](#)
- mqn_singleserver_fcfs, [35](#)
- mqn_singleserver_ps, [37](#)
- node types
 - Cache, [97](#)
 - ClassSwitch, [38](#)
 - Fork, [48](#)
 - Join, [48](#)
 - Place, [124](#)
 - Transition, [124](#)
- objective function, [297](#)
- open queueing network, [2](#)
- opt_bisection_sizing, [301](#)
- opt_decomposition, [302](#)
- opt_load_balancing, [303](#)
- opt_lqn_host_demand, [304](#)
- opt_pareto_frontier, [306](#)
- opt_population_sizing, [307](#)
- opt_robust_sizing, [308](#)
- opt_sensitivity_report, [309](#)
- opt_server_sizing, [309](#)
- opt_service_rate, [310](#)
- oqn_basic, [3](#)
- oqn_bmmapt, [5](#)
- oqn_cs_routing, [5](#)
- oqn_fourqueues, [7](#)
- oqn_mapt, [9](#)
- oqn_mapt_service, [9](#)
- oqn_mmapt, [10](#)
- oqn_multichain_cs, [10](#)
- oqn_nhpp, [11](#)
- oqn_oneline, [12](#)
- oqn_trace_driven, [13](#)
- oqn_vsinks, [13](#)
- parameter inference, [286](#)
- Pareto frontier, [300](#), [301](#)
- pas_closed_tandem_fig6, [203](#)
- pas_compatibility_5class, [204](#)
- pas_cyclic_ctmc_vs_bruteforce, [204](#)
- pas_cyclic_normconst_bruteforce, [205](#)
- pas_mmk, [206](#)
- pas_nc_sampling, [207](#)
- pas_saturation, [207](#)
- pas_selfloop, [208](#)
- pass and swap, [203](#)
- Place node, [124](#)
- polling_exhaustive_det, [175](#)
- polling_exhaustive_exp, [176](#)
- polling_gated, [177](#)
- polling_klimited, [178](#)
- prio_hol_closed, [43](#)
- prio_hol_open, [45](#)
- prio_identical, [46](#)
- prio_psprio, [47](#)
- QMLE, [286](#)
- random environment, [190](#), [222](#)
- renv_basic, [191](#)
- renv_container_terminal, [192](#)
- renv_fourstages_repairmen, [193](#)
- renv_genqn, [194](#)
- renv_lqn_twostages, [195](#)
- renv_map_fallback, [195](#)
- renv_node_breakdown, [196](#)
- renv_rotterdam_blending, [197](#)
- renv_threestages_repairmen, [197](#)
- renv_twostages_repairmen, [198](#)
- response time distribution, [180](#)
- retrieval_chain, [110](#)
- retrieval_default, [111](#)
- retrieval_ps, [113](#)
- retrieval_routing, [114](#)
- retrieval_simple, [116](#)
- reward models, [209](#)
- rewardModel_aggregation, [209](#)
- rewardModel_mmlk, [210](#)
- rewardModel_multiclass, [211](#)
- rewardModel_templates, [212](#)

round-trip fixture, [282](#)
routing matrix, [1](#)
runnable entry, [1](#)

scheduling strategies
 HOL, [43](#)
script structure, [1](#)
sdroute_closed, [170](#)
sdroute_jsq, [171](#)
sdroute_krzesinski, [171](#)
sdroute_multibranch, [173](#)
sdroute_open, [173](#)
sdroute_twoclasses_closed, [174](#)
serial routing, [1](#)
server sizing, [297](#), [301](#)
slot, [273](#)
slotted time, [273](#)
snc_delay_quantile, [188](#)
snc_tandem_multiclass, [189](#)
solution methods
 bisection, [298](#)
 block coordinate descent, [301](#)
 differential evolution, [297](#)
 epsilon-constraint method, [300](#)
 Extended Kalman Filter, [286](#)
 flow-equivalent server, [139](#)
 maximum likelihood, [286](#)
 MCMC, [286](#)
 QMLE, [286](#)
 stochastic network calculus, [187](#)
 variational inference, [286](#)
solver_custom, [221](#)
solver_custom_analyzer, [221](#)
solvers
 CTMC, [279](#)
spn_basic_closed, [124](#)
spn_basic_open, [125](#)
spn_closed_fourplaces, [126](#)
spn_closed_twoplaces, [127](#)
spn_colored_gspn, [128](#)
spn_fluid_dae, [128](#)
spn_fourmodes, [129](#)
spn_inhibiting, [130](#)
spn_lpbounds, [131](#)
spn_open_sevenplaces, [132](#)
spn_pareto_service, [133](#)
spn_productform_nc, [134](#)
spn_queueing_place, [134](#)
spn_twomodes, [135](#)
state probability, [158](#)
state-dependent routing, [169](#)
statepr_aggr, [158](#)
statepr_aggr_large, [159](#)
statepr_allprobs_fcfs, [160](#)
statepr_allprobs_ps, [161](#)
statepr_sys_aggr, [162](#)
statepr_sys_aggr_large, [163](#)
statepr_sys_marg, [164](#)
stochastic network calculus, [187](#)
stochastic Petri net, [124](#), [281](#)
switchover times, [179](#)
switchover_basic, [179](#)

test_spn_nrm_open, [136](#)
topology figure, [1](#)
Transition node, [124](#)

variational inference, [286](#)

warm start, [165](#)
wf_aph, [137](#)
wf_branch, [137](#)
wf_complex, [137](#)
wf_erlang, [137](#)
wf_loop, [138](#)
wf_parallel, [138](#)
wf_serial, [138](#)
wire key, [285](#)
worked example, [1](#)
workflow model, [137](#)
workload scenario, [300](#)